

# Forty-eight hours with the Kourovka Notebook: an experiment in agentic mathematical research

Codex gpt-6-astra  
OpenAI

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Independent mathematical review: Claude Opus[1m] (Anthropic), via Claude Code

## Abstract

We report an experiment in mathematical research and independent review by two language models. OpenAI’s GPT-6 Astra, running through Codex, had 48 hours to investigate questions listed as open in the supplied Kourovka Notebook, using GAP, exact programs and mathematical literature. The resulting archive contains 46 entries classified as complete solution candidates, together with partial theorems, rediscoveries and bounded searches. We present explicit counterexamples, structural arguments and finite certificates, with their hypotheses and imported ingredients. We compare eleven entries with contemporary solutions, identifying shared arguments, different constructions and stronger alternatives. Anthropic’s Claude Opus[1m], running through Claude Code, subsequently reviewed all 46 entries, verified a majority in detail, and independently reconstructed many finite examples and calculations. Its two reports and the exchange with Codex confirmed substantial portions of the work, corrected interpretations and attribution, and led to a stronger counterexample with a globally principal fixed divisor. The paper presents both the mathematical portfolio and the research-and-review process, with exact records and verbatim correspondence in the appendices.

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## 1 Introduction

The Kourovka Notebook [65] collects open problems in group theory and neighboring areas, ranging from concrete questions about finite groups to structural and logical problems about infinite ones. We present counterexamples, theorems and algorithms covering 46 problem entries or specified subparts, partial results on nine questions, and rediscoveries and deductions from earlier work, obtained in a 48-hour experiment with an AI research agent. The archive includes finite certificates and computational checks. A second language model subsequently reviewed the mathematical work independently. Version 2 compares eleven entries with contemporary solutions brought to our attention by the Notebook’s editor.

Michael Kinyon and Josef Urban conceived and oversaw the experiment. They gave OpenAI’s GPT-6 Astra, running through Codex, an explicit 48-hour deadline and requested local resource limits of generally 20 CPU cores and 100 GB of memory. With access to GAP, further software and the literature, the agent could choose which questions listed as open in that edition to pursue. The aim was to evaluate current AI and automated reasoning capabilities on open research problems under a shared budget: selecting questions, interpreting hypotheses, developing arguments and checking the resulting claims.

The experiment has two phases. During 10–12 September 2026, Codex selected problems, developed arguments and checked its own work. After manuscript preparation, Anthropic’s

Claude Opus[1m], running through Claude Code, independently reviewed the paper and then responded to the authors’ reply and supporting material. Claude examined all 46 entries on the deadline candidate list, verified a majority in detail, and independently rebuilt many of the finite constructions and calculations. This substantial external review strengthens the evidence for the mathematical results. Its corrections, confirmations and exchanges with Codex form a second part of the experiment, described in Section 3.

## 1.1 Mathematical outcomes

Table 1 highlights selected results; the full inventory follows in Section 2. Four representative arguments appear in Section 5; further proofs, partial results and prior-work deductions follow in Sections 6–8. Each proof states its hypotheses and imported ingredients.

Table 1: Selected mathematical outcomes. The last two rows distinguish a post-review strengthening and partial results.

| Problem         | Result                                                                                                                                            | Section         |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| 21.106          | A parameter-free formula defines two elements generating an infinite subgroup of the residually finite integral Heisenberg group.                 | 5.1             |
| 21.68           | A semi-abelian group of order 2,592 has a non-monomial irreducible representation of degree eight.                                                | 5.2             |
| 20.108          | A centerless group of order 605 has an element of order four in its multiple-holomorph quotient.                                                  | 5.3             |
| 19.56           | A coprime-order inequality on commutators of prime-power inputs forces a finite group to be soluble.                                              | 5.4             |
| 19.61–<br>19.62 | Carpet closure and derived-carpet assertions are reduced to explicit universal integer derivations.                                               | 6.2.7,<br>6.2.6 |
| 14.72           | A smooth affine surface has a smooth fixed divisor but a singular cyclic quotient; the review led to a globally principal fixed divisor.          | 6.2.3           |
| 20.100          | A distinct-powers theorem for torsion-free groups, a finite abelian reduction, and certified cases $n = 4, 5, 6, 7$ ; arbitrary $n$ remains open. | 7.1             |

The number 46 records the problem entries classified as complete solution candidates at the deadline. It counts specified subparts once per entry, and some entries share a construction or a substantial imported theorem. The inventory in Section 2 gives the covered parts and convention-sensitive hypotheses. Subsequent review provides evidence of correctness; novelty is a separate question. The comparison in Appendix N identifies overlapping work on eleven entries, including the previously acknowledged 21.106. Some constructions agree closely; others differ or yield stronger results. We assign no discovery priority from the available chronology, and do not infer that the other 35 entries are new. Known results and deductions retain their prior-work status; the qualitative computability assertion in 16.9 is now explicitly identified as a consequence of older work. The nine partial-result questions treated in Section 7 are 20.100, 20.89/19.83, 21.121(b), 20.124, 17.118, 21.114, 19.108, 18.120 and 6.47. Here 20.89/19.83 is one repeated question, and 21.121(a) is already among the 46 entries. That section also records bounded searches on five other questions.

## 1.2 Motivation and reading routes

The design follows the use of explicit time and resource budgets in ATP competitions and human mathematical contests such as the International Mathematical Olympiad [49]. Urban’s specific inspiration was the MPTPChallenge, designed in 2006, twenty years before this experiment [110]: attempt related problems under a shared budget and reuse discoveries across them. Here the challenge concerns open research problems, with freedom to select

questions, consult sources and choose computational tools. We omit a comprehensive survey of the rapidly growing work on learning-guided proving and LLM-based AI for mathematics.

The same experimenters subsequently ran Anthropic’s Claude Fable 5.1 on the same machine, task, 48-hour budget and requested local resource limits [26]. Appendix M gives a preliminary mathematical review and comparison of the two efforts.

After the inventory, AI and automated reasoning readers can continue to Sections 3–4 and 9. Mathematical readers can select proofs from the inventory or begin with Section 5.

## 2 The mathematical portfolio: scope and index

The table indexes all 46 deadline candidates and their proofs, preserving the covered subparts and original descriptions. The external assessment is described in Section 3; the additional column highlights scope qualifications and shared ingredients. The arguments are grouped by subject in Section 6. The historical entry for 19.62 also groups 19.63 with it; the latter’s square-closedness criterion was already published before the experiment, as explained in Section 6.2.6. This grouping contributes one ledger entry. An em dash in the review column means no additional note here, not that the proof has no imported ingredients. The count records deadline coverage; the independent review does not by itself establish discovery priority. Version 2 adds individual attribution notes for eleven entries, compared in Appendix N. The other entries have not thereby been established as new. For 16.9, qualitative computability is a prior consequence; the displayed construction gives the explicit reflection-length reduction and polynomial operation bound.

| No.    | Parts | Deadline description                                                           | Scope / dependency note                                                                                                 | Sec.  |
|--------|-------|--------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|-------|
| 21.106 | all   | A parameter-free definable two-element set generating an infinite subgroup     | Heisenberg work by Zhang-Li and a different positive formula by Jayadevan acknowledged; priority unassigned.            | 5.1   |
| 21.121 | a     | The p-Jordan infimum need not be attained                                      | Different contemporary construction works for every prime in a locally finite triangular group.                         | 6.2.5 |
| 21.68  | all   | A semi-abelian non-monomial group of order 2592                                | Kida supplies the order-96 input; Rizzoli and Jayadevan give the same extension and character obstruction.              | 5.2   |
| 16.87  | a,b   | Maximal test rank at every finite rank in a nonabelian nonperiodic variety     | —                                                                                                                       | 6.4.1 |
| 10.35  | all   | A torsion-free complex linear group not residually rational in the same degree | —                                                                                                                       | 6.2.1 |
| 16.28  | a     | A closed set whose every power greater than one is nonclosed                   | Our reducible set uses the algebraic closure of $F_5(t)$ ; a different contemporary example also contains the identity. | 6.2.2 |
| 14.22  | all   | Failure of the stated finite isolated-radical presentation                     | Our coefficients are infinitely generated; a different contemporary example has two-generated polycyclic coefficients.  | 6.4.2 |

| No.    | Parts | Deadline description                                                                     | Scope / dependency note                                                                                                                            | Sec.  |
|--------|-------|------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 16.20  | all   | A finite modular nondistributive dominion lattice                                        | Dominions allow the ambient group to lie outside the target quasivariety.                                                                          | 6.4.3 |
| 17.33  | all   | Infinite axiomatic rank of the Klein bottle quasivariety                                 | —                                                                                                                                                  | 6.4.4 |
| 17.101 | all   | Every group representation has a homogeneous extension                                   | HNN-module method shared with contemporary work, which additionally obtains faithfulness.                                                          | 6.5.8 |
| 18.76  | all   | A nonsplit additive division-ring extension                                              | Contemporary characteristic-zero construction is isomorphic up to inversion of the additive kernel.                                                | 6.5.5 |
| 18.92  | a,b   | Non-algebraic and non-modular complete lattices of formations                            | Printed definition: an intersection-closed family with a largest member; joins are internal.                                                       | 6.4.5 |
| 20.108 | a,b   | A centerless group of order 605 with a multiple-holomorph quotient element of order four | Tsang identifies the order-605 group; the explicit normalizer argument is given here.                                                              | 5.3   |
| 20.90  | all   | A two-generator infinite profinite CA-group mapping onto A5                              | The example is CA, hence CN: its nonidentity centralizers are abelian.                                                                             | 6.3.3 |
| 11.116 | all   | Finite subgroup-lattice dimension for the specified Chernikov groups                     | —                                                                                                                                                  | 6.3.9 |
| 14.72  | all   | A smooth affine surface with a smooth fixed divisor and a singular cyclic quotient       | Original: smooth Cartier divisor. Post-review invariant open: globally principal reduced divisor.                                                  | 6.2.3 |
| 21.40  | all   | Virtual nilpotence of rational linear groups with finitely many automorphism orbits      | Root-and-trace proof and quantitative bound shared with contemporary work.                                                                         | 6.2.4 |
| 21.60  | all   | An order-twelve counterexample to the semiperfectness criterion                          | —                                                                                                                                                  | 6.1.8 |
| 4.55   | all   | Non-unique indecomposable projective decompositions over $\mathbb{Z}_-(5)[3.A7]$         | Different rank-88 PSL2(11) example in contemporary work; see the v2 attribution after the proof.                                                   | 6.1.7 |
| 21.107 | all   | A countable resolvable group without an expansive sequence                               | —                                                                                                                                                  | 6.5.4 |
| 15.89  | all   | An infinite four-regular Cayley graph with invertible adjacency on all functions         | Same graph and inverse in contemporary work, with the stronger full punctured point spectrum.                                                      | 6.5.3 |
| 16.9   | all   | An exact cubic-time minimum-palindrome-factorization algorithm in free groups            | Qualitative computability and cancellation recurrence are prior ingredients; explicit reduction and cubic algorithm shared with contemporary work. | 6.5.1 |
| 18.18  | all   | Non-enumerability of both the cofinite finite-group theory and its complement            | —                                                                                                                                                  | 6.5.2 |

| No.   | Parts | Deadline description                                                                                           | Scope / dependency note                                                                       | Sec.   |
|-------|-------|----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|--------|
| 19.56 | all   | Solubility from the restricted coprime-commutator order condition                                              | Imports hyperfocal, minimal-simple-group structure and Baer–Suzuki theorems.                  | 5.4    |
| 20.92 | a     | The printed brace average is pre-Lie for every uniformly sufficiently large prime                              | —                                                                                             | 6.2.9  |
| 9.47  | all   | A height-four scattered compact space excluded from the stated E-spaces                                        | —                                                                                             | 6.5.6  |
| 9.45  | all   | An explicit criterion for orthogonal bases in every dimension                                                  | —                                                                                             | 6.5.7  |
| 10.62 | all   | A three-generated perfect periodic group with one involution class and odd pairwise products                   | Imports Amelio’s geometric construction, including a selected odd relation; shared with 4.75. | 6.3.1  |
| 10.32 | all   | An explicit logarithmic degree bound for symmetric-group power-word universality                               | —                                                                                             | 6.1.6  |
| 15.65 | all   | Nonrationality of the stated classical cyclic and semisimple limits                                            | Imports the Fulman–Neumann–Praeger series; the source-index discrepancy remains explicit.     | 6.2.10 |
| 15.92 | all   | Continuum many infinite alternating-residual quotients of every triangle group $\Delta(2,3,r)$ , $r>6$         | Imports Conder’s cycle and parity data; the diagram determines 11 or 13.                      | 6.3.4  |
| 4.75  | all   | A periodic group with Sylow two-subgroups of order two and noncentral involutions modulo the odd-order radical | Uses the same infinite group and geometric dependency as 10.62.                               | 6.3.2  |
| 9.4   | all   | No finite quasigroup generates the ordinary isotope variety of a nontrivial finite group                       | —                                                                                             | 6.4.6  |
| 12.40 | all   | Unbounded Brauer-degree p-parts with the group-order p-part fixed                                              | Negative answer: no finite-valued bound of the requested form exists.                         | 6.1.5  |
| 13.42 | all   | A free nilpotent group with a $\mathbb{Z}[t]$ -completion that is not locally nilpotent                        | Unrestricted tensor completion; scalar torsion is allowed.                                    | 6.3.6  |
| 14.26 | all   | Closure of the torsion-free nilpotent quasivariety under restricted wreath products with $\mathbb{Z}$          | —                                                                                             | 6.3.7  |
| 15.76 | b     | Innerness of category automorphisms for full varieties of groups of fixed derived length                       | —                                                                                             | 6.4.7  |
| 16.38 | all   | Periodicity of subgroups contained in a product of periodic subgroups in a soluble group                       | —                                                                                             | 6.3.5  |
| 19.93 | all   | Unbounded lower p-central factors in the specified two-generator finite p-groups                               | —                                                                                             | 6.1.3  |
| 20.1  | all   | The insoluble totally two-closed group $J1$ times $C13$ with Fitting subgroup $C13$                            | —                                                                                             | 6.1.4  |

| No.    | Parts | Deadline description                                                               | Scope / dependency note                                                  | Sec.                  |
|--------|-------|------------------------------------------------------------------------------------|--------------------------------------------------------------------------|-----------------------|
| 19.62  | all   | Every derived elementary carpet is closed                                          | Shared carpet inputs; the grouped 19.63 criterion was already published. | <a href="#">6.2.6</a> |
| 19.61  | all   | Every carpet closure is a carpet with unchanged derived carpet                     | Universal integer derivations and their checkers accompany the source.   | <a href="#">6.2.7</a> |
| 21.76  | all   | Irreducible closed noncompletable nets of every order in every odd characteristic  | —                                                                        | <a href="#">6.2.8</a> |
| 11.115 | all   | Proper-power normal closures with equal commutator subgroups                       | —                                                                        | <a href="#">6.3.8</a> |
| 20.122 | a,b,c | Minimum-order intersections outside the Fitting subgroup in a soluble wreath group | —                                                                        | <a href="#">6.1.1</a> |
| 17.39  | all   | Unbounded numbers of system normalizers, with fixed Frattini order three           | —                                                                        | <a href="#">6.1.2</a> |

**Answers retained under prior-work status.** The following entries in Section 8 answer the printed questions but remain outside the historical candidate count. They are shown here to make that classification visible, not to increase 46.

| Problem | Prior-work status                                                                                                                 |
|---------|-----------------------------------------------------------------------------------------------------------------------------------|
| 21.132  | Deduction from graded radical and Golod constructions.                                                                            |
| 21.42   | Deduction of self-similarity from the positive-grading theorem and Mathieu’s lattice criterion.                                   |
| 20.33   | Effective universal-group construction from standard embeddings, with oracle bookkeeping; Mikaelian improves the generator bound. |

### 3 Independent review and mathematical dialogue

The external reviewer was **Anthropic’s Claude Opus[1m]** (`claude-opus-5[1m]`), with **xhigh** reasoning, running through **Claude Code**. The organizers supplied this identification and configuration. Claude reviewed the manuscript as a separate model from a different provider, wrote its own checking programs, and discussed its findings with Codex. We credit Claude with the independent mathematical review, computational reconstructions, source checks and the suggestion that led to the strengthened surface example below.

#### 3.1 What was checked

In the first report, Claude compared all 46 candidate entries with the Notebook and read the proofs, verifying a majority in full detail. It also examined the partial results and prior-work deductions. Much of the finite checking was reconstructed from the mathematical descriptions before the authors’ programs were available to the reviewer. The resulting checks therefore provided evidence beyond rerunning the same implementation. They included finite groups and their characters, permutation and matrix constructions, polynomial identities, word algorithms and source-dependent estimates.

Some of the most substantial confirmations concern the carpet results 19.61–19.62. Claude independently rebuilt Chevalley constants and used its own saturation computation to recover all 104 derived-carpet targets and all 300 square-completion targets. It also

checked the 19 displayed derivations. For the main examples, it reconstructed the order-2,592 non-monomial group and the order-605 multiple-holomorph example in GAP. These confirmations support both the displayed arguments and their finite computational inputs.

After receiving the authors’ reply and accessible source material, Claude conducted a second round. It replayed the documented carpet checks, verified the order-512 inputs used for 21.114, and checked the finite cycle data for 15.92. Its final assessment remained acceptance after revision, with increased confidence in the mathematical content. The first report found no false theorem among the claims it could settle; the follow-up resolved several items that had initially lacked sufficient accessible evidence. The reports give the assessment of each entry.

### 3.2 What changed through discussion

The exchange improved the mathematics as well as its presentation. Three examples illustrate the different roles of review.

**A stronger counterexample: 14.72.** The original surface had a smooth fixed Cartier divisor, but the reduced divisor was not globally principal. Claude identified the distinction and suggested removing a suitable orbit of fibres. Codex worked out the invariant open  $x^2 + 1 \neq 0$ , on which  $f = y - (1 + i)x$  generates the fixed ideal while the quotient singularity survives. Section 6.2.3 proves this strengthening and credits the suggestion. Codex checked the new argument during revision; the exchanged reports concern the original surface.

**An objection corrected: 16.28.** Claude’s first text extraction lost the algebraic-closure bar in  $\overline{\mathbb{F}_5(t)}$ , leading to an objection to the field used in the example. The authors pointed to the original notation, and Claude checked the rendered page and withdrew the objection. The argument did not need repair. The exchange also replaced speculation about proposers’ intentions with explicit convention statements for other entries.

**Evidence made accessible.** Claude confirmed the prime-cycle input to 15.92 once the finite data were available, and identified the missing small input for the fourth ancillary carpet checker. The revision supplies that file and explains the choice of prime. Claude also identified contemporaneous work on 21.106, now acknowledged alongside the proof. These changes improve what readers can check and how they can assess attribution.

The review substantially increases confidence in the arguments examined. Its independence concerns the reviewing model and the separately written checks; it is not a proof of discovery independence, human specialist endorsement or formal verification. One sizeable verification input, the  $n = 7$  certificate for 20.100, remains outside the public repository, so the public logs cannot support a fresh complete replay of that case. We preserve the reports and replies verbatim in Appendices I–L, including withdrawn assertions. Appendix G records the changes.

### 3.3 Contemporary solutions identified during revision

On 18 September, Evgeny Khukhro drew our attention to other AI-assisted work on questions in the paper. The full list overlaps eleven deadline entries. He reported that group theorists had confirmed alternative solutions for 21.68, 16.28(a) and 15.89; many other drafts remained unverified. This information concerns those alternatives, and does not constitute specialist review of our whole portfolio.

For version 2, Codex read the eleven cited papers, compared their constructions and hypotheses with our proofs, checked additional primary references, and carried out targeted

algebraic and GAP checks. Appendix N records the results and the available chronology. This later comparison is separate from Claude’s earlier assessment. In particular, the reported Lean formalizations in the alternative papers have not been built here. We thank Khukhro for the information and acknowledge the authors alongside the affected proofs.

## 4 How the research proceeded

### 4.1 A shared research budget

Kinyon and Urban asked Codex to solve as many previously unsolved Notebook problems as possible in 48 hours, using GAP and any further useful software, while keeping plans, proofs and frequent local commits. The supplied twenty-first edition was updated through September 2026. The agent used GPT-6 Astra with `xhigh` reasoning, GAP 4.16.1, exact Python and C/C++ programs, and internet access to mathematical sources. It worked within requested local limits of generally 20 CPU cores and 100 GB of memory. The research phase used one language-model agent; parallel computational workers handled searches and certificate checks. Claude’s independent review came later.

The shared budget allowed the agent to choose problems opportunistically and to reuse constructions. Written state preserved promising leads, proof drafts, failed approaches and unfinished checks between work sessions. Nearly two hundred local commits record the progression. The original instruction and the exact configuration, counts and accounting appear in Appendices A–B.

### 4.2 From a lead to an argument

The first retained lead was the definability counterexample to 21.106. While developing short symbolic arguments, the agent launched finite screens for other questions. Literature frequently supplied a productive starting point: Kida’s order-96 configuration led to the non-monomial semi-abelian group, and an exception in Tsang’s computational coverage led to the multiple-holomorph example. Other questions shared deeper inputs, notably Amelio’s geometric construction for 10.62 and 4.75. The proofs identify which steps are imported and which deductions the experiment supplied.

Validation changed the direction of the work. In the graph-symmetry search for 21.52, a permutation normalizing an inner-automorphism action was initially treated as evidence for a group automorphism. The agent replaced that weaker test by constructing an actual automorphism and checking its action. A program can compute its specified predicate correctly while that predicate fails to express the mathematical question. This example explains the value of reviewing the specification as well as the calculation.

Source checks also changed the classification of results. The agent’s argument for 17.34 was ultimately covered by an existing strong-amalgamation theorem. A small construction for 13.19 likewise became a rediscovery after an earlier result was found. Those arguments remain in Section 8, with their prior-work status, because they help explain both the mathematics and the course of the experiment.

### 4.3 Long computations and partial theorems

For 20.100, a structural reduction made each fixed distinct-powers case amenable to a finite certificate. Generating the  $n = 7$  certificate was only an intermediate step: checking it took approximately 26 hours. Other work continued during that computation. The resulting fixed cases and torsion-free theorem are presented in Section 7.1; the unrestricted question for arbitrary  $n$  remained open.

Almost Engel elements and  $p$ -Jordan bounds supplied further substantial partials. Some broad finite searches ended with useful coverage but no complete answer. This variety of outcomes is why we retain the proofs and their boundaries, rather than report only a solved-problem total. Appendix C gives the dated sequence, the larger search counts and the distinction between generation, checking and closeout. The research record ends with internal review; the separate model review in Section 3 tests the resulting mathematical work.

## 5 Representative mathematical arguments

The four examples range from an elementary definability construction to a solubility criterion using finite-group structure. Their proofs state the ingredients they import and explain the role of the finite checks. Throughout,  $[x, y] = x^{-1}y^{-1}xy$ .

### 5.1 A finite definable set generating an infinite subgroup

Following Conte and Petschick [28], a formula  $\varphi(x)$  is *concise* in a class of groups if every finite truth set in that class generates a finite subgroup. Petschick's Problem 21.106 asks whether every first-order formula with one free variable is concise in residually finite groups. Here formulas may have alternating quantifiers; restricting to group words or to existential formulas gives different questions.

**Theorem 5.1.** *There is a parameter-free first-order formula whose truth set in the integral Heisenberg group is  $\{z, z^{-1}\}$ , where  $z$  generates its infinite cyclic center. In particular, the formula is not concise in residually finite groups.*

*Proof.* Use the following abbreviations in the language of groups:

$$\begin{aligned} \text{Cent}(z) : & \quad \forall t [z, t] = 1, \\ P(u) : & \quad \forall z (\text{Cent}(z) \implies \exists t [u, t] = z), \\ Q(u, v) : & \quad \forall g \exists h \exists k (g = hk \wedge [h, u] = 1 \wedge [k, v] = 1). \end{aligned}$$

Define

$$\varphi(x) : \quad \exists u \exists v (x = [u, v] \wedge P(u) \wedge P(v) \wedge Q(u, v)).$$

After renaming bound variables and expanding the abbreviations, this is a parameter-free formula with exactly one free variable. All its quantifiers range over group elements.

Write the integral Heisenberg group as

$$H = \mathbb{Z}^3, \quad (a, b, c)(d, e, f) = (a + d, b + e, c + f + ae).$$

Matrix multiplication, or direct calculation, gives

$$(a, b, c)^{-1} = (-a, -b, ab - c), \quad [(a, b, c), (d, e, f)] = (0, 0, ae - db).$$

Thus  $Z(H) = \{(0, 0, c) : c \in \mathbb{Z}\} = \langle z \rangle$  with  $z = (0, 0, 1)$ . Reduction of coordinates modulo  $m$  is a homomorphism to a finite group of order  $m^3$ . Choosing  $m$  larger than the absolute value of a nonzero coordinate separates any nonidentity element, so  $H$  is residually finite.

Let  $\pi : H \rightarrow \mathbb{Z}^2$  be projection onto the first two coordinates. For  $u = (a, b, c)$ , the third coordinates of  $[u, t]$  form the ideal  $a\mathbb{Z} + b\mathbb{Z}$ . Consequently

$$H \models P(u) \iff \gcd(a, b) = 1.$$

If  $(a, b)$  is primitive, the solutions of  $as - br = 0$  in  $(r, s) \in \mathbb{Z}^2$  are precisely the multiples of  $(a, b)$ . For example, choose  $\alpha a + \beta b = 1$  and put  $k = \alpha r + \beta s$ ; the determinant relation gives  $ka = r$  and  $kb = s$ . Hence

$$\pi(C_H(u)) = \mathbb{Z}(a, b),$$

and the centralizer is the full inverse image of this subgroup.

Suppose  $P(u)$  and  $P(v)$  hold, with  $\pi(u) = (a, b)$  and  $\pi(v) = (d, e)$ . Both centralizers contain  $Z(H)$ , so their set product is all of  $H$  exactly when their projected subgroups sum to  $\mathbb{Z}^2$ . Indeed, a factorization after projection can be lifted, and its central error absorbed into the second factor. It follows that

$$H \models Q(u, v) \iff \mathbb{Z}(a, b) + \mathbb{Z}(d, e) = \mathbb{Z}^2 \iff ae - db = \pm 1.$$

Thus any value satisfying  $\varphi$  is  $z$  or  $z^{-1}$ . Conversely, the ordered pairs  $((1, 0, 0), (0, 1, 0))$  and  $((0, 1, 0), (1, 0, 0))$  witness the two values. Therefore  $H_\varphi = \{z, z^{-1}\}$ , but  $\langle H_\varphi \rangle \cong \mathbb{Z}$ .  $\square$

The counterexample is already two-generated, torsion-free, and nilpotent of class two. It is residually a finite  $p$ -group for each prime  $p$ , by reduction modulo powers of  $p$ . The elements used to describe witnesses do not occur as parameters in the formula. No assertion that first-order truth commutes with finite quotients is needed. The existential-formula theorem of Conte and Petschick does not apply to this alternating formula. The experimental controls checked commutator arithmetic and finite interpretations, while the proof above establishes the infinite statement.

Contemporaneous work recorded in a public repository credits Lily Zhang with a counterexample and proof and Evan Li with a Heisenberg simplification; it also supplies a Lean formalisation[115]. Our first local proof commit is dated 10 September 2026, 21:05:23 UTC; the commit adding those Lean files is dated 11 September, 10:54:21 UTC. These are artifact timestamps, not discovery or verified upload times, and do not establish priority. We have not built that Lean development. The retrieval index in Appendix F provides a limited record of source access, not a test of independence.

**Version 2 attribution.** Jayadevan [51] gives a different positive formula with the same two-element truth set in the integral Heisenberg group. Over any commutative ring it defines the central units. This is additional to the Zhang–Li work acknowledged above; see Appendix N for the comparison and chronology.

## 5.2 A semi-abelian group which is not monomial

A finite group is *monomial* if each irreducible complex representation is induced from a one-dimensional representation of a subgroup. In the definition used in Problem 21.68, semi-abelian groups are obtained along a subgroup chain by abelian semidirect extensions and quotients. The construction below uses only split extensions with abelian kernels.

**Theorem 5.2.** *There is a semi-abelian group of order 2,592 with a non-monomial irreducible complex representation of degree eight.*

*Proof.* Let  $E$  be the even-weight subspace of  $\mathbb{F}_2^4$  and let  $A_4$  permute its coordinates. Put  $K = E \rtimes A_4$ , with multiplication  $(v, g)(w, h) = (v + g(w), gh)$ . In  $K$ , set

$$z = (1111, 1), \quad x = (1010, (12)(34)), \quad y = (1001, (13)(24)), \quad r = (0000, (234)).$$

Direct multiplication gives

$$x^2 = y^2 = (xy)^2 = z, \quad z^2 = 1, \quad r^3 = 1, \quad r x r^{-1} = y, \quad r y r^{-1} = xy.$$

The eight elements  $1, z, x, zx, y, zy, xy, zxy$  are distinct: their permutation parts form the Klein four group, with two distinct vectors above each part. Thus  $\langle x, y \rangle$  is quaternion of order eight and  $H = \langle x, y, r \rangle$  has order 24. Its abelianization is  $C_3$ : in an abelian quotient the conjugation relations give  $x = y = xy$ , whence  $x = y = 1$ . In particular,  $H$  has no subgroup of index two.

The group  $H$  has an irreducible complex representation  $\rho$  of degree two. To see this explicitly, represent  $x, y$  by the quaternion units  $i, j$  and  $r$  by  $(-1 - i - j - k)/2$ . The latter has order three and conjugates  $i$  to  $j$  and  $j$  to  $k = ij$ . The usual two-dimensional complex representation of the quaternions is irreducible:  $i$  has the two coordinate eigenlines and  $j$  interchanges them.

Let  $\Omega = K/H$ , a set of four left cosets, and put

$$B = \left\{ b \in \mathbb{F}_3^\Omega : \sum_{\omega \in \Omega} b_\omega = 0 \right\}, \quad G = B \rtimes K.$$

Then  $|G| = 27 \cdot 96 = 2,592$ . The chain

$$1 < C_3 < A_4 < K < G$$

exhibits the required abelian split extensions. Fix  $\omega_0 = H$  and a primitive cube root of unity  $\zeta$ . The character  $\lambda(b) = \zeta^{b_{\omega_0}}$  has stabilizer exactly  $H$  in  $K$ . Indeed, the four coordinate characters on  $B$  are distinct: for distinct coordinates  $u, v$ , choose a third coordinate  $w$  and take  $b_u = 1, b_w = -1$ , with all other coordinates zero. Hence the inertia subgroup of  $\lambda$  in  $G$  is  $I = B \rtimes H$ .

The representation  $\tau(b, h) = \lambda(b)\rho(h)$  of  $I$  is well-defined because  $H$  fixes  $\lambda$ . Let  $V = \text{Ind}_I^G \tau$ . Its restriction to  $B$  has four distinct weights, each of multiplicity two. The group  $G$  permutes their weight spaces transitively, and the stabilizer acts irreducibly on each such space. Every invariant subspace splits into  $B$ -weight spaces, using the averaging projections for the finite abelian group  $B$ . Therefore  $V$  is irreducible and has dimension eight.

Suppose  $V \cong \text{Ind}_L^G \mu$  with  $\mu$  one-dimensional. Then  $[G : L] = 8$ . Since  $B$  is a normal 3-group,  $[B : B \cap L] = [BL : L]$  is a power of three dividing eight; thus  $B \leq L$ . The character  $\mu|_B$  must be one of the four weights of  $V|_B$ . After conjugation, assume it is  $\lambda$ . A linear character is invariant under conjugation by its own domain, so  $L$  stabilizes  $\lambda$ . Thus  $L \leq I$  and

$$[I : L] = [G : L]/[G : I] = 8/4 = 2.$$

Because  $B \leq L$ , this gives an index-two subgroup of  $I/B \cong H$ , a contradiction.  $\square$

Kida's paper [66] provided the small order-96 configuration and posed the monomiality conjecture. The abelian extension and the character obstruction are the proposed additional argument in the experimental archive. The proof does not rely on a SmallGroups identifier. That identifier and separate finite checks are useful for reproduction, not for the logical existence of the example.

**Version 2 attribution.** Rizzoli [92] and Jayadevan [53] give the same order-2,592 construction and index-two obstruction, in separate papers. Kida's starting configuration is shared prior input. We credit both accounts without assigning discovery priority; see Appendix N.

### 5.3 An order-four element in a multiple holomorph quotient

For a group  $G$ , write  $\text{Hol}(G)$  for the normalizer of its left regular action in the permutation group on its underlying set, and  $\text{NHol}(G)$  for the normalizer of  $\text{Hol}(G)$ . Tsang's

Problem 20.108(a,b) asks whether a centerless group, and specifically a finite one, can have  $T(G) = \text{NHol}(G)/\text{Hol}(G)$  which is not elementary abelian of exponent two.

**Theorem 5.3.** *For  $G = (C_{11} \times C_{11}) \rtimes C_5$ , where a generator of  $C_5$  acts by  $\text{diag}(3, 9)$  over  $\mathbb{F}_{11}$ , the group  $G$  is centerless and  $T(G)$  contains an element of order four.*

*Proof.* Put  $\lambda = 3 \in \mathbb{F}_{11}$ , of order five, and represent  $G$  by triples with multiplication

$$(x, y, k)(x', y', k') = (x + \lambda^k x', y + \lambda^{2k} y', k + k'),$$

where  $k, k' \in \mathbb{Z}/5\mathbb{Z}$ . Commuting with all translations forces a central element to have  $k = 0$ ; commuting with  $(0, 0, 1)$  then forces  $x = y = 0$ . Thus  $Z(G) = 1$ .

The unique Sylow 11-subgroup  $V = \mathbb{F}_{11}^2$  is characteristic. An automorphism induces a matrix  $M$  on  $V$  and multiplication by  $u \in (\mathbb{Z}/5\mathbb{Z})^\times$  on  $G/V$ . Compatibility requires

$$M \text{diag}(\lambda, \lambda^2) = \text{diag}(\lambda^u, \lambda^{2u})M.$$

Equality of the eigenvalue sets gives  $\{u, 2u\} = \{1, 2\}$  modulo five, so  $u = 1$  and  $M$  is diagonal. The image of a quotient generator can have any translation part. Conversely, the resulting maps

$$(x, y, k) \mapsto (ax + b(1 - \lambda^k), dy + e(1 - \lambda^{2k}), k), \quad a, d \neq 0,$$

are automorphisms.

A normalizer of the left regular group, after a translation making it fix the identity, is an automorphism. Consequently the full holomorph consists exactly of the permutations

$$h(x, y, k) = (Ax + B + C\lambda^k, Dy + E + F\lambda^{2k}, k + j), \quad (*)$$

with  $A, D \neq 0$  and all remaining parameters arbitrary. Define

$$\theta(x, y, k) = (y, \lambda^{-k}x, 2k), \quad \theta^{-1}(X, Y, K) = (\lambda^{3K}Y, X, 3K).$$

Direct composition gives

$$\begin{aligned} (\theta h \theta^{-1})(X, Y, K) &= (DX + E + F\lambda^K, \\ &\quad \lambda^{-j}AY + \lambda^{-j}C + \lambda^{-j}B\lambda^{2K}, K + 2j). \end{aligned}$$

This again has form (\*). Since the holomorph is finite, the resulting containment is equality, so  $\theta$  normalizes it. Moreover,

$$\theta^2(x, y, k) = (\lambda^{-k}x, \lambda^{-2k}y, -k), \quad \theta^4 = 1.$$

Every holomorph element acts on the final coordinate by a translation, whereas  $\theta^2$  acts by negation. These are different functions on  $\mathbb{Z}/5\mathbb{Z}$ . Therefore  $\theta^2 \notin \text{Hol}(G)$  and the coset of  $\theta$  has order exactly four.  $\square$

Tsang's paper [108] singled out this group among exceptions to its computational coverage. The proof above uses that lead but not the paper's general classification argument. The experimental checks compare the explicit holomorph with GAP's automorphism group and evaluate the normalizer identity separately. Determining all of  $T(G)$  is unnecessary for the stated conclusion.

## 5.4 Solubility from restricted coprime commutators: Problem 19.56

For finite  $G$ , let

$$X(G) = \{[x, y] : |x|, |y| \text{ are powers of distinct primes}\}.$$

Suppose  $|ab| \geq |a||b|$  for  $a, b \in X(G)$  of coprime orders. Then  $G$  is soluble. In fact it suffices to impose the inequality when the outputs  $a, b$  also have prime-power orders. The prime-power restrictions on the *inputs* are the additional issue here, compared with nearby commutator criteria [81, 11].

First, a primary element  $a \in X(G)$  normalizing a subgroup  $R$  of different prime-power order must centralize  $R$ . For  $z \in R$ , the element  $b = [a, z]$  lies in  $R \cap X(G)$ , while  $ab = a^z$ . If  $b \neq 1$ , the hypothesis gives  $|a| = |ab| \geq |a||b| > |a|$ , a contradiction. This adapts the normalizer argument in [81], with the required input orders checked explicitly.

For  $P \in \text{Syl}_p(G)$ , let  $O^p(G)$  be the  $p$ -residual, the intersection of normal subgroups with  $p$ -group quotient. The hyperfocal theorem in the finite-group form of [19, Lemma 2.2] gives

$$P \cap O^p(G) = \langle [x, g] : x \in Q \leq P, g \in N_G(Q), (|g|, p) = 1 \rangle.$$

Decompose  $g$  into its commuting primary components inside  $\langle g \rangle$ . They still normalize  $Q$ , and  $[x, uv] = [x, v][x, u]^v$  expresses each generator as a product of elements of  $P \cap X(G)$ . Conversely, every  $X(G)$ -value vanishes in a  $p$ -group quotient because at least one input does. Hence

$$P \cap O^p(G) = \langle P \cap X(G) \rangle.$$

For perfect  $G$ , its  $p$ -group quotient is trivial, so these primary  $X(G)$ -values generate every Sylow subgroup.

Take a nonsoluble counterexample of least order. The condition passes to subgroups, so every proper subgroup is soluble and  $G = G'$ . Its soluble radical  $R$  is contained in every maximal subgroup: otherwise  $G = RM$  makes  $G/R$  soluble. Thus  $R \leq \Phi(G)$ . The Frattini subgroup of a finite group is nilpotent: the Frattini argument for any of its Sylow subgroups gives  $G = \Phi(G)N_G(P)$ , forcing  $N_G(P) = G$ . Thus  $R = \Phi(G)$ , and  $G/R$  is minimal nonabelian simple.

For a Sylow subgroup  $R_p$  of  $\Phi(G)$ , the normalizer observation makes every primary  $X(G)$ -value of another prime centralize  $R_p$ . The generation result makes its centralizer contain all the corresponding Sylow subgroups of  $G$ . That normal centralizer has  $p$ -power index, and perfection forces it to be  $G$ . It follows that

$$\Phi(G) = Z(G), \quad S = G/Z(G) \text{ is minimal simple.}$$

The order condition has not been passed to a quotient at any point.

We use the standard local consequence of Thompson's minimal-simple classification recorded in [11, Proposition 18]: a finite minimal simple group contains a nontrivial elementary abelian 2-subgroup  $V$  with a nontrivial action by an odd-order subgroup normalizing it. Choose an element of that subgroup acting nontrivially and decompose it into primary components; at least one component  $\bar{y}$ , of odd prime-power order, acts nontrivially. This weaker primary-order choice suffices here. The archive also supplies explicit local matrices for all five minimal-simple families, giving an odd-prime-order choice.

The preimage of  $V$  in  $G$  has central derived subgroup, hence is nilpotent of class at most two. Its unique Sylow 2-subgroup  $P$  maps onto  $V$ . Choose a primary lift  $y$  of  $\bar{y}$  by taking the appropriate primary component of any lift; it normalizes  $P$ . For some  $u \in P$ ,

$$a = [u, y] \in P \cap X(G)$$

has nonidentity image  $\bar{a} \in V$ , an involution.

The Baer–Suzuki theorem, in its usual  $p$ -element form recalled in [43], gives a conjugate  $\bar{a}^g$  such that  $\langle \bar{a}, \bar{a}^g \rangle$  is not a 2-group, since  $O_2(S) = 1$ . Two involutions generate a dihedral group. Their product has order divisible by an odd prime  $r$ , and its appropriate power  $c$  has order  $r$  and is inverted by  $\bar{a}$ .

The preimage of  $\langle c \rangle$  is abelian, as its quotient modulo the central kernel is cyclic. Its unique Sylow  $r$ -subgroup  $R_r$  maps onto  $\langle c \rangle$  and is normalized by  $a$ , in the preimage of the dihedral group. The normalizer observation therefore makes  $a$  centralize  $R_r$ . In the quotient this says that  $\bar{a}$  centralizes  $c$ , contradicting its inversion action on an element of odd order. This proves the theorem.

The hyperfocal theorem, minimal-simple local structure, and Baer–Suzuki theorem are imported dependencies. The normalizer and central-extension strategy has precedents in the cited works; the required replacement is generation and witness construction with coprime primary inputs. Source: `research/19.56-proof.md`.

## 6 Further mathematical arguments

The remaining 42 candidate arguments are organized by subject below. The four representative proofs in Section 5 are not repeated. Each entry states the relevant scope and imported ingredients; Section 2 provides the complete problem index.

### 6.1 Finite groups and their representations

Finite constructions, word maps, modular representations and subgroup structure.

#### 6.1.1 20.122(a)–(c): minimum intersections outside the Fitting subgroup

For nilpotent  $A, B, C \leq G$ , the question defines  $\text{Min}_G(A, B, C)$  using all inclusion-minimal intersections  $A \cap B^x \cap C^y$ , and  $\text{min}_G(A, B, C)$  using those of minimum order. Both can lie outside  $F(G)$ , even for soluble  $G$ . The distinction matters: the experiment’s earlier smaller example addressed only inclusion-minimal intersections. Zenkov’s earlier examples separate the analogous notions for pairs of subgroups [114]. The argument below computes the full triple-intersection family for a different group.

Let  $\Omega = \mathbb{F}_3^2$ ,  $o = 0$ , and let  $H \cong D_8$  be the eight signed coordinate permutations. It is generated by  $d(x, y) = (-x, y)$  and  $s(x, y) = (y, x)$ . Put

$$Q = \mathbb{F}_3^2 \rtimes H, \quad E = C_2^\Omega = \langle z_w : w \in \Omega \rangle, \quad G = E \rtimes Q.$$

In its natural action on  $\Omega \times \mathbb{F}_2$ , a base element flips selected fibers and  $Q$  permutes their indices. This soluble group has order  $512 \cdot 72 = 36864$ . Take

$$Z = \langle z_o \rangle, \quad A = Z \times H, \quad B = C = G_{(o,0)}.$$

The orders of  $A, B, C$  are 16, 2048, 2048, so all are nilpotent. The definition permits  $B = C$ .

The two points in each fiber have the same stabilizer,

$$B_w = \langle z_t : t \neq w \rangle \rtimes Q_w.$$

Transitivity shows that these nine distinct groups are exactly the conjugates of  $B$ , accounting for base-group conjugation as well. For each nonzero  $w$ ,  $H_w$  is an order-two reflection group  $R_L$ , where  $L = \langle w \rangle$ . The four lines give four distinct reflections: sign change of either

coordinate, interchange, and negative interchange. Stabilizers of nonzero vectors on distinct lines intersect trivially. It follows that all 81 ordered pairs of conjugates give precisely:

| $u, v$                            | $A \cap B_u \cap B_v$ | order | pairs |
|-----------------------------------|-----------------------|-------|-------|
| $u = v = o$                       | $H$                   | 8     | 1     |
| exactly one is $o$ , other on $L$ | $R_L$                 | 2     | 16    |
| both nonzero on $L$               | $Z \times R_L$        | 4     | 16    |
| nonzero on different lines        | $Z$                   | 2     | 48    |

Membership in  $B_o$  kills  $Z$ ; all other  $B_w$ 's retain it. The remaining condition is to fix the specified vectors in  $H$ , which proves every table entry. There are ten distinct intersections, none trivial. The five groups  $Z, R_L$  are exactly both the minimum-order and the inclusion-minimal members. The reflections generate  $H$ , so

$$\min_G(A, B, B) = \text{Min}_G(A, B, B) = A.$$

Finally  $F(G) = E$ . First  $O_2(Q) = 1$ : a normal 2-subgroup centralizes the normal translation subgroup  $\mathbb{F}_3^2$ , whose centralizer in  $Q$  is itself. Also  $C_G(E) = E$ , because an affine part centralizing all independent base flips fixes all nine indices, and the action is faithful. For any normal nilpotent subgroup of  $G$ , its characteristic odd Sylow subgroups centralize  $E$ , and hence are trivial. It is therefore a 2-group whose image in  $Q$  is trivial, so lies in  $E$ . Conversely  $E$  is normal abelian. The reflection  $d \in A$  is outside  $E$ , proving failure of all assertions (a)–(c). In fact  $A \cap B_o \cap B_{(0,1)} = \langle d \rangle$  is already a minimum-order intersection with an outside witness.

The argument is elementary and determines the entire conjugacy and intersection family; it is not an extrapolation from a sample. The related affine stabilizer pattern has prior antecedents recorded in `research/20.122-minimum-source-audit.md`. The proof and independent enumeration packet are `research/20.122-minimum-proof.md` and `results/20.122-minimum-packet.json`.

### 6.1.2 17.39: no uniform number of system normalizers

For every odd prime  $p$  and  $d \geq 1$ , there is a supersoluble metabelian group  $G$  of order  $2p^{2d+1}$ , with trivial hypercenter, for which the least number of system normalizers having trivial intersection is exactly  $d+1$ . Its Frattini subgroup has order  $p$ . Fixing  $p = 3$  disproves a uniform bound even at fixed Frattini order.

Write  $V = \mathbb{F}_p^d$ , and define

$$P = V \times V \times \mathbb{F}_p, \quad (u, v, z)(a, b, c) = (u + a, v + b, z + c + u \cdot b).$$

These are upper unitriangular block matrices of block sizes  $1, d, 1$ . The automorphism  $\alpha(u, v, z) = (u, -v, -z)$  has order two. Set  $G = P \rtimes \langle t \rangle$ , with  $x^t = \alpha(x)$ . The commutator formula is

$$[(u, v, z), (a, b, c)] = (0, 0, u \cdot b - a \cdot v),$$

so  $Z(P) = P' = \{(0, 0, z)\} =: Z$ . The element  $t$  inverts  $Z$ , and every element outside  $P$  does too. Thus  $Z(G) = 1$ , and its entire hypercenter is trivial. The subgroup  $\{(0, v, z)\}$  is abelian normal with abelian quotient, making  $G$  metabelian. Coordinate flags in the two copies of  $V$  in  $G/Z$ , preceded by  $Z$ , give a normal series of prime-order cyclic factors, so  $G$  is supersoluble.

A Sylow system consists here of the normal Sylow  $p$ -subgroup  $P$  and a Sylow 2-subgroup. Its system normalizer is a conjugate of

$$D = N_G(\langle t \rangle) = C_G(t) = \{(u, 0, 0) : u \in V\} \times \langle t \rangle.$$

All such conjugates occur, by Sylow conjugacy, and  $G = DP$  allows conjugators in  $P$ . Represent an element of  $G$  as  $(u, v, z, e)$ , meaning  $(u, v, z)t^e$ ,  $e = 0, 1$ . For  $g = (a, b, c)$  and  $q = c - a \cdot b$ , direct multiplication gives

$$\begin{aligned} D^g = D_{b,q} &= \{(u, 0, u \cdot b, 0) : u \in V\} \\ &\cup \{(u, -2b, -2q - u \cdot b, 1) : u \in V\}. \end{aligned}$$

Every pair  $b, q$  occurs, and different pairs give different subgroups. Thus there are exactly  $p^{d+1}$  system normalizers.

For  $k$  arbitrary normalizers  $D_{b_i, q_i}$ , their intersection with  $P$  consists of

$$\{(u, 0, u \cdot b_1, 0) : u \cdot (b_i - b_1) = 0, 2 \leq i \leq k\}.$$

At most  $k - 1$  independent equations are imposed on  $V$ ; for  $k \leq d$  this intersection is nontrivial. Conversely take  $D_{0,0}$  and  $D_{e_i,0}$  for all  $d$  standard basis vectors. Their common part in  $P$  is trivial. Already the first two have no common element outside  $P$ , whose second coordinates would have to be both 0 and  $-2e_1$ . Their full intersection is therefore trivial, proving the exact minimum.

To locate the example relative to the known Frattini-free positive case [6], note that  $\Phi(G) = Z$ . Indeed  $P^p = 1$ , by

$$(u, v, z)^m = (mu, mv, mz + \binom{m}{2}u \cdot v),$$

so  $\Phi(P) = P'P^p = Z$ . If a maximal subgroup  $M$  of  $G$  missed this normal subgroup, then  $G = M\Phi(P)$  and  $P = (P \cap M)\Phi(P)$ , contrary to the nongenerator property of  $\Phi(P)$ . Thus  $Z \leq \Phi(G)$ . On the other hand  $G/Z \cong V \times (V \rtimes C_2)$ , with inversion on the second copy. The coordinate-hyperplane subgroups in either copy, extended by the other factors, and the index-two subgroup  $V \times V$ , are maximal with trivial total intersection. Their preimages give  $\Phi(G) \leq Z$ . The fixed nontrivial Frattini subgroup therefore explains why the prior positive result does not apply. See `research/17.39-proof.md`.

### 6.1.3 19.93: unbounded lower $p$ -central width without a wreath quotient

Write  $P_1(Q) = Q$ ,  $P_{i+1}(Q) = P_i(Q)^p[P_i(Q), Q]$ . For every fixed prime  $p \geq 7$  and integer  $K$ , we construct a finite two-generator  $p$ -group with no quotient  $C_p \wr C_p$  and some factor  $P_i/P_{i+1}$  of order greater than  $p^K$ .

First write  $C_p \wr C_p = \mathbb{F}_p^p \rtimes \langle t \rangle$  and let  $s(v)$  be the sum of the coordinates of  $v$ . The map  $(v, t^a) \mapsto (s(v), a)$  is onto  $\mathbb{F}_p^2$ . The element  $(v, t^a)$  has  $p$ -th power one exactly when  $s(v)a = 0$ : for  $a \neq 0$  its  $p$ -th power has constant vector  $s(v)$ , and for  $a = 0$  it is always one. If  $u, v$  generate the wreath product and  $u^p = v^p = 1$ , their images must lie on different coordinate axes, so  $(uv)^p \neq 1$ . Consequently the three relations

$$x^p = y^p = (xy)^p = 1 \tag{\dagger}$$

exclude this quotient for a group generated by  $x, y$ .

Consider the complete filtered algebra

$$A = \mathbb{F}_p \langle\langle X, Y \rangle\rangle / \overline{(X^p, Y^p, (X + Y + XY)^p)}.$$

Let  $A_n$  be its degree filtration,  $a_n = \dim A_n/A_{n+1}$ , and  $H(t) = \sum a_n t^n$ . All three relations have initial degree  $p$ ; in the last relation the degree- $p$  part  $(X + Y)^p$  is the sum of all distinct length- $p$  words and is nonzero. The complete filtered Golod–Shafarevich inequality [37, Theorem 2.6] states coefficientwise that

$$(1 - 2t + 3t^p) \frac{H(t)}{1 - t} \geq \frac{1}{1 - t}.$$

At  $t = 2/3$  the polynomial factor is at most  $-115/729 < 0$ , so  $H(2/3)$  diverges. The denominator  $1 - t$  is retained: these relations are not homogeneous.

For  $N \geq 2$ , let  $R_N = A/A_N$  and  $M_N = A_1/A_N$ . The finite group  $1 + M_N$  is a  $p$ -group, because  $M_N$  is nilpotent and  $|1 + M_N| = |M_N|$ . Set

$$Q_N = \langle 1 + X, 1 + Y \rangle \leq 1 + M_N.$$

The defining relations give  $(\dagger)$ , excluding the wreath quotient. Reduction to  $A_1/A_2 \cong \mathbb{F}_p^2$  shows that  $Q_N$  requires exactly two generators. The natural map  $\mathbb{F}_p[Q_N] \rightarrow R_N$  is onto: its image contains  $X, Y$ , and every truncated series is a polynomial. For its augmentation ideal  $I_N$ , the map kills  $I_N^N$ ; hence

$$\dim R_N \leq \dim \mathbb{F}_p[Q_N]/I_N^N. \quad (\dagger)$$

Here is an elementary bound on the right side if all lower  $p$ -central widths were bounded. For any finite  $p$ -group  $Q$ , put  $b_i = \dim_{\mathbb{F}_p} P_i/P_{i+1}$ , and choose basis lifts  $g_{ij}$ , ordered by increasing  $i$ . The series terminates, and successive quotient normal forms give unique products  $\prod g_{ij}^{e_{ij}}$ ,  $0 \leq e_{ij} < p$ . Expanding in this fixed order shows that  $\prod (g_{ij} - 1)^{e_{ij}}$  span  $\mathbb{F}_p[Q]$ . For the augmentation ideal  $I$ ,

$$g - 1 \in I^i \quad (g \in P_i).$$

This follows inductively from  $h^p - 1 = (h - 1)^p$  and

$$[h, g] - 1 = h^{-1}g^{-1}((h - 1)(g - 1) - (g - 1)(h - 1)).$$

Thus only products with  $\sum i e_{ij} < N$  are needed modulo  $I^N$ . If every  $b_i \leq K$ , and  $c_n$  are the coefficients of

$$P_K(t) = \prod_{i \geq 1} (1 + t^i + \dots + t^{(p-1)i})^K,$$

then  $\dim \mathbb{F}_p[Q]/I^N \leq \sum_{n < N} c_n$ . This product converges for  $0 < t < 1$ , being bounded by  $\prod_{i \geq 1} (1 - t^i)^{-K}$ ; the logarithms are bounded by  $K \sum_i t^i / (1 - t)$ .

If every  $Q_N$  had every  $b_i \leq K$ ,  $(\dagger)$  would give  $\sum_{n < N} a_n \leq \sum_{n < N} c_n$  for all  $N$ . In particular  $a_n \leq \sum_{j \leq n} c_j$ , and therefore

$$H(t) \leq \frac{P_K(t)}{1 - t} < \infty \quad (0 < t < 1),$$

contradicting divergence at  $2/3$ . This proves the claimed unboundedness, already for  $(p, d) = (7, 2)$ . The finite groups are obtainable from algebra truncations, but the proof gives no practical size bound. It excludes wreath quotients of  $Q_N$  itself, exactly as asked, not quotients of every subgroup. See `research/19.93-proof.md`.

#### 6.1.4 20.1: a totally 2-closed insoluble group with nontrivial Fitting subgroup

The example is  $J_1 \times C_{13}$ . The imported input is that  $J_1$  is totally 2-closed [4, Theorem 1.1 and Proposition 4.8]. We give the product argument for arbitrary faithful actions, including intransitive ones.

For  $X \leq \text{Sym}(\Omega)$ , its 2-closure consists exactly of permutations  $s$  such that each ordered pair  $(\alpha, \beta)$  has an  $x \in X$  agreeing with  $s$  at both points. In particular  $s$  preserves every point orbit. For a finite group, total closure for finite faithful sets implies the same for arbitrary sets: choose a finite invariant faithful subset  $\Omega_0$ , using one moved point for each nonidentity group element. The restriction of  $s$  agrees with a unique group element there. Applying finite closure to  $\Omega_0 \cup \alpha^X$  for every  $\alpha$  forces the same element everywhere.

**Lemma 6.1.** *If finite groups  $A, B$  are totally 2-closed and have coprime orders, then  $A \times B$  is totally 2-closed.*

*Proof.* In any faithful action,  $\alpha^A \cap \alpha^B = \{\alpha\}$ . Indeed,  $\alpha^a = \alpha^b$  makes  $ab^{-1}$  fix  $\alpha$ . Since  $a, b$  commute and have coprime orders, the Chinese remainder theorem expresses  $a$  as a power of  $ab^{-1}$ , so  $a$  fixes  $\alpha$ .

Let  $\Delta_B$  be the set of  $B$ -orbits. The induced  $A$ -action is faithful: an element fixing every such orbit moves each  $\alpha$  within  $\alpha^A \cap \alpha^B$ , so fixes every point. Similarly  $B$  acts faithfully on  $\Delta_A$ . If  $s$  is in the 2-closure of  $A \times B$ , it preserves both orbit equivalence relations, since each is a union of ordered-pair orbits. For two  $B$ -orbits, choose representatives and apply the pair criterion; the witnessing element  $ab$  induces the  $A$ -element  $a$  on the pair of blocks. Thus  $s$  induces an element of the 2-closure of  $A$  on  $\Delta_B$ , and total closure gives one  $a_0 \in A$  for all blocks. Similarly one  $b_0 \in B$  gives its action on  $\Delta_A$ . The permutation  $s(a_0b_0)^{-1}$  fixes every orbit of both factors setwise, hence every singleton intersection, and is the identity.  $\square$

A prime cyclic group is totally 2-closed: in a faithful action choose a point with trivial stabilizer; pairing it with every other point forces one common group element to realize a 2-closure permutation. Now

$$|J_1| = 175560 = 2^3 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 19$$

is coprime to 13, so the lemma proves the claim about closure. The group is insoluble because it has the simple quotient  $J_1$ . Every normal nilpotent subgroup projects trivially to  $J_1$ , whereas the central  $C_{13}$  is nilpotent. Thus

$$F(J_1 \times C_{13}) = C_{13} \neq 1.$$

The supplied question imposes no direct-indecomposability restriction. Prior related results include the nilpotent product case of Chen–Ren and coprime transitive-action decompositions [24, 1]; the lemma above handles all faithful actions of the stated factors. The  $J_1$  theorem retains its classification and computational dependencies. See `research/20.1-proof.md`.

### 6.1.5 12.40: unbounded Brauer-degree parts at fixed Sylow order

**Proposition 6.2.** *For every  $k \geq 3$ , some odd positive integer  $a$  gives an irreducible 2-Brauer character  $\varphi$  of  $G = \text{PSL}_6(5^a)$  with*

$$|G|_2 = 8192, \quad \varphi(1)_2 = 2^k.$$

*Thus no finite-valued bound  $\varphi(1)_p \leq f(|G|_p)$  exists, already for the fixed prime  $p = 2$ .*

*Proof.* Bray’s projective-point module [16, Main Theorem] has an absolutely irreducible heart in characteristic two of dimension

$$F(q) = \frac{q^6 - 1}{q - 1} - 2 = q^5 + q^4 + q^3 + q^2 + q - 1.$$

Here  $q = 5^a$ . To recall why the assertion applies, let  $P$  be the projective permutation module over an algebraically closed field of characteristic two,  $U$  its constant line, and  $W$  its augmentation kernel. Its even permutation degree puts  $U \subseteq W$ . Perfectness of  $\mathrm{SL}_6(q)$  excludes a trivial submodule of  $W/U$ : its preimage would be an extension of two trivial modules, which splits, contradicting the one-dimensional invariant space of  $P$ . The permutation form makes  $W/U$  self-dual, so there is no trivial quotient either. Each nontrivial simple factor has dimension at least  $q^5 - 1$ : on the elementary abelian unipotent subgroup  $\mathbb{F}_q^5$  it has a nontrivial weight, and the Levi subgroup  $\mathrm{SL}_5(q)$  permutes all nontrivial weights transitively. The unipotent subgroup normally generates  $\mathrm{SL}_6(q)$ , so it cannot act trivially on a nontrivial simple factor. Since  $2(q^5 - 1) > F(q)$ , there is only one nontrivial factor. Absence of trivial submodules and quotients now makes the heart simple. Scalars fix projective points, so it descends to  $\mathrm{PSL}_6(q)$ .

For odd  $a$ ,  $q \equiv 5 \pmod{8}$  and  $q \equiv -1 \pmod{3}$ ; hence  $\gcd(6, q - 1) = 2$ . The order formula is

$$|G| = \frac{q^{15}}{2} \prod_{i=2}^6 (q^i - 1).$$

The five factors have 2-adic valuations 3, 2, 4, 2, 3. Their sum, less the central factor of two, is 13, independently of  $a$ .

It remains to obtain arbitrarily prescribed degree valuations at actual prime powers. Repeated squaring gives  $v_2(5^{2^j} - 1) = j + 2$ , so for  $k \geq 3$ ,

$$5^{2^{k-2}} \equiv 1 + 2^k \pmod{2^{k+1}}.$$

For odd  $x$ ,  $F'(x)$  is odd, and the binomial theorem gives

$$F(x + 2^k) \equiv F(x) + 2^k \pmod{2^{k+1}}.$$

Therefore, whenever odd  $a$  satisfies  $2^k \mid F(5^a)$ , exactly one of  $a$  and  $a + 2^{k-2}$  gives divisibility by  $2^{k+1}$ ; the other gives valuation exactly  $k$ . Start with  $a = 1$ , since  $F(5) = 3904$  is divisible by eight, and continue using the divisible choice. The unused choice at each step supplies the required odd exponent and exact valuation.  $\square$

The representation family and its irreducibility are prior mathematics; the fixed group-order valuation and unbounded degree valuation are the deduction proposed here. No assertion about ordinary character degrees is being made. See `research/12.40-proof.md`; finite module and integer controls corroborate the formulas without replacing the arbitrary- $k$  argument.

### 6.1.6 10.32: a logarithmic bound for symmetric-group power words

**Proposition 6.3.** *Let  $r, s \neq 0$ , with at least one odd, and  $m = \prod_{\ell \mid rs} \ell$ , the product of the distinct prime divisors of  $|rs|$ . The word  $x^r y^s$  is universal on  $S_n$  whenever*

$$n \geq \max\{1126, 16 \log m\}.$$

*More generally, every real  $c > 8$  admits  $N_0(c)$  for which  $n \geq \max\{N_0(c), c \log m\}$  suffices.*

*Proof.* We use two permutation-covering results. First, if two classes in  $S_d$  have  $h, k$  cycles, counting fixed points, and

$$h + k + 1 \leq L \leq d, \quad L \equiv h + k - 1 \pmod{2},$$

some representatives have product an  $L$ -cycle. This is Boccara's positive-genus three-partition result in the form reproved by Song–Xu–Ye [102, Propositions 1.4 and 2.1].

Second, if  $u, v \geq 4$  and  $\lfloor 3n/4 \rfloor + 1 \leq u + v \leq n$ , every even permutation is a product of two permutations each consisting of a disjoint  $u$ -cycle and  $v$ -cycle [17, Lemma 5].

Suppose an odd  $q$  satisfies  $3n/8 + 1 < q \leq n/2$ . Every odd  $g \in S_n$  is a product of a  $2q$ -cycle and two disjoint  $q$ -cycles. For a transposition  $g$ , choose the two  $q$ -cycles with its letters in different supports; multiplying their inverse by  $g$  merges them. Otherwise let  $s_0$  be the support size,  $t$  its number of nontrivial cycles, and  $e = s_0 - t$ , an odd integer at least three. Restrict to an invariant subset of size  $d = \max\{s_0, 2q\}$ . The class of  $g$  there has  $h = t + d - s_0$  cycles; the class of two  $q$ -cycles has  $k = d - 2q + 2$ . If  $s_0 \leq 2q$ , then  $h + k + 1 = 2q - e + 3 \leq 2q$ . Otherwise, using  $t \leq s_0/2$ ,

$$h + k + 1 = s_0 + t - 2q + 3 \leq 3n/2 - 2q + 3 < 2q.$$

Also  $h + k - 1 = 2d - e - 2q + 1$  is even. The first covering result, followed by inversion of its second factor and simultaneous conjugation, gives the assertion. Removing the unneeded fixed points before applying it is essential.

Now let  $q \geq 5$  be prime in that interval with  $q \nmid rs$ , and suppose  $r$  is odd. For odd targets the factors just obtained have orders  $2q, q$ ; for even targets the second covering result with  $u = v = q$  supplies two factors of order  $q$ . Their orders are coprime to  $r, s$ , respectively, so taking inverse exponents modulo those orders gives  $g = x^r y^s$ . If  $s$  is the odd exponent, reverse the argument's factor order using  $U^s V^r = V^r (V^{-r} U V^r)^s$ . Negative exponents cause no change.

It remains to find a prime in the interval missing from  $m$ . Let  $\vartheta(x) = \sum_{\ell \leq x} \log \ell$ . Rosser–Schoenfeld [93, Theorem 4, (3.14), and Theorem 9] give

$$\vartheta(x) > x \left(1 - \frac{1}{2 \log x}\right) \quad (x \geq 563), \quad \vartheta(y) < 1.01624y \quad (y > 0).$$

For  $n \geq 1126$ ,  $\log(n/2) > 6$ , and  $1.01624 < 49/48$ . Consequently

$$\vartheta(n/2) - \vartheta(3n/8 + 1) > \frac{29n}{384} - \frac{49}{48} > \frac{n}{16}.$$

If all primes in the interval divided  $m$ , this would contradict  $\log m \leq n/16$ . The explicit bound follows. The same source gives  $\vartheta(x) = x + o(x)$ , so the interval's prime-log sum is  $n/8 + o(n)$ , eventually exceeding  $n/c$  for any  $c > 8$ . This proves the second statement.  $\square$

If both exponents are odd, every degree works: every permutation is a product of two involutions, by reversing each cycle. An exponent 1 or  $-1$  also makes universality immediate. The stated constants are sufficient, not optimal. The candidate contribution is this symmetric-group deduction from the credited covering and prime estimates, rather than new versions of those imported theorems. See `research/10.32-proof.md` and `results/10.32-summary.json`.

### 6.1.7 Nonunique decompositions of projectives: Problem 4.55

Let  $G = 3.A_7$ , the perfect central cover of order 7560, and  $R = \mathbb{Z}_{(5)}$ . There is an  $RG$ -projective module of  $R$ -rank 55,440 with two indecomposable decompositions having no isomorphism class of summand in common. This addresses uniqueness itself, beyond the earlier failure of the condition that indecomposable endomorphism rings be local.

We first give the descent facts used. If an  $R$ -free  $G$ -lattice  $L$  has projective reduction modulo five, its restriction to a Sylow 5-subgroup  $S$  is free: lift a free basis over the local algebra  $\mathbb{F}_5 S$ , apply Nakayama's lemma to the resulting  $RS$ -map, and use equal  $R$ -ranks for injectivity. Averaging over a transversal, with  $[G : S]$  invertible in  $R$ , splits  $RG \otimes_{RS} L \rightarrow L$ .

Thus  $L$  is projective. Also, two finite projective  $RG$ -modules with isomorphic reductions are isomorphic: projectivity lifts the reduction map, Nakayama gives surjectivity, and equal  $R$ -ranks give injectivity.

Suppose a finite projective  $\mathbb{Z}_5G$ -module  $M$  has generic fiber obtained from a rational  $G$ -module  $V$ . In their identified  $\mathbb{Q}_5$ -space, approximate a  $\mathbb{Z}_5$ -basis of  $M$  by rational vectors modulo  $5M$ . They form both a  $\mathbb{Z}_5$ -basis of  $M$  and a  $\mathbb{Q}$ -basis of  $V$ . The  $G$ -matrices in this basis have entries in  $\mathbb{Q} \cap \mathbb{Z}_5 = R$ . Their  $R$ -span is a stable lattice completing to  $M$ , and is projective by the preceding reduction criterion.

The finite character input is the following part of the 5-modular decomposition data for 3.A7 [80, 18]. Use four Frobenius pairs of absolute Brauer characters, denoted  $A, B, C, D$ , of respective absolute degrees 3, 6, 18, 21. Their  $\mathbb{F}_5$ -simple modules have dimensions 6, 12, 36, 42, and their projective covers have dimensions 90, 60, 90, 90. For one faithful central character, the relevant rows are

| GAP ordinary index | degree | $A$ | $B$ | $C$ | $D$ |
|--------------------|--------|-----|-----|-----|-----|
| 10                 | 6      | 0   | 1   | 0   | 0   |
| 16                 | 21     | 0   | 0   | 0   | 1   |
| 18                 | 21     | 1   | 0   | 1   | 0   |
| 20                 | 24     | 1   | 0   | 0   | 1   |
| 22                 | 24     | 0   | 1   | 1   | 0   |

and the conjugate rows have indices 11, 17, 19, 23, 21. These indices use `CharacterTable` ("3.A7") and its 5-modular table in GAP 4.16.1 with the character-table library `CTblLib` version 1.3.11. All other rows are zero in these columns. The ordinary Galois orbits are  $\{10, 11\}, \{16, 17\}, \{18, 19\}, \{20, 21, 22, 23\}$ . Indices refer to the retained GAP table, not a numbering across all covers in the printed tables.

Brauer reciprocity shows that a completed projective whose reduction has cover multiplicities  $(a, b, c, d)$  has generic ordinary multiplicities  $b, d, a + c$  on the first three paired orbits, and  $a + d, b + c$  on the two halves of the degree-24 orbit. Its character is rational-valued exactly when

$$a + d = b + c. \quad (1)$$

This is necessary for every rationally defined projective with this support.

Consider the four rays

$$r_1 = (1, 1, 0, 0), \quad r_2 = (1, 0, 1, 0), \quad r_3 = (0, 1, 0, 1), \quad r_4 = (0, 0, 1, 1).$$

Each gives a completed projective, since idempotents lift over the complete ring  $\mathbb{Z}_5G$ , and each satisfies (1). Multiplication of its generic character by  $N = 168$  gives a rational form without assuming Schur index one. Indeed, for an ordinary Galois orbit  $\mathcal{O}$ , the sum of its primitive complex central idempotents lies in  $\mathbb{Q}G$ , and  $\mathbb{Q}Ge_{\mathcal{O}}$  has character  $\sum_{\chi \in \mathcal{O}} \chi(1)\chi$ . Since 168 is divisible by 6, 21, 24, the desired multiple is a nonnegative sum of these rational characters. Semisimple  $\mathbb{Q}_5G$ -modules are determined by their characters after scalar extension, so the descent argument gives projectives  $L_i$  over  $RG$  with reduction vectors  $168r_i$ .

Now  $r_1 + r_4 = r_2 + r_3$ , and reduction detects isomorphism of projectives. Hence

$$L_1 \oplus L_4 \cong L_2 \oplus L_3.$$

The common rank is  $168(150 + 180) = 55,440$ . Every nonzero direct summand of  $L_1$  has reduction supported on  $A, B$ ; finite-dimensional Krull–Schmidt and (1) force its vector onto the positive ray through  $r_1$ . The same holds for each of the other three rays. They meet only at zero. Thus no nonzero summand on one side can be isomorphic to a summand on

the other. Splitting successively terminates by positive finite  $R$ -rank, giving incompatible indecomposable refinements.

The character data are a genuine dependency: the four-ray calculation alone would not construct modules. The archive reconstructs the ordinary table from the actual group and the faithful decomposition rows from modular representations, in addition to comparison with published tables. The  $L_i$  themselves are not asserted to be indecomposable. Source: `research/4.55-proof.md`.

**Version 2 attribution.** Jayadevan [59] gives a substantially smaller example over  $\mathbb{Z}_{(11)}[\mathrm{PSL}_2(\mathbb{F}_{11})]$ : a rank-88 module with two decompositions into indecomposable rank-44 summands. Both arguments use completion and rational-character obstructions to descent, with different groups and character data. The comparison and our new GAP check are recorded in Appendix N.

### 6.1.8 Grothendieck classes and semiperfectness: Problem 21.60

Let  $G = C_3 \rtimes C_4 = \langle a, b : a^3 = b^4 = 1, bab^{-1} = a^{-1} \rangle$ ,  $R = \mathbb{Z}_{(2)}$ , and  $k = \mathbb{F}_2$ . Every simple  $kG$ -module is the reduction of a simple rational module, but  $RG$  is not semiperfect. Thus the nonnegative-combination criterion in the question does not suffice.

The central involution  $b^2$  acts trivially on a simple  $kG$ -module: its difference from the identity is nilpotent with a nonzero invariant kernel. Every such module factors through  $S_3$ . In  $kS_3$ , the central idempotent  $e = 1 + a + a^2$  cuts out  $kC_2$ , whose only simple module is  $k$ . The complementary four-dimensional algebra is  $M_2(k)$ , via

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

These satisfy the relations, and  $I, A, B, AB$  are linearly independent, proving the identification. Its only simple module is the natural module  $V$ . This lifts to the rational representation with matrices

$$A_{\mathbb{Q}} = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}, \quad B_{\mathbb{Q}} = \begin{pmatrix} 1 & -1 \\ 0 & -1 \end{pmatrix}.$$

The  $a$ -matrix has irreducible polynomial  $X^2 + X + 1$ , proving rational simplicity, and the stable lattice  $R^2$  reduces to  $V$ . The trivial module lifts trivially. Hence every finite-dimensional  $kG$ -module has Grothendieck class a nonnegative combination of reductions of simple rational modules, by a composition series.

For the obstruction to semiperfectness, let  $E = \mathbb{Q}(\omega)$ ,  $\omega^2 + \omega + 1 = 0$ , and form the quaternion algebra

$$D = E + Ej, \quad j^2 = -1, \quad ju = \bar{u}j.$$

The norm of  $u + vj$  is  $u\bar{u} + v\bar{v}$ . For  $u = x + y\omega$ , its summand is  $x^2 - xy + y^2$ ; the norm is positive on nonzero elements. The conjugate divided by the norm gives an inverse, so  $D$  is a division algebra. The order  $O = R[\omega] + R[\omega]j$  is a quotient of  $RG$  by  $a \mapsto \omega, b \mapsto j$ . Its reduction is  $M_2(k)$ , since  $\omega, j$  map to the matrices  $A, B$  above and their four basis images are independent.

Moreover  $J(O) = 2O$ . For every  $y \in O$ , the norm of  $1 - 2y$  is congruent to one modulo  $2R$ , and quaternion conjugation preserves  $O$ ; the inverse formula shows  $1 - 2y$  is a unit. The radical unit criterion gives  $2O \subseteq J(O)$ , and the semisimplicity of  $O/2O$  gives the reverse inclusion. The nontrivial idempotents of  $M_2(k)$  cannot lift to  $O$ , because a subring of a division ring has only idempotents zero and one. Thus  $O$  is not semiperfect.

A quotient of a semiperfect ring is semiperfect: its quotient by its Jacobson radical is a quotient of the original semisimple quotient; idempotents lift first to that semisimple ring, then to the original ring, and finally to the quotient. Consequently  $RG$  is not semiperfect either. The coefficient ring is the rational localization, not the 2-adic completion.

The question is Conjecture 9 in the retained version of [63]. The example also conflicts with its Proposition 3.3 as written. The extension-lifting step there cannot hold for arbitrary chosen lattices: with trivial  $C_2$ -actions,

$$\mathrm{Ext}_{RC_2}^1(R, R) = 0, \quad \mathrm{Ext}_{kC_2}^1(k, k) = k.$$

Directly, an extension with fixed trivial submodule and quotient has generator matrix  $\begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$ ; its square forces  $2t = 0$  over  $R$ , whereas  $t = 1$  gives a nonsplit extension over  $k$ . This concerns the cited version and step, and does not import the proposition as a valid premise. Source: `research/21.60-proof.md`.

## 6.2 Linear groups and algebraic structures

Linear representations, algebraic actions and polynomial structure.

### 6.2.1 Residual rationality in a fixed degree: Problem 10.35

The same representation degree occurs on both sides of this question. Consider

$$U(z) = \begin{pmatrix} 1 & z \\ 0 & 1 \end{pmatrix}, \quad T = \mathrm{diag}(2i, 2), \quad G = \langle U(1), T \rangle \leq \mathrm{GL}_2(\mathbb{C}).$$

Since  $TU(z)T^{-1} = U(iz)$ , its unique normal forms are  $U(z)T^k$ ,  $z \in \mathbb{Z}[i]$ ,  $k \in \mathbb{Z}$ . Uniqueness follows from the lower diagonal entry  $2^k$  and then the upper-right entry. This is  $\mathbb{Z}^2 \rtimes \mathbb{Z}$ , with the generator acting by a quarter turn, and is torsion-free because its kernel  $\mathbb{Z}^2$  and quotient  $\mathbb{Z}$  are torsion-free.

**Lemma 6.4.** *Let  $F$  have characteristic zero and contain no square root of  $-1$ . If  $X, Y \in \mathrm{GL}_2(F)$  satisfy*

$$[X, YXY^{-1}] = I, \quad Y^2XY^{-2} = X^{-1},$$

*then  $X^2 = I$ .*

*Proof.* Assume  $X^2 \neq I$ . A scalar  $X$  contradicts the second relation, so  $X$  is nonscalar. Conjugacy to its inverse gives  $\det(X)^2 = 1$ . If  $\det(X) = -1$ , equality of traces gives  $\mathrm{tr} X = 0$ , and Cayley–Hamilton gives  $X^2 = I$ . Thus  $\det(X) = 1$  and  $X^{-1} = (\mathrm{tr} X)I - X$ .

The centralizer algebra of a nonscalar 2-by-2 matrix is  $E = F[I, X]$ . Indeed, a vector  $v$  with  $v, Xv$  independent is cyclic, and a commuting operator is determined by its value on  $v$ . Now  $YXY^{-1}$  is a nonscalar member of  $E$ , so it too generates  $E$  with  $I$ . Conjugation by  $Y$  therefore normalizes  $E$  and fixes its scalar subspace. On the one-dimensional quotient  $E/FI$  it acts by a scalar  $q \in F^\times$ . Its square sends the nonzero class of  $X$  to its negative, so  $q^2 = -1$ , a contradiction.  $\square$

Put  $a = U(1)$ . The relations of the lemma hold for  $a, T$  in  $G$ . Consequently every homomorphism  $\phi : G \rightarrow \mathrm{GL}_2(\mathbb{Q})$ , and even every homomorphism into  $\mathrm{GL}_2(\mathbb{R})$ , kills the fixed nonidentity element  $a^2 = U(2)$ . This disproves the required residual property. If the image is torsion-free it kills  $a$  itself, and hence the whole normal subgroup  $U(\mathbb{Z}[i])$ . The common scalar factor in  $T$  is essential to torsion-freeness:  $T^4 = 16I$ . The example does embed in  $\mathrm{GL}_3(\mathbb{Q})$ , so it does not address a version allowing the target degree to increase. Source: `research/10.35-proof.md`.

### 6.2.2 A closed set with no closed higher power: Problem 16.28(a)

Let  $K = \overline{\mathbb{F}_5(t)}$ ,  $H = \mathrm{SL}_2(K)$ , and  $Y = \{A \in H : \mathrm{tr} A = 1\}$ . This is a closed conjugacy class: the polynomial  $X^2 - X + 1$  has distinct roots, and a  $\mathrm{GL}_2(K)$ -conjugating matrix can be made to have determinant one by multiplying it by a scalar.

The square  $Y^2$  omits  $-I$ , since  $AB = -I$  would give  $B = -A^{-1}$  and  $\mathrm{tr} B = -1$ . Nevertheless, for  $s \neq 0, -1$ , put

$$u = \frac{s}{1+s}, \quad w = u(1-u) - 1, \quad A_s = \begin{pmatrix} u & 1 \\ w & 1-u \end{pmatrix}, \quad D_s = \mathrm{diag}(s, s^{-1}).$$

Both  $A_s$  and  $A_s^{-1}D_s$  have determinant one and trace one; for the latter the trace is  $(1-u)s + us^{-1} = 1$ . Thus  $D_s \in Y^2$ . The curve  $s \mapsto D_s$  shows that  $-I = D_{-1}$  is in its closure, and  $Y^2$  is not closed.

In the connected reductive group  $G = H \times K^\times$ , take the closed set

$$X = (Y \times \{1\}) \cup \{(I, t)\}.$$

Since  $(I, t)$  is central,

$$X^n = \bigcup_{k=0}^n Y^k \times \{t^{n-k}\}.$$

The powers of  $t$  are distinct. For every  $n \geq 2$ , intersection with the closed fiber  $H \times \{t^{n-2}\}$  is therefore exactly  $Y^2 \times \{t^{n-2}\}$ , which is not closed in that fiber. Hence no  $X^n$ ,  $n > 1$ , is closed.

This answers part (a) as printed. It uses a reducible  $X$  and a field with an element of infinite multiplicative order; neither is excluded by the statement. The same class  $Y$  also gives the negative answer to part (b), since  $Y^2$  contains the open set  $\mathrm{tr}(C)^2 \neq 4$  and omits  $-I$ . The experiment classified that part as already implied by prior class-product results, so it is excluded from this candidate's counted coverage. Source: `research/16.28-proof.md`.

**Version 2 attribution.** Jayadevan [58] gives a different triangular-matrix example in  $\mathrm{GL}_3(\overline{\mathbb{F}_5(u)})$ . His closed set contains the identity and has one diagonal matrix omitted from every higher power but lying in each closure. Both constructions use characteristic five and a transcendental marker; see Appendix N.

### 6.2.3 A smooth fixed divisor and a singular quotient: Problem 14.72

Over  $\mathbb{C}$ , let  $X \subset \mathbb{A}^4$  have coordinates  $(x, y, u, v)$  and equations

$$y^2 = x^3 - x, \quad yu = xv, \quad (x^2 - 1)u = yv.$$

The automorphism  $\sigma(x, y, u, v) = (-x, iy, iu, v)$  has order four. The curve  $C : y^2 = x^3 - x$  is smooth and irreducible. The opens  $x \neq 0$  and  $x^2 - 1 \neq 0$  cover  $X$ . On the first, eliminate  $v = yu/x$ ; on the second, eliminate  $u = yv/(x^2 - 1)$ . They identify  $X$  locally with the product of an open subset of  $C$  and an affine line, proving it is a smooth irreducible affine surface without hidden components.

The full fixed scheme of  $\langle \sigma \rangle$  is  $x = y = u = 0$ , with  $v$  free. It is a smooth affine line. On the second chart its ideal is generated by  $y$ , because  $x = y^2/(x^2 - 1)$  and  $u = yv/(x^2 - 1)$ . It is therefore a smooth effective Cartier divisor  $D$ . The equation  $x = 0$  defines  $D$  set-theoretically but defines  $2D$  as a Cartier divisor. The reduced divisor is not globally principal on  $X$ : on the zero section its class is the nonzero order-two class of  $(0, 0) - \infty$  on the elliptic curve. The restriction below supplies a global equation.

Nevertheless the quotient is singular. Put

$$a = x^2, \quad d = xu^2 - v^2.$$

These and  $v$  are invariant, and the defining equations give  $xd = u^2$  and  $(a - 1)d = v^2$ . We claim

$$\mathbb{C}[X]^{\langle \sigma \rangle} = \mathbb{C}[a, v, d]/(v^2 - (a - 1)d).$$

After inverting  $a$ , a basis of the coordinate ring consists of  $x^s y^b u^c$ ,  $s \in \mathbb{Z}$ ,  $b = 0, 1$ ,  $c \geq 0$ , with weight  $i^{2s+b+c}$ . The invariant monomials have respectively the forms

$$x^s u^{2h} = a^{(s+h)/2} d^h, \quad x^s y u^{2h+1} = a^{(s+h+1)/2} v d^h.$$

Invariance makes the displayed exponents integral. The monomials  $a^r d^h$ ,  $a^r v d^h$  have distinct images in that Laurent basis, so there are no further relations. After inverting  $a - 1$ , the other chart has invariant ring  $\mathbb{C}[a, (a - 1)^{-1}, v]$ , since the invariant ring of the curve is  $\mathbb{C}[x^2]$ . Here  $d = v^2/(a - 1)$ . The two invariant principal opens cover, so the identifications prove the global assertion. One can also clear powers of  $a$  and  $a - 1$  and use their comaximality to see that no invariant is omitted.

At  $(a, v, d) = (1, 0, 0)$  the local equation is  $v^2 = (a - 1)d$ . The local dimension is two and the cotangent dimension three, so the quotient is not regular. This point comes from the orbit of  $(1, 0, 0, 0)$ , whose stabilizer is the proper subgroup  $\langle \sigma^2 \rangle$ . The full fixed divisor does not control this additional stabilizer.

**Post-review strengthening.** Following the reviewer's suggestion to remove suitable fibres, replace  $X$  by the invariant affine open

$$U = X \setminus V(x^2 + 1), \quad f = y - (1 + i)x.$$

Then the full fixed scheme on  $U$  is the smooth *principal* divisor  $V(f)$ . Indeed, in  $\mathbb{C}[U]$ ,

$$f(y + (1 + i)x) = x(x - i)^2,$$

and  $x - i$  is a unit because  $(x - i)(x + i) = x^2 + 1$  is a unit. Hence  $x \in (f)$ , then  $y \in (f)$ , and the defining relation  $u = x^2 u - yv$  gives  $u \in (f)$ . The reverse containment is immediate, so  $(f) = (x, y, u)$ , with reduced quotient ring  $\mathbb{C}[v]$ . Localization at the invariant  $a + 1$  commutes with taking invariants:

$$\mathbb{C}[U]^{\langle \sigma \rangle} = (\mathbb{C}[a, v, d]/(v^2 - (a - 1)d))[(a + 1)^{-1}].$$

The singular point  $(1, 0, 0)$  survives, as does the whole fixed divisor  $a = 0$ . Thus  $U$  answers the question negatively even when “regular hypersurface” requires a single global equation for the reduced fixed scheme. This restriction was obtained during revision; it is not part of the deadline construction.

Both constructions use cyclic order four; neither asserts a prime-order variant or an action free outside the fixed divisor. The original source is `research/14.72-proof.md`; the restriction is proved above.

#### 6.2.4 Finitely many automorphism orbits in a rational linear group

For Problem 21.40, let  $G \leq \mathrm{GL}_n(\mathbb{Q})$  have finitely many orbits under its abstract automorphism group. Then  $G$  has a normal torsion-free nilpotent subgroup of class at most  $n - 1$  and index at most  $(2n + 1)^{n^2}$ .

Fix  $g \in G$ . Two members of  $g, g^2, g^4, \dots$  are in one orbit. Thus some  $h = g^{2^i}$  and automorphism  $\alpha$  satisfy  $\alpha(h) = h^r$  for  $r = 2^{j-i} > 1$ . The matrices  $\alpha^{-k}(h)$  are  $r^k$ -th roots of  $h$ . We first show this forces every eigenvalue of  $h$  to be a root of unity.

For an eigenvalue  $\lambda$  of  $h$ , choose an eigenvalue  $\mu_k$  of its  $r^k$ -th root with  $\mu_k^{r^k} = \lambda$ ; its degree over  $\mathbb{Q}$  is at most  $n$ . Choose an integer  $D > 0$  clearing the entries of  $h$  and let  $M \geq 1$  bound the absolute values of all its eigenvalues. Every conjugate of  $\mu_k$  has absolute value at most  $M$ . Also

$$(D\mu_k)^{r^k} = D^{r^k-1}(D\lambda)$$

is an algebraic integer, so  $D\mu_k$  is one by transitivity of integrality. Algebraic integers of bounded degree with all conjugates bounded form a finite set: their monic minimal polynomials have integer coefficients bounded by the elementary symmetric-function estimates. Thus some  $\mu_k = \mu_l$ ,  $k < l$ , giving  $\mu_k^{r^l-r^k} = 1$ . It follows that  $\lambda$  is a root of unity, and then the same is true for each eigenvalue of  $g$ .

Consequently  $\text{tr}(g)$  is a rational algebraic integer of absolute value at most  $n$ , so  $\text{tr}(G) \subseteq \{-n, \dots, n\}$ . Let  $A = \text{span}_{\mathbb{Q}} G$ , a unital algebra of dimension  $d \leq n^2$ , and define its trace radical

$$R = \{a \in A : \text{tr}(ab) = 0 \text{ for all } b \in A\}.$$

Cyclicity of trace makes  $R$  a two-sided ideal. For  $a \in R$ , all  $\text{tr}(a^s)$  vanish. Newton's identities and Cayley–Hamilton give  $a^n = 0$ .

In fact  $R^n = 0$ . On  $V = \mathbb{Q}^n$ , the chain  $V \supseteq RV \supseteq R^2V \supseteq \dots$  cannot stabilize at a nonzero module  $W = RW$ . If it did, choose a smallest finite  $A$ -generating list  $w_1, \dots, w_s$  of  $W$  and express  $w_s = \sum_i a_i w_i$  with  $a_i \in R$ . Since  $a_s$  is nilpotent,  $1 - a_s$  is invertible, allowing deletion of  $w_s$ , a contradiction. Thus there are at most  $n$  strict dimension drops, and faithfulness gives  $R^n = 0$ .

The group  $1 + R$  is nilpotent of class at most  $n - 1$ , using  $[1 + R^a, 1 + R^b] \subseteq 1 + R^{a+b}$ . It is torsion-free: a finite-order unipotent matrix in characteristic zero has minimal polynomial dividing both  $(X - 1)^n$  and the squarefree polynomial  $X^m - 1$ , and is the identity. Set  $U = G \cap (1 + R)$ , a normal subgroup. Choose a basis  $g_1, \dots, g_d$  of  $A$  from  $G$ . The tuples

$$(\text{tr}(gg_1), \dots, \text{tr}(gg_d))$$

take at most  $(2n + 1)^d$  values. Equality of tuples for  $g, h$  is equivalent to  $g - h \in R$ , hence to  $gh^{-1} \in U$ . They distinguish exactly the cosets of  $U$ , proving the index bound. No finite-generation hypothesis or matrix realization of the abstract automorphisms was used. Source: `research/21.40-proof.md`.

**Version 2 attribution.** Jayadevan [54] gives the same bounded-root and trace-radical argument, including the bound  $(2n + 1)^d$ . He separately states the finite-trace-set version and the number-field consequence obtained by restriction of scalars. These are overlapping arguments with some different elementary steps; see Appendix N.

### 6.2.5 An unattained 2-Jordan exponent: Problem 21.121(a)

A group  $\Gamma$  is  $p$ -Jordan if some constants  $J, e$  have the following property: every finite  $H \leq \Gamma$  has a normal abelian  $p'$ -subgroup of index at most  $J|H_{(p)}|^e$ , where  $H_{(p)}$  is a Sylow  $p$ -subgroup. The question asks whether the infimum of the admissible exponents must itself be admissible.

**Proposition 6.5.** *There is a countable 2-Jordan group with exponent*

$$e_0 = \frac{\log 24}{\log 8}$$

*for which no constant works at exponent  $e_0$ .*

*Proof.* For  $k \geq 1$ , put  $n = 2^k - 1$  and index the coordinates of  $\mathbb{F}_2^n$  by nonzero  $v \in \mathbb{F}_2^k$ . Use the simplex code

$$C_k = \{(a \cdot v)_{v \neq 0} : a \in \mathbb{F}_2^k\}.$$

It has dimension  $k$ . We need a small intersection estimate: if  $L$  is spanned by  $r$  nonzero vectors with disjoint supports and  $c = \dim(C_k \cap L)$ , then

$$2^c \leq r + 1. \quad (2)$$

Indeed, a basis of the intersection corresponds to  $c$  independent linear forms on  $\mathbb{F}_2^k$ , so its coordinate columns include all  $2^c - 1$  nonzero columns in  $\mathbb{F}_2^c$ . Vectors in  $L$  are constant on each of the  $r$  disjoint supports and zero elsewhere, giving at most  $r$  different nonzero columns.

Let  $S = Q_8 \rtimes C_3$ , where a generator of  $C_3$  cycles  $i, j, ij$  in  $Q_8$ . Identify the center of  $Q_8^n$  with  $\mathbb{F}_2^n$ , and form

$$G_k = S^n / C_k, \quad P_k = Q_8^n / C_k, \quad T_k = C_3^n.$$

The code is central and fixed by  $T_k$ , so

$$G_k = P_k \rtimes T_k, \quad |G_k| = 24^n / 2^k, \quad |P_k| = 8^n / 2^k.$$

Write  $Z_0$  for the image of  $Z(Q_8^n)$ . Then  $V = P_k / Z_0 = \bigoplus_{i=1}^n V_i$ , with each  $V_i$  two-dimensional over  $\mathbb{F}_2$ ; the  $i$ -th  $C_3$  factor acts irreducibly on  $V_i$  and trivially on the other coordinates.

We claim that every  $H \leq G_k$  contains a normal abelian odd-order subgroup  $A_0$  for which, with  $u = |H_{(2)}|$ ,

$$[H : A_0] = u3^r, \quad u \geq \frac{8^r}{r+1} \quad (3)$$

for some integer  $r \geq 0$ . Put  $U = H \cap P_k$ , its normal Sylow 2-subgroup. A Sylow 3-subgroup  $A$  complements  $U$ , simply by comparing orders. Conjugating into the Sylow subgroup  $T_k$ , we may assume  $H = U \rtimes A$  with  $A \leq T_k$ . Set  $A_0 = C_A(U)$ ; it is normal abelian of odd order, since it centralizes  $U$  and  $A$  is abelian. Write  $|A/A_0| = 3^r$ .

Let  $W$  be the image of  $U$  in  $V$ . The kernel of the action of  $A$  on  $W$  is exactly  $A_0$ . To check the nontrivial inclusion, an element  $a$  in that kernel satisfies  $x^a = xz_x$ , with  $z_x \in U \cap Z_0$ , for every  $x \in U$ . It fixes  $z_x$ , and  $a^3 = 1$ , so  $x = x^{a^3} = xz_x^3 = xz_x$ , whence  $z_x = 1$ .

The representation of  $A$  on  $W$  is completely reducible. Over  $\mathbb{F}_4$ , its commuting operators diagonalize, since  $X^3 - 1$  has distinct roots. Nontrivial irreducibles over  $\mathbb{F}_2$  therefore have dimension two and correspond to pairs  $\{\alpha, -\alpha\}$  of nonzero homomorphisms  $\alpha : A \rightarrow \mathbb{F}_3$ . The occurring characters span a space of dimension  $r$ , because their common kernel is  $A_0$ . Choose  $r$  independent occurring characters and nonzero vectors  $w_j$  in their respective isotypic components of  $W$ . Their coordinate supports are disjoint: a coordinate  $V_i$  belongs to only one character pair.

Choose  $a_j \in A$  acting nontrivially on the character of  $w_j$  and lift  $w_j$  to  $x_j \in U$ . In each nonzero quaternion coordinate,  $w_j$  and  $w_j^{a_j}$  are distinct nonzero vectors of  $Q_8/Z(Q_8)$ , so the commutator of their lifts is the central involution. Thus  $[x_j, x_j^{a_j}]$  is the image of the binary support vector  $d_j$  of  $w_j$ . The  $d_j$  are disjoint and independent before quotienting by  $C_k$ . If  $L$  is their span, (2) shows that their images span a space of dimension at least  $r - \log_2(r+1)$  in  $[U, U] \leq U \cap Z_0$ . Also  $\dim W \geq 2r$ . Consequently

$$u = |W| |U \cap Z_0| \geq 2^{2r} 2^{r-c} \geq 8^r / (r+1).$$

Together with  $[H : A_0] = |U| |A/A_0|$ , this proves (3), including  $r = 0$ .

For  $e > e_0 > 1$ , it follows that

$$\frac{[H : A_0]}{|H_{(2)}|^e} \leq (r+1)^{e-1} \left( \frac{3}{8^{e-1}} \right)^r.$$

The right-hand side has a finite supremum over  $r \geq 0$ , independent of  $k$  and  $H$ , since its exponential base is less than one.

On the other hand, every normal odd-order subgroup of  $G_k$  centralizes  $P_k$ : their commutator is in their trivial intersection. Its image in  $T_k$  must then act trivially on  $V$ , because inner automorphisms from  $P_k$  act trivially on  $P_k/Z_0$ . The action of  $T_k$  is faithful, so that normal subgroup is trivial. Hence the smallest permissible index for  $G_k$  itself is  $|G_k|$ , and

$$\frac{|G_k|}{|P_k|^{e_0}} = 2^{k(e_0-1)} \longrightarrow \infty.$$

Finally take  $\Gamma = *_{k \geq 1} G_k$ . Every finite subgroup of a free product is conjugate into a factor. One proof uses the free-product tree: a finite group fixes the center of the finite subtree spanned by an orbit, and the action has no edge inversions. Vertex stabilizers are trivial or conjugate to factors. The uniform estimates therefore give every exponent  $e > e_0$  for  $\Gamma$ , while its embedded factors rule out  $e_0$ . A smaller admissible exponent would imply admissibility of  $e_0$ , so the infimum is exactly  $e_0$ .  $\square$

The use of a free product is material; a direct sum would introduce additional finite products not covered by the argument. No finite generation assumption appears in the question. The construction does not resolve part (b). The archive contains GAP checks on all subgroup conjugacy classes for  $k = 1, 2$ , and direct code and quaternion controls; the uniform argument above supplies the unbounded quantifiers. Source: `research/21.121a-proof.md`.

**Version 2 attribution.** Jayadevan [57] gives a different construction for every prime  $p$ : a locally finite metabelian subgroup of  $\mathrm{GL}_2(\overline{\mathbb{F}_p})$  whose admissible exponents are exactly  $(1, \infty)$ . His scalar orders come from cyclotomic values at factorial indices. Our free-product construction has a different infimum and concerns  $p = 2$ ; see Appendix N.

### 6.2.6 19.62: the derived carpet is closed

Let  $\Phi$  be a reduced crystallographic root system and  $K$  a commutative ring with identity. A carpet  $A = (A_r)_{r \in \Phi}$  is a family of additive subgroups of  $K$  satisfying all Chevalley conditions

$$c_{ij,rs} a^i b^j \in A_{ir+js} \quad (a \in A_r, b \in A_s, i, j > 0, ir + js \in \Phi).$$

Here  $r, s$  are distinct and nonopposite, and  $c_{ij,rs}$  is the absolute value of the corresponding integral commutator constant. Signs do not affect additive subgroup membership. Write  $\partial A = B$  for the family additively generated by these parameters at each destination root. Then  $B \subseteq A$  and  $B$  is itself a carpet: a rule applied to elements of  $B$  is also a rule applied to elements of  $A$ , so its value belongs to the appropriate subgroup of  $B$ .

The candidate proves the stronger inclusion

$$A_p^2 B_{-p} \subseteq B_p \quad (p \in \Phi), \tag{4}$$

where  $A_p^2 = \{a^2 : a \in A_p\}$ . Consequently  $B_p^2 B_{-p} \subseteq B_p$ . Nuzhin's already published square-closedness criterion then says that  $B$  is closed:

$$\langle x_r(B_r) : r \in \Phi \rangle \cap x_p(K) = x_p(B_p).$$

The criterion, and hence the answer to Problem 19.63, predates the experiment [85, Theorem 2]. The ledger's grouping of 19.62 with 19.63 must not be read as a new solution of the latter.

Substantial cases of Problem 19.62 were also known. Nuzhin's 2023 discussion records the answer for type  $A_\ell$ ; the earlier triple-product inclusion quoted there, together with its Theorem 1, also gives the simply laced types  $D_\ell, E_\ell$ . Its Corollary 2 treats  $B_\ell, C_\ell, F_4$  when  $\gcd(\text{char } K, 2) = 1$ , and  $G_2$  when  $\gcd(\text{char } K, 6) = 1$  [84, Section 3]. The integer derivations below apply without these characteristic restrictions; the already covered cases are not new answers.

**Reduction to finite integer derivations.** Fix  $x \in A_p$ . It suffices to prove

$$cx^2u^iv^j \in B_p \quad \text{whenever} \quad ir + js = -p, \quad u \in A_r, \quad v \in A_s, \quad (*)$$

since multiplication by  $x^2$  distributes over every additive expression for an element of  $B_{-p}$ . The intersection of  $\Phi$  with the plane spanned by  $r, s$  is an irreducible rank-two root system, of type  $A_2, B_2$ , or  $G_2$ . Every root string between its roots stays in that plane. Thus the restricted carpet has precisely the required absolute Chevalley constants, and any derived parameter in the restriction belongs to the ambient  $B$ . No commutation of restriction with the entire operation  $\partial$  is assumed.

The following finite proof calculus establishes (\*). Initial nodes are independent variables with specified membership in  $A_r$ . A Chevalley rule applied to two earlier nodes gives  $cf^ig^j \in B_{ir+js} \subseteq A_{ir+js}$ . If two applications give integer multiples  $m_1w, m_2w$  of the same monomial at the same root, their integer linear combinations give  $\gcd(m_1, m_2)w \in B$ . Each final coefficient must divide the requested coefficient. These operations use no division in  $K$  and remain valid in characteristics two and three.

There are 6, 20, and 78 distinct requests in types  $A_2, B_2, G_2$ , respectively, after coincident requests are combined by taking the gcd of their coefficients. Short certificates contain 655 nodes in total, at most nine per request, including initial nodes. Relabeling the carpet by Weyl transformations reduces the display to 19 requests, one destination root of each length. This is transport of a universal implication; it does not assert that the particular carpet is Weyl invariant. The ancillary checker verifies the absolute-rule invariance, the coverage by those representatives, and all 104 requests directly. The 19 representative derivations are printed in Appendix E.

For illustration, take  $G_2$  with short simple root  $a$ , long simple root  $b$ , and  $p = 2a + b$ . For  $x \in A_p, u \in A_a, v \in A_{-3a-b}$ , the rules give

$$3xu \in B_{3a+b}, \quad xv \in B_{-a}, \quad x^2v \in B_{a+b}.$$

Applying the rules again gives  $3x^2uv \in B_p$  from the first two expressions and  $2x^2uv \in B_p$  from the third and  $u$ . Their difference gives  $x^2uv \in B_p$ . This illustrates why retaining both small integer coefficients can prove an assertion that an argument dividing by a structure constant would miss.

**A direct proof of the known closedness criterion.** We retain the following alternative argument as part of the experiment's mathematical record. Assume  $A_r^2 A_{-r} \subseteq A_r$  for every root and work in the integral Chevalley Lie algebra tensored with  $K$ . Put

$$M = \sum_r A_r e_r + \sum_r \langle A_r A_{-r} \rangle_{\mathbb{Z}} h_r.$$

Root coordinates in this additive group are independent; in particular  $M \cap Ke_r = A_r e_r$ . The group  $M$  is a Lie subring. Root-root brackets use the  $(1, 1)$  carpet condition or, for opposite

roots, the displayed Cartan summands. For a Cartan generator  $uvh_r$  and a root  $s \neq \pm r$ , Jacobi expresses their bracket as two successive root brackets, each allowed by the carpet rules. For  $s = r$  or  $-r$ , polarization supplies the required coefficient:

$$2uvw = (u + w)^2v - u^2v - w^2v \in A_r \quad (u, w \in A_r, v \in A_{-r}).$$

The Cartan part commutes with itself. This also shows that  $M$  is exactly the Lie subring generated by the root summands.

Every  $x_s(t)$ ,  $t \in A_s$ , preserves  $M$ . On  $ue_r$  with  $r \neq \pm s$ , its root-string terms have coefficients  $C_{i1, sr}t^i u$ , covered by the carpet rules. It fixes  $e_s$ , while

$$ue_{-s} \mapsto ue_{-s} + tuh_s - t^2ue_s$$

uses the square assumption. The inverse generator gives equality. If a root  $p$  admits a root  $q$  with  $\langle p, q^\vee \rangle = \pm 1$ , all generating root elements send  $h_q$  into  $h_q + M$ . Thus  $x_p(t)$  in the carpet group forces  $\pm te_p \in M$ , and hence  $t \in A_p$ . Such a  $q$  exists in simply laced systems of rank at least two by adjacency; in  $B_n$ ,  $n \geq 3$ , use  $e_1 + e_2$  for a short root  $e_1$ , and  $e_2 + e_3$  for a long root  $e_1 + e_2$ . The same choices cover  $F_4$ , and the same-length hexagons cover  $G_2$ .

For  $C_n$ , including  $B_2 = C_2$  and rank one, use natural symplectic matrices indexed by  $1, \dots, n, -1, \dots, -n$ . The root vectors are

$$E_{ij} - E_{-j, -i}, \quad E_{i, -j} + E_{j, -i}, \quad E_{i, -i},$$

and their negative-root counterparts. Together with  $H_i = E_{ii} - E_{-i, -i}$  they form a free Chevalley basis over  $K$ . Every root vector  $N$  satisfies  $N^2 = 0$ . Choose a diagonal matrix  $D$  with entries  $d_i \in \{0, 1\}$  at  $i$  and  $1 - d_i$  at  $-i$ . Then

$$[D, e_r] = \lambda_r e_r, \quad \lambda_{e_i - e_j} = d_i - d_j, \quad \lambda_{e_i + e_j} = d_i + d_j - 1, \quad \lambda_{2e_i} = 2d_i - 1.$$

For any prescribed  $p$ , the  $d_i$  can make  $\lambda_p = \pm 1$ . Since  $NDN = 0$ ,

$$(I + tN)D(I - tN) = D - t\lambda_r N.$$

Each initial root generator sends  $D$  into  $D + M$ , and invariance of  $M$  extends this to every word. The formula detects  $t \in A_p$  for a new root element  $I + te_p$ . This also works in every standard Chevalley isogeny form: an equality of root words gives equality of actions on the abstract Chevalley Lie algebra. The quotient of their natural symplectic lifts centralizes that algebra. Commuting with  $E_{i, -i}$  and  $E_{-i, i}$  forces a matrix to be scalar on each two-coordinate block and zero between blocks; the short-root matrices force the scalars to agree. Such a quotient is therefore scalar over any commutative ring and commutes with  $D$ . The same detection argument applies. This proves the criterion without dividing by two.

The retained proof and audits are `research/19.62-rank-two-proof.md`, `research/19.62-closedness-criterion.md`, and `results/19.62-rank-two-packet.json`. The portable ancillary files include the proof calculus, target coverage checks, and integer certificates.

### 6.2.7 19.61: root closure preserves the derived carpet

Use the notation of Section 6.2.6. For a carpet  $A$ , put

$$H = \langle x_r(A_r) : r \in \Phi \rangle, \quad D_r = \{t \in K : x_r(t) \in H\}.$$

The candidate asserts that  $D$  is a carpet and  $\partial D = \partial A$ , over every commutative ring. Troyanskaya previously proved carpet closedness of this root closure in odd characteristic, with characteristic divisible by three also excluded in type  $G_2$  [107]. The argument here uses Nuzhin's known square criterion and a universally valid finite adjoining lemma. Priority of the stronger statement is not established by those source comparisons.

**Lemma 6.6.** *Let  $B = \partial A$ , fix  $a \in A_r, b \in A_{-r}$ , and replace  $A_r$  by  $A_r + \mathbb{Z}a^2b$ , leaving all other entries unchanged. The resulting family  $A'$  is a carpet and  $\partial A' = B$ .*

*Proof.* Consider an ordered Chevalley rule whose first input changes from  $x$  to  $x + na^2b$ . Its parameter expands as

$$c(x + na^2b)^i u^j = \sum_{k=0}^i c \binom{i}{k} n^k x^{i-k} a^{2k} b^k u^j.$$

The  $k = 0$  term lies in  $B_{ir+js}$  by definition. For every other term, the required universal implication is

$$c \binom{i}{k} x^{i-k} a^{2k} b^k u^j \in B_{ir+js}, \quad x, a \in A_r, \quad b \in A_{-r}, \quad u \in A_s. \quad (5)$$

The variables  $x, a$  are independent even though their root labels coincide. Multiplication by any integer  $n^k$  is allowed in the additive subgroup  $B_{ir+js}$ . The analogous substitution into the second input is checked separately, retaining the constant of the original ordered rule; reversing an ordered source pair in  $G_2$  can change its absolute constant. A nontrivial rule has distinct nonopposite source roots, so only one input subgroup changes.

The same root-plane argument as in Section 6.2.6 reduces (5) to types  $A_2, B_2, G_2$ . The ancillary certificates apply exactly the integer proof calculus explained there. Their complete sizes are

| type  | targets | nodes | largest derivation |
|-------|---------|-------|--------------------|
| $A_2$ | 12      | 72    | 6                  |
| $B_2$ | 48      | 380   | 12                 |
| $G_2$ | 240     | 2739  | 19                 |
| total | 300     | 3191  | 19                 |

Every noninitial node lies in  $B$  by a carpet application and, where needed, a Bezout combination of two such applications. The last coefficient divides the coefficient in (5). Thus each certificate is a finite integer algebraic proof valid under every substitution in  $K$ . They prove all 300 implications. For example, in type  $A_2$  with  $p = r + s$ , the only changed first-input term is  $a^2bu$ . The rules give  $au \in B_p$ , then  $abu \in B_s$ , and then  $a^2bu \in B_p$ . The longer certificates perform the same operation where coefficients two and three intervene.

The coverage checker reconstructs the requests by repeated sparse polynomial multiplication, independently of the binomial formula used by the target generator. It covers both inputs of all ordered rules: 416 substitutions and 600 nonconstant terms, giving the 300 requests after coincident requests are combined by gcd. The checked rule tables and the complete derivations accompany this paper; their search procedure is not a mathematical premise.

Expansion now gives  $\partial A' \subseteq B$ ; monotonicity gives the reverse inclusion. Since  $B \subseteq A \subseteq A'$ , every defining parameter of  $A'$  lies in its designated entry. This is precisely the carpet condition.  $\square$

Starting with  $C^{(0)} = A$ , define

$$C_r^{(n+1)} = C_r^{(n)} + \langle u^2v : u \in C_r^{(n)}, v \in C_{-r}^{(n)} \rangle_{\mathbb{Z}}.$$

Each  $C^{(n)}$  is a carpet with derived family  $B$ . Indeed, adjoin any finite collection of the displayed generators one at a time. Its parameters remain available after earlier adjoinings, so the lemma applies throughout. These finite extensions form a directed family whose union is  $C^{(n+1)}$ . A carpet condition involves finitely many elements, and a derived parameter

uses only two input elements. Hence directed union preserves both carpet membership and equality of the derived family.

Put  $C_r = \bigcup_n C_r^{(n)}$ . Again  $C$  is a carpet and  $\partial C = B$ . Any  $u \in C_r, v \in C_{-r}$  lie in a common stage, so  $u^2v$  lies in the next stage. Thus  $C_r^2 C_{-r} \subseteq C_r$ , and the known square criterion makes  $C$  closed. Since  $A \subseteq C$ , one has  $H \subseteq E(\Phi, C)$ ; closedness yields

$$A \subseteq D \subseteq C.$$

The root-group addition law makes each  $D_r$  additive. Every commutator parameter formed from  $D$  belongs to  $\partial C = B \subseteq A \subseteq D$ , proving that  $D$  is a carpet. Finally monotonicity gives

$$B = \partial A \subseteq \partial D \subseteq \partial C = B.$$

The constructed square completion  $C$  need not equal  $D$ ; the proof only requires the displayed sandwich. It does not assert that the adjoined root elements already belong to  $H$ . In rank one there are no nonopposite commutator conditions and all derived families are zero, so the conclusion is immediate. See `research/19.61-proof.md` and `results/19.61-square-completion-packet.json`.

### 6.2.8 21.76: closed irreducible nets without a diagonal completion

For every field  $k$  and every  $n \geq 3$ , set

$$R = k[x, y], \quad K = k(x, y), \quad J = yR, \quad \sigma_{12} = \sigma_{21} = kx + J, \quad \sigma_{ij} = J \text{ otherwise } (i \neq j).$$

This gives an irreducible closed elementary net over  $K$  which cannot be completed by diagonal additive groups. In particular it supplies every odd characteristic requested.

Recall that an elementary net requires  $\sigma_{ik}\sigma_{kj} \subseteq \sigma_{ij}$  for pairwise distinct indices; irreducibility here means every level is nonzero. All levels contain  $J$ , and at least one of  $(i, k), (k, j)$  is outside the two exceptional positions, so every required product lies in  $RJ = J$ . Thus it is an irreducible elementary net.

We first check the rank-one pair. In  $\mathrm{SL}_2(k(x))$ , let

$$H = \left\langle U(a) = \begin{pmatrix} 1 & ax \\ 0 & 1 \end{pmatrix}, V(b) = \begin{pmatrix} 1 & 0 \\ bx & 1 \end{pmatrix} : a, b \in k \right\rangle.$$

Its intersection with either elementary root subgroup has parameters exactly  $kx$ . To see this, combine adjacent factors of the same kind and remove zero parameters. Any remaining word is alternating with all parameters nonzero. If its length  $m \geq 2$  is even and it starts with  $U$ , choosing all  $m$  off-diagonal terms gives the unique degree- $m$  contribution in entry  $(1, 1)$ , nonzero because  $E_{12}E_{21} \cdots E_{21} = E_{11}$ . If  $m$  is odd, a degree- $(m-1)$  contribution omits one factor. Omitting the last gives  $E_{11}$ , omitting the first gives  $E_{22}$ , and omitting any interior factor gives zero by adjacent equal matrix units. Thus entry  $(1, 1)$  has nonzero leading term of degree  $m-1$ . For a word starting with  $V$ , interchange the indices. No such word is an elementary transvection, whose diagonal entries are both one. Length one and zero give exactly the stated intersections. This is a special case of Koibaev's prior closed-pair theorem [67]; its needed special case has just been proved.

Let  $E(\sigma)$  be generated by  $I + aE_{ij}$  with  $a \in \sigma_{ij}$ . All these matrices and their inverses lie in  $\mathrm{SL}_n(R)$ . If  $I + aE_{ij} \in E(\sigma)$ , initially with  $a \in K$ , its matrix entry therefore gives  $a \in R$ . Reduction at  $y = 0$ , now a legitimate ring map on  $R$ , sends  $E(\sigma)$  to  $\mathrm{diag}(H, I_{n-2})$ . At an ordinary position this forces  $a \bmod y = 0$ ; at either exceptional position the rank-one result gives  $a \bmod y \in kx$ . In each case  $a \in \sigma_{ij}$ , proving closedness. No evaluation homomorphism on the entire field  $K$  was used.

A diagonal completion would force  $x^2 \in \sigma_{11}$  from the two exceptional levels, and then  $x^3 \in \sigma_{11}\sigma_{12} \subseteq \sigma_{12}$ . Reduction modulo  $y$  would give  $x^3 \in kx$ , impossible for transcendental  $x$ . Thus no completion exists.

The archive also gives a one-variable characteristic-three example. In  $\mathbb{F}_9 = \mathbb{F}_3[j]$ ,  $j^2 = -1$ , the two elementary matrices with upper parameter 1 and lower parameter  $j$  generate a group of order 120, established by an explicit finite coset table. Its intersections with the order-nine root groups therefore have parameters  $\mathbb{F}_3$  and  $j\mathbb{F}_3$ . Using exceptional levels  $\mathbb{F}_3 + t\mathbb{F}_9[t]$ ,  $j\mathbb{F}_3 + t\mathbb{F}_9[t]$  and all other levels  $t\mathbb{F}_9[t]$  repeats the reduction argument over  $\mathbb{F}_9(t)$ ; a completion would force  $j$  into the first exceptional level. The full table is `research/21.76-finite-table.md`. This supplementary finite certificate is not needed for the two-variable proof, which is independent of the carpet candidates in Problems 19.61–19.62. See `research/21.76-proof.md`.

### 6.2.9 20.92(a): the printed brace average for large primes

For a left brace write  $a * b = a \circ b - a - b$ . The question asks whether the specific operation

$$a \cdot b = - \sum_{i=0}^{p-2} 2^{-i} ((2^i a) * b)$$

is pre-Lie on every  $k$ -dimensional  $\mathbb{F}_p$ -brace when  $p$  is sufficiently large relative to  $k$ . The following bound is coarse but uniform.

**Proposition 6.7.** *For  $k \geq 2$ , put  $E = k^{2k-2}$  and  $D = (k-1)E$ . The displayed operation is bilinear and pre-Lie if  $p > k$  and  $\text{ord}_p(2) > D$ . In particular  $p > \max\{k, 2^D\}$  suffices. For  $k = 1$  every odd prime works, with zero product.*

*Proof.* Let  $V = (A, +)$ , and set  $\lambda_a(b) = -a + a \circ b$ . The brace identities make

$$\Gamma = \left\{ G_a = \begin{pmatrix} \lambda_a & a \\ 0 & 1 \end{pmatrix} : a \in A \right\}$$

a regular affine group of order  $p^k$ . The  $p$ -group  $\lambda(A)$  has a full invariant flag with trivial action on successive quotients: the orbit equation gives a nonzero fixed vector in every nonzero  $\mathbb{F}_p$ -module of a finite  $p$ -group, and one repeats on the quotient. In a suitable basis  $\Gamma \leq \text{UT}_{k+1}(\mathbb{F}_p)$ . Write  $J$  for the strictly upper triangular matrix algebra. Then  $J^{k+1} = 0$ , and truncated exponential and logarithm are inverse when  $p > k$ .

The subgroup assertion of Lazard's correspondence [72, pp. 178–180] implies that  $\mathfrak{g} = \log \Gamma$  is a Lie subring of  $J$ , hence an  $\mathbb{F}_p$ -linear Lie subalgebra of dimension  $k$ . Projection  $\pi$  onto the translation column is injective on  $\mathfrak{g}$ : zero translation in  $X$  gives zero translation in  $\exp X$ , and regularity forces  $\exp X = 1$ . Thus  $\pi$  is an isomorphism, and

$$\mathfrak{g} = \left\{ X_x = \begin{pmatrix} L_x & x \\ 0 & 0 \end{pmatrix} : x \in V \right\}$$

for a linear map  $x \mapsto L_x$ , all  $L_x$  strictly upper triangular. Lie closure gives

$$[L_x, L_y] = L_{L_x y - L_y x}.$$

Therefore  $x \triangleright y = L_x y$  is a bilinear pre-Lie product; the equation is precisely the symmetry of its associator.

The translation of  $\exp X_x$  is the polynomial

$$W(x) = \sum_{j=0}^{k-1} \frac{L_x^j x}{(j+1)!}.$$

It has degree at most  $k$ , constant term zero, and linear term  $x$ . Regularity makes  $W : V \rightarrow V$  bijective. If  $\Omega$  denotes its inverse as a function, then  $\lambda_a = \exp L_{\Omega(a)}$ . We need a polynomial representative of  $\Omega$  of degree independent of  $p$ ; finite-set interpolation would not provide that.

Define the polynomial

$$C(x, y) = \pi \log(\exp X_x \exp X_y).$$

Every surviving associative monomial has length at most  $k$ , so  $\deg C \leq k$ . Truncated BCH is a Lie polynomial over  $\mathbb{F}_p$  in this class. Consequently, after scalar extension to the polynomial ring, Lie closure gives the formal identities

$$X_{C(x,y)} = \log(\exp X_x \exp X_y), \quad W(C(x, y)) = W(x) + \exp(L_x)W(y). \quad (*)$$

This step uses a polynomial identity, not an inference from equality on finite-field points.

Put  $V_i = \langle e_1, \dots, e_{k+1-i} \rangle$  for  $1 \leq i \leq k+1$ . For every  $x$ ,  $L_x V_i \subseteq V_{i+1}$ . Thus for  $r \in V_i$ ,

$$W(r) - r \in V_{i+1}, \quad (\exp L_x - I)W(r) \in V_{i+1}.$$

These inclusions hold after any scalar extension. Starting with  $x_0(a) = 0$ , set

$$r_{i-1}(a) = a - W(x_{i-1}(a)), \quad x_i(a) = C(x_{i-1}(a), r_{i-1}(a)).$$

Inductively  $r_{i-1} \in V_i$ , because  $(*)$  changes the residual to  $r_{i-1} - \exp(L_{x_{i-1}})W(r_{i-1}) \in V_{i+1}$ . Hence  $W(x_k(a)) = a$  formally. Since  $x_1(a) = a$  and every subsequent step multiplies the previous degree bound by at most  $k^2$ ,  $\deg x_k \leq E$ . On  $V$ ,  $x_k = \Omega$ . It has zero constant term and linear term  $a$ , the latter by comparison in  $W(x_k(a)) = a$ .

It follows that

$$P(a) = \exp(L_{\Omega(a)}) - I = \sum_{j=1}^D P_j(a)$$

has a polynomial representative of degree at most  $D$ , with homogeneous linear part  $P_1(a) = L_a$ . If  $\text{ord}_p(2) > D$ , then

$$\sum_{i=0}^{p-2} 2^{i(j-1)} = \begin{cases} -1 & j = 1, \\ 0 & 2 \leq j \leq D. \end{cases}$$

The stated average is therefore  $L_a b = a \triangleright b$  in the original additive coordinates. The scalar 2 need not be a primitive root. Finally, if  $p > 2^D$ , an order  $d \leq D$  would force  $p \mid 2^d - 1 < p$ , a contradiction. In dimension one,  $\lambda(A) \leq \mathbb{F}_p^\times$  is a  $p$ -group and is trivial.  $\square$

The affine correspondence, residual correction, and weighted coefficient extraction have prior antecedents, especially Trappeniens [106, Proposition 5.5 and Section 6]. The additional step here is the dimension-only polynomial bound using the common triangular flag. It does not assume the stronger condition  $L_{V_i} V_j \subseteq V_{i+j}$ , or strong nilpotence of the brace. Part (b), the correspondence itself, is recorded as prior work in the Notebook. The retained proof and exact controls are `research/20.92a-proof.md` and `results/20.92a-summary.json`; the controls include actual braces which are not right nilpotent.

### 6.2.10 15.65: nonrational classical-group limits

For the full families  $U(n, q)$ ,  $\mathrm{Sp}(2m, q)$ ,  $O(2m+1, q)$ ,  $O^+(2m, q)$ , and  $O^-(2m, q)$ , both limiting proportions asked about are nonrational functions of  $q$ . For symplectic and orthogonal groups the assertion holds separately on even and on odd prime powers. The Notebook calls a matrix separable when its *minimal* polynomial is squarefree; this is semisimplicity. The enumeration source uses separable for squarefree *characteristic* polynomial. We distinguish the two and prove nonrationality for both, as well as for cyclic matrices. This is a statement about functions of field size, not about irrationality of an individual probability.

All enumeration and limit-existence formulas below are those of Fulman–Neumann–Praeger (FNP) [39]. The proposed additional deduction is their analytic continuation in inverse field size and the obstruction to rational interpolation. Let  $t = 1/q$ . The count polynomials for unitary factors are

$$\begin{aligned} N_d(q) &= \frac{1}{d} \sum_{a|d} \mu(a)(q^{d/a} + 1) & (d \text{ odd}), \\ M_1(q) &= (q^2 - q - 2)/2, \\ M_d(q) &= \frac{1}{2d} \sum_{a|d} \mu(a)(q^{2d/a} - q^{d/a}) & (d > 1 \text{ odd}), \\ M_d(q) &= \frac{1}{2d} \sum_{a|d} \mu(a)q^{2d/a} & (d \text{ even}). \end{aligned}$$

For symplectic and orthogonal factors fix  $e = 1$  for even  $q$  or  $e = 2$  for odd  $q$ , and use

$$\begin{aligned} J_d(q) &= \frac{1}{2d} \sum_{\substack{a|d \\ a \text{ odd}}} \mu(a)(q^{d/a} + 1 - e), \\ K_1(q) &= (q - e - 1)/2, \\ K_d(q) &= \frac{1}{2}N(q; d) & (d > 1 \text{ odd}), \\ K_d(q) &= \frac{1}{2}(N(q; d) - J_{d/2}(q)) & (d \text{ even}), \\ N(q; d) &= \frac{1}{d} \sum_{a|d} \mu(a)q^{d/a} & (d > 1). \end{aligned}$$

These are FNP Lemmas 1.3.12 and 1.3.16. In particular

$$N_1(1/t) = 1 + 1/t, \quad J_1(1/t) = (1/t + 1 - e)/2.$$

For  $0 < |t| < 1$ , their explicit formulas give bounds by a constant times  $|t|^{-d}$  for  $N_d, J_d, K_d$ , and  $|t|^{-2d}$  for  $M_d$ . For instance, bounding the number of divisors by  $d$  suffices. These estimates will justify the infinite products as analytic functions.

**The analytic obstruction.** Suppose a holomorphic function  $a(t)$  has a zero of multiplicity  $h > 0$  at  $b \neq 0$ ,  $P(t)$  is holomorphic and nonconstant near  $b$ , and  $K(t)$  is a nonzero single-valued meromorphic germ there. Then a branch of  $K(t) \exp(P(t) \log a(t))$  cannot continue a rational germ. One circuit around  $b$  multiplies it by  $\exp(2\pi i h P(t))$ . Rational single-valuedness would make this identically one on an open set, forcing  $P' = 0$ . An integer-order zero or pole of  $K$  cannot cancel this monodromy.

Each germ constructed below has a finite nonzero value at  $t = 0$  and agrees with the indicated probability at sufficiently small positive  $t = 1/q$ . If a rational function of  $q$  agreed on all powers of a fixed prime, its transform in  $t$  would have no pole at zero and would agree with the germ by the identity theorem. The analytic obstruction therefore excludes even that restricted form of rational interpolation.

**Cyclic limits.** Put

$$A(u) = \frac{1+u-u^2}{1+u}, \quad B(u) = \frac{1-u+u^2}{1-u} = \frac{1+u^3}{1-u^2}.$$

FNP Theorems 2.1.9 and 2.2.9 give

$$C(U; q) = (1+t) \prod_{\substack{d \geq 1 \\ d \text{ odd}}} A(t^d)^{N_d(1/t)} \prod_{d \geq 1} B(t^{2d})^{M_d(1/t)},$$

$$C(\mathrm{Sp}; q) = \prod_{d \geq 1} A(t^d)^{J_d(1/t)} B(t^d)^{K_d(1/t)}.$$

Powers mean exponentials of logarithms initially vanishing at zero. The first  $A$ -factor has a simple zero at  $\alpha = (1 - \sqrt{5})/2$ . Remove it. On  $|t| < 3/4$ , every remaining  $A$ -factor has degree  $d \geq 2$ , and

$$\left| \frac{t^{2d}}{1+t^d} \right| \leq \frac{(3/4)^4}{1-(3/4)^2} = \frac{81}{112} < 1.$$

The logarithms of all  $B$ -factors are single-valued on  $|t| < 1$  by the second expression for  $B$ . Their logarithms vanish to order at least  $2d$ , or  $4d$  after substitution  $t^{2d}$ . Multiplication by the count polynomials therefore leaves summable bounds by constant multiples of  $|t|^d$  or  $|t|^{2d}$ , uniformly on compact subdiscs. All apparent singularities at zero are removable. The remaining product is consequently holomorphic and nonvanishing on  $|t| < 3/4$ . The first exponents  $N_1(1/t)$  and  $J_1(1/t)$  are nonconstant, so the obstruction at  $\alpha$  proves nonrationality. In the unitary case the real local exponent is  $1+1/\alpha = \alpha$ , also giving a direct noninteger-order proof. FNP Theorem 2.3.11 gives

$$C(O_{\mathrm{odd}}; q) = (1-t)C(\mathrm{Sp}; q), \quad C(O^{\pm}; q) = \frac{(1-t)^e + 1}{2} C(\mathrm{Sp}; q).$$

These nonzero rational multipliers transfer the conclusion to all full orthogonal families.

**A zero needed for the literal separability question.** Define holomorphic functions on  $|u| < 1$  by

$$\mathcal{U}(u) = \sum_{m \geq 0} \frac{u^{m^2}}{\prod_{i=1}^m (1 - (-u)^i)}, \quad \mathcal{L}(u) = \sum_{m \geq 0} \frac{u^{m^2}}{\prod_{i=1}^m (1 - u^i)}.$$

On  $|u| \leq r < 1$ , denominator products are bounded below by  $\prod_{i \geq 1} (1 - r^i) > 0$ , proving normal convergence. Set  $a_m(r) = r^{m^2} / \prod_{i=1}^m (1 - r^i)$ . For  $m \geq 1$ ,

$$\frac{a_{m+1}(r)}{a_m(r)} \leq \frac{r^3}{1-r^2}, \quad \sum_{m \geq 1} a_m(3/7) \leq \frac{210}{253} < 1.$$

Hence both functions are nonzero on  $|u| \leq 3/7$ . On the negative real axis,  $\mathcal{U}(-r) = 1 - a_1(r) + a_2(r) - \dots$ . For  $r \leq 3/5$  these positive terms decrease. Alternating bounds give positivity for  $r \leq 1/2$ , a lower bound  $9/56$  at  $1/2$ , and the following negative upper bound at  $3/5$ :

$$1 - a_1 + a_2 - a_3 + a_4 = -\frac{1128797279}{26656000000} < 0.$$

Thus there is a first negative zero  $b \in (-3/5, -1/2)$ , of finite positive multiplicity, with  $\mathcal{U}(t) > 0$  for  $b < t \leq 0$ . Also  $\mathcal{L}(-r) > 1 - r/(1+r) > 0$  for  $0 < r \leq 13/20$ : its alternating term ratios are bounded by  $r^3/(1-r^2) < 1$ . No simplicity or exact location of  $b$  is needed.

Put

$$E(u) = (1 - u)\mathcal{U}(u), \quad D(u) = (1 - u)\mathcal{L}(u).$$

They are nonzero on  $|u| \leq 3/7$ , and both are  $1 + O(u^2)$ . Let  $SS$  denote semisimplicity, namely squarefree minimal polynomial. FNP Theorems 3.1.13 and 3.1.15 give

$$\begin{aligned} SS(U; q) &= (1 + t) \prod_{\substack{d \geq 1 \\ d \text{ odd}}} E(t^d)^{N_d(1/t)} \prod_{d \geq 1} D(t^{2d})^{M_d(1/t)}, \\ SS(\mathrm{Sp}; q) &= (1 - t)^e \mathcal{P}(t)^e T_e(t), \\ T_e(t) &= \prod_{d \geq 1} E(t^d)^{J_d(1/t)} D(t^d)^{K_d(1/t)}, \\ \mathcal{P}(t) &= 1 + \sum_{m \geq 1} \frac{t^{m(2m+1)}}{\prod_{i=1}^m (1 - t^{2i})}. \end{aligned}$$

The function  $\mathcal{P}$  is single-valued holomorphic on the unit disc, with value one at zero. Take  $R = 13/20$ . All arguments of factors of degree at least two lie in  $|u| \leq R^2 = 169/400 < 3/7$ . Their logarithms and the count-polynomial bounds thus give normally convergent, nonvanishing remainders on  $|t| < R$ . For  $T_e$ , separate also the degree-one  $D(t)$ -factor. It has a holomorphic logarithm near  $[b, 0]$ , by the positivity just proved; multiplication by  $K_1(1/t)$  is removable at zero since  $\log D(t) = O(t^2)$ . The only required branching at  $b$  comes from  $E(t)$ , with nonconstant exponent  $N_1(1/t)$  or  $J_1(1/t)$ . The prefactors can have only integer-order zeros there. The obstruction proves both nonrationality claims.

For the orthogonal case put

$$\mathcal{Q}(t) = 1 + \sum_{m \geq 1} \frac{t^{m(2m-1)}}{\prod_{i=1}^m (1 - t^{2i})}.$$

It is likewise holomorphic on the unit disc, with value one at zero. The reciprocal orders of the two even-dimensional full orthogonal groups sum to the terms of  $\mathcal{Q}(1/q) - 1$ . FNP Theorem 3.1.18 gives

$$\begin{aligned} SS(O_{\mathrm{odd}}; q) &= (1 - t)^e \mathcal{Q}(t)^{e-1} \mathcal{P}(t) T_e(t), \\ SS(O^{\pm}; q) &= \frac{(1 - t)^e}{2} (\mathcal{Q}(t)^e + (e - 1) \mathcal{P}(t)^2) T_e(t). \end{aligned}$$

The extra factors are single-valued and not identically zero: at zero they equal 1 and  $e/2$ . Their possible finite-order zeros cannot cancel the monodromy of  $E(t)^{J_1(1/t)}$ . This proves the literal Notebook assertion for all the stated families.

**The other meaning of separability.** For completeness let  $RS$  mean squarefree characteristic polynomial, and set  $V(u) = (1 - u)(1 + 2u)/(1 + u)$ . FNP Theorems 2.1.3, 2.2.3 and 2.3.4 give

$$\begin{aligned} RS(U; q) &= (1 + t) \prod_{\substack{d \geq 1 \\ d \text{ odd}}} V(t^d)^{N_d(1/t)}, \\ RS(\mathrm{Sp}; q) &= (1 - t)^e \prod_{d \geq 1} V(t^d)^{J_d(1/t)}, \\ RS(O_{\mathrm{odd}}; q) &= RS(\mathrm{Sp}; q), \\ RS(O^{\pm}; q) &= 2^{e-2} RS(\mathrm{Sp}; q). \end{aligned}$$

The first  $V$ -factor vanishes at  $-1/2$ . All higher factors have zero-free logarithms on  $|t| < 13/20$ , and the same estimates prove normal convergence. The first exponents are nonconstant, so the branching obstruction applies even when their values at  $-1/2$  happen to be integers. This proves nonrationality for the source's convention too.

The symplectic semisimple formula uses  $M^*(q; d)$ , as in FNP Theorem 3.1.15 and its underlying product; an intermediate display in Lemma 3.1.9(e) prints  $M^*(q; 2d)$ . The archive records that index discrepancy and a separate coefficient discrepancy. No numerical coefficient table is a premise here. The distinction between the dimension variable of FNP's generating functions and our field-size variable  $t$  is essential. See `research/15.65-proof.md`, `research/15.65-classical-proof.md`, and their source audits.

### 6.3 Infinite groups and residual properties

Periodic constructions, profinite examples, words and infinite permutation quotients.

#### 6.3.1 10.62: a periodic perfect group generated by involutions

**Proposition 6.8.** *For every sufficiently large odd prime  $p$ , there is an infinite three-generated perfect group  $G$  of exponent  $2p$ , with no subgroup of order four, whose involutions form one conjugacy class and generate  $G$ . Any two distinct involutions have product of order  $p$ .*

The deep existence input is Amelio's geometric quotient construction [3, Proposition 6.1, Theorem 6.7, Remark 6.8]. We use its construction as well as its final theorem: the first quotient may impose a prescribed eligible primitive odd-power relation. This feature is needed to eliminate the quotient of order two. The cone-off and small-cancellation estimates underlying that result are imported, not reproved by the experiment.

*Proof.* Start with

$$F = \langle a, b, c \mid a^2 = b^2 = c^2 = 1 \rangle = C_2 * C_2 * C_2$$

acting by left multiplication on its trivalent Cayley tree with unit-length edges. Vertex stabilizers are trivial; a nontrivial point stabilizer is the order-two group interchanging an edge's endpoints. A finite subgroup fixes the center of a finite invariant subtree and therefore has order at most two. The action is acylindrical: the number of elements moving one point at most  $R$  is bounded by the size of a uniform vertex ball of radius  $R + 1$ . This also shows that elliptic subgroups are finite. The translations  $ab, ac$  have different endpoint pairs, so the action is non-elementary. Every nonelliptic element translates by at least two, and the injectivity radius is exactly two.

Amelio calls an action tame when there is no subgroup of order four and each maximal loxodromic subgroup has finite normal subgroup of order at most two. Here an axis stabilizer acts faithfully on a simplicial line, since it fixes no vertex nontrivially; it is cyclic or infinite dihedral and has trivial finite normal subgroup. Thus the action is tame. The additional numerical invariants in Theorem 6.7 are finite. More precisely, its invariant  $\nu$  is at most one: if  $h$  is loxodromic and  $\langle g, h^{-1}gh \rangle$  is elliptic, then  $g \neq 1$  would force  $h$  to fix the unique midpoint fixed by  $g$ , a contradiction. Thus  $\tau = \max\{\nu, 3\} = 3$ . Regard the tree as  $\delta$ -hyperbolic for  $0 < \delta < \min\{1/L_S, 1/100\}$ , with  $L_S$  the source's constant. Every short element used in its overlap invariant  $\Omega$  is then elliptic. The  $13\delta$ -enlargement of its almost-fixed set lies within  $17\delta$  of its unique midpoint. Distinct midpoints are at least one apart, while a non-elementary tuple contains two distinct nontrivial elliptic elements. Hence the relevant intersections are empty and  $\Omega = 0$ . Using positive  $\delta$  respects the source's strict displacement inequalities. Take  $Q$  to be all loxodromic elements, explicitly permitted as a strongly stable family by Remark 6.6.

The word  $w = abc$  is primitive in its maximal loxodromic subgroup. Its axis has periodic label pattern  $abcabc\cdots$ , whose least positive period is three; free vertex action makes its translation stabilizer exactly  $\langle w \rangle$ . An orientation reversal would identify  $abc$  with  $cba$  up to cyclic rotation, which is impossible. Thus the full axis stabilizer is  $\langle w \rangle$ .

Choose an initial scale  $\lambda > 0$  sufficiently small that  $3\lambda \leq L_S\delta_1$ , together with the initialization conditions in Theorem 6.7. Remark 6.8 permits this smaller scale at the cost of increasing the critical exponent. The new injectivity radius is still  $2\lambda > 0$ ; the required lower bound tends to zero as the critical exponent grows. All choices are made before choosing  $p$ . Explicitly, the source's Remark 6.8 illustrates the extra condition  $2\lambda \leq L_S\delta_1$ ; the word  $abc$  requires  $3\lambda \leq L_S\delta_1$  instead. The permission to decrease  $\lambda$  is the part we use. For a sufficiently large odd prime  $p$ , use the exponent family  $\{p\}$ . The primitive cyclic subgroup  $\langle abc \rangle$  is eligible for the first family; start with it and extend to a maximal admissible selection, choosing one representative for each eligible maximal loxodromic subgroup up to conjugacy. Proposition 6.1 kills  $w^p$ . Neither the extra relation nor the scale  $3\lambda$  is asserted as part of the headline statement of Theorem 6.7.

The theorem constructs epimorphisms

$$F = F_0 \twoheadrightarrow F_1 \twoheadrightarrow F_2 \twoheadrightarrow \cdots \twoheadrightarrow G$$

with infinite limit, embedding the initial elliptic subgroups, and with every initial loxodromic element mapping to an element of order dividing  $2p$  or to an initial elliptic image. Consequently  $a, b, c$  survive as involutions generating  $G$ , every element of  $G$  has order dividing  $2p$ , and

$$(abc)^p = 1. \tag{****}$$

Each  $F_i$  is tame and has no subgroup of order four. This property passes to the limit here: a finite subgroup of  $G$  has finitely many multiplication-table relations; choose lifts of its elements, and a stage at which all those relations hold. They define an embedding into that stage, because composition to  $G$  is the original inclusion. Thus  $G$  also has no subgroup of order four.

Let  $x, y$  be distinct involutions of  $G$ , including any not identified beforehand with original involutions. Put  $t = xy$ . If  $t$  had even order  $d$ , then  $z = t^{d/2}$  would be an involution centralized by  $x$ . Also  $x \notin \langle t \rangle$ , since otherwise  $y$  would lie there too and a cyclic group has only one involution. Thus  $\langle x, z \rangle \cong C_2^2$ , a contradiction. Hence  $|t| = p$ . With  $k = (p+1)/2$ ,

$$t^{-k}xt^k = xt^{2k} = y,$$

so all involutions are conjugate. There are at least two, since otherwise  $G$  would be cyclic of order two. The exponent is exactly  $2p$ .

In the abelianization all three generating involutions have the same image  $\epsilon$ , of order dividing two. Relation (\*\*\*\*) gives  $3p\epsilon = 0$ , and oddness implies  $\epsilon = 0$ . Thus  $G$  is perfect, and in particular has no subgroup of index two.  $\square$

No numerical threshold for  $p$ , finite presentation, or simplicity claim follows here. The new candidate is the application with the selected odd relation; the geometric construction is Amelio's. Finite dihedral and abelianization controls in `results/10.62-summary.json` check only the elementary deductions. They do not construct this infinite group. The complete parameter and source audit is `research/10.62-review.md`.

### 6.3.2 4.75: noncentral involutions modulo the odd-order radical

Use the same group  $G$  as in Section 6.3.1; this is a second consequence of one geometric construction. Every 2-subgroup has exponent two because  $G$  has exponent  $2p$ . It is therefore

abelian, and the absence of a subgroup of order four forces its order to be at most two. The maximal 2-subgroups are exactly the cyclic groups generated by involutions, and are locally cyclic. No infinite-group Sylow theorem is required.

For a periodic group,  $O_{2'}(G)$  denotes the largest normal subgroup all of whose elements have odd order. It exists: if  $A, B$  are normal odd-order-element subgroups, any involution of  $AB$  has trivial image in  $AB/A$  and would lie in  $A$ , a contradiction. Periodicity then makes all elements of  $AB$  odd-order. The directed union of finite products gives the largest such subgroup. Write it as  $O$ . Every involution  $i$  has nontrivial image in  $G/O$ .

If  $iO$  were central, all conjugates of  $i$  would have the same image. They generate  $G$ , so  $G/O = \langle iO \rangle$  would be cyclic. It is also a quotient of the perfect group  $G$ , and hence perfect, forcing it to be trivial. This contradicts  $i \notin O$ . Thus every involution has noncentral image in  $G/O_{2'}(G)$ , answering 4.75 negatively even with Sylow 2-subgroups of order two. Determination of the radical itself is unnecessary. See `research/4.75-proof.md`.

### 6.3.3 Infinite non-prosoluble profinite CN-groups: Problem 20.90

The question asks for an infinite, topologically finitely generated profinite group with pronilpotent centralizers of nonidentity elements, which is not prosoluble. The archive gives two constructions, counted as one answer. The first has abelian centralizers.

Let  $R = \mathbb{F}_4[[t]]$ , with  $\omega^2 + \omega + 1 = 0$ , and put

$$X = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & 1 \\ 1 & \omega + t \end{pmatrix}, \quad G = \overline{\langle X, Y \rangle} \leq \mathrm{SL}_2(R).$$

The congruence topology makes  $\mathrm{SL}_2(R)$  profinite, hence so is its closed subgroup  $G$ , which is two-generated by definition. Writing  $z = \omega + t$ , define  $P_0 = 0, P_1 = 1, P_{n+1} = zP_n + P_{n-1}$ . Then

$$Y^n = \begin{pmatrix} P_{n-1} & P_n \\ P_n & P_{n+1} \end{pmatrix}.$$

The polynomial  $P_{n+1}$  is monic of degree  $n$  for  $n \geq 1$ . Thus no positive power of  $Y$  is the identity and  $G$  is infinite.

Reduction modulo  $t$  is onto  $\mathrm{SL}_2(\mathbb{F}_4)$ . To check this directly, write  $x, y$  for the reduced matrices and  $U(a) = \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix}, L(a) = \begin{pmatrix} 1 & 0 \\ a & 1 \end{pmatrix}$ . Matrix multiplication gives

$$yxy^{-1} = L(1), \quad d = y^{-1}xy^{-2} = \mathrm{diag}(\omega, \omega^2).$$

Conjugation by  $d$  multiplies the parameters of  $U$  and  $L$  by  $\omega^2$  and  $\omega$ , respectively. Addition of parameters therefore supplies all elementary matrices, which generate  $\mathrm{SL}_2(\mathbb{F}_4)$  by row reduction. The commutators

$$dU(a)d^{-1}U(a)^{-1} = U(\omega a), \quad dL(a)d^{-1}L(a)^{-1} = L(\omega^2 a)$$

show that this nontrivial finite group is perfect. Hence the continuous quotient is nonsoluble, and  $G$  is not prosoluble. This quotient is the usual group  $A_5$ .

Regard  $G$  inside  $\mathrm{SL}_2(\mathbb{F}_4((t)))$ . In characteristic two, a scalar determinant-one matrix is the identity. Every nonidentity  $g \in G$  is therefore nonscalar. The cyclic-vector argument of Section 6.2.1 identifies its matrix centralizer with the commutative algebra  $\mathbb{F}_4((t))[g]$ . Thus  $C_G(g)$  is abelian. It is closed, hence an abelian profinite group and in particular pronilpotent. This field-level centralizer argument is classical; its characteristic-two form appears in [38]. We do not claim it as new.

Thus the example is a CA-group (its nonidentity centralizers are abelian), hence a CN-group (those centralizers are pronilpotent). It is not necessary that finite quotients be CN-groups. Indeed, the archive's image modulo  $t^2$  has order 240 and contains the nonidentity

scalar  $(1+t)I$ , whose centralizer is the whole nonsoluble image. The theorem concerns centralizers in the inverse-limit group itself.

**A characteristic-zero companion.** Let  $O = \mathbb{Z}_2[\omega] \subset K = \mathbb{Q}_2(\omega)$  with the same quadratic relation, and take  $H = \mathrm{SL}_2(O)/\{I, -I\}$ . The four matrices  $U(1), U(\omega), L(1), L(\omega)$  topologically generate  $\mathrm{SL}_2(O)$ : their powers generate parameters in the dense subring  $\mathbb{Z} + \mathbb{Z}\omega$ , and elementary row reduction works over the local ring  $O$ . The image of  $U(1)$  has infinite order, and reduction gives the nonsoluble quotient  $\mathrm{SL}_2(\mathbb{F}_4)$ .

For a nonidentity  $h \in H$ , a lift  $g \in \mathrm{SL}_2(O)$  is nonscalar. Its strict centralizer  $C_{\mathrm{SL}_2(O)}(g)$  is abelian by the same matrix lemma. The preimage of  $C_H(h)$  consists of  $s$  with  $sgs^{-1} = g$  or  $-g$ ; the possible sign is a homomorphism with that strict centralizer as kernel. If  $\mathrm{tr} g \neq 0$ , only the positive sign occurs, so the projective centralizer is abelian.

If  $\mathrm{tr} g = 0$ , its reduction  $\bar{g}$  cannot be the identity. Otherwise  $g = I + 2A$  gives  $\mathrm{tr} A = -1$ , and

$$1 = \det(I + 2A) = 1 + 2 \mathrm{tr} A + 4 \det A$$

would give  $2 \det A = 1$  in  $O$ . Thus  $\bar{g}$  is a nonidentity unipotent of  $\mathrm{SL}_2(\mathbb{F}_4)$ , with centralizer of order four (the matrices  $U(a)$  after a change of basis). The strict centralizer has reduction a 2-group and kernel in the pro-2 congruence subgroup, whose successive quotients embed in  $M_2(\mathbb{F}_4)^+$ . It is pro-2. The extension by a sign group and subsequent quotient by  $\{I, -I\}$  are also pro-2. Every projective centralizer is therefore pronilpotent. Sources: `research/20.90-proof.md` and `research/20.90-2adic-proof.md`.

### 6.3.4 15.92: continuum many alternating-residual triangle quotients

For every  $r > 6$ , the triangle group

$$\Gamma_r = \langle x, y \mid x^2 = y^3 = (xy)^r = 1 \rangle$$

has continuum many pairwise nonisomorphic infinite quotients in which every nonidentity element survives an epimorphism onto a finite alternating group. The proof combines Conder's finite permutation constructions with the following separating-kernel argument.

**Lemma 6.9.** *Suppose a finitely generated group  $\Gamma$  has epimorphisms  $\theta_i : \Gamma \rightarrow A_{n_i}$ , with  $n_i \rightarrow \infty$ , and one element  $v$  whose image is always nontrivial and moves at most a fixed number  $S$  of points. Then  $\Gamma$  has continuum many pairwise nonisomorphic infinite alternating-residual quotients.*

*Proof.* By diagonal subsequence selection, arrange that each element of the countable group  $\Gamma$  is eventually always trivial or eventually always nontrivial under  $\theta_i$ . The eventually trivial elements form a normal subgroup  $R$ ; let  $G = \Gamma/R$ , with  $1 \neq \bar{v} \in G$ . The action on a nonprincipal ultraproduct of the underlying finite sets is faithful: a word eventually nontrivial has a sequence of moved points. The support of  $\bar{v}$  has at most  $S$  elements, since  $S+1$  distinct moved ultraproduct points would contradict the coordinate support bound on an ultrafilter-large set.

Let  $N = \langle \bar{v}^G \rangle$ . On the union  $\Omega_*$  of all translates of  $\mathrm{supp}(\bar{v})$ , the group  $N$  acts faithfully by finitary permutations. There are at most  $S$   $G$ -orbits. Because  $N \trianglelefteq G$ , its orbits within each  $G$ -orbit have equal size. Thus a finite  $M$  bounds all finite  $N$ -orbit sizes. Let  $\eta$  be the action of  $N$  on the union of its finite orbits, and put  $L = \ker \eta$ . The exponent of  $\eta(N)$  divides  $M!$ .

Suppose  $\phi : G \rightarrow A_m$  is onto,  $m \geq 5$ , and  $\phi(\bar{v}) \neq 1$ . Then  $\phi(N) = A_m$ . Set  $H = \ker(\phi|_N)$ , a finite-index normal subgroup of  $N$ . Every  $H$ -orbit inside an infinite  $N$ -orbit

is infinite, since the latter is a finite union of translates of the former. For  $x \in L$  and  $y \in N$ , there is  $h \in H$  moving  $\text{supp}(x)$  disjoint from  $\text{supp}(y)$ . Indeed  $H$  is locally finite as a finitary permutation group. For finite sets  $A = \text{supp}(x)$ ,  $B = \text{supp}(y)$ , choose finitely many elements of  $H$  so that the finite subgroup they generate has every orbit of  $a \in A$  larger than  $|A||B|$ . A uniform element of this finite subgroup avoids all equalities  $h(a) = b$  by the union bound. Empty supports are immediate. Thus  $h x h^{-1}$  commutes with  $y$ , and  $\phi(h) = 1$  gives  $\phi(x) \in Z(\phi(N)) = 1$ . Therefore  $A_m$  is a quotient of  $\eta(N)$ , of exponent dividing  $M!$ . For  $m$  at least an odd prime greater than  $M$ , this is impossible because  $A_m$  contains a cycle of that prime length. All sufficiently large alternating quotients of  $G$  consequently kill  $\bar{v}$ .

For sufficiently large  $i$ ,  $\theta_i$  cannot factor through  $G$ . Hence  $\theta_i(R) \neq 1$ , and normality and simplicity give  $\theta_i(R) = A_{n_i}$ . Inductively select indices  $i_j$  and elements  $r_j \in R$  so that

$$\theta_{i_k}(r_j) \neq 1 \iff k = j.$$

At step  $j$ , choose  $i_j$  beyond the cutoffs for all preceding  $r_k$ , with  $|A_{n_{i_j}}| > \prod_{k < j} |A_{n_{i_k}}|$ . The subgroup  $L_j = R \cap \bigcap_{k < j} \ker \theta_{i_k}$  has index in  $R$  at most this product, so it cannot be killed by the surjection  $\theta_{i_j}|_R$ . Choose  $r_j \in L_j$  with nontrivial image. Since  $r_j \in R$ , all sufficiently late maps kill it, supplying the next cutoff.

For each infinite  $I \subseteq \mathbb{N}$ , put  $K_I = \bigcap_{j \in I} \ker \theta_{i_j}$ . The witnesses give  $r_j \notin K_I$  exactly when  $j \in I$ ; hence there are continuum many different kernels. Each quotient embeds in the corresponding product of alternating groups with onto coordinates, and is infinite because their orders are unbounded. For each fixed countable quotient isomorphism type, there are only countably many epimorphisms from a finitely generated  $\Gamma$ , as images of a finite generating tuple determine a map. The continuum many kernels therefore give continuum many types.  $\square$

We specify the finite input, which is stronger than mere existence of alternating quotients. Write  $r = h + 6d$ ,  $h \in \{7, 8, 9, 10, 11, 12\}$ ,  $d \geq 0$ , and choose

$$p = \begin{cases} 11 & h = 7, 8, 11, \\ 13 & h = 9, 10, 12. \end{cases}$$

Conder's thesis [27, Chapter 5, pp. 117–120] constructs arbitrarily large actions of the extended triangle group with generators  $x_n, y_n, t_n$  satisfying

$$x_n^2 = y_n^3 = (x_n y_n)^r = t_n^2 = (x_n t_n)^2 = (y_n t_n)^2 = 1.$$

The parity choice makes  $\langle x_n, y_n \rangle = A_n$ , including when  $r$  is even. In these same actions  $w_n = x_n y_n t_n$  has one  $p$ -cycle, no other cycle length divisible by  $p$ , and all cycle lengths divide  $720720 = 2^4 3^2 5 7 11 13$ . The displayed choice of 11 or 13 is the length of the unique exceptional prime cycle in the respective diagram; it is not a free choice. These cycle and parity facts are read from the construction, not inferred from its headline generation theorem. The finite transcription, diagram reconstruction and 150 cycle controls are indexed in `results/15.92-summary.json` in the public snapshot (Appendix F); the external follow-up reports replaying both controls.

The relations give  $w_n^2 = x_n y_n x_n y_n^{-1}$ . The even integer  $E = 720720/p$  is coprime to  $p$  and divisible by every other occurring cycle length, so  $w_n^E$  is a single nontrivial  $p$ -cycle. Thus the one fixed word

$$v_r = (x y x y^{-1})^{32760} \quad (p = 11), \quad v_r = (x y x y^{-1})^{27720} \quad (p = 13)$$

has nontrivial support at most 13 in all the selected alternating images of the ordinary triangle group. The lemma proves the result. The presentation printed in the Notebook omits

the final = 1; that literal group also surjects onto  $\Gamma_r$ , so the same quotients answer it. See `research/15.92-proof.md` and its source-data controls. Neither residual alternatingness of  $\Gamma_r$  itself under the chosen family nor independence of arbitrary quotient kernels is assumed.

### 6.3.5 16.38: periodic subgroups in a set product

**Proposition 6.10.** *If  $G$  is soluble and  $A, B \leq G$  are periodic, every subgroup contained in the set  $AB$  is periodic.*

*Proof.* First, periodic soluble groups are locally finite. A finitely generated one has finite quotient by its last nontrivial derived subgroup, by induction on derived length. That subgroup is finitely generated by Schreier's theorem, and abelian periodic, hence finite.

We need a cocycle observation. Let a locally finite group  $C$  act on a torsion-free abelian group  $D$ , and let  $\delta(cd) = \delta(c) + c\delta(d)$ . The image of  $\delta$  cannot contain both  $t$  and  $2t$  for  $t \neq 0$ . Indeed, choose preimages  $c_1, c_2$ , and put  $H = \langle c_1, c_2 \rangle$ , which is finite. In  $\mathbb{Q} \otimes D$ , averaging gives

$$v = |H|^{-1} \sum_{h \in H} \delta(h), \quad \delta(h) = v - hv.$$

Average an inner product on the finite-dimensional real span of  $Hv$ . The vectors  $v, v-t, v-2t$  then have equal norm. Expanding the last two equalities yields  $2\langle v, t \rangle = \|t\|^2$  and  $4\langle v, t \rangle = 4\|t\|^2$ , so  $t = 0$ . This averaging is over the finite group  $H$ , not over all of  $C$ .

Suppose now that  $D \trianglelefteq E$  is torsion-free abelian and  $A, B \leq E$  are locally finite. For  $\pi : E \rightarrow E/D$ , let  $C = \pi(A) \cap \pi(B)$ . The preimages of  $C$  in  $A, B$  map isomorphically to  $C$ , since their intersections with  $D$  are trivial. Write their inverse sections as  $a, b$ . In  $\pi^{-1}(C) = D \rtimes a(C)$ , put  $b(c) = \delta(c)a(c)$ ; then  $\delta$  is a cocycle and  $C$  is locally finite. A factorization lying in  $D$  must have inverse quotient coordinates, and hence

$$D \cap AB = \{a(c)b(c)^{-1} : c \in C\} = -\delta(C).$$

The observation shows that this set cannot contain both  $g$  and  $g^2$  for any  $1 \neq g \in D$ .

It suffices to prove that  $\langle g \rangle \subseteq AB$  implies that  $g$  has finite order. Induct on the derived length of  $G$ . The abelian case is immediate. Otherwise let  $D$  be its last nontrivial derived subgroup. Induction in  $G/D$  gives  $g^r \in D$  for some  $r > 0$ . The torsion subgroup  $T$  of  $D$  is normal in  $G$ , and  $D/T$  is torsion-free abelian. Apply the preceding observation in  $G/T$  to  $g^r T$  and  $g^{2r} T$ , both of which belong to the appropriate set product. It gives  $g^r \in T$ , proving the claim.  $\square$

The set  $AB$  itself can contain elements of infinite order. For example, let  $D = \bigoplus_{i \geq 1} \mathbb{Z}$ , let  $C = \bigoplus_{i \geq 1} C_2$  act by coordinate sign changes, and set  $\delta(c)_i = c_i$  for  $c_i \in \{0, 1\}$ . This is a cocycle. In  $D \rtimes C$ , the two complements

$$A = \{(0, c)\}, \quad B = \{(\delta(c), c)\}$$

are elementary abelian, whereas  $D \cap AB = -\delta(C)$  contains nonzero infinite-order elements and excludes their doubles. The theorem concerns subgroups contained in  $AB$ . It also does not assert the locally soluble version of Problem 9.65. See `research/16.38-proof.md`.

### 6.3.6 13.42: a tensor completion that is not locally nilpotent

Let  $A = \mathbb{Z}[t]$  and  $N = F(x, y)/\gamma_4 F(x, y)$ . Its unrestricted tensor  $A$ -completion is not locally nilpotent. The Notebook refers to Myasnikov–Remeslennikov's foundational paper [82]. Here we use the power, conjugation, commuting-product axioms and tensor-completion universal

property as restated in [83, Definitions 1–2]. The universal category allows scalar torsion and does not restrict targets to nilpotent groups. This distinction is part of the claim.

Let  $L$  be the rational BCH group of the Lie algebra with basis  $X, Y, U, V, W$ , brackets

$$[X, Y] = U, \quad [X, U] = V, \quad [Y, U] = W,$$

and all other basic brackets zero. Coordinates  $(a, b, c, d, e)$  denote  $\exp(aX + bY + cU + dV + eW)$ , and  $\beta(a, b, c, d, e) = b$  is a homomorphism  $L \rightarrow (\mathbb{Q}, +)$ . These are finite polynomial operations: realize the Lie algebra in the free associative algebra on  $X, Y$  truncated above degree three and use  $\exp Z = 1 + Z + Z^2/2 + Z^3/6$ . For  $l = (a, b, c, d, e)$ ,  $h = (A, B, C, D, E)$ , BCH gives the first three coordinates of  $l^h = h^{-1}lh$  as

$$(a, b, c + aB - bA).$$

It also gives the commuting criterion

$$[l, h] = 1 \iff aB - bA = aC - cA = bC - cB = 0.$$

For example, equality of the degree-two coordinates in the two products first gives  $aB - bA = 0$ ; then equality in degree three gives the other two minors. Thus commuting elements have proportional triples  $(a, b, c)$ ,  $(A, B, C)$ , including the zero cases.

Let  $M = \mathbb{Q}[\mathbb{Q}]$  additively, with basis  $T^q$ ,  $q \in \mathbb{Q}$ . Give it the right  $L$ -action  $v^l = vT^{\beta(l)}$ , and define

$$F(a, b, c, d, e) = \begin{cases} aT^{c/a} & b = 0, a \neq 0, \\ 0 & \text{otherwise.} \end{cases}$$

The conjugation formula, coordinate powers, and proportional-triple criterion prove, respectively,

$$\begin{aligned} F(l^h) &= F(l)T^{\beta(h)}, & F(l^n) &= nF(l), \\ F(lh) &= F(l) + F(h) \quad ([l, h] = 1), & F(l)T^{\beta(l)} &= F(l). \end{aligned}$$

For the third assertion, the defining function of a triple is homogeneous over  $\mathbb{Q}$ , hence additive on any line. It is not asserted to be additive on arbitrary pairs.

Form the right-action semidirect product with multiplication

$$H = L \ltimes M, \quad (l, v)(h, w) = (lh, vT^{\beta(h)} + w),$$

write  $i(v) = (1, v)$ , and put  $f(l, v) = i(F(l))$ . Then

$$f(g^h) = f(g)^h, \quad [g, f(g)] = 1, \quad f(gh) = f(g)f(h) \text{ if } [g, h] = 1,$$

and, for all integers  $n, m$ ,

$$f(g^n f(g)^m) = f(g)^n.$$

The last identity follows by projecting to  $L$ . If  $g, h$  commute, equivariance also gives  $[f(g), h] = [g, f(h)] = 1$ .

For  $P(t) \in \mathbb{Z}[t]$ , define

$$g^P = g^{P(0)} f(g)^{P'(0)}.$$

This supplies all the  $A$ -group axioms. Unit, zero and addition follow from  $[g, f(g)] = 1$ . If  $P(0) = a, P'(0) = b, Q(0) = c, Q'(0) = d$ , then

$$(g^P)^Q = g^{ac} f(g)^{bc+ad} = g^{PQ},$$

using  $f(g^P) = f(g)^a$ . Equivariance proves the conjugation axiom. For commuting  $g, h$ , the cross-commutations just established allow rearrangement to give  $(gh)^P = g^P h^P$ . Thus  $H$

is an  $A$ -group in the required unrestricted category. It is torsion-free as an ordinary group, being an extension of torsion-free groups, although its scalar action factors through  $A/(t^2)$ .

Set  $x = (\exp X, 0)$ ,  $y = (\exp Y, 0)$ . They generate a group of class at most three, so the map from the free nilpotent group  $N$  extends by the tensor-completion universal property to an  $A$ -homomorphism  $N^A \rightarrow H$ . But  $x^t = i(T^0)$ , and direct multiplication gives

$$[i(v), y] = i(v(T - 1)).$$

For every  $r \geq 1$ , the image of  $[x^t, y, \dots, y]$ , with  $r$  copies of  $y$ , is  $i((T - 1)^r) \neq 1$ : its coefficient at  $T^r$  is one. The fixed pair  $x^t, y$  therefore generates a nonnilpotent subgroup of  $N^A$ , proving the claim. The map need not be onto  $H$ , and injectivity of  $N \rightarrow H$  is unnecessary. Completion within nilpotent  $A$ -groups would be a different construction. See `research/13.42-proof.md`.

### 6.3.7 14.26: restricted wreath products in a nilpotent quasivariety

Let  $\mathcal{C}$  be all torsion-free nilpotent groups, with no fixed class bound. If  $G \in Q(\mathcal{C})$ , then  $G \wr \mathbb{Z} \in Q(\mathcal{C})$  for the restricted wreath product. This is the direct wreath product of the question: Budkin's definition uses finite-support functions [20, Section 1].

A finite set  $B \ni 1$  locally embeds in a group if an injective identity-preserving map preserves every product staying in  $B$ . Since  $\mathcal{C}$  is closed under finite direct products,

$$G \in Q(\mathcal{C}) \iff \text{every finite } B \subseteq G, 1 \in B, \text{ locally embeds in some } C \in \mathcal{C}.$$

Indeed, write down the finite multiplication table of  $B$ . For each unequal pair, some realization in  $\mathcal{C}$  must separate it, or the corresponding quasiidentity would be violated by  $G$ . The product of these finitely many realizations separates all pairs. Conversely, a quasiidentity violation involves only finitely many intermediate word values, which a local embedding would preserve.

We construct the needed local models. Fix  $c, R \geq 1$  and a finite nonempty set  $I \subseteq \mathbb{Z}$  of size  $M$ , with diameter at most  $R$ . Choose  $m \geq 2R(c - 1) + M - 1$ , a rational vector space with basis  $a_0, \dots, a_m$ , and a nilpotent derivation determined by  $D(a_j) = a_{j+1}$ ,  $D(a_m) = 0$ . Put

$$f(z) = \sum_{j=0}^m z^j a_j / j!.$$

In its symmetric algebra let  $J$  be the ideal generated by all coefficients in  $z$  of  $f(z)f(z + d)$ ,  $1 \leq d \leq R$ , and form

$$\mathcal{A} = \text{Sym}\langle a_0, \dots, a_m \rangle / (J + \text{Sym}^{>c}).$$

This is finite-dimensional, commutative, and graded by ordinary degree. The ideal is  $D$ -stable because  $D(f(z)f(z + d)) = \frac{d}{dz}(f(z)f(z + d))$ . On the degree-truncated algebra,  $D$  raises the total index of each monomial, bounded by  $cm$ , so  $D^{cm+1} = 0$ , also after quotienting. Thus  $\alpha = \exp D$  is an automorphism, with

$$\alpha^s f(i) = f(i + s), \quad f(i)f(j) = 0 \quad (0 < |i - j| \leq R). \quad (\text{a})$$

For each  $1 \leq k \leq c$ , the vectors  $f(i)^k$ ,  $i \in I$ , are linearly independent in  $\mathcal{A}$ . To prove it, a linear functional on  $\text{Sym}^k$  corresponds to a symmetric polynomial  $Q(z_1, \dots, z_k) = \ell(f(z_1) \cdots f(z_k))$  of degree at most  $m$  in each variable. Conversely the coefficients of any such polynomial prescribe  $\ell$ , with factorial denominators. If  $Q$  vanishes when  $z_2 = z_1 + d$ ,  $1 \leq d \leq R$ , the functional kills the degree- $k$  part of  $J$ . Coefficient extraction and Vandermonde

spanning by the other  $f(z_j)$ 's justify this for every quadratic generator multiplied by an arbitrary degree- $k - 2$  term. Define

$$\Delta_k = \prod_{a < b} \prod_{d=1}^R ((z_a - z_b)^2 - d^2), \quad C_k = \prod_{a < b} \prod_{d=1}^R (-d^2).$$

If  $L_i$  is the degree- $(M - 1)$  Lagrange polynomial on  $I$ , then

$$Q_i = \frac{\Delta_k}{C_k} L_i \left( \frac{z_1 + \cdots + z_k}{k} \right)$$

has the required degree bound in each variable, kills those relations, and has  $Q_i(j, \dots, j) = \delta_{ij}$ . Empty products handle  $k = 1$ ; the cutoff adds no relation in these degrees. This proves independence.

Let  $\mathcal{A}_+$  be the positive-degree ideal,  $U$  the upper unitriangular group of matrix size  $(c + 1) \times (c + 1)$  with strict entries in  $\mathcal{A}_+$ , and

$$H = U \rtimes \langle t \rangle, \quad tut^{-1} = \alpha(u).$$

For  $g \in \text{UT}_{c+1}(\mathbb{Q})$ , define  $\rho_i(g)$  by the off-diagonal entries

$$\rho_i(g)_{ab} = g_{ab} f(i)^{b-a} \quad (a < b).$$

Superdiagonal distances add, so this is a homomorphism, and the independence above makes it injective for  $i \in I$ . Equation (a) gives  $\alpha \rho_i = \rho_{i+1}$ . For distinct positions in  $I$ , the corresponding strictly upper matrices multiply to zero in either order. Thus

$$\prod_{i \in I} \rho_i(g_i) = 1 + \sum_{i \in I} (\rho_i(g_i) - 1)$$

and the independence of each  $f(i)^k$  makes this direct-product map injective.

The group  $H$  is torsion-free nilpotent. On each successive superdiagonal quotient of  $U$ , conjugation by  $U$  is trivial and  $t$  acts as  $\alpha$ . Writing  $\delta = \alpha - 1$ , we have  $\delta^{cm+1} = 0$ . Insert the images of successive powers of  $\delta$  in every superdiagonal quotient. Their inverse images give a central series, of length at most  $c(cm + 1) + 1$  including  $H/U$ . Torsion projects trivially to  $H/U = \mathbb{Z}$ ; in  $U$ , a positive power multiplies the first nonzero superdiagonal by a positive integer, so cannot kill it over  $\mathbb{Q}$ .

Now take a finite subset  $S \ni 1$  of  $N \wr \mathbb{Z}$ , with  $N \in \mathcal{C}$ . The finitely many base coordinates generate a finitely generated torsion-free nilpotent group, which embeds in  $\text{UT}_{c+1}(\mathbb{Q})$  for some  $c$ . This classical embedding input follows from Jennings' augmentation-separation theorem [62, Theorem 5.2]: a finite-dimensional quotient  $\mathbb{Q}[N]/\Delta^s$  is faithful, and the  $\Delta$ -filtration makes the action unitriangular. Choose  $I$  to contain all supports in  $S$  and all shifted supports needed in products  $uv = w$  within  $S$ ; take  $I = \{0\}$  if empty. With  $R$  bounding its diameter, the map

$$(f, n) \mapsto \left( \prod_i \rho_i(f(i)) \right) t^n$$

preserves those products: conjugation by  $t^n$  shifts the indices, different positions commute, and equal positions multiply correctly. It is injective by the quotient  $H/U$  and the direct-product independence above. Hence every such  $S$  locally embeds in  $\mathcal{C}$ .

Finally, if  $G \in Q(\mathcal{C})$ , collect from a finite  $S \subseteq G \wr \mathbb{Z}$  all base coordinates and all coordinate products needed by its finite multiplication table. Locally embed this finite set in a group  $N \in \mathcal{C}$ . Coordinate replacement, leaving the integer exponents unchanged, locally embeds  $S$  into  $N \wr \mathbb{Z}$ , where the preceding construction applies. The local criterion proves  $G \wr \mathbb{Z} \in Q(\mathcal{C})$ . The model and its nilpotency class depend on  $S$ ; no global embedding of the wreath product in a nilpotent group is asserted. See `research/14.26-proof.md`.

### 6.3.8 11.115: equal commutator subgroups from a proper power

Let  $X = F(a, b)$ , fix  $m \geq 2$ , and put

$$N = \langle\langle a^m \rangle\rangle_X, \quad T = \ker(X \longrightarrow C_m), \quad a \mapsto 1, \quad b \mapsto 0.$$

Then  $1 < N < T < X$ :  $a^m$  is nontrivial,  $a \notin T$ , and  $b \in T \setminus N$ , as the exponent-sum map to  $\mathbb{Z}$  in  $b$  shows. In particular  $N$  is not maximal in  $X$ . Nevertheless

$$[T, N] = [X, N].$$

To prove this, set  $K = [T, N]$ . Both  $T, N$  are normal in  $X$ , so  $K \trianglelefteq X$ . In  $X/K$ , the image of  $a^m$  commutes with  $b \in T$ , and it already commutes with  $a$ . It is therefore central. Its normal closure, the image of  $N$ , is central too, giving  $[X, N] \leq K$ . The reverse inclusion is immediate. This disproves the proposed strict containment.

More explicitly,

$$K = \langle\langle [a^m, b] \rangle\rangle_X, \quad X/K = \langle a, b \mid [a^m, b] = 1 \rangle.$$

This quotient is  $F(s_0, \dots, s_{m-1}) \rtimes \mathbb{Z}$ , where a generator of  $\mathbb{Z}$  cyclically permutes the  $s_i$ 's: send  $a$  to that generator and  $b$  to  $s_0$ . The inverse map uses  $s_i = a^i b a^{-i}$ ; the final cyclic step is just  $[a^m, b] = 1$ . Thus  $N/K \cong \mathbb{Z}$  is central and  $X/N \cong C_m * \mathbb{Z}$ . The common subgroup  $K$  is nontrivial because  $[a^m, b]$  is a nonempty reduced word, and  $K < N$  by exponent sum in  $a$ . For the optional stronger condition that  $T$  itself not be maximal, take  $m = 4$  and insert the kernel of the map to  $C_2$ .

The supplied question imposes no torsion-free quotient or primitive-element assumption. Those extra conditions would exclude this example. The short proof is self-contained; its intended scope and historical priority still warrant outside scrutiny. The full packet is `research/11.115-proof.md`.

### 6.3.9 Order dimension of Chernikov subgroup lattices: Problem 11.116

Suppose  $D \trianglelefteq G$ ,  $[G : D] = m < \infty$ , and

$$D = C_{p_1^\infty} \times \cdots \times C_{p_r^\infty},$$

with distinct primes  $p_i$  and  $r \geq 1$ . Then the order dimension of the entire subgroup lattice of  $G$  is at most  $2rm$ .

Here is a coordinate proof. Write  $C_{p^\infty}$  additively as  $\mathbb{Z}[1/p]/\mathbb{Z}$ . Its unique terminating base- $p$  expansion  $x = \sum_{i \geq 1} d_i p^{-i}$ ,  $0 \leq d_i < p$ , defines a bijection to  $\mathbb{N}_0$  by

$$b_p(x) = \sum_{i \geq 1} d_i p^{i-1}.$$

A coset of the subgroup of order  $p^n$  has all digits beyond  $n$  fixed. Its image is an integer interval  $[ap^n, (a+1)p^n - 1]$ ; the whole group maps to  $\mathbb{N}_0$ . Inclusion is therefore determined by two endpoint inequalities, left endpoints ordered in reverse and right endpoints in the usual order.

Every subgroup of  $D$  is the product of its primary components: Chinese remaindering recovers each component as an integer multiple of an element. Thus the product of these bijections sends every coset of a subgroup of  $D$  to a nonempty box. Its inclusion order is determined by  $2r$  endpoint coordinates. Include the empty set by giving it left endpoints  $+\infty$  and right endpoints  $-\infty$ .

Fix representatives  $t_q$  of the  $m$  cosets of  $D$ . For  $H \leq G$ , its fiber

$$S_q(H) = \{d \in D : dt_q \in H\}$$

is empty or a coset of  $H \cap D$ . If  $d_0$  lies in the fiber, multiplying  $dt_q$  by  $(d_0 t_q)^{-1}$  proves that the fiber is  $(H \cap D)d_0$ . The fibers partition  $H$ , and  $H \leq K$  exactly when every fiber of  $H$  is contained in the corresponding fiber of  $K$ . This embeds the subgroup lattice into a product of  $2rm$  chains.

For clarity, a poset embedded in  $d$  chains has order dimension at most  $d$ : use the full coordinate lexicographic order to break ties, and for each coordinate order first by that coordinate. All  $d$  orders extend the poset, and any failure of comparison is detected in at least one coordinate. Their intersection is exactly the original order.

A Chernikov group has such a finite-index divisible subgroup. The question's exclusion of two quasicyclic factors for the same prime ensures the required distinct primes. Finite groups have finite subgroup lattices and cause no exception. The extension need not split. Source: `research/11.116-proof.md`.

## 6.4 Varieties, quasivarieties and formations

Universal constructions and the effects of the stated category or lattice convention.

### 6.4.1 Maximal test rank in a nonabelian variety: Problem 16.87

A test set in a group is a set whose pointwise stabilizer among endomorphisms consists of automorphisms. The test rank is the least size of such a set. Both parts of the question have a negative answer for the variety

$$\mathcal{M}_p = \{G : [[x, y], z] = 1, [x, y]^p = 1\},$$

where  $p$  is any prime. It is nonabelian because it contains  $\text{UT}_3(\mathbb{F}_p)$ , and nonperiodic because it contains  $\mathbb{Z}$ .

**Proposition 6.11.** *The relatively free group  $G_r$  of rank  $r \geq 1$  in  $\mathcal{M}_p$  has test rank  $r$ .*

*Proof.* Its abelianization is  $\mathbb{Z}^r$ : the generator maps to and from  $\mathbb{Z}^r$  give inverse maps on abelianization, since  $\mathbb{Z}^r$  belongs to the variety. Denote the resulting map by  $v : G_r \rightarrow \mathbb{Z}^r$ . If  $S$  has fewer than  $r$  elements, choose a primitive nonzero integral row vector  $w$  annihilating all  $v(s)$ ,  $s \in S$ , by solving the corresponding homogeneous linear equations over  $\mathbb{Q}$ . Then  $f = wv : G_r \rightarrow \mathbb{Z}$  is onto and vanishes on  $S$ .

Each free generator  $x_i$  has central  $p$ -th power, because  $[x_i^p, g] = [x_i, g]^p = 1$ . Choose  $i$  with  $w_i \neq 0$  and put  $c = x_i^{p \cdot \text{sign}(w_i)}$ . The map

$$\phi(g) = gc^{f(g)}$$

is an endomorphism by centrality of  $c$ , and fixes  $S$  pointwise. But  $f(\phi(G_r)) = (1 + p|w_i|)\mathbb{Z}$  is proper in  $\mathbb{Z}$ , so  $\phi$  is not surjective. Thus no smaller set is a test set. The  $r$  free generators themselves form one. For  $r = 1$  the same argument excludes the empty set.  $\square$

The free groups here have torsion in their commutator subgroups; nonperiodicity of the variety does not mean that its free groups are torsion-free. No such torsion-free hypothesis is printed in the problem. The proof uses only central powers and integer linear algebra. Source: `research/16.87-proof.md`.

### 6.4.2 A coefficient-support obstruction: Problem 14.22

Write the additive rational group multiplicatively as  $G = \{a_q : q \in \mathbb{Q}\}$ , with  $a_q a_r = a_{q+r}$ . It is torsion-free and linear via  $a_q \mapsto U(q)$  in  $\mathrm{SL}_2(\mathbb{Q})$ . The solution set of  $S = \{[x, a_1] = 1\}$  is all of  $G$ , and is irreducible in the group-equation Zariski topology. Indeed, evaluating a word  $w \in G * \langle x \rangle$  at  $x = a_t$  gives  $a_{m(w)t+b(w)}$ , where  $m(w) \in \mathbb{Z}$  is its exponent sum and  $b(w) \in \mathbb{Q}$  its coefficient sum. An equation therefore has zero, one, or all points as solutions. Proper closed sets in this one-variable topology are finite, and  $G$  is infinite.

Let  $T$  be any finite system equivalent to  $S$ . Every word in  $T$  has  $m(w) = b(w) = 0$ . Choose  $d > 0$  such that all its coefficients lie in the cyclic group  $C = \langle a_{1/d} \rangle$ . Form

$$K = G *_C (C \times \langle x \rangle)$$

and the natural epimorphism  $\pi : G * \langle x \rangle \rightarrow K$ . Each word of  $T$  vanishes in the second factor, since both exponent sums are zero. The group  $K$  is torsion-free: normal form in an amalgamated free product conjugates a torsion element into a factor, and both factors are torsion-free. Thus  $\ker \pi$  is a normal isolated subgroup containing  $T$ , and contains the least such subgroup  $\sqrt{T}$ .

But  $[x, a_{1/(2d)}]$  vanishes at every point of  $G$ , so belongs to  $\mathrm{Rad}_G(T)$ . Its image in  $K$  is a reduced alternating word of length four:  $x \notin C$  and  $a_{1/(2d)} \notin C$ . It is nontrivial by normal form. Hence

$$\sqrt{T} < \mathrm{Rad}_G(T)$$

for every finite equivalent  $T$ . The radical is isolated because it is an intersection of kernels of maps into a torsion-free group, which also justifies the displayed inclusion.

The coefficient group is not finitely generated, as permitted by the printed question. The separating quotient need only be torsion-free; isolatedness does not mean unique roots. In particular, commuting with a square does not imply commuting with its root in that quotient.

For completeness, the archive also avoids the possible concern that the intended geometry uses nonabelian coefficients. Take  $G = \mathbb{Q} * \langle b \rangle$ , represented faithfully by  $a_q \mapsto U(q)$ ,  $b \mapsto \begin{pmatrix} 1 & 0 \\ t & 1 \end{pmatrix}$  over  $\mathbb{Q}(t)$ . For a cyclically reduced alternating word, the trace has nonzero leading coefficient  $\prod_j q_j n_j$ , where the nonzero rational syllables are  $q_j$  and the  $b$ -syllables are  $b^{n_j}$ . Thus the representation is faithful. The solution set of  $[x, a_1] = 1$  is the factor  $\mathbb{Q}$ , since free factors are malnormal.

The coordinate group is  $(\mathbb{Q} \times \langle x \rangle) * \langle b \rangle$ . To verify this and irreducibility directly, represent  $x$  by  $U(z)$  with  $z$  a second indeterminate. The same leading-coefficient argument is faithful. Every nontrivial word remains nontrivial under all but finitely many rational specializations  $z = q$ : a nonzero coefficient polynomial has only finitely many roots. These evaluations discriminate the coordinate group and make every proper relative algebraic subset of the solution set finite.

For a finite equivalent system, choose a common denominator for all rational syllables in its coefficients. Its words already vanish in  $(C \times \langle x \rangle) * \langle b \rangle$ , which embeds in the coordinate group by normal form. They therefore vanish in the torsion-free group

$$(\mathbb{Q} *_C (C \times \langle x \rangle)) * \langle b \rangle,$$

while  $[x, a_{1/(2d)}]$  again survives. This gives the same strict radical obstruction with non-abelian linear coefficients. Source: `research/14.22-proof.md`.

**Version 2 attribution.** Jayadevan [55] establishes the radical obstruction over a two-generated torsion-free polycyclic subgroup of  $\mathrm{SL}_3(\mathbb{Z})$ . His Laurent-polynomial and bilinear-

extension construction is different and removes the infinite-generation feature of our coefficient groups. Both use normal isolated closure in the ordinary coefficient free product; see Appendix N.

### 6.4.3 A modular nondistributive dominion lattice: Problem 16.20

For  $H \leq A$  and a class  $\mathcal{N}$  of target groups, the dominion consists of those  $a \in A$  on which every pair of homomorphisms from  $A$  into a member of  $\mathcal{N}$  agree whenever they agree on  $H$ . The literature convention allows  $A \notin \mathcal{N}$ ; see [99]. Here is the smaller of the archive's two constructions under that convention.

Let

$$U_1 = \mathrm{SL}_2(5), \quad U_2 = \mathrm{SL}_2(7), \quad U_3 = \mathrm{SL}_2(13), \quad A = (U_1 \times U_2 \times U_3) / \langle z_1 z_2 z_3 \rangle,$$

where  $z_i = -I$ , and take  $H = 1$ . These standard perfect central covers have simple quotients  $S_i = \mathrm{PSL}_2(p_i)$  of orders 60, 168, 1092. No one of these three orders divides another. The center  $W = Z(A) \cong C_2^2$  has as its three nonzero vectors the images  $v_i$  of the  $z_i$ . The group has order 44,029,440.

Every normal subgroup  $R$  of  $A$  is  $A_I K$ , where  $A_I$  is the product of the factors indexed by some  $I \subseteq \{1, 2, 3\}$ , and

$$K = R \cap W, \quad \langle v_i : i \in I \rangle \leq K \leq W.$$

To see this, its image in the product of the three simple quotients is a product of factors. For any occurring factor,  $[R, U_i]$  projects onto  $S_i$ . A normal subgroup of  $U_i$  projecting onto  $S_i$  is all of  $U_i$ : if  $U_i = JZ(U_i)$ , perfectness gives  $U_i \leq J$ . Removing those factors leaves precisely a central subgroup.

The key fact is that every homomorphism  $f : A \rightarrow A/R$  kills  $R$ . Project its restriction on  $U_i$  to each surviving  $S_j$ . For  $j \neq i$  that map is trivial. Indeed, every proper normal subgroup of  $U_i$  is central, so every nontrivial image has order divisible by  $|S_i|$ , which cannot divide  $|S_j|$ . Thus a removed factor has central image and is killed by perfectness. A surviving factor maps into its own image  $D_i$  times the center; perfectness puts its image in  $[D_i Z(A/R), D_i Z(A/R)] = D_i$ . When  $v_i \in K$ , this  $D_i$  is  $S_i$ , and every homomorphism  $U_i \rightarrow S_i$  kills  $z_i$ . Every nonzero vector of  $K$  is some  $v_i$ , so  $f$  kills  $K$  as well as the removed factors, proving the fact.

Let  $q(\mathcal{C})$  denote the quasivariety generated by  $\mathcal{C}$ , and put  $\mathcal{M} = q(\{A/R : R \trianglelefteq A\})$ . For a finite domain  $A$ , the condition defining membership of an element in a dominion is a finite quasiidentity in the target: use two variables for each element of  $A$ , impose both multiplication tables and agreement on  $H$ , and conclude agreement at that element. It follows that the dominion for  $q(\mathcal{C})$  is exactly the intersection of the dominions for the generating targets.

For  $H = 1$ , a dominion is simply an intersection of kernels of maps into the target class, by comparison with the trivial map. It is therefore normal. For each  $R$ , all maps into  $A/R$  kill  $R$ , while the canonical quotient has kernel exactly  $R$ . The preceding transfer gives

$$\mathrm{dom}_{q(A/R)}^A(1) = R.$$

Every normal subgroup occurs, and the entire dominion family is the normal subgroup lattice of  $A$ . That lattice is modular by the Dedekind identity, and its interval  $[1, W]$  is the five-element diamond  $M_3$ , so it is not distributive.

The larger archived example also handles the extra convention that every admissible target quasivariety contain  $A$ . It uses seven covers  $\mathrm{SL}_2(p)$ ,  $p = 7, 11, 13, 17, 19, 23, 29$ , with

their central involutions mapped onto all nonzero vectors of  $\mathbb{F}_2^3$ , and a fixed central line  $H$ . Under that convention the dominions are exactly the five subspaces between  $H$  and  $\mathbb{F}_2^3$ . The smaller construction above suffices for the documented convention of the printed question. Source: `research/16.20-proof.md`.

#### 6.4.4 The Klein bottle quasivariety: Problem 17.33

Let  $K = \mathbb{Z} \rtimes \mathbb{Z}$  with  $(u, k)(v, l) = (u + (-1)^k v, k + l)$ , the Klein bottle group. Its quasivariety  $q(K)$  cannot be axiomatized by quasiidentities with any fixed finite bound on the number of variables.

For  $n \geq 2$ , put  $E = \mathbb{F}_2^n$ , and index  $m = n(2^n - 1)$  integer coordinates by pairs  $(\chi, j)$ , where  $0 \neq \chi \in E^*$  and  $1 \leq j \leq n$ . Let  $D_e$  multiply coordinate  $(\chi, j)$  by  $(-1)^{\chi(e)}$ . Define

$$\Gamma_n = \{(e, c) \in E \times \mathbb{Z}^m : c_{\chi, j} \equiv e_j \pmod{2}\}, \quad (e, c)(f, d) = (e + f, c + D_e d).$$

Parity makes this a subgroup of the affine semidirect product. Its translation subgroup  $N = \{(0, 2z) : z \in \mathbb{Z}^m\}$  has quotient  $E$ .

The group is torsion-free. If  $e \neq 0$ , choose a nonzero  $\chi$  vanishing on  $e$ , possible because  $n \geq 2$ , and  $j$  with  $e_j = 1$ . Then the  $(\chi, j)$ -coordinate of  $(e, c)^2 = (0, (1 + D_e)c)$  is nonzero. A nonidentity translation also has nonzero square, and any torsion element would have square one since its square lies in the torsion-free group  $N$ . On the other hand, the abelianization of  $\Gamma_n$  is finite. Every translation basis vector is inverted by some lift of  $e_i$ , since  $\chi_i = 1$  for some  $i$ , so  $2N \leq \Gamma'_n$ . The quotient by  $2N$  is finite.

Every homomorphism  $\Gamma_n \rightarrow K$  is trivial: composition with  $K \rightarrow \mathbb{Z}$  is trivial by finite abelianization, and the image is then in the translation subgroup  $\mathbb{Z}$ , where the same argument applies again.

In contrast, the inverse image  $\Gamma_L$  of every proper subspace  $L < E$  embeds in  $K^m$ . Choose  $0 \neq \psi \in E^*$  vanishing on  $L$ , and define, with coefficients  $\chi_j \in \{0, 1\}$ ,

$$\ell_\chi(e, c) = \sum_j \chi_j c_{\psi, j}, \quad F_{\chi, j}(e, c) = (c_{\chi, j}, \ell_\chi(e, c)).$$

The integer-valued  $\ell_\chi$  is additive on  $\Gamma_L$ , because  $D_e$  fixes the  $\psi$ -coordinates for  $e \in L$ , and it has parity  $\chi(e)$ . Thus every  $F_{\chi, j}$  is a homomorphism into  $K$ . Their product is injective: all its first coordinates determine  $c$ , and the parities of  $c$  determine  $e$ . In particular, every subgroup generated by fewer than  $n$  elements belongs to  $q(K)$ .

It remains to justify that  $\Gamma_n \notin q(K)$ , without assuming a residual characterization that might require additional hypotheses. The group is finitely presented. Explicit generators are translations  $t_{\chi, j}$  and lifts  $g_i = (e_i, c_i)$  with  $(c_i)_{\chi, j} = \delta_{ij}$ . A finite presentation is given by commuting translations and the relations

$$\begin{aligned} g_i t_{\chi, j} g_i^{-1} &= t_{\chi, j}^{(-1)^{\chi_i}}, \\ g_i^2 &= \prod_{\chi: \chi_i=0} t_{\chi, i}, \\ g_i g_j &= \left( \prod_{\chi} t_{\chi, i}^{\chi_j} t_{\chi, j}^{-\chi_i} \right) g_j g_i \quad (i < j). \end{aligned}$$

These hold by the affine multiplication law. They collect every word to a translation monomial followed by  $g_1^{\epsilon_1} \cdots g_n^{\epsilon_n}$ ,  $\epsilon_i \in \{0, 1\}$ . Its affine image is the identity only if all  $\epsilon_i$  and then all translation exponents vanish, proving the presentation.

For such a presentation, the quasiidentity asserting that its finitely many relations imply that a chosen nontrivial generator is one holds in  $K$ , since every homomorphism  $\Gamma_n \rightarrow K$  is

trivial. It fails in  $\Gamma_n$ . Thus  $\Gamma_n \notin q(K)$ . If quasiidentities in at most  $r$  variables axiomatized  $q(K)$ , take  $n > r$ . Every tuple of at most  $r$  elements of  $\Gamma_n$  lies in a subgroup of  $q(K)$ , so all those quasiidentities hold in  $\Gamma_n$ , a contradiction. Source: `research/17.33-proof.md`.

#### 6.4.5 Complete lattices of formations: Problem 18.92

The question defines a complete lattice of formations as an intersection-closed nonempty family with a largest member. Joins are internal to that family. Under this definition both requested examples follow from prime sets.

Let  $\mathcal{P}$  be the set of primes and let  $\mathcal{F}_S$  be the finite groups whose order has all prime divisors in  $S \subseteq \mathcal{P}$ . This is a formation: quotients and finite subdirect products introduce no new prime divisor. Moreover

$$\mathcal{F}_S \subseteq \mathcal{F}_T \iff S \subseteq T, \quad \bigcap_i \mathcal{F}_{S_i} = \mathcal{F}_{\cap_i S_i}.$$

The cyclic group of order  $p$  witnesses the reverse implication in the first assertion.

For part (a), take  $\Theta = \{\mathcal{F}_S : S \text{ finite}\} \cup \{\mathcal{F}_{\mathcal{P}}\}$ . Its joins are union of prime sets when that union is finite and the top otherwise. Every nonbottom element fails to be compact. Choose a prime  $q$  in its set and enumerate  $\mathcal{P} \setminus \{q\}$ . The formations for successive finite initial segments have join the top in  $\Theta$ , whereas no finite subjoin contains the cyclic group of order  $q$ . The bottom is therefore the only compact element, and  $\Theta$  is not algebraic.

For part (b), use the five formations with prime sets

$$\emptyset, \quad \{2\}, \quad \{2, 3\}, \quad \{5\}, \quad \mathcal{P}.$$

They are closed under intersections, and their order is the pentagon  $N_5$ . Explicitly, writing  $A = \mathcal{F}_{\{2\}}, B = \mathcal{F}_{\{2,3\}}, C = \mathcal{F}_{\{5\}}$ , we have  $A < B$  but

$$A \vee (C \wedge B) = A, \quad (A \vee C) \wedge B = B.$$

Thus the lattice is not modular.

Both examples contain the formation of all finite groups as top, so the empty-intersection convention causes no problem. These statements concern the chosen families; they do not assert either property for the full lattice of all formations. Source: `research/18.92-proof.md`.

#### 6.4.6 9.4: finite generators of group-isotope varieties

Use ordinary quasigroups in the three-operation, one-sorted signature  $(*, \backslash, /)$ . If a nontrivial finite group generates a variety  $\mathcal{M}$ , the class  $IM$  of all isotopes of groups in  $\mathcal{M}$  cannot be generated by any finite quasigroup, or by any finite family of finite quasigroups. The same obstruction applies to quasivarieties and pseudovarieties.

Define terms  $t_0(x, y) = y$ ,  $t_{j+1}(x, y) = x * t_j(x, y)$ . Every quasigroup of size at most  $n$  satisfies

$$t_{n!}(x, y) = y, \tag{***}$$

because each left translation is a permutation whose cycle lengths divide  $n!$ . Identities persist under each of the three generation operations just listed.

Choose a subgroup  $C_p$  of the nontrivial finite generating group. For any  $N \geq 1$ , let  $m = N + 1$ ,  $A = \mathbb{F}_p^m$ , and let  $T$  cycle its standard basis  $e_0, \dots, e_{m-1}$ . The operation

$$a * b = a + T(b), \quad a \backslash c = T^{-1}(c - a), \quad c / b = c - T(b)$$

defines a quasigroup isotopic to  $A$ ; the isotopy is  $(\text{id}, T, \text{id})$ . Only a subgroup and finite direct powers of the original group were used, so this finite quasigroup belongs to  $\mathcal{IM}$  in all three interpretations. At  $x = 0, y = e_0$ , however,

$$t_N(0, e_0) = T^N(e_0) = e_N \neq e_0.$$

Taking  $N = n!$  contradicts (\*\*\*) for every proposed finite family of finite generators.

There is also a local-finiteness obstruction. On  $A = \bigoplus_{i \in \mathbb{Z}} \mathbb{F}_p e_i$ , for the same prime  $p$  chosen above, use the bilateral shift and the same operation. This infinite quasigroup is generated by  $e_0$ :  $e_0 \setminus e_0 = 0$ , the terms  $0 * e_i$  and  $0 \setminus e_i$  move in both directions, and  $a + b = a * (0 \setminus b)$  forms all finite sums. A variety generated by a finite algebra  $Q$  is locally finite, since its free algebra on  $k$  generators has at most  $|Q|^{|Q|^k}$  term functions.

The interpretation is essential. In a three-sorted reversible automaton,  $x * (x * y)$  need not be well typed. Gvaramiya distinguishes these objects from ordinary quasigroups, and states the ordinary isotope variety theorem in [44, Section 3, Theorem 8]; compare also [70]. The argument concerns ordinary algebraic generation, as in the supplied statement, and makes no assertion about generation defined to include closure under isotopy. Its simplicity makes independent checking of the intended convention and priority particularly valuable. See `research/9.4-proof.md`.

#### 6.4.7 15.76(b): automorphisms of categories of free soluble groups

Let  $\mathcal{S}_d$  be the full variety of groups of derived length at most  $d \geq 1$ . Every automorphism of its category of finite-rank free groups is naturally isomorphic to the identity. This includes the full metabelian variety. It does not assert the same for every proper subvariety of soluble groups: the proper metabelian subvariety of exponent four studied by Fernandes–Tsurkov provides a counterexample [7]. The reduction of category automorphisms to verbal operations is an established method, used in the nilpotent predecessor [109]. The additional step here is the affine coefficient induction for the full soluble varieties.

We first recall the affine coefficient model. For an absolutely free group  $F = F(x_1, \dots, x_n)$ , a normal subgroup  $N$ , and  $Q = F/N$ , map  $x_i$  to  $(\bar{x}_i, e_i)$  in

$$Q \ltimes \mathbb{Z}[Q]^n, \quad (q, U)(r, V) = (qr, U + qV).$$

Its kernel is  $[N, N]$ . Indeed the covering graph of the rose corresponding to  $N$  has vertices  $Q$  and edges  $q \rightarrow q\bar{x}_i$ . The two affine coordinates of a word are its endpoint and signed integral edge chain. A loop has zero chain exactly when its class in the first homology of this graph is zero, which is its image in  $N/[N, N]$ . Finite support makes the argument valid for infinite covering graphs.

For a word  $w(x, y)$ , write its affine value as

$$w((a, U), (b, V)) = (w(a, b), f(a, b)U + g(a, b)V).$$

The boundary of its edge chain gives the ordered group-ring identity

$$f(a, b)(a - 1) + g(a, b)(b - 1) = w(a, b) - 1. \quad (\text{b})$$

We prove by induction on  $d$  that a normalized associative word in  $\mathcal{S}_d$ , meaning

$$w(x, 1) = x, \quad w(1, y) = y, \quad w(w(x, y), z) = w(x, w(y, z)),$$

must equal  $xy$  or  $yx$  as an identity. For  $d = 1$ , exponent sums suffice. For  $d \geq 2$ , induction and, if necessary, reversal of the two arguments allow us to assume  $w(a, b) = ab$  in  $\mathcal{S}_{d-1}$ . Let

$Q$  be free in that variety on  $a, b, c$ . The affine group  $Q \ltimes \mathbb{Z}[Q]^3$  belongs to  $\mathcal{S}_d$ . Associativity, evaluated at its three module basis vectors, gives

$$f(ab, c)f(a, b) = f(a, bc), \quad g(ab, c) = g(a, bc)g(b, c).$$

Set  $h(t) = f(1, t) \in \mathbb{Z}[t, t^{-1}]$ . The first equation gives  $f(b, c)h(b) = h(bc)$ ; putting  $c = b^{-1}$  shows that  $h$  is a Laurent unit with  $h(1) = 1$ . Hence  $h = t^k$  for an integer  $k$ , and

$$f(a, b) = (ab)^k a^{-k}.$$

Only the units of the ordinary cyclic Laurent ring are used: highest and lowest powers in a product show that a unit is  $\pm t^k$ . Likewise, using  $q(t) = g(t, 1)$ , the second equation gives

$$g(a, b) = (ab)^l b^{-l}$$

for an integer  $l$ . Map (b) to the free abelian quotient, with variables  $X, Y$ . It becomes

$$Y^k(X - 1) + X^l(Y - 1) = XY - 1.$$

Differentiate in  $X$  and set  $X = 1$ , obtaining  $Y^k = l + (1 - l)Y$ . Coefficient comparison gives exactly  $(k, l) = (0, 1)$  or  $(1, 0)$ .

In the first case  $f = 1, g = a$ , the affine coefficients of  $xy$ . Apply the kernel model to  $F(x, y)$  with  $N = F^{(d-1)}$ : equality of affine values proves  $w = xy$  modulo  $F^{(d)}$ . In the second case  $f = aba^{-1}, g = 1$ ; equation (b) forces  $aba^{-1} = b$  in  $Q$ . For  $d \geq 3$ ,  $Q$  maps onto the nonabelian metabelian group  $S_3$ , so this is impossible. For  $d = 2$ ,  $Q$  is abelian and these are exactly the affine coefficients and scalar coordinate of  $yx$ . The same kernel model gives  $w = yx$ . Undoing any initial reversal proves the induction.

We finish the categorical reduction explicitly. Rank one is recognized by its infinite commutative endomorphism monoid: rank zero has a one-element monoid, and ranks at least two have noncommuting basis projection and interchange endomorphisms. Coproducts of rank-one objects then recognize every finite rank. Conjugate the given functor by chosen object isomorphisms to fix objects, and by basis automorphisms to fix the maps  $\iota_i : F_1 \rightarrow F_n$  selecting the basis. Their images form a coproduct diagram and hence a free basis, justifying the second step. At  $F_1$  this normalization is the identity.

For  $a \in F_n$ , let  $f_a : F_1 \rightarrow F_n$  send the generator to  $a$ , and set  $s_n(a) = \Psi(f_a)(x)$ . These are bijections fixing the basis and identity. Functoriality gives, for every homomorphism  $u : F_n \rightarrow F_m$ ,

$$\Psi(u) = s_m u s_n^{-1}.$$

With  $w = s_2(xy)$ , it also gives  $s_n(ab) = w(s_n(a), s_n(b))$ . Thus  $w$  transports a group law and is normalized associative on all free objects, hence throughout  $\mathcal{S}_d$ . The word result applies. If  $w = xy$ , every  $s_n$  is a basis-fixing homomorphism and is the identity. If  $w = yx$ , let  $\eta_n$  invert each basis generator as an ordinary automorphism, and let  $j_n$  be the inversion set map. Then  $s_n = j_n \eta_n$ . Every group homomorphism commutes with the inversion maps, so  $\Psi(u) = \eta_m u \eta_n^{-1}$ , an inner functor. Undoing the earlier object and basis conjugations supplies a natural isomorphism for the original functor on every object and morphism. The argument works equally with all named free objects or a skeleton; rank zero contributes its unique maps. The affine kernel model is the classical Magnus method, proved here from graph homology. No general group-ring unit theorem or classification of automorphisms of free soluble groups is imported. See `research/15.76-proof.md`.

## 6.5 Algorithms and neighboring structures

Word algorithms, logical decidability, graphs, rings, topology and linear algebra.

### 6.5.1 Computing minimum palindromic length: Problem 16.9

In the free group  $F_n$  with specified basis  $x_1, \dots, x_n$ , let  $\text{pl}(w)$  be the minimum number of palindromes whose product is  $w$ , with  $\text{pl}(1) = 0$ . The following algorithm uses  $O(L^3)$  arithmetic operations and  $O(L^2)$  integer storage for an input word of length  $L$ . Its stored integers have magnitude  $O(L)$ .

Let  $\tau(x_i) = x_i^{-1}$ , and form  $H = F_n \rtimes \langle t \rangle$ , with  $t^2 = 1$  acting by  $\tau$ . The substitutions  $s_0 = t, s_i = x_i t$  give the presentation

$$H = \langle s_0, \dots, s_n : s_i^2 = 1 \text{ for all } i \rangle.$$

The subgroup  $F_n$  is its even-parity subgroup. An element  $p \in F_n$  is a palindrome exactly when  $\tau(p) = p^{-1}$ , hence when  $pt$  is an involution. Every nonidentity involution of this free product is conjugate to an  $s_i$ , by cyclic reduction, and has odd parity. Thus the involutions, or reflections, are precisely the elements  $pt$  for palindromes  $p$ .

If  $r(h)$  is reflection length in  $H$ , then

$$\text{pl}(w) = \min\{r(w), r(wt)\}. \quad (6)$$

Indeed, a product  $w = p_1 \cdots p_k$  gives

$$wt^k = (p_1 t)(\tau(p_2)t) \cdots (\tau^{k-1}(p_k)t).$$

Conversely, write a reflection factorization of  $wt^j$ ,  $j = 0, 1$ , as  $(q_1 t) \cdots (q_m t)$ . Parity gives  $m \equiv j \pmod{2}$ , so  $w = q_1 \tau(q_2) \cdots \tau^{m-1}(q_m)$ ; each factor is a palindrome. This proves both inequalities in (6).

For a word  $V$  on the involution alphabet, let  $d(V)$  be the minimum number of unmatched positions in a noncrossing partial pairing of equal letters. This is unchanged by insertion or deletion of two adjacent equal letters. Pairing the inserted letters gives one inequality. For the reverse, start with a pairing before deletion. If the deleted positions are paired together, remove that pair. If neither has a partner, remove them. If just one has another partner, leave that partner unmatched. If both have other partners, pair those partners to each other. Their labels agree, and noncrossing is preserved: the old arcs joined along the short deleted interval give a new arc disjoint from the others. None of these repairs increases the number unmatched.

It follows from free-product reduction that  $d$  depends only on the represented element. Moreover  $d(h) = r(h)$ . A conjugate  $vs_i v^{-1}$  has a pairing leaving only its central letter unmatched, so a product of  $k$  reflections has  $d \leq k$ . Conversely, delete the unmatched letters of a pairing. The remaining fully paired word reduces to the identity by successively removing innermost pairs. Deleting one letter uses the identity

$$X s_i Y = (X s_i X^{-1})(XY).$$

Successive deletions express the original word as a product of that many reflections. This proves equality. The general Coxeter deletion principle is prior work of Dyer [36]; the special case needed here has been proved directly.

Replace each  $x_i$  in the input by  $i0$  and each  $x_i^{-1}$  by  $0i$ , obtaining  $W$  of length at most  $2L$ . Then

$$\text{pl}(w) = \min\{d(W), d(W0)\}.$$

For half-open intervals of  $V = W0$ , set  $D(i, i) = 0$  and compute

$$D(i, j) = \min \left\{ \begin{array}{l} 1 + D(i+1, j), \\ \min_{\substack{i < k < j \\ V_i = V_k}} (D(i+1, k) + D(k+1, j)) \end{array} \right\}.$$

The minimum over an empty set is  $+\infty$ . The first option leaves position  $i$  unmatched. Pairing it with  $k$  separates the remaining positions into the inside and the following interval, by noncrossing. This exhausts all possibilities, proving the recurrence. There are  $O(L^2)$  intervals with  $O(L)$  choices each.

Stored minimizing choices recover the unmatched positions, whose successive deletions recover reflection factors  $r_i$ . Put  $q_i = r_i t$ , rewrite their even-parity words in the  $x_i$ , and apply  $\tau$  to every second factor as above. This gives a minimum palindrome factorization. The total output length is  $O(L^2)$ , since each conjugating prefix has length  $O(L)$  and there are  $O(L)$  factors. Bit complexity adds the usual  $O(\log(L+1))$  overhead for the stored integers; encoding generator indices is separate. This computes group palindromic length, allowing cancellation between factors, not merely a partition of the reduced word. Source: `research/16.9-proof.md`.

**Version 2 attribution.** Jayadevan [60] gives the same reflection-length reduction and cubic-time algorithm, with explicit reconstruction and output bounds. His prior-work discussion also identifies a qualitative answer already implied by Dahmani–Guirardel[30, Theorem 4]: with  $\tau(x_i) = x_i^{-1}$ , decide  $P_1 \cdots P_k = w$ ,  $\tau(P_j)P_j = 1$  for  $k \leq |w|$ . The twists generate a finite subgroup of the outer automorphism group. The first solvable system gives the minimum, and enumeration then finds its factors. The interval recurrence is the cancellation-norm method of Brandenbursky–Gal–Kędra–Marcinkowski[15, Section 2.I and the 2023 erratum], specialized to involutions. Thus qualitative computability and the recurrence are prior ingredients; the explicit palindrome reduction and its assembled quantitative algorithm are shared with contemporary work. See Appendix N.

### 6.5.2 The cofinite theory of finite groups: Problem 18.18

Let  $T_{\text{fin}}$  consist of group-language sentences true in every finite group, and  $T_{\text{cof}}$  those true in all but finitely many finite groups up to isomorphism. The latter is equivalent to truth in all sufficiently large finite groups. Both  $T_{\text{cof}}$  and its complement fail to be computably enumerable. The input theorem is Mal’cev’s undecidability of  $T_{\text{fin}}$  [75]: finite countermodels can be enumerated, so  $T_{\text{fin}}$  is co-computably-enumerable and cannot also be computably enumerable.

For a sentence  $\phi$ , effectively relativize every quantifier to the definable subgroup

$$C(x; a, b) : \quad xa = ax \wedge xb = bx,$$

renaming bound variables first. Write the resulting formula as  $\phi^C(a, b)$ . Formula induction gives its truth exactly when  $C_G(a, b) \models \phi$ . Then  $P_\phi = \forall a \forall b \phi^C(a, b)$  satisfies

$$\phi \in T_{\text{fin}} \iff P_\phi \in T_{\text{cof}}.$$

The forward direction is immediate. For the reverse, if finite  $H \not\models \phi$ , use  $H \times S_n$ ,  $n \geq 3$ , and two generators of the symmetric factor. Their common centralizer is  $H \times 1$ , giving countermodels of unbounded order. This proves the first non-enumerability assertion.

For the complement we need a fixed first-order sentence  $\sigma$  whose finite models are exactly the  $S_n$ ,  $n \geq 5$ . The following explicit interpretation supplies it. Choose an involution  $t \neq 1$ , let  $C_t$  be its conjugacy class, and let  $Q(\mathbf{u})$  say that four entries of  $\mathbf{u}$  are distinct members of  $C_t$  and pairwise noncommuting. Put

$$\begin{aligned} M(c, \mathbf{u}) : \quad & c \in C_t \wedge \bigwedge_{i=1}^4 (c = u_i \vee cu_i \neq u_i c), \\ E(\mathbf{u}, \mathbf{v}) : \quad & \forall c \in C_t (M(c, \mathbf{u}) \leftrightarrow M(c, \mathbf{v})). \end{aligned}$$

On  $Q$ -tuples,  $E$  is equality of the incidence sets and hence an equivalence relation. Call these classes vertices. This is abbreviation only: vertex quantifiers use four group variables and vertex equality uses  $E$ . Require that there be at least five vertices; that each  $c \in C_t$  be incident with exactly two vertices; that every pair of distinct vertices have exactly one common incident  $c$ ; that

$$E(cuc^{-1}, u) \iff \neg M(c, u);$$

and that the common centralizer of  $C_t$  be trivial. These are finitely many fixed first-order assertions; their conjunction, existentially quantifying  $t$ , defines  $\sigma$ .

In a finite model, conjugation acts faithfully on the finite vertex set: an element fixing all vertices fixes their unique edges and therefore all  $C_t$ . Each  $c$  preserves its incident pair, moves both endpoints by the displayed assertion, and fixes every other vertex. Its action is that pair's transposition. Every pair occurs, so the faithful image is the full symmetric group. Conversely, in  $S_n$ , take  $t$  a transposition. Four distinct pairwise intersecting two-element subsets have a common point: the only alternative for three is a triangle, which admits no fourth distinct edge meeting all three. The sets defined by  $M$  are therefore exactly the stars at points. Every point supplies a four-tuple when  $n \geq 5$ . All the assertions hold, including trivial centralizer because transpositions generate centerless  $S_n$ . This proves the claim about  $\sigma$ .

We also need every finite  $H$  as a common centralizer in *every sufficiently large*  $S_n$ . Write  $h = |H|$ , enumerate  $H = \{h_0 = 1, h_1, \dots, h_{h-1}\}$ , and on  $\Omega = H \times \{0, \dots, h-1\}$  define

$$a(g, j) = (g, j + 1 \bmod h), \quad b(g, j) = (gh_j, j).$$

The fixed points of  $b$  form precisely level zero. A commuting permutation must therefore have form  $(g, j) \mapsto (u(g), j)$  by commutation with  $a$ ; commutation with  $b$  gives  $u(gh_j) = u(g)h_j$ , hence  $u(g) = u(1)g$ . The common centralizer is exactly the left regular copy of  $H$ . The action of  $\langle a, b \rangle$  is transitive, by moving between levels to realize all right multiplications.

For  $n \geq 2h^2 + 3$ , add a disjoint orbit of  $k = n - h^2 > h^2$  points, extending  $a, b$  by a  $k$ -cycle and a transposition of consecutive points. Their action there is  $S_k$ , with trivial centralizer. The two orbit sizes differ, so no commuting permutation exchanges them. The total common centralizer in  $S_n$  is therefore  $H$ , as required.

Finally form the effective sentence

$$R_\phi = (\neg\sigma) \vee \exists a \exists b \neg\phi^C(a, b).$$

If  $\phi \in T_{\text{fin}}$ , the second disjunct is false in every finite group, and  $R_\phi$  fails in every  $S_n$ ,  $n \geq 5$ . If a finite  $H$  refutes  $\phi$ , all nonsymmetric groups satisfy the first disjunct, and every sufficiently large symmetric group satisfies the second by the construction. Thus

$$\phi \in T_{\text{fin}} \iff R_\phi \notin T_{\text{cof}}.$$

Computable enumerability of the complement would give it for  $T_{\text{fin}}$ , a contradiction. The syntactic transformation does not search for  $H$ ; the witness is needed only for its correctness proof. No asymptotic-density interpretation of *almost all* is asserted. Source: `research/18.18-proof.md`.

### 6.5.3 An invertible adjacency operator on all functions: Problem 15.89

Let  $G = C_3 * C_2 = \langle a, b : a^3 = b^2 = 1 \rangle$  and

$$S = \{aba, aba^2, a^2ba, a^2ba^2\}.$$

Its Cayley graph, with edges  $g \sim gs$ , is simple and four-regular: free-product normal form gives four distinct nonidentity words, and  $S = S^{-1}$ . It is connected because, for  $s_1 = aba$ ,  $s_2 = a^2ba$ ,  $s_1s_2^{-1} = a^2$  and  $a^2s_1a^2 = b$ . The group is infinite by the distinct reduced words  $(ab)^n$ , and left translation is vertex-transitive.

On the space  $\mathbb{C}^G$  of all functions, put  $(R_gf)(x) = f(xg)$ ,  $U = R_a + R_{a^2}$ , and  $B = R_b$ . Then  $R_gR_h = R_{gh}$ , and

$$U^2 = U + 2I, \quad U^{-1} = (U - I)/2, \quad B^2 = I.$$

The adjacency operator is exactly  $A = UBU$ , hence

$$A^{-1} = \frac{1}{4}(U - I)B(U - I).$$

Both products with  $A$  equal the identity. All these expressions are finite sums of translations, so they act on arbitrary, even unbounded, functions. There is no support or convergence condition. In particular zero is not an eigenvalue, answering the question negatively.

The factorization has the familiar zig-zag form: a move within a triangle, across a matching edge, and within the next triangle. The general operation is due to [91]; the invertibility of finite factors also occurs in [21]. The proposed contribution is this particular infinite simple vertex-transitive example and its use for the printed question, not the graph-product operation. Source: `research/15.89-proof.md`.

**Version 2 attribution.** Jayadevan [61] gives the same graph and inverse, and proves the stronger statement that its point spectrum on all complex functions is exactly  $\mathbb{C} \setminus \{0\}$ . He also identifies the graph with the established three-arc graph of the cubic tree. The particular example is therefore shared with contemporary work; Appendix N discusses the additional argument.

#### 6.5.4 A resolvable group without an expansive sequence: Problem 21.107

As usual, the open sets in the definition of an expansive sequence are understood to be nonempty: every such open set must meet all sufficiently late terms of a sequence of pairwise disjoint finite subsets. Take a free ultrafilter  $\mathcal{U}$  on  $\mathbb{N}$ , obtained by extending the cofinite filter, and the countable Boolean group  $B = [\mathbb{N}]^{<\omega}$  under symmetric difference. For  $A \in \mathcal{U}$ , put  $H_A = [A]^{<\omega}$  and use these subgroups as an identity neighborhood base. Intersections are  $H_A \cap H_C = H_{A \cap C}$ , so this defines a group topology. It is Hausdorff by deleting a coordinate on which two elements differ, and nondiscrete because every  $A \in \mathcal{U}$  is infinite.

The sets

$$D_k = \{s \in B : v_2(|s| + 1) = k\}, \quad k \geq 0,$$

partition  $B$  into dense subsets. Given  $x + H_A$  and  $k$ , choose an arbitrarily large  $r = 2^k(2q + 1) - 1 \geq |x|$ , and add to  $x$  a subset of  $A \setminus x$  of size  $r - |x|$ . This gives a point of the coset in  $D_k$ .

Suppose  $(F_n)$  is any sequence of pairwise disjoint finite subsets of  $B$ . Infinitely many empty terms already preclude expansiveness. Otherwise discard the finitely many empty terms and the at most one term containing the identity, leaving an infinite sequence of nonempty finite families of nonempty supports. Put  $S_n = \{\max s : s \in F_n\}$ . For every  $r$ , only finitely many  $S_n$  meet  $\{1, \dots, r\}$ , since there are just  $2^r - 1$  nonempty supports with maximum at most  $r$  and the  $F_n$  are disjoint. Choose a subsequence with  $\max S_{n_j} < \min S_{n_{j+1}}$ . Let  $E$  be the union of its even-indexed  $S_{n_j}$ . Either  $E \in \mathcal{U}$  or its complement is in  $\mathcal{U}$ . Whichever belongs to  $\mathcal{U}$  misses all  $S_{n_j}$  of the opposite parity. The corresponding open subgroup  $H_A$  therefore misses infinitely many  $F_{n_j}$ , a contradiction.

Thus countable resolvability does not imply existence of an expansive sequence. Only an ordinary free ultrafilter is used, with no selectivity assumption. The underlying filter topology is standard; the dense partition and obstruction are given explicitly. Source: `research/21.107-proof.md`.

### 6.5.5 A nonsplit additive division-ring extension: Problem 18.76

Let  $F = F(a, b)$ ,  $L = F(c, d)$ , and  $G = F \times L$ . Free groups and their direct products admit two-sided invariant orders. The Mal'cev–Neumann theorem therefore embeds  $kG$  in a division ring for  $k = \mathbb{Q}$  or any prime field; see [94, 14]. Let  $D$  be its division closure there. This is countable, and  $G \leq D^\times$ . Only elements from different free factors are assumed to commute.

Define a derivation  $\delta : F \rightarrow kF$  by

$$\delta(xy) = \delta(x) + x\delta(y), \quad \delta(a) = 0, \quad \delta(b) = 1.$$

The prefix-sum formula on reduced words gives it explicitly; contributions from adjacent inverse letters cancel, with  $\delta(b^{-1}) = -b^{-1}$ . In  $N = (D, +) \rtimes F$ , with  $(r, x)(s, y) = (r + xs, xy)$ , the maps

$$\gamma_c(r, x) = (cr, x), \quad \gamma_d(r, x) = (dr + \delta(x), x)$$

are automorphisms. The derivation law proves the homomorphism identity; multiplication by  $c^{-1}$  and the map  $(r, x) \mapsto (d^{-1}(r - \delta(x)), x)$  give their inverses. Freeness of  $L$  extends them to an action, giving the actual group  $E = N \rtimes_\gamma L$ .

Its quotient by  $(D, +)$  is  $G = F \times L$ , and the conjugation action on the kernel is left multiplication in  $D$ . To check this and nonsplitting, define similarly  $\eta : L \rightarrow kL$  by  $\eta(c) = 0, \eta(d) = 1$ . Then

$$(r, x, h)(s, y, l) = (r + xhs + x\eta(h)\delta(y), xy, hl).$$

This follows by composing the generator automorphisms; the two derivations take values in commuting factor algebras.

If a section existed, its values on  $a, b, c, d$  would have arbitrary kernel coordinates  $u, v, w, z \in D$ . Put  $A_0 = a - 1, B_0 = b - 1, C_0 = c - 1, D_0 = d - 1$ . The four cross-factor commutations would imply

$$A_0w - C_0u = 0, \quad A_0z - D_0u = 0, \quad B_0w - C_0v = 0, \quad B_0z - D_0v = 1_D.$$

Here  $1_D$  is a nonzero additive kernel element. Since  $A_0, C_0$  are nonzero and invertible, the first three equations give

$$w = A_0^{-1}C_0u, \quad z = A_0^{-1}D_0u, \quad v = B_0A_0^{-1}u.$$

The left side of the fourth is then

$$B_0A_0^{-1}D_0u - D_0B_0A_0^{-1}u = 0,$$

because the two free factors commute, contradicting  $1_D \neq 0$ . Thus the extension does not split, in any of the stated characteristics. Its description as a nested semidirect product does not split this particular extension by  $(D, +)$ . Source: `research/18.76-proof.md`.

**Version 2 attribution.** Jayadevan [52] gives the same free-factor and division-ring obstruction in characteristic zero, using the cup product of the two derivations. The cocycles differ by a coboundary and a sign; Appendix N gives the resulting isomorphism, which reverses the additive kernel. Our statement also permits prime positive characteristic; the total group is torsion-free in characteristic zero.

### 6.5.6 9.47: a scattered compact space excluded by the E-topology

The E-topology on the closed subgroups  $L(G)$  of a locally compact Hausdorff group is the full Vietoris topology, with subbasic sets

$$D_1(O) = \{H : H \subseteq O\}, \quad D_2(O) = \{H : H \cap O \neq \emptyset\}$$

for arbitrary open  $O \subseteq G$ . In particular its upper subbasis is more extensive than that of the Chabauty topology. A *clopen selector* on  $X$  assigns a clopen neighborhood  $U_x$  to every  $x$ , with

$$x \in U_y, y \in U_x \implies x = y, \quad y \in U_x \implies U_y \subseteq U_x.$$

We prove that a compact Hausdorff space embeds in the E-space of closed noncompact subgroups of some locally compact group exactly when it has such a selector. A known height-four compact space fails this test. The ingredients are Protasov's neighborhood lemma [90, 68] and the Dow–Watson example [35]; their use here does not make either ingredient new.

**Lemma 6.12.** *If  $\mathcal{F} \subseteq L(G)$  is compact and  $H \in \mathcal{F}$  is noncompact, then  $\{K \in \mathcal{F} : K \subseteq H\}$  is a neighborhood of  $H$  in  $\mathcal{F}$ .*

*Proof.* The space  $L(G)$  is Hausdorff: if  $g \in H \setminus K$ , choose an open  $V \ni g$  with  $\overline{V} \cap K = \emptyset$ , and use  $D_2(V)$  and  $D_1(G \setminus \overline{V})$ . Suppose the conclusion fails, so  $H$  is in the closure of  $\mathcal{A} = \{K \in \mathcal{F} : K \not\subseteq H\}$ . For each neighborhood  $W$  of  $H$  in  $\mathcal{F}$ , let

$$E_W = \bigcup_{K \in \mathcal{A} \cap W} (K \setminus H).$$

This set is not relatively compact. Otherwise its closure  $C$  is compact; choose  $h \in H \setminus (C \cup C^{-1}C)$  and an identity neighborhood  $V$  so that  $Vh$  misses both sets. Some  $K \in \mathcal{A} \cap W$  meets  $Vh$ , at  $x$ . Since  $Vh \cap C = \emptyset$ ,  $x \in H$ . For  $z \in K \setminus H$ , both  $z, zx$  belong to  $C$ , contradicting  $CVh \cap C = \emptyset$ .

Fix a compact identity neighborhood  $U$ . Inductively choose  $K_i \in \mathcal{A}$ ,  $x_i \in K_i \setminus H$ , and a compact neighborhood  $V_i$  of  $x_i$  disjoint from  $H$ , with

$$K_i \cap V_j = \emptyset \ (j < i), \quad x_i U \cap x_j U = \emptyset \ (j < i).$$

The first condition restricts  $K_i$  to an open neighborhood  $W_i$  of  $H$ ; the second follows by choosing  $x_i \in E_{W_i}$  outside the compact union  $\bigcup_{j < i} x_j U U^{-1}$ . Local compactness supplies  $V_i$ . The set  $\mathcal{B} = \{K_i\}$  is discrete, because  $D_2(\text{int } V_i)$  meets it in finitely many points, which Hausdorff separation reduces to  $K_i$ .

Also  $D = \{x_i\}$  is closed discrete in  $G$ : choose open  $Q \ni 1$  with  $Q^{-1}Q \subseteq U U^{-1}$ ; every translate  $gQ$  contains at most one  $x_i$ . If  $K \in \overline{\mathcal{B}} \setminus \mathcal{B}$ , it cannot meet  $\text{int } V_i$ , whose associated lower neighborhood meets  $\mathcal{B}$  only finitely. Thus  $K \cap D = \emptyset$ , but the open upper neighborhood  $D_1(G \setminus D)$  contains no  $K_i$ , a contradiction. Hence  $\mathcal{B}$  is closed as well as discrete, impossible inside compact  $\mathcal{F}$ .  $\square$

If every member of  $\mathcal{F}$  is noncompact, set  $U_H = \{K \in \mathcal{F} : K \subseteq H\}$ . It is closed, since its complement is  $D_2(G \setminus H)$ . At each  $K \in U_H$  the lemma gives a neighborhood contained in  $U_K \subseteq U_H$ , so it is also open. Subgroup inclusion supplies the selector axioms.

Here is the counterexample, including the obstruction. For every  $f : \omega \rightarrow \omega$ , choose an infinite  $A_f \subseteq \omega$  so that distinct  $A_f$ 's meet finitely. Such a family is obtained by coding finite initial segments of the distinct binary branches  $0^{f(0)}10^{f(1)}1 \dots$ . Start with  $Y = (\omega + 1) \times \omega$ , a disjoint union of convergent sequences. Write  $u_n = (\omega, n)$ ,  $v_{m,n} = (m, n)$ , and set

$$B_f = \{u_n : n \in A_f\} \cup \{v_{m,n} : n \in A_f, m > f(n)\}.$$

Adjoin a point  $a_f$  with basic neighborhoods consisting of  $a_f$  and the part of  $B_f$  in rows  $n \geq N$ . Call the resulting space  $Z$ . These neighborhoods are compact: after covering  $a_f$ , only finitely many compact row tails remain. Almost disjointness separates different  $a_f$ 's; omission of a row separates them from points of  $Y$ . Thus  $Z$  is locally compact Hausdorff, and these compact neighborhoods are clopen. The infinite set  $\{a_f\}$  is closed discrete, so  $Z$  is noncompact. Let  $X = Z \cup \{\infty\}$  be its one-point compactification. Its successive derived sets are

$$X' = \{u_n\} \cup \{a_f\} \cup \{\infty\}, \quad X'' = \{a_f\} \cup \{\infty\}, \quad X''' = \{\infty\}, \quad X'''' = \emptyset.$$

It is consequently scattered of height four.

Suppose  $X$  had a selector. Choose  $t(n)$  so that  $v_{m,n} \in U_{u_n}$  whenever  $m > t(n)$ . For any  $f$  with  $f(n) > t(n)$  for all  $n$ , the neighborhood  $U_{a_f}$  contains  $u_n$  for all sufficiently large  $n \in A_f$ . Transitivity makes it contain the infinite set

$$D_f = \{v_{t(n)+1,n} : n \in A_f, n \geq N\}.$$

This is closed discrete in  $Z$ : each row meets it at most once, the neighborhood based on  $B_f$  misses it, and every other  $a_g$  has a neighborhood omitting the finitely many common rows. Thus no compact subset of  $Z$  contains  $D_f$ , so  $\infty \in \overline{D_f} \subseteq U_{a_f}$ . There are infinitely many such  $f$ , for example  $f_j(n) = t(n) + j$ ,  $j \geq 1$ . Every neighborhood of  $\infty$ , including  $U_\infty$ , contains all but finitely many  $a_{f_j}$ , since compact subsets of  $Z$  meet the closed discrete set  $\{a_f\}$  finitely. For some  $j$ , both  $a_{f_j} \in U_\infty$  and  $\infty \in U_{a_{f_j}}$ , contradicting antisymmetry. The necessary condition above now answers 9.47 negatively.

For completeness, the condition is also sufficient. Given a compact  $X$  with selector, take the discrete abelian group

$$G = \mathbb{Z} \oplus \bigoplus_{y \in X} C_2, \quad H_x = \mathbb{Z} \oplus \bigoplus_{\{y: x \notin U_y\}} C_2.$$

Each  $H_x$  is closed and noncompact, and antisymmetry makes  $x \mapsto H_x$  injective. Membership of any fixed element in  $H_x$  is a finite intersection of clopen conditions, proving continuity for the lower subbasis. If  $z \in U_x$ , transitivity implies  $H_z \subseteq H_x$ ; this proves continuity for the upper subbasis. A continuous injection from compact  $X$  to Hausdorff  $L(G)$  is an embedding. The full archive proof is `research/9.47-proof.md`.

### 6.5.7 9.45: a coordinate test for orthogonal lattice bases

For  $a \in \mathbb{Q}^n$ , let  $S(a) = \mathbb{Z}^n + \mathbb{Z}a$ . Subtracting an integer vector, write

$$a = r/m, \quad 0 \leq r_j < m, \quad \gcd(m, r_1, \dots, r_n) = 1.$$

The following finite test uses  $O(mn)$  integer arithmetic operations; no polynomial bound in the binary length of  $m$  is asserted.

For each  $1 \leq k < m$ , center every coordinate of  $kr \bmod m$  in  $[-m/2, m/2]$ . At a half-coordinate retain both signs; discard a coset with more than two such ties. Among the resulting at most four vectors  $w$ , retain those with first nonzero coordinate positive which satisfy, with  $Q(w) = \sum_j w_j^2$ ,

$$Q(w) \mid mw_j \quad \text{for every } j, \quad Q(w) \mid \sum_j r_j w_j. \quad (**)$$

Call their list  $\mathcal{W}$ , and let  $z$  count the zero coordinates of  $r$ .

**Proposition 6.13.** *The lattice  $S(a)$  has an orthogonal basis if and only if  $z + |\mathcal{W}| = n$ . In that case a basis is  $\{e_j : r_j = 0\} \cup \{w/m : w \in \mathcal{W}\}$ .*

*Proof.* For a Euclidean lattice  $L$ , a vector  $b \neq 0$  spans an orthogonal rank-one direct summand exactly when  $c_b = b/\|b\|^2 \in L^*$ : subtract the integer multiple  $(x \cdot c_b)b$  from each  $x \in L$ . Call such a vector splitting. For splitting vectors  $b, v$ , both

$$\frac{b \cdot v}{\|b\|^2}, \quad \frac{b \cdot v}{\|v\|^2}$$

are integers. Their product is the squared cosine. If the lines differ, it is less than one and hence zero; if they coincide, the ratio and its reciprocal are integers, so  $v = \pm b$ . All splitting lines are therefore mutually orthogonal. There are at most  $n$ , and if there are  $n$ , the orthogonal expansion of every  $x \in L$  has integer coefficients, proving that their generators form a basis.

Now  $L = S(r/m)$  contains  $\mathbb{Z}^n$ , so  $L^* \subseteq \mathbb{Z}^n$ . For a splitting vector write  $c = c_b$ ,  $d = \|c\|^2$ . Then  $c \in \mathbb{Z}^n$ ,  $d$  is a positive integer, and  $b = c/d$ . If  $d = 1$ ,  $b = \pm e_j$ , which splits precisely when  $r_j = 0$ . If  $d \geq 2$ , then  $2|c_j| \leq d$  in every coordinate, so  $|b_j| \leq 1/2$ . If  $h$  coordinates attain equality, then  $hd^2/4 \leq d$ , giving  $h \leq 2$ . Moreover  $\|b\|^2 = 1/d < 1$ , so  $b \notin \mathbb{Z}^n$ . The gcd condition shows that  $r/m$  has order exactly  $m$  modulo  $\mathbb{Z}^n$ . Thus every remaining splitting vector, up to its unique first-positive sign, occurs in the centered enumeration.

For a listed  $w$ ,  $b = w/m$  lies in  $L$ , and  $c_b = mw/Q(w)$ . Membership in  $L^*$  says that its coordinates and its scalar product with  $r/m$  are integers, which is exactly (\*\*). Different cosets or tie choices give different vectors. The test therefore lists all splitting lines exactly once, proving the result.  $\square$

Half-coordinate signs matter: for  $a = (1/6, 1/6)$ , the vector  $(3, -3)$  passes (\*\*) while  $(3, 3)$  fails. For  $a = (1/2, 1/2, 1/2)$ , there are no splitting lines. A three-dimensional positive example is  $a = (1, 5, 2)/6$ , with basis

$$(1, -1, 2)/6, \quad (1, -1, -1)/3, \quad (1, 1, 0)/2.$$

The archive also proves that when  $m = p^k$ , at most two coordinates of  $r$  can be nonzero in an orthogonal example. Indeed an integral orthogonal basis matrix  $C$  of  $L^*$  has determinant  $p^k$ , Smith form  $\text{diag}(1, \dots, 1, p^k)$ , and diagonal  $CC^t$  whose entries are powers of  $p$ . The rank inequality modulo  $p$  gives  $\text{rank}(CC^t) \geq n - 2$ . All but at most two row norms are therefore one, forcing the other residue coordinates to vanish.

Bardakov already solved the two-dimensional case and gave a higher-dimensional sufficient condition [8]. Normalized-dual arguments and orthogonal decomposition are prior methods; compare [23, 45]. The candidate contribution is this particular cyclic-coset criterion in arbitrary dimension. The archive `research/9.45-proof.md` contains the prime-power classification; `results/9.45-summary.json` binds independent finite controls.

### 6.5.8 Homogeneous extensions of representations: Problem 17.101

Fix a commutative ring  $k$  with identity. A representation is a left  $kG$ -module  $V$  with its acting group  $G$ ; an embedding is injective on both sorts. The relevant homogeneity extends isomorphisms of subrepresentations whose group is finitely generated and whose module is finitely generated over that group's algebra. The coefficient ring remains fixed; compare the category in [89].

Given any set of partial isomorphisms  $(f_i, \theta_i) : (U_i, A_i) \rightarrow (U'_i, B_i)$  in  $(V, G)$ , form the multiple HNN extension

$$H = \langle G, t_i : t_i a t_i^{-1} = \theta_i(a), a \in A_i, \text{ all } i \rangle.$$

The HNN embedding theorem embeds  $G$  in  $H$  [46]. Define

$$M = kH \otimes_{kG} V, \quad E = \bigoplus_i kH \otimes_{kA_i} U_i, \quad \delta_i(h \otimes u) = h \otimes u - ht_i^{-1} \otimes f_i(u),$$

and  $W = M / \text{im } \delta$ . The map  $\delta_i$  is well-defined because  $at_i^{-1} = t_i^{-1}\theta_i(a)$  and  $f_i$  is equivariant.

We verify that no vector of  $V$  is lost. The HNN Bass–Serre tree has vertices  $hG$  and edges  $hA_i$  with endpoints  $hG, ht_i^{-1}G$ . Its being a tree follows from HNN normal form: a reduced closed path would be a reduced stable-letter word lying in  $G$ . As  $k$ -modules,  $M$  is the direct sum of one copy of  $V$  at each vertex, and  $E$  the direct sum of the edge modules. Each edge maps injectively into both incident vertex modules, using the original inclusion and  $f_i$ .

If  $\delta(z)$  were supported at a single vertex  $v_0$  for a nonzero  $z \in E$ , the finitely many nonzero edge components of  $z$  would form a finite forest with an edge. It has a leaf other than  $v_0$ . The single incident nonzero edge contributes a nonzero vector at that leaf, by injectivity, a contradiction. Hence  $\text{im } \delta$  intersects each vertex module trivially, and  $V \hookrightarrow W$ . This argument uses no semisimplicity, purity, or finite-dimensionality.

In  $W$ , the edge relation gives  $t_i u = f_i(u)$ . The automorphism of  $(W, H)$  with components

$$w \mapsto t_i w, \quad h \mapsto t_i h t_i^{-1}$$

therefore extends the prescribed partial isomorphism. Now repeat the construction at stages  $0, 1, 2, \dots$ , each time including all isomorphisms between finitely generated subrepresentations, and take the union. The domain and range of any such isomorphism in the union lie in a common stage: finitely many group and module generators suffice. A stable letter at the next stage realizes it, and its action continues to be an automorphism of the union. This proves homogeneity.

If  $\kappa = \max(\aleph_0, |k|, |G|, |V|)$ , there are at most  $\kappa$  relevant finite tuples and maps at each stage; induction, quotients, and the countable union preserve that bound. Thus a countable input over a countable ring has a countable homogeneous extension. Source: `research/17.101-proof.md`.

**Version 2 attribution.** Jayadevan [56] uses the same HNN-induced-module construction, forest argument and cardinal bound. Over a nonzero ring, he also obtains a faithful extension by adjoining a regular module at intermediate stages. His account supplies further references for the HNN module construction. Appendix N separates the shared argument from this additional property.

## 7 Partial results and bounded computations

The distinct-powers problem, almost Engel elements and  $p$ -Jordan bounds led to substantial theorems while leaving their full questions open. We give the reductions and proofs, specifying the remaining cases. A final collection records additional constructions and bounded searches. These partial results are outside the 46-entry candidate count.

### 7.1 20.100: distinct powers, reduction, and four finite proofs

For a positive integer  $n$ , assume  $G$  has no element of order  $2, \dots, n+1$ . The question asks whether every  $n$ -element set can be ordered as  $a_1, \dots, a_n$  with  $a_1, a_2^2, \dots, a_n^n$  pairwise distinct. The identity is allowed. Sun’s Conjecture 4.1 is this assertion; his Theorem 1.4 covers torsion-free abelian groups and  $n \leq 3$  [103]. The experiment obtained a torsion-free theorem without commutativity, a reduction to finite abelian groups, an explicit finite counterexample bound for fixed  $n$ , and complete certificates for  $n = 4, 5, 6, 7$ .

**Proposition 7.1.** *Any finite list of infinite-order elements, with repetitions allowed, can be assigned prescribed distinct positive integer exponents so that the resulting powers are distinct.*

*Proof.* On the cyclic subgroups generated by the list, nontrivial intersection is an equivalence relation: intersect the two finite-index subgroups of the middle infinite cyclic group to prove transitivity. In each of the finitely many classes choose a nonidentity common power  $c$ . Write  $c = a^{k_a}$ ,  $k_a \neq 0$ , for each entry in that class and give it length  $1/|k_a|$ . If  $a^i = b^j$ , then the entries lie in the same class, and raising to the integer  $k_a k_b$  gives  $c^{i k_b} = c^{j k_a}$ . Infinite order implies  $i/|k_a| = j/|k_b|$ . Sort the entire list by nondecreasing positive length and assign the exponents in increasing order. The products of exponent and length strictly increase, so no equality is possible. One additional identity element can be assigned any unused exponent.  $\square$

For fixed  $n$ , the question for all groups is equivalent to the question for finite abelian groups. To see the nontrivial direction, partition the finite-order nonidentity entries according to the cyclic subgroups they generate. All prime divisors of their orders exceed  $n + 1$ ; otherwise  $G$  has an element of forbidden prime order. Thus  $\langle a^i \rangle = \langle a \rangle$  for  $1 \leq i \leq n$ . Distinct cyclic groups cannot contribute equal powers in this exponent range. Represent the entries in each cyclic group in a separate abstract cyclic coordinate. This preserves exactly their relevant power equalities. Replace all infinite-order entries by distinct nonidentity elements in one additional cyclic coordinate of prime order greater than  $n + 1$ , and represent the identity by the identity. An affirmative finite abelian result assigns the exponents. Keep the assignments on finite-order entries; reassign the remaining exponents to the infinite-order entries using the proposition. Finite- and infinite-order powers cannot coincide. This proves the equivalence, without embedding  $G$  in an abelian group.

There is also a finite size bound. Let  $p_n$  be the least prime greater than  $n + 1$ . A counterexample at  $n \geq 2$ , if one exists, exists in a finite abelian group of order at most

$$p_n n^{n-1}.$$

Indeed, from a finite abelian counterexample choose a collision for every exponent permutation and impose those relations  $ix_a = jx_b$  on a free module over  $S = \mathbb{Z}[1/P]$ , where  $P$  is the product of primes at most  $n + 1$ . Its distinguished elements remain distinct by mapping to the counterexample, and every permutation fails. In the multigraph of relations, a spanning tree in a component of  $k$  vertices expresses each variable as an  $S$ -unit multiple of one generator. A nontree edge either gives the zero relation or gives an annihilator, up to an  $S$ -unit, equal to the difference of the two coefficient products around a simple cycle. A nonzero difference has absolute value less than  $n^k$ . The component is therefore free of rank one over  $S$ , or cyclic finite of order less than  $n^k$ , after stripping inverted primes. If every component is finite, the total order is at most  $n^n < p_n n^{n-1}$ . Otherwise the finite components involve at most  $n - 1$  variables; replace all infinite-order variables by a separate  $C_{p_n}$  as above. A good assignment would lift back to a contradiction, and the resulting order is at most  $p_n n^{n-1}$ . This is decidability for each fixed  $n$ , not a uniform proof in  $n$ .

**The certificate is a proof of an all-group assertion.** A node consists of an integer relation matrix  $R$  on  $n$  distinguished variables, interpreted over  $S$ . It asserts that no counterexample tuple satisfies those relations. There are two rules. A leaf proves that a coordinate difference belongs to the  $S$ -row span of  $R$ , forcing two entries equal. A branch chooses one exponent permutation and treats every one of its  $\binom{n}{2}$  possible collisions. For each collision it points to an earlier proved node, with a bijective variable relabeling, and proves that every relation of that child follows from the parent relations and the collision. The children need

not describe exact quotients; the indicated implications suffice. The final node has no relations. Induction along earlier-node references proves that no counterexample exists in any  $S$ -module, hence in any eligible finite abelian group and, by the reduction, in any eligible group.

The producer represents connected components by rational weights and annihilators and uses gcd operations to merge collisions. The GAP checker instead uses integer Smith normal form: it verifies  $UMV = D$ , unimodularity of  $U, V$ , and diagonality of  $D$ , then tests divisibility after removing exactly the inverted prime factors and requires zero in all free coordinates. This checks the mathematical implications without trusting the producer's normalization. A separate data-only parser checks the certificate grammar, excluding executable instructions that could alter the checker.

| $n$ | Nodes      | Collision cases | Relation implications |
|-----|------------|-----------------|-----------------------|
| 4   | 227        | 1,056           | 3,953                 |
| 5   | 10,493     | 87,570          | 426,345               |
| 6   | 265,104    | 3,245,190       | 19,094,383            |
| 7   | 42,891,332 | 786,577,344     | 5,447,182,055         |

For  $n = 7$ , six GAP workers partitioned identifiers by residue modulo six. Each read all literal states but checked its assigned nodes; together their disjoint classes cover the proof, with every child identifier smaller than its parent. All six recorded actual process exit zero. The completed aggregate was written at 13:08:05 UTC on 12 September, about 26 hours after launch. The complete compressed certificate occupies 1,271,256,410 bytes. The six 14 GiB workspace ceilings are configuration limits, not a measurement of peak memory. These historical completion records have been inspected in the manuscript phase; the 26-hour verification has not been represented as a fresh paper-phase replay.

As a hypothesis control, allowing order five at  $n = 4$  yields the four nonidentity elements of  $C_5$ . In additive coordinates modulo five, distinct nonzero outputs would have product  $-1$ , whereas the input product times the product of exponents is 1. This is a counterexample to the weakened assumption and is stored as such. Sun's higher- $n$  small cyclic searches are a separate kind of evidence from these universal fixed- $n$  certificates. The unrestricted conjecture remains open in this investigation. Detailed proofs, grammar and receipts are indexed by `research/20.100-reduction.md` and `research/20.100-n7-report.md`.

## 7.2 20.89/19.83: almost Engel closure under additional hypotheses

An element  $a$  is almost Engel in  $G$  if one finite set  $E(a)$  contains  $[x, {}_n a]$  for every sufficiently large  $n$ , for each  $x \in G$ . The entry time may depend on  $x$ . The question asks whether these elements form a subgroup of an arbitrary linear group. A finite sink inside  $\langle a_1, \dots, a_r \rangle$  does not by itself provide a sink for a product of the  $a_i$  relative to all of  $G$ . The following deductions supply that ambient quantifier in specified cases; the general positive-characteristic case remains unresolved.

**The local reduction.** In every characteristic, a subgroup generated by finitely many elements almost Engel in  $G$  is finite-by-nilpotent. Here is the reduction and its imported inputs. Restrict to the finitely generated linear group  $X = \langle a_1, \dots, a_r \rangle$ , which is residually finite. Choose a finite-index normal subgroup  $H$  avoiding the nonidentity members of their finite sinks. Then each  $a_i$  is Engel in  $H\langle a_i \rangle$ : for  $h \in H$  eventual commutators lie both in  $H$  and in its avoided sink, and  $[ha_i^k, {}_n a_i] = [h, {}_n a_i]^{a_i^k}$ . Gruenberg's theorem identifies the Engel elements of a linear group with its Hirsch–Plotkin radical. Hence  $[H, a_i] \subseteq R = \text{HP}(H)$ . In  $X/R$ , all  $a_i$  centralize  $H/R$ ; its centralizer has a central subgroup of finite index, namely

its intersection with  $H/R$ . The locally nilpotent linear group  $R$  is soluble. Consequently  $X$  is virtually soluble. Shumyatsky's Theorem 3.3 now makes  $X$  finite-by-nilpotent [101]; the residual-finiteness and linear-radical inputs are recorded also in [111]. No product closure has been assumed in this deduction.

Let  $S$  be the set in question and  $L = \langle S \rangle$ . Every finitely generated subgroup of  $L$  is finite-by-nilpotent, by collecting the finitely many members of  $S$  used in its words. Call this property  $\mathcal{P}$ . In any group with  $\mathcal{P}$ , the torsion elements form a locally finite characteristic subgroup: finitely many torsion elements generate a finite-by-nilpotent group whose nilpotent quotient is generated by finitely many torsion elements, hence finite. This proves closure as well as local finiteness.

The unique minimal finite sink consists of the periodic points of  $x \mapsto [x, a]$ . Each such point belongs to  $\langle a^G \rangle$ . Express its finitely many points using finitely many conjugates of  $a$ , including  $a$  itself. Their subgroup is finite-by-nilpotent by the local reduction. In its nilpotent quotient a periodic repeated commutator must be trivial, so the entire minimal sink lies in the finite kernel. Thus its generated subgroup is finite. It need not be normal in the ambient group. In particular, for a torsion-free linear group every almost Engel element is Engel, proving closure there in every characteristic.

**Two lifting facts.** If  $X/D$  is locally nilpotent for some finite normal  $D$ , one can choose  $D$  characteristic: take one of minimum order. For an automorphism  $\sigma$ , the quotient by  $D \cap \sigma(D)$  embeds in the product of the two locally nilpotent quotients. That product is locally nilpotent, as are its subgroups. Minimality forces  $D = \sigma(D)$ .

Suppose  $X$  has  $\mathcal{P}$ ,  $N \triangleleft X$  is torsion-free, and  $X/N$  is finite-by-(locally nilpotent). Choose a finite characteristic kernel in  $X/N$  and let  $R$  be its preimage. The torsion subgroup  $K = T(R)$  is locally finite and injects into the finite group  $R/N$ , so is finite. For a finitely generated  $Y \leq X$ , a sufficiently late lower central term is finite by  $\mathcal{P}$ , and a sufficiently late term lies in  $R$  since  $X/R$  is locally nilpotent. A common later term therefore lies in  $K$ . Thus  $X/K$  is locally nilpotent. The construction preserves any group of automorphisms preserving  $N$ .

**Characteristic zero.** The group  $L$  contains no nonabelian free subgroup, since such a group cannot have  $\mathcal{P}$ . The characteristic-zero form of the Tits alternative applies without finite generation and makes  $L$  virtually soluble [105, Theorem 1]. Over an algebraic closure of the field let  $A$  be its Zariski closure and  $U$  the unipotent radical of  $A^\circ$ . The connected component is soluble, and  $A^\circ/U$  is a torus; the quotient  $A/U$  is linear algebraic [79]. Since  $L \triangleleft G$ , the subgroup  $N = L \cap U$  is normal in  $G$ . In characteristic zero it is torsion-free: the finite matrix logarithm gives  $\log(u^m) = m \log(u)$ . The group  $L/N$  is therefore linear and virtually abelian.

Shumyatsky's Lemma 3.2 gives the following prior consequence: in a virtually abelian group, almost Engel elements form a subgroup. Indeed it makes the normal closure of each such element finite-by-(locally nilpotent); choose its finite kernel characteristic as above and use the Hirsch–Plotkin radical after factoring out the finitely many kernels needed for a product. The images of  $S$  generate  $L/N$ , so every element of  $L/N$  is almost Engel there. Only now apply Shumyatsky's Theorem 4.2: an almost Engel linear group is finite-by-hypercentral, hence finite-by-(locally nilpotent). The torsion-free lifting fact gives a finite  $G$ -normal subgroup  $K \leq L$  with  $L/K$  locally nilpotent. For  $l \in L, x \in G$ , normality gives  $[x, l] \in L$ . The two elements  $[x, l]K, lK$  generate a nilpotent subgroup, so  $[x, {}_n l] \in K$  eventually. Thus  $K$  is one ambient sink for every  $l$ , proving  $S = L$ . The minimum-kernel argument makes  $K$  characteristic in  $L$ , and hence in  $G$ .

This proves closure for every characteristic-zero linear group, without generation or solubility assumptions on  $G$ . It does not make the full torsion subgroup of  $L$  finite; an infinite central torsion subgroup is possible. Both the unrestricted Tits step and the torsion-free unipotent kernel require characteristic zero in this argument.

**Two further characteristic-free cases.** If all finite subgroups of  $G$  have order at most  $b$ , the locally finite torsion subgroup  $K = T(L)$  has order at most  $b$ :  $b + 1$  of its elements would generate a larger finite subgroup. Property  $\mathcal{P}$  then gives  $L/K$  locally nilpotent, and the same commutator argument proves  $S = L$ , with  $K$  a characteristic common sink. If  $G$  has a torsion-free subgroup of index  $d$ , every finite subgroup acts freely on its  $d$  cosets, so its order divides  $d$ ; in particular  $|K| \mid d$ . This is not automatic in positive characteristic: the group generated by  $\text{diag}(t, 1)$  and  $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  over  $\mathbb{F}_p(t)$  contains elementary abelian subgroups of orders  $p^r$  for all  $r$ .

For an upper triangular group, order matrix entries by decreasing superdiagonal distance and refine its unipotent radical into one-entry additive factors  $B_s$ . Conjugation acts on the factor at  $(i, j)$  by the character  $\chi_s(g) = g_{jj}/g_{ii}$ . Define

$$P = \bigcap_{B_s \text{ infinite}} \ker \chi_s.$$

If multiplication by  $\chi - 1 \neq 0$  on an additive subgroup of a field has one finite eventual sink, a nonzero orbit first shows that  $\chi - 1$  has finite multiplicative order. All points are then periodic and must lie in the sink, so the additive subgroup is finite. This proves necessity of membership in  $P$ . Conversely  $P$  has a finite normal series whose factors are finite or  $P$ -central, including its abelian diagonal quotient. The Hall–Baer finite-central-series lemma makes it finite-by-nilpotent [111]. A finite lower-central term  $K = \gamma_{c+1}(P)$  is normal in  $G$ ; since  $[G, P] \subseteq P$ ,  $[x, {}_{c+1}g] \in K$  for  $x \in G, g \in P$ . This proves that  $P$  is exactly the almost Engel subgroup, with a uniform commutator-length bound in this triangular case.

**Normal subgroups of algebra unit groups.** Let  $A$  be a finite-dimensional associative unital algebra over an infinite field  $F$ , and  $G \triangleleft A^\times$ . Every almost Engel element of  $G$  is Engel. For  $a \in G$ , all but finitely many  $c_t = 1 + ta$  are units:  $\det(I + tL_a)$ , in the faithful left regular representation, is a nonzero polynomial with constant term one. Each such  $c_t$  commutes with  $a$  and normalizes  $G$ , hence permutes its minimal sink  $E(a)$ . If  $c_s^{-1}ec_s = c_t^{-1}ec_t = f$  for  $s \neq t$ , then

$$e + sea = f + saf, \quad e + tea = f + taf.$$

Subtracting gives  $ea = af$ , and substitution gives  $e = f$ ; thus  $e$  commutes with  $a$ . Finiteness of  $E(a)$  and infinitely many parameters give such a repetition for every  $e$ . The commutator map consequently sends the whole sink to 1; since it permutes the minimal sink, that sink is  $\{1\}$ . Gruenberg’s theorem now proves the subgroup assertion. This includes normal subgroups of  $\text{GL}_n(F)$  and  $\text{SL}_n(F)$ . Over  $\mathbb{F}_q$ , the same argument gives  $|E(a)| \geq q - \dim_F A + 1$  for a nontrivial minimal sink; in a matrix subalgebra it sharpens to  $q - s(a) + 1$ , where  $s(a)$  counts the distinct eigenvalues in  $\mathbb{F}_q$ .

Normality supplies the parameter conjugations. An arbitrary finite  $S_3 \leq \text{GL}_2(\mathbb{Q})$  has non-Engel almost Engel elements, so this hypothesis cannot simply be dropped. Proofs, exact finite controls, and the precise classical imports are indexed by `research/20.89-character-proof.md`, `research/20.89-residual-reduction.md`, `research/20.89-triangular-proof.md`, `research/20.89-bounded-torsion-proof.md`, and `research/20.89-normal-algebra-proof.md`.

### 7.3 21.121(b): sharp soluble bounds and permanence

Use the definition of admissible  $p$ -Jordan exponents from Section 6.2.5. The unresolved part asks for a universal upper bound three. The following arguments prove exponent two under a soluble-subgroup hypothesis, give a nonsoluble example with exact exponent strictly between two and three, and preserve the entire set of admissible exponents under specified operations. The definition and prior general framework are those in [48, 25].

**A characteristic abelian subgroup lemma.** If a finite group  $X$  has an abelian subgroup of index at most  $j$ , it has a characteristic abelian subgroup of index at most  $j^2$ . Recall the Chermak–Delgado argument in this form. For  $H \leq X$  put  $m(H) = |H||C_X(H)|$ . The product-size formula and centralizer inclusions give

$$m(H)m(K) \leq m(H \cap K)m(\langle H, K \rangle).$$

Thus the intersection  $D$  of all subgroups of maximum measure also has maximum measure and is characteristic. Its centralizer has maximum measure as well, since  $D \subseteq C_X(C_X(D))$ . Minimality of the intersection implies  $D \subseteq C_X(D)$ , so  $D$  is abelian. For the given abelian  $A$ ,

$$|X||D| \geq m(D) \geq m(A) \geq |A|^2 \geq |X|^2/j^2,$$

proving the bound. The lemma does not force  $D \subseteq A$ . It will be applied inside a  $p'$ -group when prime-to- $p$  order is required.

An abelian  $p'$ -subgroup  $C \leq \text{Aut}(P)$ , for a nontrivial finite  $p$ -group  $P$ , satisfies

$$|C| \leq |P/\Phi(P)| - 1.$$

Indeed the kernel of  $\text{Aut}(P) \rightarrow \text{Aut}(P/\Phi(P))$  is a  $p$ -group: it acts freely on the  $|\Phi(P)|^d$  tuples lifting a fixed basis of  $P/\Phi(P)$ , because each lift tuple generates  $P$ . Thus  $C$  acts faithfully on  $\mathbb{F}_p^d$ . Complete reducibility and commutativity embed it in a product of multiplicative groups of finite fields of orders  $p^{d_i}$ , with  $\sum d_i = d$ . Their orders have product at most  $p^d - 1$ .

**The soluble theorem.** Suppose every finite  $p'$ -subgroup of  $\Gamma$  has a normal abelian subgroup of index at most  $J$ , and every finite subgroup of  $\Gamma$  is soluble. For each finite  $G \leq \Gamma$  there is a characteristic abelian  $p'$ -subgroup  $B$  with

$$[G : B] \leq J^3 |G_{(p)}|^2.$$

First let  $R \neq 1$  be finite soluble with  $O_{p'}(R) = 1$ . Its Fitting subgroup  $F(R) \neq 1$  is a  $p$ -group and  $C_R(F(R)) \subseteq F(R)$ . For the latter standard fact, if the centralizer modulo  $Z(F(R))$  were nontrivial, choose a minimal normal elementary abelian subgroup there. Its preimage is a normal class-two nilpotent subgroup strictly exceeding the centralizer's intersection with  $F(R)$ , contradicting the Fitting theorem. A Hall  $p'$ -subgroup  $H$  therefore acts faithfully on  $F(R)$ . Choose  $A \triangleleft H$  abelian of index at most  $J$  and apply the coprime-action bound:

$$|R| = |R_{(p)}||H| \leq J|R_{(p)}|(|R_{(p)}| - 1).$$

Hall-subgroup existence here is the soluble Hall theorem, obtainable by induction with Schur–Zassenhaus.

For general  $G$ , set  $N = O_{p'}(G)$  and  $R = G/N$ . Then  $O_{p'}(R) = 1$ ; the same  $J$  works for its  $p'$ -subgroups, whose full preimages are  $p'$ -groups. The characteristic lemma gives  $B \text{ char } N$  abelian with  $[N : B] \leq J^2$ . Since  $N \text{ char } G$ , also  $B \text{ char } G$ , and the preceding estimate proves the assertion, with  $R = 1$  handled directly. If every finite subgroup instead has a

soluble subgroup of index at most  $d$ , take its core, of index at most  $d!$ . The bound becomes  $d!J^3|G_{(p)}|^2$ , with  $B$  normal in  $G$ .

When every finite subgroup has normal Sylow  $p$ -subgroup  $P$ , a sharper bound is available without solubility. Write  $G = P \rtimes H$  by Schur–Zassenhaus, choose  $A \triangleleft H$  abelian with  $[H : A] \leq J$ , and put  $B = C_A(P)$ . Both  $P$  and  $H$  normalize  $B$ , so  $B \triangleleft G$ . For  $P \neq 1$ ,

$$[G : B] \leq J|P|(|P/\Phi(P)| - 1) < J|P|^2;$$

for  $P = 1$ , use  $A$ . Exponent two is sharp for every  $p$ : the locally finite affine group  $\text{AGL}_1(\overline{\mathbb{F}}_p)$  has these hypotheses with  $J = 1$ . Its subgroups  $\text{AGL}_1(\mathbb{F}_q)$ ,  $q = p^n$ , have order  $q(q - 1)$  and no nontrivial normal  $p'$ -subgroup. Such a subgroup would centralize the normal translations, whose centralizer is the translation group itself. Hence their least normal abelian  $p'$ -indices are  $q(q - 1)$ , excluding every exponent less than two.

**An exact nonsoluble exponent.** Let  $\Gamma = \bigoplus_{\mathbb{N}} A_5$ , the finite-support direct product, and  $\rho = \log(60)/\log(4)$ . Then its 2-Jordan exponent is exactly  $\rho$ , attained with constant one. For a subgroup  $H \leq A_5^n$ , separate coordinates with full projection from those with proper projection. The first projection  $T$  is a product of diagonal simple factors, hence  $A_5^r$ ; induction proves this by projecting off the last coordinate, whose kernel is either  $A_5$  or trivial. In the latter case a surjection  $A_5^r \rightarrow A_5$  is supported on exactly one factor because its factor images commute and are normal. The second projection  $B$  is soluble, since all proper subgroups of  $A_5$  are soluble. Goursat's argument gives a common quotient of  $T$  and  $B$ ; it is perfect and soluble, hence trivial. Thus  $H = T \times B$ . Every odd-order subgroup of  $B$  is abelian, since its coordinate projections are cyclic subgroups of orders 1, 3, 5. The soluble theorem with  $J = 1$  gives a normal abelian odd-order  $D \leq B$  of index at most  $|B_{(2)}|^2$ . Therefore

$$[H : 1 \times D] \leq 60^r |B_{(2)}|^2 \leq (4^r |B_{(2)}|)^{\rho} = |H_{(2)}|^{\rho}.$$

Every finite subgroup of  $\Gamma$  has finite support, so this is a uniform bound. Full powers  $A_5^r$  have no nontrivial abelian normal subgroup and exclude every smaller exponent. Since  $16 < 60 < 64$ , one has  $2 < \rho < 3$ . This example is not virtually soluble: a soluble subgroup of  $A_5^r$  has index at least  $5^r$ .

**Preserving exponents.** Suppose  $e$  is admissible for  $\Gamma$  with constant  $J$ . For any finite  $G \leq \Gamma$ , apply the characteristic lemma inside  $N = O_{p'}(G)$ . The original abelian normal subgroup lies in  $N$ , giving  $[G : N] \leq J|G_{(p)}|^e$ , and the  $p'$ -subgroup bound gives a characteristic abelian  $D \leq N$  of index at most  $J^2$ . Thus

$$[G : D] \leq J^3 |G_{(p)}|^e.$$

Replacing normal by characteristic in the definition does not change its admissible exponents or attainment. If  $\Lambda \triangleleft \Gamma$  and finite subgroups of  $\Gamma/\Lambda$  have order at most  $b$ , apply this form to  $G \cap \Lambda$ ; its characteristic subgroup is normal in  $G$ . For a real  $e$ , the constant  $bJ^3 \max(1, b^{-e})$  works in  $\Gamma$ . Restriction gives the reverse implication. Taking a core proves the same invariance for any finite-index subgroup.

More generally, assume a fixed generalized  $p$ -Jordan constant  $J_0$  for  $\Gamma$ , and let  $N \triangleleft \Gamma$  be finite. Then  $\Gamma$  and  $\Gamma/N$  have exactly the same admissible exponents. Downward passage uses finite full preimages. For upward passage, take finite  $G \leq \Gamma$ ,  $K = G \cap N$ , and the preimage  $M$  of a normal abelian  $p'$ -subgroup in  $G/K$ . Put  $C = C_M(K)$ . It is normal in  $G$ ,  $[M : C] \leq |N|!$ , and  $C' \subseteq K \subseteq C_G(C)$ ; so  $C$  has class at most two. The soluble theorem supplies  $D \text{ char } C$ , abelian of  $p'$ -order, with  $[C : D] \leq J_0^3 |K_{(p)}|^2$ . If the quotient bound has constant  $J$ , then

$$[G : D] \leq J J_0^3 |N|! \max(1, |N|^{2-e}) |G_{(p)}|^e.$$

This preserves the exponent and its attainment. For a finite normal  $p$ -kernel the generalized hypothesis follows from the quotient bound, since  $p'$ -subgroups inject into the quotient. For other finite kernels it is necessary: the infinite extraspecial group at a prime  $q \neq p$  has central kernel  $C_q$  and abelian quotient, but contains finite extraspecial groups with unbounded least abelian index.

These results leave the unrestricted bound three unresolved. The full arguments and constants are in `research/21.121b-soluble-proof.md`, `research/21.121b-normal-sylow-proof.md`, `research/21.121b-a5-proof.md`, `research/21.121-permanence-proof.md`, and `research/21.121-finite-kernels.md`.

## 7.4 20.124: a Rota–Baxter operator in every infinite free rank

The question asks for a weight-one Rota–Baxter operator on a nonabelian free group with image its derived subgroup. The experiment constructs one in every infinite rank. The bijective-cocycle framework is prior work [9, 42]; the construction and identity can be given directly.

For an infinite cardinal  $\kappa$ , partition a free basis into levels  $X_0, X_1, \dots$ , each of size  $\kappa$ . Let  $F_m$  be free on levels through  $m$ . Both  $F_m$  and  $F'_m$  have cardinality  $\kappa$ ; for the lower bound on  $F'_m$ , use the distinct commutators with one fixed basis element. Choose a surjection  $X_{m+1} \rightarrow F'_m$  and define an endomorphism  $\phi$  by those maps, killing  $X_0$ . It lowers levels, so each element is killed by an iterate. Moreover  $\phi(F) = \bigcup_m F'_m = F'$ .

For any locally terminating endomorphism of any group put

$$D(g) = g\phi(g)^{-1}, \quad T(g) = g\phi(g)\phi^2(g) \cdots$$

Each product is finite for its particular  $g$ . Cancellation gives  $D(T(g)) = g = T(D(g))$ , so these are inverse set bijections. Define  $B(g) = \phi(T(g))$ , whence  $B(D(u)) = \phi(u)$ . For  $g = D(u), h = D(v)$ ,

$$gB(g)hB(g)^{-1} = uv\phi(uv)^{-1} = D(uv).$$

Therefore

$$B(gB(g)hB(g)^{-1}) = \phi(uv) = B(g)B(h),$$

the required identity, and  $\text{im } B = \text{im } \phi = F'$ . If  $\phi(x) = w$ , the explicit preimage is  $B(xw^{-1}) = w$ . The descendant multiplication is carried from the original free group by  $D$ . In finite rank an endomorphism image is finitely generated whereas  $F'$  has infinite rank. This limits this particular method; it does not rule out another Rota–Baxter map. See `research/20.124-infinite-rank-proof.md`.

## 7.5 17.118: characteristic exponent subgroups

Let a finite  $p$ -group  $G$  have an exponent- $p$  subgroup  $H$  of index  $p$ . The question asks for a characteristic exponent- $p$  subgroup of index bounded solely in terms of  $p$ . The experiment proves bounds  $p^{c+1}$  if  $H$  has class at most  $c$ , and  $p^c$  if  $G$  has class at most  $c$ . The optimal universal bounds for  $p = 2, 3$  are 4, 27.

Here is the class argument. Intersect all automorphic images of  $H$ , obtaining characteristic exponent- $p$  subgroup  $K$  with  $G/K \cong \mathbb{F}_p^r$ . Choose  $r$  images  $H_i$  defining independent coordinate hyperplanes and lifts  $x_i$  of the dual coordinate basis. Together with generators of  $K$ , they generate  $G$ , and omitting  $x_i$  leaves a subgroup inside  $H_i$ . For a class- $c$  group with this deletion property,  $r > c$  forces exponent  $p$ . To justify deletion without assuming it is an endomorphism of  $G$ , form the free class- $c$  group and impose  $w^p = 1$  for every word omitting a distinguished  $x_i$ . The deletion maps descend as commuting idempotents  $e_i$ . Every  $p$ th power lies in  $\bigcap_i \ker e_i$ . On the weight- $k$  lower-central factor,  $k < r$ , the product  $\prod_i (1 - e_i)$

kills each generating bracket, since it omits some  $x_i$ . It is the identity on the common kernel. Induction thus places that kernel in  $\gamma_r = 1$ . This proves the deletion assertion and the bound for  $G$ . If only  $H$  has class  $c$  and  $r > c + 1$ , every generating commutator of weight  $c + 1$  omits one  $x_i$ , hence vanishes in its  $H_i$ . So  $G$  has class at most  $c$ , and the same argument applies.

For  $p = 2$ ,  $H$  is abelian, giving 4. For  $p = 3$ , Laffey's theorem says that a finite 3-group not of exponent three has at most a  $7/9$  proportion of cube roots of one [71]. The union of the independent hyperplanes  $H_i$  has proportion  $1 - (2/3)^r$ . At  $r \geq 4$  this exceeds  $7/9$ . Thus  $r \leq 3$ , proving 27.

For every prime  $p$ , a uniform bound must be at least  $p^p$ . Take the associative  $\mathbb{F}_p$ -algebra with basis all words without repeated letters in  $p$  letters  $x_0, \dots, x_{p-1}$ , including the empty word, and multiply by concatenation when the supports are disjoint and by zero otherwise. Let  $J$  be its positive-degree ideal and  $G = 1 + J$ . Then  $J^{p+1} = 0$ . If  $u$  has linear part  $\sum a_i x_i$ ,

$$(1 + u)^p = 1 + \left( \prod_i a_i \right) \Omega,$$

where  $\Omega \neq 0$  is the sum of the distinct length- $p$  basis words. The preimage of any coordinate hyperplane has index  $p$  and exponent  $p$ , with class exactly  $p - 1$ . The subgroup  $K = 1 + J^2$  has exponent  $p$  and index  $p^p$ . It is intrinsically characterized by

$$K = \{g : (gh)^p = h^p \text{ for every } h \in G\}.$$

For a nonzero linear part  $a$ , choose  $b_j = 0$  where  $a_j \neq 0$  and  $b_i \neq -a_i$  for all other coordinates; then  $\prod b_i = 0 \neq \prod (a_i + b_i)$ , proving the characterization. Cycling the letters induces a  $p$ -cycle on  $\mathbb{F}_p^p$ . Any nonzero invariant subspace contains a nonzero constant vector, by taking the last nonzero iterate of  $\sigma - 1$ . That vector has a lift with nontrivial  $p$ th power. The image of a characteristic exponent- $p$  subgroup must therefore be zero, forcing it inside  $K$ . This proves sharpness for  $p = 2, 3$  and the general lower bound. It does not give unbounded index at a fixed  $p$ . See `research/17.118-partial-proof.md`; the bounded-class parameter is not known here to depend only on  $p \geq 5$ .

## 7.6 Other proved restrictions and a lower-bound construction

For Problem 21.114, write  $a(X) = |X/X'|$ ; a finite group is weakly ab-maximal if  $a(H) \leq a(X)$  for all  $H \leq X$ . The prior general framework is due to Lisi–Sabatini [74]. A central-product construction in the experiment yields order 8192 and derived length three, so a universal derived-length bound, if it exists, is at least three. Let  $P$  have cyclic central subgroup  $Z$  of order  $p^k$  with  $Q = P/Z$  weakly ab-maximal. Form the Heisenberg group  $E$  on  $(\mathbb{Z}/p^k\mathbb{Z})^r \times (\mathbb{Z}/p^k\mathbb{Z})^r \times \mathbb{Z}/p^k\mathbb{Z}$ , where  $r \geq 1$ , with central cross term  $u \cdot v'$ , and identify  $E' = Z(E)$  with  $Z$  in the central product  $G$ . Then  $|G| = |P|p^{2rk}$ ,  $a(G) = a(Q)p^{2rk}$ , and its derived length is  $\max(2, \text{dl } P)$ . For  $H \leq G$ , enlarge to  $HZ$ , which can only increase  $a(H)$ . In  $H/Z \leq Q \times V$ , let  $U = (H/Z) \cap V$  and let  $T$  be the projection on  $Q$ . Write  $|Z/(H' \cap Z)| = p^t$ . Then

$$a(H) \leq p^t |U| a(T) \leq p^t |U| a(Q).$$

The reduction of  $U$  modulo  $p^t$  is isotropic in the nondegenerate alternating pairing on  $(\mathbb{Z}/p^t\mathbb{Z})^{2r}$ . Character orthogonality gives its size at most  $p^{rt}$ , even when it is not a free module. Hence  $|U| \leq p^{2rk-rt}$ , proving  $a(H) \leq a(G)$ . Literal finite permutation models supply  $P$  of order 512, derived-series orders 512, 32, 2, 1, and central  $Z$  of order four, with  $Q$  of order 128 and  $a(Q) = 16$ . Taking  $r = 1$  gives the stated group and  $a(G) = 256$ . The exact input checks enumerate every subgroup of the two retained  $Q$ -models; they are recorded in

`research/21.114-central-products.md`. That file also proves that central extensions of these particular outputs still have derived length at most three. Iteration of this construction does not establish unboundedness.

For Problem 19.108, let  $p$  be odd and  $|P| = p^{10}$ , with abelian normal subgroup  $A$  of order at least  $p^6$ . Then for every irreducible complex  $\chi$  and every  $\chi(x) \neq 0$ , one has  $|x| \mid |P|/\chi(1)^2$ . Wilde's prior results cover degree at most  $p^3$ , groups of order at most  $p^9$ , and give, for a faithful violating character, a conjugate of  $\langle x \rangle$  intersecting it trivially [112]. If  $\chi$  has nontrivial kernel, the order-at-most- $p^9$  theorem in the quotient lifts the bound: the order of  $x$  is at most its quotient order times the kernel order. Thus a violation requires faithful  $\chi$  of degree  $p^4$ . The bound  $\chi(1) \leq [P : A]$  forces  $|A| = p^6$ , and equality in Clifford theory gives  $\chi = \lambda^P$ . Nonvanishing puts  $x$  in the normal subgroup  $A$ . The disjoint conjugate cyclic subgroups then force  $|x| = p^3$  and  $A = \langle x \rangle \times \langle x \rangle^g \cong C_{p^3}^2$ . The dual orbit of  $\lambda$  has size  $p^4$ . Such an orbit of a  $p$ -subgroup of  $\text{GL}_2(\mathbb{Z}/p^3\mathbb{Z})$  is invariant under translation by  $p^2\mathbb{F}_p^2$ . To see this, its image modulo  $p$  has size at most  $p$ . If its image modulo  $p^2$  had size  $p^3$ , the congruence kernel would supply  $I + pM_1, I + pM_2$  with independent directions at a point. Their  $p$ th powers are  $I + p^2M_i$  modulo  $p^3$ , forcing full  $p^2$ -fibers and an orbit of size  $p^5$ , a contradiction. The image modulo  $p^2$  thus has size at most  $p^2$ ; the size  $p^4$  forces every fiber full. Translation in a coordinate where  $x$  is primitive multiplies the orbit character sum by a nontrivial  $p$ th root of unity, so it vanishes, the final contradiction. Nonsplit extensions are included. Groups with no such large abelian normal subgroup remain uncovered; see `research/19.108-partial.md`.

For Problem 18.120, suppose  $G = AB$ ,  $A$  abelian and  $B$  of class at most two. Put  $C = (B')^G = \langle (B')^a : a \in A \rangle$  and  $K = \langle A, B' \rangle = AC$ . If  $C$  is abelian,  $B_0 = K \cap B$  is abelian and  $K = AB_0$ . Indeed an element  $ac \in K \cap B$  centralizes  $B'$ ; as  $c$  does too,  $a$  centralizes  $B'$ , and commutativity of  $A$  makes it centralize all its  $A$ -conjugates. Two such elements commute. The product assertion follows by writing an element of  $K$  in  $AB$ . If  $G$  has class at most four, conjugates of elements of  $B' \subseteq \gamma_2(G)$  differ from them by elements of  $\gamma_3(G)$ ; all cross brackets have weight at least five, while  $B'$  is abelian. Thus  $C$  is abelian. This proves the class-four case without finiteness or  $A \cap B = 1$ . The defect-two subnormal case is also a direct deduction from McCann's earlier lemma [77]; see `research/18.120-partial.md`.

For Problem 6.47, if a binary word defines another group operation on a class-two group  $G$ , collection and the two identity laws give  $x * y = xy[x, y]^k$ . This is associative, with unchanged inverses and powers and with new commutator  $[x, y]_* = [x, y]^{2k+1}$ . Collect an identity in the free class-two group as a product of unary powers and binary commutator powers. Setting all but one, and then all but two, variables equal to one shows that each collected factor is separately an identity of  $G$ . Unchanged powers and the commutator formula transfer them all to the new group, which therefore lies in  $\text{var}(G)$ . This is a direct proof of a restricted case within the subject already studied by Cooper [29], without a novelty claim; see `research/6.47-class-two.md`.

## 7.7 Bounded exclusions with no general conclusion

The following completed searches are useful negative evidence, with the stated coverage boundaries.

- For Problem 20.21, structural reductions and catalogue exclusions through order 1536 force any counterexample to have order at least 3072. All 26,760 retained kernel pairs at order 1536 were separated from their multiplication tables; catalogue and subgroup-enumeration coverage remain dependencies.
- For Problem 19.20, all 91,774 nonabelian groups through order 511 gave no equality

case and 469 reverse inequalities. This does not decide the arbitrary-group equality question.

- For Problem 21.113, 2,750 ordinary character tables provided 9,850 table/prime cases. Of the corresponding modular cases, 7,573 were available and 2,277 were unavailable. No counterexample appeared in the available cases.
- For Problem 21.99, the retained coverage includes all 4,722 transitive groups of degrees 2–20; direct word-certificate checks cover the 86 groups of degrees 2–8. These are different scopes, not interchangeable verification counts.
- For Problem 18.43, exact matrix-trace fingerprints screened all 3,933,931,043 positive binary necklaces of lengths 1–36, with no identity candidate. A future collision would require symbolic certification; larger lengths remain outside the search.

The corresponding reports and scripts are indexed in `reports/FINAL_REPORT.md` and `reports/STATUS.md`. The absence of a witness in these finite domains is not a proof for unbounded orders, dimensions, or word lengths.

## 8 Rediscoveries and deductions from prior work

Earlier theorems settled several questions encountered during the run, sometimes after the agent had developed a separate argument. The examples below include strong amalgamation, character and coset constructions, self-similar nilpotent groups and effective universal groups. We retain the proofs and deductions with their sources, distinguishing an explicit prior answer from a consequence requiring an additional argument. These results remain outside the deadline candidate count.

### 8.1 13.19: a small subdirect-quotient counterexample

The question asks whether  $Q/H$  must be a regular  $p$ -group when  $H \triangleleft Q$  and both are subdirect in the same finite direct product. An earlier construction of Kearnes–Mayr–Ruškuc already implies a negative answer [64]. Specifically, their Section 6 has a finite perfect group  $G = HM$ , with  $M$  a 2-group and normal  $N \leq M$ , satisfying  $[N, N] > [G, N, N]$ . In their construction with  $d = 2$ , Theorem 4.2 gives  $S = \Gamma_1(G, N) \leq G^4$  and  $T = \Gamma_3(G, N) = \gamma_3(S)$ , with  $S/T$  nonabelian of class two. Both contain the diagonal  $G$ , hence are subdirect. All nondiagonal generators lie in  $M^4$ ; since the diagonal is killed in  $S/T$ , this quotient is a finite 2-group. A nonabelian 2-group is irregular. This is the prior implication used to remove 13.19 from the count.

The experiment’s smaller example is transparent. Let  $D = \text{UT}_3(\mathbb{F}_2)$ , writing its elements as

$$d(a, b, c)d(a', b', c') = d(a + a', b + b', c + c' + ab').$$

It is dihedral of order eight. In  $D^4$  put

$$Q = \{(d(a_i, b_j, c_{ij}))_{1 \leq i, j \leq 2} : a, b \in \mathbb{F}_2^2, C \in M_2(\mathbb{F}_2)\}.$$

Its multiplication on the eight coordinates is

$$(a, b, C)(a', b', C') = (a + a', b + b', C + C' + a(b')^T),$$

so  $|Q| = 256$ . The map

$$\phi(a, b, C) = d(a_1 + a_2, b_1 + b_2, \sum_{i,j} c_{ij})$$

is onto and multiplicative, since the sum of the entries of  $a(b')^\top$  is  $(a_1 + a_2)(b'_1 + b'_2)$ . Its kernel  $H$  has order 32 and contains the entire diagonal  $D$ . Both  $H$  and  $Q$  thus project onto all four factors, with  $Q/H \cong D$ . For  $s = d(1, 0, 0)$ ,  $t = d(0, 1, 0)$ , one has  $s^2 = t^2 = 1$ , but  $(st)^2 = d(0, 0, 1) \neq 1$ . Every derived-subgroup square is trivial, so no correction allowed in the regularity identity can repair this failure. The original question permits  $p = 2$ .

There is also a direct universal construction: every finite  $p$ -group  $K$  is such a quotient when the factors may vary. Let  $m = |K|$ , choose generators  $g_1, \dots, g_d$ , and let  $I$  be the augmentation ideal in  $\mathbb{F}_p[K]$ . Induction on  $|K|$  gives  $I^m = 0$ : choose central  $z$  of order  $p$ ; the kernel of the map to  $\mathbb{F}_p[K/\langle z \rangle]$  is  $(z - 1)\mathbb{F}_p[K]$ , whose  $p$ th power is zero, and use induction in the quotient. Let  $A$  be the nonunital free associative algebra on symbols  $x_{ij}$ ,  $1 \leq i \leq m, 1 \leq j \leq d$ , truncated at degree  $m$ . In  $1 + A$  take  $Q = \langle 1 + x_{ij} \rangle$ . Substitution  $x_{ij} \mapsto g_j - 1$  maps  $Q$  onto  $K$ . For each  $i$ , delete the  $i$ th block of symbols and let  $G_i$  be the image of  $Q$ . These maps jointly embed  $Q$ : each surviving monomial has length less than  $m$ , omits some block, and retains its coefficient when that block is deleted. The kernel  $H$  of the map onto  $K$  projects onto every  $G_i$ , since  $(1 + x_{kj})(1 + x_{ij})^{-1} \in H$  projects to  $1 + x_{kj}$  for  $k \neq i$ . This proves the assertion, including simultaneous subdirectness. The trivial  $K$  is handled by  $Q = H = C_p$ . The construction is not claimed to be small or new. See `research/13.19-proof.md`, `research/13.19-universality.md`, and the source comparison in `research/13.19-review.md`.

## 8.2 17.34: a bounded-class separation proof using PBW

Let  $\mathcal{N}_c$  be the full quasivariety of torsion-free nilpotent groups of class at most  $c$ . If  $H \leq G \in \mathcal{N}_c$  is divisible, then its dominion is exactly  $H$ . The fixed-class strong amalgamation theorem of d'Elbée–Müller–Ramsey–Siniora already implies this [33, Theorem 4.35]. Use the lower-central filtration on the rational Lie algebra of the Malcev completion of  $G$ , and its intersection filtration on the subalgebra corresponding to  $H$ . These are compatible Lazard series even if the latter is not the intrinsic lower-central series of the subalgebra. Strong amalgamation separates the two copies outside  $H$ . Finite-dimensional amalgamation suffices in arbitrary rank: every finite set of constraints lies in a finitely generated, hence finite-dimensional, nilpotent Lie subalgebra; compactness then supplies the full pair of embeddings.

Here is the retained alternative proof. For a rational Lie algebra  $L$  of class at most  $c$  and subalgebra  $\mathfrak{h}$ , let  $U = U(L)$ ,  $I$  its augmentation ideal, and  $J = U\mathfrak{h}$ , a left ideal. Set

$$M = I/(J + I^{c+1}), \quad d(x) = x + J + I^{c+1}.$$

This is a left  $U$ -module and  $d([x, y]) = xd(y) - yd(x)$ . Choose a basis of  $L$  adapted both to the lower-central filtration and to  $\mathfrak{h}$ , weighting a basis element by its largest filtration index. Order the basis with the complementary vectors before the  $\mathfrak{h}$ -vectors. PBW monomials of weight at least  $n$  span exactly  $I^n$ . One inclusion follows since a vector of weight  $w$  lies in  $I^w$ ; the other follows by reordering, because brackets do not decrease total weight. Also  $J$  is spanned exactly by ordered monomials containing a nonempty terminal  $\mathfrak{h}$ -block: multiplication on the right by an  $\mathfrak{h}$ -vector only reorders that block. Thus the complementary single-letter vectors all survive independently in  $M$ , while the  $\mathfrak{h}$ -vectors vanish. This proves  $\ker d = \mathfrak{h}$ , in any dimension.

In  $P = L \ltimes M$ , with  $M$  abelian, the two maps

$$f(x) = (x, 0), \quad g(x) = (x, d(x))$$

are injective Lie homomorphisms agreeing exactly on  $\mathfrak{h}$ . Induction gives  $\gamma_r(P) \subseteq \gamma_r(L) + I^{r-1}M$ . Since  $I^c M = 0$ , the class remains at most  $c$ . The numerator  $I$  is essential: replacing it by  $U$  would generally allow an unwanted extra nilpotency step.

Embed  $G$  in its rational Malcev completion and use the BCH correspondence, which preserves the class bound and works in arbitrary rank [5]. Unique roots make the divisible subgroup  $H$  a rational Lie subalgebra there. The BCH group of  $P$  lies in  $\mathcal{N}_c$ , and the restrictions of  $f, g$  to  $G$  agree exactly on  $H$ . This is precisely the required dominion separation. The result concerns the full  $\mathcal{N}_c$ , with no assertion for every proper subquasivariety. The class-two predecessor is Shakhova's explicit 2015 result [100]. PBW separation in unrestricted Lie algebras is also prior work of Bergman [13]. See `research/17.34-proof.md` and `research/17.34-known-consequence.md`.

### 8.3 21.115: Fourier support and class-two linearization

Sambale's preprint of 8 September 2026, two days before the experiment, proves the general complement bound for unions of cosets, including mixed left and right cosets [96]. The experiment first developed the following restricted arguments, then found that resolution and excluded the problem from its candidate count. The earlier Sambale–Tărnăuceanu paper treated several special cases [97]; the character-product method below also appears in Szegedy's work [104].

If cosets  $g_i H_i$ ,  $1 \leq i \leq n$ , do not cover a finite abelian group  $A$ , choose  $x_0$  outside their union. Characters separate points in  $A/H_i$ , so choose  $\chi_i$  trivial on  $H_i$  with  $\chi_i(x_0) \neq \chi_i(g_i)$ . The nonzero function

$$f(x) = \prod_{i=1}^n (\chi_i(x) - \chi_i(g_i))$$

vanishes on the union and is a linear combination of at most  $2^n$  distinct characters. For a nonzero combination  $f = \sum_{\chi \in T} a_\chi \chi$ , orthogonality and Cauchy–Schwarz give

$$|A| \sum |a_\chi|^2 = \|f\|_2^2 \leq |\text{supp } f| \|f\|_\infty^2 \leq |\text{supp } f| |T| \sum |a_\chi|^2.$$

Thus the complement has at least  $|A|/2^n$  elements.

The same deduction applies to a class-two finite group whose derived subgroup has odd order. On its underlying set define  $x + y = xy[x, y]^{-1/2}$ , taking the unique half only in the central odd-order derived subgroup. Class-two commutator identities show this is abelian: both triple sums equal  $xyz[x, y]^{-1/2}[x, z]^{-1/2}[y, z]^{-1/2}$ . Every original subgroup is additive, since the half of its commutator is a power of that commutator. For fixed  $g$ ,  $T_g(x) = [g, x]$  is an additive central-valued map with  $T_g^2 = 0$ ; hence  $L_g = 1 + \frac{1}{2}T_g$  is an additive automorphism, and  $gx = g + L_g(x)$ . Every original coset is therefore an affine coset in the same abelian group. Applying the Fourier argument proves the bound. This avoids any assumption that the whole group has odd order, but does not cover every class-two 2-group. The general theorem is credited to the earlier preprint, not inferred from these restricted cases. See `research/21.115-partial-proof.md`.

### 8.4 16.60: a tensor proof of an existing character bound

The full affirmative answer was stated in the May 2026 preprint of Yerrapati–Dixit–Shukla [113, Theorem 3.1]. For a finite group  $G$  and any automorphism  $\alpha$ , the number of elements inverted by  $\alpha$  is at most  $T(G) = \sum_{\chi \in \text{Irr}(G)} \chi(1)$ . The following proof supplies the needed estimate without treating a character value on a product as a product of character values.

For a unitary irreducible representation  $\rho$  on  $V$ , put

$$E = \frac{1}{|G|} \sum_{g \in G} \rho(g) \otimes \rho(\alpha(g)), \quad F(v \otimes w) = w \otimes v.$$

The operator  $E$  is the orthogonal projection onto invariants of  $V \otimes (V \circ \alpha)$ . Its rank is  $\dim \text{Hom}_G(V^*, V \circ \alpha)$ , at most one by Schur's lemma. The trace identity  $\text{tr}((A \otimes B)F) = \text{tr}(AB)$  gives

$$\iota_\alpha(\chi) = \frac{1}{|G|} \sum_g \chi(g\alpha(g)) = \text{tr}(EF), \quad |\iota_\alpha(\chi)| \leq \text{rank } E \leq 1.$$

The inequality uses an orthonormal basis of  $\text{im } E$  and unitarity of  $F$ ;  $E$  need not commute with  $F$ . The regular-character identity now yields

$$|\{g : \alpha(g) = g^{-1}\}| = \sum_\chi \chi(1) \iota_\alpha(\chi) \leq T(G).$$

No involutivity of  $\alpha$  is needed. Alternatively, every inverted element lies in  $\text{Fix}(\alpha^2)$ , where the restriction of  $\alpha$  is involutive. The classical twisted indicator bound there and  $T(H) \leq T(G)$  for  $H \leq G$  imply the same result; the latter inequality follows by restricting the direct sum of all irreducible  $G$ -modules and using Frobenius reciprocity. The source audit identifies an unexplained scalar Cauchy–Schwarz passage in its Lemma 2.19; this tensor argument establishes the needed statement independently of that passage, without assigning priority to the answer. See `research/16.60-known-consequence.md`.

## 8.5 21.132: a graded radical quotient with finite centre

The experiment ultimately classified this as a deduction from prior radical theorems, rather than a new solution. Start with a two-generated, positively graded, infinite-dimensional nil  $\mathbb{F}_p$ -algebra  $A$ , with finite-dimensional graded pieces, from the graded Golod construction [37]. Let  $L(A)$  be its Levitzki radical, the largest locally nilpotent ideal. The centreless quotient  $B = A/L(A)$  is the construction used by Sereda–Sozutov [98]. The radical is proper, since otherwise finite generation would make  $A$  nilpotent. It is homogeneous by the classical graded-radical theorem explicitly recorded in [47]. Hence  $B$  remains positively graded with finite-dimensional pieces. It is infinite-dimensional: otherwise positive grading makes it nilpotent, contradicting  $L(B) = 0$ .

If  $z$  commutes with all of  $B$ , then  $z$  is nilpotent and the ideal  $B^1z$ , using the unitization, is nilpotent. It must be zero, so  $z = 0$ . Let  $H = \langle 1 + b_1, 1 + b_2 \rangle \leq 1 + B$ . It is a  $p$ -group, since each of its elements is one plus a nilpotent element. Its centre is trivial: a central  $1 + b$  commutes with the algebra generators, so  $b$  is central in  $B$ . It is infinite, since a finite  $H$  would have finite-dimensional linear span containing the algebra generators and hence all of  $B$ . The truncations  $B/B_{\geq n}$  give finite  $p$ -group quotients separating every element, by finite homogeneous support. Thus  $H$  is residually finite. Adjoin a one-dimensional zero-multiplication algebra  $\mathbb{F}_p z$ . The subgroup generated by one plus the three algebra generators is  $H \times C_p$ , with centre  $C_p$ . It meets the Notebook's algebraic definition of a Golod group, for every prime  $p$ , and is at most three-generated. The grading is what proves residual finiteness; it is not inferred from an arbitrary quotient preserving that property. The exact deduction under the question number was not located in a prior paper, but its inputs and classification are explicit. See `research/21.132-prior-consequence.md`.

## 8.6 21.42: positive gradings and self-similar actions

Every three-generated torsion-free nilpotent group of class three is self-similar, giving a negative answer to the existence question. Dekimpe–Igodt–Pouseele proved the positive-grading assertion for the corresponding Lie algebras [32]; the exact positive form is restated in [31]. Mathieu's criterion transfers it to the original lattice, not merely an unspecified

finite-index subgroup [76, Proposition 6]. The rational Lie algebra of the Malcev completion is generated by the logarithms of the three group generators, so these results apply.

The mechanism is instructive. Positive weights give dilations by  $q^j$  in weight  $j$ . For a suitable integer  $q > 1$  the dilation  $\delta$  preserves the given lattice  $G$ , with finite-index image. Its inverse is a virtual endomorphism  $\delta(G) \rightarrow G$ . If a subgroup  $K \subseteq \delta(G)$  is invariant under this inverse, every  $\delta^{-n}(\log k)$  remains in the discrete lattice and tends to zero. Thus  $k = 1$ . The virtual endomorphism has trivial normal core and gives a faithful recurrent action on the coset tree. There is no prescribed degree or finite-state requirement in the printed question. The archive also retains an independent classification argument giving positive weights for the three-generator class-three Lie algebras, with weight choices  $(1, 1, 2)$ ,  $(2, 2, 1)$ , and  $(3, 4, 2)$  in the nonhomogeneous cases. That proof is a rediscovery of the grading assertion; the lattice step remains an imported theorem. See `research/21.42-known-consequence.md`.

## 8.7 20.33: an effective universal group, with oracle bookkeeping

For any oracle  $X$ , there is a three-generated  $X$ -computably-enumerably presented group  $G_X$  containing every finitely generated group with an  $X$ -c.e. presentation. This is a standard embedding consequence; Mikaelian's explicit universal words improve three generators to two [78].

Dovetail all oracle programs enumerating relators on each finite alphabet, ignoring ill-formed outputs. Take the free product  $U_X$  of all their presented groups. It has an  $X$ -c.e. presentation on a countable computable alphabet  $g_1, g_2, \dots$ , and each factor embeds. For any such countably presented  $U$ , in  $U * F(a, b)$  the elements  $q_i = a^i b a^{-i}$  freely generate a subgroup, as part of a Schreier basis. The elements  $g_i q_i$  also freely generate: the retraction killing  $U$  detects any freely reduced relation among them. The HNN extension

$$\langle U, a, b, t \mid t^{-1} q_i t = g_i q_i \ (i \geq 1) \rangle$$

therefore embeds  $U$ , and its relations express  $g_i = t^{-1} q_i t q_i^{-1}$ . Eliminating the  $g_i$  gives a three-generator presentation by effective substitution in the old relators. No word-problem decision is required, and the same oracle suffices.

Conversely, a finitely generated subgroup of an  $X$ -c.e. presented group has such a presentation: enumerate products of conjugates of enumerated relators and compare their freely reduced words with substitutions in the subgroup generators. Dovetailing enumerates exactly the subgroup kernel. Finite free products and adding finitely many relators preserve  $X$ -c.e. presentability as well. Thus both directions of the Notebook formulation follow; in the forward direction take one copy of  $G_X$  and the empty extra relator set. Decidable word problem and finite presentation are not additional requirements. See `research/20.33-known-consequence.md`.

## 8.8 20.8 and 20.11: free-group deductions with the exact quantifiers

For Problem 20.8, the Lei-Zhang construction supplies a counterexample [73]. In  $F(a, b)$  put

$$u = (a^{-1}b)^2, \quad v = (a^{-2}b^2)^2, \quad H = \langle a, u, v \rangle, \quad K = \langle u, aua^{-1}, v, a^2va^{-2} \rangle.$$

The ranks are 3, 4, respectively. The homomorphism on the free basis of  $H$  sending  $a$  to  $b$  and fixing  $u, v$  differs from inclusion but fixes  $K$ : commutativity of  $c_i = a^{-i}b^i$  with  $c_i^2$  gives  $a^i c_i^2 a^{-i} = b^i c_i^2 b^{-i}$ . To check the additional intermediate-subgroup condition, identify  $H$  with  $F(t, x, y)$ . The core graph of  $\langle x, txt^{-1}, y, t^2yt^{-2} \rangle$  has vertices 0, 1, 2, a  $t$ -path between them,  $x$ -loops at 0, 1, and  $y$ -loops at 0, 2. Its rank is four. Of its five vertex partitions, no identification and  $0 = 2$  give rank-four folded images; the other three give the full three-loop

rose. The image in the core graph of any intermediate subgroup is one of these. A rank-three image forces that subgroup to be all of  $H$ . Thus every proper intermediate subgroup has rank at least four, including infinitely generated ones. The ambient rank is two, which the question permits. Full folded graphs are retained in `research/20.8-known-consequence.md`.

For Problem 20.11, Jaikin-Zapirain’s prior rigidity theorem [50] implies all four assertions, even when the initial subgroup  $F$  and ambient free group  $H$  are not finitely generated. Let  $\mathcal{C}$  be the nonempty family of overgroups of  $F$  in  $H$  of least finite rank  $r$ . For  $G_1, G_2 \in \mathcal{C}$ ,  $G_1$  is compressed in  $\langle G_1, G_2 \rangle$ , hence inert by Corollary 1.2. Their intersection has rank at most  $r$ , and thus exactly  $r$  by minimality. Apply Corollary 1.5 to this now finitely generated intersection to show that the join also has rank  $r$ . A chain union has rank at most  $r$ : otherwise  $r + 1$  members of a free basis generate a free factor lying in one chain member, forcing its rank too large. Zorn’s lemma and pairwise joins give a greatest member. For a least member, work inside one countable  $G_0$ . A countable subfamily has the same intersection as all of  $\mathcal{C}$ , by choosing a subgroup excluding each element of  $G_0$  outside that intersection. Finite intersections form a descending sequence of rank  $r$ . Any finite neighborhood in the intersection’s Schreier graph eventually appears unchanged in those graphs. A finite subgraph of Betti number greater than  $r$  is therefore impossible. The intersection has rank  $r$  and is the least member. These limit steps are needed to preserve the printed quantifiers. See `research/20.11-known-consequence.md`.

## 8.9 19.9, 16.14, and 19.48: further prior answers

For Problem 19.9, both answers were already explicit in the pre-experiment GroupTheory50 preprint [41]. The archive’s alternative deduction uses  $G = \mathrm{Sp}(10, \mathbb{Z})$ . Its nonabelian tensor square is its universal central extension, identified with the pullback of the real topological universal cover in this stable rank. Deligne’s theorem makes that extension non-residually finite [34, 12]. It is finitely generated: lifts of generators generate any perfect central extension of a finitely generated group. Residual finiteness of finitely generated linear groups then excludes linearity over every field.

For braid groups  $B_n$ ,  $n \geq 4$ , their abelianization  $\mathbb{Z}$  splits the tensor square as  $\mathbb{Z} \times (B_n \wedge B_n)$ . The exterior square has kernel  $H_2(B_n) = C_2$  over  $B'_n$  [10, 2]. In the rational Clifford algebra with  $e_i^2 = 1/2$  and  $e_i e_j = -e_j e_i$ , put  $u_i = e_i - e_{i+1}$ . Then  $u_i^2 = 1$ , adjacent generators satisfy the braid relation, and distant generators anticommute. The induced map from the free presentation sends all relators to central signs and detects the nontrivial Hopf-kernel element  $[x_1, x_3]$  as  $-1$ . Combining it with the commutator image embeds the exterior square in  $B'_n \times A_n^\times$ . Left multiplication on the  $2^n$ -dimensional Clifford algebra and the faithful Lawrence–Krammer representation [69] give a faithful tensor-square representation of dimension  $1 + \binom{n}{2} + 2^n$ . This detects the central kernel explicitly; it does not assume that every finite central extension of a linear group is linear. See `research/19.9-proof.md`.

For Problem 16.14, centrality of all involutions makes  $E = \Omega_1(Z(G)) = \Omega_1(G)$  a maximal elementary abelian normal subgroup, of rank  $r = d(Z(G))$ . Sambale’s earlier centralizer bound gives  $|C_G(E) : \Phi(C_G(E))| \leq 2^{2r}$  [95]. Since  $C_G(E) = G$  and  $\Phi(G) = G^2$  for a finite 2-group, the requested inequality is  $d(G/G^2) \leq 2d(Z(G))$ . Products of quaternion groups attain equality. This is a specialization of the prior bound, not a new theorem; see `research/16.14-prior-result.md`.

For Problem 19.48, another pre-experiment GroupTheory50 preprint explicitly gives a nonclosed irreducible net [40]. With  $R = \mathbb{Z}$ , let  $\alpha$  be a root of  $X^3 - X - 1$  and  $T = \mathbb{Z}[\alpha]$ . Its exceptional levels are

$$A = \mathbb{Z} + \mathbb{Z}\alpha + 2\mathbb{Z}\alpha^2, \quad B = \mathbb{Z} + 2\mathbb{Z}\alpha + \mathbb{Z}(\alpha^2 - 1),$$

with other levels  $2T$ . The identity  $\alpha^{-1} = \alpha^2 - 1$  allows root transvections to produce the diagonal  $\text{diag}(\alpha, \alpha^{-1})$ , whose conjugation yields the forbidden upper parameter  $\alpha^2 \notin A$ . The common ideal  $2T$  ensures the net conditions and nonzero levels. These are  $R$ -modules, as the question requires, rather than subspaces over the fraction field. The independently found characteristic-three example remains in `research/19.48-proof.md`. The earlier posted proofs are preprints; no outside human acceptance is inferred from their availability.

### 8.10 12.69: a literal counterexample retained as a formulation issue

The printed condition permits a nonzero element of  $\mathbb{Z}G$  sending each field element into a subfield. For  $F = \mathbb{Q}(i)$ ,  $S = \mathbb{Q}$ , and  $G = \{1, \sigma\}$  with  $\sigma$  complex conjugation, the fixed element  $1 + \sigma$  acts multiplicatively as the norm  $x\sigma(x) \in \mathbb{Q}$ . Its nonnegative exponents also give zero at  $x = 0$ . Additive interpretation gives the trace and the same conclusion. The extension is proper and separable. The example  $F = \mathbb{C}$ ,  $S = \mathbb{R}$  likewise contradicts the printed claim that the uncountable analogue is affirmative. This points to a missing qualification or other formulation problem; changing from norm to trace does not repair it. The experiment retained the literal observation, proposed no invented correction, and counted no substantive resolution of the intended question. See `research/12.69-formulation-audit.md`.

## 9 What the experiment shows

The experiment produced a substantial body of mathematical work in a fixed time, and a second model was able to assess much of it by detailed reading and independent reconstruction. Explicit small examples made some arguments particularly accessible. Structural reductions and finite certificates made other claims checkable despite a much larger search behind them. The mathematical contribution of the run includes partial theorems and useful deductions from prior work, as well as candidate answers to the selected Notebook questions.

The review phase adds a different contribution. Claude confirmed many arguments and computations, exposed gaps in what had been supplied to readers, and prompted a stronger geometric example. Codex’s response also corrected mistaken objections. Preserving that exchange shows how two different language models can contribute to mathematical scrutiny: checking a result, challenging its hypotheses, consulting the original source, and revising a judgment. The resulting evidence is stronger than the producing agent’s initial self-assessment.

The scope of the experiment remains specific. The agent chose its own problems, so the 46 entries do not estimate a success rate on a random sample of open questions. Shared inputs link several answers, and prior exposure through training or literature is not fully observable. Independent checking supports correctness of the claims examined; discovery priority needs its own source assessment. The public archive also has a stated limit for the largest certificate. These qualifications accompany the mathematical statements and the review record without changing their value as inspectable results.

The eleven overlaps examined for version 2 make the distinction concrete. They range from matching arguments to different constructions with stronger hypotheses or conclusions. The preserved deadline archive records what the agent produced under the budget; the later source comparison records what we can now say about its relationship to other work. Available document and repository dates do not determine every discovery’s priority or independence. This experiment therefore supplies a research portfolio and review record, rather than a verified total of new solutions or a controlled comparison of model capabilities.

For a further experiment, the main improvements would be a predetermined comparison set, separate records for correctness and novelty, and measurement of the time needed for

independent review. Selected formalizations could strengthen the most consequential reductions and finite checks. Appendix D develops these design and measurement questions; the present paper supplies the arguments and review history on which such work can build.

**Contributions and reviewer credit.** Codex gpt-6-astra carried out the research, drafted proofs and programs, prepared the manuscript, and conducted internal checks. Michael Kinyon and Josef Urban conceived and oversaw the experiment. Anthropic’s Claude Opus[1m], through Claude Code, provided the independent mathematical review, wrote separate checking programs, examined sources, and proposed the strengthening of 14.72. We thank Claude for the detailed reports and constructive mathematical exchange. We also thank Evgeny Khukhro for identifying contemporary solutions, and acknowledge Aluna Rizzoli, Achyuth Jayadevan, Lily Zhang and Evan Li for the overlapping work discussed in this version.

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## A The instruction and experimental record

The original goal record in the JSONL and the first expanded goal directive in the HTML retain the following user-provided objective, reproduced verbatim:

Solve as many previously unsolved problems from THE KOUROVKA NOTEBOOK (docs/21tk.pdf) as you can within an overall time limit of 48 hours. The gap system is installed in gap-4.16.1 but you can also download and install further SW if useful. In general, try to limit yourself to 20 CPU cores and 100G RAM on this machine while working. Write detailed plans and reports, iterate fast and create a local git repo keeping your work and documenting your progress, commit often there (but don't try to push). Work rigorously - your solutions will be independently reviewed.

The surrounding continuation instruction told the agent to keep the full goal active across turns, use the current worktree and external state as authoritative, and base process status on observed handles and exit records. The JSONL records the continuations and context compactions described in Appendix B. The human message during setup says that GAP is now present. The initial availability issue and subsequent working installation are documented separately in the early repository records.

The original local plan, `research/PLAN.md`, divided work into overlapping selection, experiment, concentration, and review phases. It called for exact statement transcription, source checks, bounded searches, second representations, and explicit separation of partials from complete candidates. It prohibited delegation without authorization; no model sub-agents are reported in the retained research record. Independent operating-system processes were allowed for computational partitions.

The original research interval was

2026-09-10 20:56:46 UTC   to   2026-09-12 20:56:46 UTC.

The final administrative commit is

`cff2c37b9bf6b737e8ad5f7ead12291a551b5201`.

All measurements of the research Git history stop at this revision; later manuscript commits are excluded. The final research response was logged at 20:57:38 UTC, approximately 52 seconds after the deadline. We distinguish this closeout from the 48-hour interval. The supplied Notebook file has SHA-256 digest

2fcce9b98a4df10267fe120229217bfe556c70510704e311540da0cef438f911.

This distinguishes the input from later updates at the editors’ public download address.

The paper was prepared in a separate phase beginning 14 September 2026, with the same agent drafting and internally reviewing the manuscript. The model designation `gpt-6-astra` and reasoning setting `xhigh`, confirmed by the user during that phase, are corroborated by the JSONL turn-context records. The typesetting toolchain is a later addition, not software used to obtain the mathematical results. On 15 September, the organizers supplied the full JSONL, the ccusage row, author names and affiliations, and the historical motivation used in this revision. These later contributions do not change the frozen experimental outcomes.

## B Configuration, records and accounting

### B.1 Task and problem source

The full user objective is reproduced in Appendix A. It specifies an overall time limit and a research objective, rather than a list of preselected questions or a target number of outputs. The agent’s initial plan explicitly prioritized precise statements, literature checks, finite counterexamples, elementary reductions, and independent implementations of computational checks. It also required partial results and unsuccessful searches to be recorded.

The input file `docs/21tkk.pdf` has SHA-256 digest

2fcce9b98a4df10267fe120229217bfe556c70510704e311540da0cef438f911.

The agent extracted a main-body index with 1,308 entries, including all 150 entries of Issue 21. This is an index of problem statements, not a verified count of open problems. Editorial stars, solved subparts, duplicates, and literature published after a problem was posed all affect the open status of individual assertions. The source PDF, rather than its text extraction, was treated as authoritative for mathematical notation and scope.

### B.2 Agent and computational environment

The experiment used Codex 0.153.4 with `gpt-6-astra` and reasoning effort `xhigh`. The full JSONL record corroborates the settings supplied during manuscript preparation: all 139 retained turn-context records identify this model and effort, including 133 in the research window. The session metadata records the OpenAI provider and the CLI application version. These observations identify the recorded configuration, without establishing every provider-side implementation detail.

The agent worked in a persistent Linux workspace with shell access, file editing, Git, internet source retrieval, and tools for inspecting PDF pages. GAP 4.16.1 supplied the main finite-group computations. The retained scripts also use Python and C/C++ for exact arithmetic, word calculations, independent finite models, enumeration, and certificate parsing. The source and report files record which libraries or catalogues each computation depends on. Versions not retained in the experimental records cannot be reconstructed merely by inspecting the later manuscript environment.

The user initially supplied an inaccessible GAP path and repaired it during setup. The visible human message is *sorry, gap is now here*. The subsequent reports establish that GAP was available with the required packages. The initial suggestion that a replacement installation might be needed is therefore not evidence that such an installation actually occurred.

The working limits were generally 20 CPU cores and 100 GB of memory. The repository contains bounded launchers and process observations, but not a continuous, comprehensive

record of global peak resource use. In the largest retained certificate check, six GAP workers each had a 14 GiB workspace ceiling, for a combined configured ceiling of 84 GiB. A workspace ceiling is not an observed resident-memory peak. Likewise, local CPU limits do not measure the remote compute used by the language model.

The research plan prohibited delegation to other agents without authorization, and the reports record no delegated agents. Parallel GAP or C++ processes were computational workers, not separate language model researchers. The same agent developed the arguments, wrote checkers, and conducted the internal reviews. Within that research phase, *independent implementation* means a separate representation or algorithm written by the same agent. The later independent review by Claude Opus[1m] is a separate phase, described in Section 3.

### B.3 Persistence and local records

The workspace served as external memory. Plans and status files recorded the next steps; proof drafts preserved mathematical arguments; scripts and logs preserved computations; and Git recorded successive states. The research window contains 54 task-start events, 52 task-completion events, one aborted turn, and 79 context compactions. One further task completion occurs during closeout. The opening task was paused during setup; after the GAP correction at 20:58:51 UTC it resumed at 20:59:58 UTC. The wall-clock budget includes this interval. Repeated continuations and compactions are visible in the record; the work was not a single uninterrupted model response.

There were 198 commits during the 48-hour window. The first, at 21:00:37 UTC on 10 September, initialized the investigation. The 199th commit, `cff2c37`, completed the local handoff at 20:57:02 UTC on 12 September. Commit times date repository checkpoints; they do not directly date the discovery of an idea or establish when its full final scope was proved.

At `cff2c37`, the Git tree contains 4,147 tracked files with a total logical size of 2,440,677,642 bytes. This includes 430 files under `research/`, 456 under `scripts/`, and 2,755 under `results/`, as well as references, the source Notebook, and other records. Logical file size is not compressed repository size. The manuscript analysis reads this fixed revision, so later writing and review do not increase the experimental counts.

### B.4 Session records and observable operations

The full JSONL session supplied on 15 September contains 36,016 records, from 10 September 20:56:46.325 UTC through 14 September 17:05:09.077 UTC. It includes the research run, its closeout, and initial manuscript preparation. We separate these phases before counting operations or usage. Manuscript preparation begins with the new goal recorded at 14 September 14:12:56.892 UTC.

The earlier `session.html` and `session.txt` files are projections ending at research closeout. The HTML advertises 34,102 events and retains 4,439 event elements, including 3,616 token notifications. Every one of those token observations matches the corresponding JSONL line, including all cumulative and last-use fields. The full record adds individual timestamps, tool observations, configuration records and request-level usage. Its digest and the reconciliation procedure are supplied in Appendix F.

Within the research window, completed-item notifications record 4,350 command or polling observations, 676 file-change items, 357 image views, and 1,293 web-tool observations. The web count includes searches and page retrieval; it is not a count of search queries. There are also four completed sleep-tool observations. We count each completed item once by its identifier, keeping these counts separate from the outer code-tool calls that can contain several operations. Of the command observations, 251 have nonzero exit status; these

include expected no-match results and interrupted processes, and cannot be interpreted as 251 failed mathematical experiments.

The dated visible messages support the narrative below. Completed human-message items record the GAP correction during research and the model confirmation during manuscript preparation; the original objectives also occur in goal records. These counts do not measure oversight outside the recorded interaction. The derived public data omit internal reasoning and system/developer instruction bodies.

## B.5 Token and cost accounting

The JSONL contains two distinct usage ledgers. Its 3,829 unique `token_usage_record` responses sum exactly to every running `thread_token_usage` value. Of these responses, 84 match context-compaction identifiers. Removing those responses gives exactly the final cumulative `event_msg/token_count` counters, which are the counter stream used by ccusage [22]. Embedded copies of history and usage are not new requests.

The organizers supplied the ccusage row reproduced in Table 3. It agrees exactly with the second ledger for the *whole supplied session*, including manuscript preparation. Its input column counts uncached input; the renderer’s input counter includes cached input. Reasoning tokens are a subset of output, and are not added a second time [87, 88].

Table 3: Supplied ccusage row and its full-session reconciliation. The row excludes the separately recorded compaction requests.

| Counter                                               | Tokens      |
|-------------------------------------------------------|-------------|
| Uncached input                                        | 21,605,689  |
| Cached input                                          | 482,598,400 |
| Output                                                | 4,055,235   |
| Of which reasoning output                             | 1,894,719   |
| Total (uncached input plus cached input plus output)  | 508,259,324 |
| Additional input and output in 84 compaction requests | 21,569,265  |
| All recorded requests, including compactions          | 529,828,589 |

Using flat rates of USD 10 per million uncached input tokens, USD 1 per million cached input tokens, and USD 50 per million output tokens reproduces the supplied USD 901.42 estimate:

$$\frac{10(21,605,689) + 482,598,400 + 50(4,055,235)}{10^6} = 901.41704.$$

These rates match the model documentation consulted on 15 September 2026 [86]. The maximum recorded request input is 259,244 tokens, below its 272,000-token long-input threshold. We use the same flat rates throughout Table 4; this is a transparent API-equivalent reconstruction, not an invoice or an inference about the organizers’ subscription bill, discounts, or additional tool charges. The exact ccusage version and invocation were not supplied.

Thus the full-session estimate including compactions is USD 969.78, of which USD 910.61 is assigned to the 48-hour research window. Adding the independently rounded phase totals gives USD 969.79; the one-cent difference comes from rounding after aggregation. The earlier HTML total of 477,334,181 tokens equals research plus closeout usage excluding compactions. A delayed token notification straddles the deadline, so phase assignment uses request-record times, not notification times. These are logging boundaries, not exact provider execution intervals. Cached input dominates both ledgers; neither total measures distinct text read or newly generated text.

Table 4: Usage by phase, assigned by request-record timestamp. “Other” means responses other than context compactions. USD estimates are rounded independently to cents.

|                               | Research    | Closeout | Manuscript |
|-------------------------------|-------------|----------|------------|
| Other responses               | 3,515       | 5        | 225        |
| Compaction responses          | 79          | 0        | 5          |
| All input and output tokens   | 497,160,051 | 460,875  | 32,207,663 |
| Other-response estimate (USD) | 846.09      | 0.56     | 54.77      |
| Compaction estimate (USD)     | 64.52       | 0.00     | 3.85       |
| All-response estimate (USD)   | 910.61      | 0.56     | 58.62      |

## B.6 Outcome categories and checks

We distinguish five questions about an artifact:

1. Does its mathematical statement match the question being addressed?
2. Is the argument correct under its stated assumptions and imported results?
3. Do the computations establish the particular finite claims assigned to them?
4. Is the assertion already known or an immediate consequence of prior work?
5. Has someone independent of the producing agent accepted the argument?

These questions require different evidence. A digest match answers none of the mathematical questions. A finite check cannot establish an unrestricted infinite-group theorem without a proved reduction. Failure to find a paper in a literature search does not establish priority. A second implementation can expose an error in a first program while sharing the same mistaken mathematical specification.

The deadline ledger uses *complete candidate* for an argument presented as answering the recorded scope and surviving the internal checks conducted during the run. A *partial result* proves a restricted assertion while leaving the full question open. A *rediscovery* reproduces a known phenomenon or proof after independent work, whereas a *prior-result deduction* applies an identified theorem. A bounded search without a hit remains *computational evidence* for its exact range. These are reporting categories, not interchangeable levels on a single scale of confidence.

## C Detailed chronology and computation record

The chronology has two complementary sources. Commits give dated, recoverable states of the repository. The timestamped session messages give an account of what the agent believed it had established at each stage. Neither records the instant at which an idea first occurred. We therefore describe observable transitions—a proof being saved, a certificate being completed, a claim being withdrawn—rather than assigning precise discovery times to unobserved reasoning.

### C.1 Selection, short arguments, and early corrections

The initial plan favored finite counterexamples, explicit constructions, linear algebra, and questions that could admit short certificates. This was an adaptive search policy over the whole Notebook. It helps explain the eventual collection: the agent repeatedly moved to another entry when a search became expensive or the required generalization remained un-

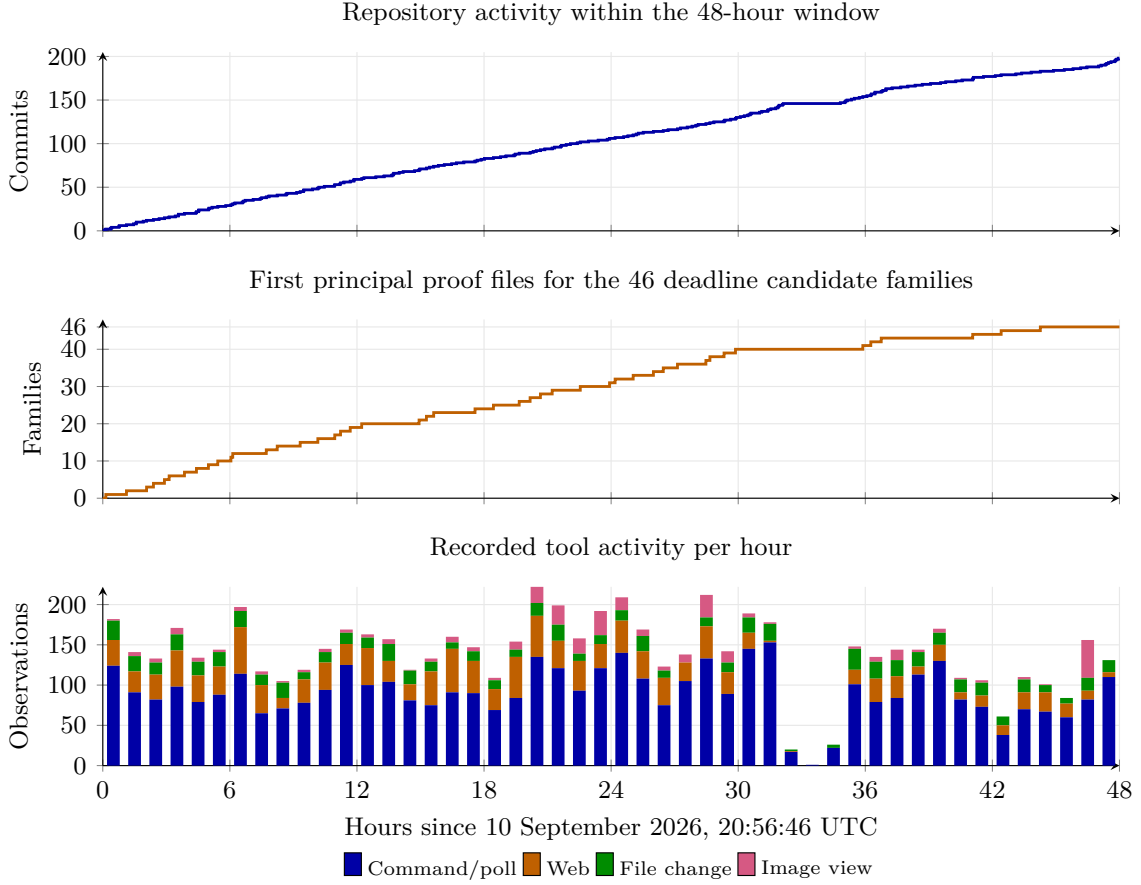


Figure 1: Repository and recorded tool activity during the research window. The upper panel counts the 198 commits within the research window; the closeout commit 16 seconds later is excluded. The middle panel retrospectively follows the first committed principal proof file for each family retained on the final candidate list. It excludes withdrawn candidates and does not measure the contemporaneous candidate count, proof completion, novelty, or outside acceptance. The lower panel counts completed command/polling, web, file-change and image-view observations in one-hour bins. It excludes context compaction and sleep operations; counts are not independent experiments. All panels are derived from the supplied CSV measurements.

clear. A problem's age was not a measure of the time spent on it in this experiment, and the initial index was not a promise to attempt every entry.

The first retained mathematical lead was Problem 21.106. Message 95 (10 September, 21:00:51 UTC) announced a possible formula selecting generators of the infinite cyclic center of the integral Heisenberg group; message 152 reported a written candidate and local commit at 21:06:04 UTC. Section 5.1 gives the argument. At the same time the agent launched finite screens for other questions, including transitive groups, character tables, and positive binary words. This pattern continued: short symbolic tasks occupied the foreground while finite searches ran in separate processes.

The early graph-symmetry search exposed a substantive validation error. For Problem 21.52, normalizing a permutation representation of inner automorphisms did not by itself certify that a permutation of involutions came from an automorphism of the original group. Message 653 records the replacement of that condition by construction of an actual group automorphism and verification of its action on every involution. The lesson concerns the mathematical predicate being checked: a correct program for a weaker predicate can still give a misleading answer to the research question.

Literature searches also changed the accounting. A graded nil-algebra construction for Problem 21.132 initially appeared on the candidate list, but was later excluded as prior work. For Problem 13.19, a small explicit quotient construction passed exact checks before a 2018 result was confirmed to imply the negative answer. Message 1155 explicitly removed it from the new-solution count while retaining the construction. The longer episode around Problem 17.34 is particularly informative. The agent wrote a PBW-based argument and ran thirteen exact controls; it first found prior coverage in nilpotency class two, then found a strong-amalgamation theorem with the full required scope. Messages 2729 and 2859 document the intervening increase and subsequent decrease in the working candidate count. These are rediscoveries or alternative deductions with potential expository value, but they cannot support a claim of new priority.

## C.2 Constructions guided by the literature

Two examples in Section 5 illustrate how source reading and explicit calculation interacted. For Problem 21.68, the order-96 configuration in Kida's work suggested an extension by an abelian group that could preserve a nonmonomial character. The resulting group has order 2592. Its membership in the class of semi-abelian groups and its failure to be monomial require different arguments; checking only one would not settle the conjecture.

For Problem 20.108, exceptions in a multiple-holomorph theorem suggested a group of order 605. The retained proof describes its entire automorphism group, its holomorph as explicit permutations, and a normalizing permutation whose coset has order four. The source supplied a productive place to look; the final certificate does not ask a reader to trust a SmallGroups identifier or an unexplained normalizer calculation. In both examples the finite checks are useful controls on conventions and coordinates, while the displayed construction explains why the claimed property holds.

Other entries depended more substantially on imported theorems. The constructions for Problems 10.62 and 4.75 share the same geometric existence theorem, and the carpet arguments for Problems 19.61 and 19.62 share structural machinery. The inventory retains separate Notebook questions, but they are not independent pieces of evidence about discovery ability. The mathematical sections identify these dependencies.

### C.3 A long computation and the meaning of completion

Problem 20.100 became the largest verification task. A structural reduction transformed a statement about arbitrary groups into an exact finite problem for each fixed value of  $n$ . Small values could then be established by certificates of integer relations, without imposing a bound on the order of the original group. Messages 6458–6763 distinguish the reduction, prototype searches, and completed checks for  $n = 4, 5, 6$ .

For  $n = 7$ , generation produced a certificate with 42,891,332 nodes. Message 10686 reports its completion and immediately notes that verification remained outstanding at 11 September, 11:08:29 UTC. Six GAP processes subsequently checked the separate parts. The final records account for 786,577,344 collision cases and 5,447,182,055 relation implications; the complete compressed certificate occupies 1,271,256,410 bytes. The retained full verification took approximately 26 hours of elapsed time. A parser restricted to the certificate's data grammar was also used, avoiding reliance on executing arbitrary certificate text as GAP code.

The last workers were still active on 12 September at 12:45 UTC. Message 28901, at 13:10:31 UTC, reports all six workers finished with zero exits and a successful aggregate check, while explicitly retaining a discrepant exit from the historical launcher. Message 29167 reports the committed proof packet at 13:31:41 UTC. These dated observations connect the long-running process records to the final mathematical claim.

This distinction between generation and checking matters. Intermediate messages report how many nodes had been checked, how many workers were still running, and which shards had completed. They do not convert partial progress into a full theorem. At the deadline the fixed cases  $n = 4, 5, 6, 7$  were recorded as established for all eligible groups. The general problem for arbitrary  $n$  remained open and contributed zero entries to the complete-candidate count.

Several other large searches ended with bounded information. The endomorphism-count investigation of Problem 19.20 checked 91,774 nonabelian groups of order at most 511, found no equality example, and found 469 cases contradicting a stronger inequality. The character-table screen for Problem 21.113 involved 9,850 cases: 7,573 had the needed modular tables and 2,277 did not. For Problem 21.99, all 4,722 transitive groups of degrees 2 through 20 in the installed catalogue were screened, with a separate direct control on 86 groups in degrees 2 through 8. The word search associated with Problem 18.43 processed 3,933,931,043 positive binary necklaces of lengths 1 through 36 without producing an identity candidate. These are coverage statements. They do not establish the unrestricted assertions.

### C.4 Late structural results and deadline review

The later work included structural progress on almost Engel elements and  $p$ -Jordan bounds. For Problems 20.89/19.83 the deadline result covers all linear groups in characteristic zero and several further classes; the unrestricted positive-characteristic question remains unresolved. For Problem 21.121(b), the archive contains bounds for soluble finite subgroups, examples with exactly computed exponents, and permanence results. The universal proposed bound of three remains unproved. These partials illustrate why a single binary solved/unsolved label would discard much of the experiment's mathematical output.

Statement checking remained necessary even when a construction was correct. For Problem 20.122, an early treatment of inclusion-minimal intersections did not settle the question about intersections of minimum order. The final construction addresses the latter condition. For Problem 15.76(b), the retained claim concerns the full varieties of fixed derived length, with a material distinction from arbitrary proper subvarieties. The fixed representation degree in Problem 10.35 and the minimal-polynomial convention in Problem 15.65 are similarly

essential parts of the claims.

The final review separated statement coverage from deeper proof review. All 46 candidate entries received a fresh comparison with the supplied PDF; 22 selected candidate proofs received additional late rereading. This is not evidence that all 46 underwent a new, exhaustive mathematical and literature audit in the final phase. The closeout audit also checked 4,231 file bindings, covering 2,242 distinct files totaling 2,008,448,358 bytes, and 92 local links. These integrity checks establish that the named artifacts existed and matched recorded digests. They do not establish the truth or novelty of their contents. The research deadline and the subsequent administrative closeout are kept separate in the chronology.

## D Measurement limits and further experimental design

The configuration and usage measurements in Appendix B identify the retained client-side record. They do not recover remote inference hardware, actual peak local resource use, provider invoices, or all historical sampling settings. The cost estimates are explicitly rate-based reconstructions; they are not measurements of billing or energy use. Human oversight outside the logged interaction is likewise not observable from the trace.

An opportunistic portfolio also limits comparisons. Counting indexed questions or mentions in a log does not recover a reliable denominator of substantive attempts. Training-data overlap and prior familiarity with constructions remain unknown. Without an ablation or a comparison run, the experiment cannot isolate the effect of persistence, tools, browsing, model choice or human oversight.

The separate Claude review provides stronger evidence of mathematical correctness than the initial internal checks. It does not remove these measurement limitations or determine discovery priority. The following protocol would make a further study more informative.

### D.1 Designing a comparative experiment

A follow-up should retain the broad research setting while improving the measurements that this run could not provide. A practical protocol would have the following components.

1. **Define the unit of evaluation in advance.** Use a dated set of problems with explicit subparts and a documented editorial status. Record every substantive attempt, including abandoned ones, with the reason for stopping. A randomized target subset or matched target sets would support comparisons that an opportunistic portfolio cannot.
2. **Separate discovery from assessment.** Maintain different fields for statement coverage, proof validity, computational verification, prior-art status, and outside review. Freeze proposed answers before specialist assessment, and preserve later corrections as new versions. Reviewers should receive the original question and the mathematical argument before seeing aggregate success claims.
3. **Record the experimental configuration.** Save model identifiers, reasoning settings, harness and package versions, continuation instructions, tool permissions, and retrieval dates. Capture raw event logs with timestamps and request identifiers, along with local CPU time and peak memory. Retain response-level usage, including compaction, and separate the research interval from subsequent writing. Keep usage estimates and billing records separate until reconciled.
4. **Treat verification as a research task.** For a search, specify the exact predicate and its relation to the question. Include a small direct implementation that checks

examples without reusing the search representation. For a large certificate, make coverage and the checking rules explicit, retain malformed-input and negative controls, and distinguish successful generation from successful verification.

5. **Use formalization selectively.** Small algebraic identities, finite certificates, and critical reduction lemmas are plausible first targets. Formalizing the entire archive at once would create a large translation project. Any formal account must also justify that its statement matches the Notebook, including quantifiers and conventions; a kernel-checked proof of a different statement would leave the original task unanswered.
6. **Measure the review burden.** Record expert time, corrections, changes in status, and which claims depend on the same imported theorem. Report a short, fully reviewed core alongside the larger exploratory archive. This would make it possible to compare output volume with the work needed to establish reliable mathematical contributions.

## E Nineteen representative carpet derivations

These tables complete the displayed finite proof of (4). A coordinate pair  $(m, n)$  denotes  $ma + nb$ , for simple roots  $a, b$ , with  $a$  short in  $B_2, G_2$ . Each initial variable lies in the specified entry of  $A$ . Every subsequent line lies in  $B$  at its root, hence also in  $A$ . An application column records a Chevalley rule on earlier lines. Two applications on the same line have the stated coefficient gcd; Bezout's identity gives the element in the second column. The displayed final element has coefficient dividing the requested coefficient. Tables use one destination root of each length. Weyl transport, checked on all absolute rules, covers the other roots. The ancillary files also check all 104 targets without this reduction.

$A_2$ , **request 3**:  $p = (1, 1)$ . Requested coefficient 1.  $X_1 \in A_{(1,1)}$ ,  $X_2 \in A_{(-1,0)}$ ,  $X_3 \in A_{(0,-1)}$ .

| Line  | Element         | Root      | Applications |
|-------|-----------------|-----------|--------------|
| $z_1$ | $X_1$           | $(1, 1)$  | initial      |
| $z_2$ | $X_2$           | $(-1, 0)$ | initial      |
| $z_3$ | $X_1 X_2$       | $(0, 1)$  | $z_2 z_1$    |
| $z_4$ | $X_3$           | $(0, -1)$ | initial      |
| $z_5$ | $X_1 X_3$       | $(1, 0)$  | $z_4 z_1$    |
| $z_6$ | $X_1^2 X_2 X_3$ | $(1, 1)$  | $z_5 z_3$    |

$B_2$ , **request 6**:  $p = (1, 1)$ . Requested coefficient 1.  $X_1 \in A_{(1,0)}$ ,  $X_2 \in A_{(1,1)}$ ,  $X_3 \in A_{(-2,-1)}$ .

| Line  | Element         | Root       | Applications |
|-------|-----------------|------------|--------------|
| $z_1$ | $X_1$           | $(1, 0)$   | initial      |
| $z_2$ | $X_2$           | $(1, 1)$   | initial      |
| $z_3$ | $X_3$           | $(-2, -1)$ | initial      |
| $z_4$ | $X_2^2 X_3$     | $(0, 1)$   | $z_3 z_2^2$  |
| $z_5$ | $X_1 X_2^2 X_3$ | $(1, 1)$   | $z_4 z_1$    |

$B_2$ , **request 7**:  $p = (1, 1)$ . Requested coefficient 1.  $X_1 \in A_{(1,1)}$ ,  $X_2 \in A_{(0,-1)}$ ,  $X_3 \in A_{(-1,0)}$ .

| Line  | Element         | Root      | Applications |
|-------|-----------------|-----------|--------------|
| $z_1$ | $X_1$           | $(1, 1)$  | initial      |
| $z_2$ | $X_2$           | $(0, -1)$ | initial      |
| $z_3$ | $X_1^2 X_2$     | $(2, 1)$  | $z_2 z_1^2$  |
| $z_4$ | $X_3$           | $(-1, 0)$ | initial      |
| $z_5$ | $X_1^2 X_2 X_3$ | $(1, 1)$  | $z_4 z_3$    |

$B_2$ , **request 8**:  $p = (2, 1)$ . Requested coefficient 1.  $X_1 \in A_{(0,1)}$ ,  $X_2 \in A_{(2,1)}$ ,  $X_3 \in A_{(-1,-1)}$ .

| Line  | Element           | Root       | Applications |
|-------|-------------------|------------|--------------|
| $z_1$ | $X_1$             | $(0, 1)$   | initial      |
| $z_2$ | $X_2$             | $(2, 1)$   | initial      |
| $z_3$ | $X_3$             | $(-1, -1)$ | initial      |
| $z_4$ | $X_2 X_3$         | $(1, 0)$   | $z_3 z_2$    |
| $z_5$ | $X_1 X_2^2 X_3^2$ | $(2, 1)$   | $z_1 z_4^2$  |

$B_2$ , **request 9**:  $p = (2, 1)$ . Requested coefficient 1.  $X_1 \in A_{(2,1)}$ ,  $X_2 \in A_{(0,-1)}$ ,  $X_3 \in A_{(-1,0)}$ .

| Line  | Element           | Root      | Applications |
|-------|-------------------|-----------|--------------|
| $z_1$ | $X_1$             | $(2, 1)$  | initial      |
| $z_2$ | $X_2$             | $(0, -1)$ | initial      |
| $z_3$ | $X_3$             | $(-1, 0)$ | initial      |
| $z_4$ | $X_1 X_3$         | $(1, 1)$  | $z_3 z_1$    |
| $z_5$ | $X_1^2 X_2 X_3^2$ | $(2, 1)$  | $z_2 z_4^2$  |

$B_2$ , **request 10**:  $p = (2, 1)$ . Requested coefficient 2.  $X_1 \in A_{(2,1)}$ ,  $X_2 \in A_{(-1,0)}$ ,  $X_3 \in A_{(-1,-1)}$ .

| Line  | Element          | Root       | Applications |
|-------|------------------|------------|--------------|
| $z_1$ | $X_1$            | $(2, 1)$   | initial      |
| $z_2$ | $X_2$            | $(-1, 0)$  | initial      |
| $z_3$ | $X_1 X_2$        | $(1, 1)$   | $z_2 z_1$    |
| $z_4$ | $X_3$            | $(-1, -1)$ | initial      |
| $z_5$ | $X_1 X_3$        | $(1, 0)$   | $z_4 z_1$    |
| $z_6$ | $2X_1^2 X_2 X_3$ | $(2, 1)$   | $2z_5 z_3$   |

$G_2$ , **request 19**:  $p = (2, 1)$ . Requested coefficient 1.  $X_1 \in A_{(1,0)}$ ,  $X_2 \in A_{(2,1)}$ ,  $X_3 \in A_{(-3,-1)}$ .

| Line  | Element         | Root       | Applications                              |
|-------|-----------------|------------|-------------------------------------------|
| $z_1$ | $X_1$           | $(1, 0)$   | initial                                   |
| $z_2$ | $X_2$           | $(2, 1)$   | initial                                   |
| $z_3$ | $3X_1 X_2$      | $(3, 1)$   | $3z_2 z_1$                                |
| $z_4$ | $X_3$           | $(-3, -1)$ | initial                                   |
| $z_5$ | $X_2 X_3$       | $(-1, 0)$  | $z_4 z_2$                                 |
| $z_6$ | $X_2^2 X_3$     | $(1, 1)$   | $z_4 z_2^2$                               |
| $z_7$ | $X_1 X_2^2 X_3$ | $(2, 1)$   | $z_5 z_3$ ; $2z_6 z_1$ ; $\gcd(3, 2) = 1$ |

$G_2$ , **request 20**:  $p = (2, 1)$ . Requested coefficient 1.  $X_1 \in A_{(0,1)}$ ,  $X_2 \in A_{(2,1)}$ ,  $X_3 \in A_{(-1,-1)}$ .

| Line  | Element           | Root       | Applications                              |
|-------|-------------------|------------|-------------------------------------------|
| $z_1$ | $X_1$             | $(0, 1)$   | initial                                   |
| $z_2$ | $X_2$             | $(2, 1)$   | initial                                   |
| $z_3$ | $X_3$             | $(-1, -1)$ | initial                                   |
| $z_4$ | $X_1 X_3$         | $(-1, 0)$  | $z_3 z_1$                                 |
| $z_5$ | $2X_2 X_3$        | $(1, 0)$   | $2z_3 z_2$                                |
| $z_6$ | $2X_1 X_2 X_3$    | $(1, 1)$   | $2z_2 z_4$                                |
| $z_7$ | $3X_2^2 X_3$      | $(3, 1)$   | $3z_3 z_2^2$                              |
| $z_8$ | $3X_1 X_2^2 X_3$  | $(3, 2)$   | $z_7 z_1$                                 |
| $z_9$ | $X_1 X_2^2 X_3^2$ | $(2, 1)$   | $z_3 z_8$ ; $2z_5 z_6$ ; $\gcd(3, 8) = 1$ |

$G_2$ , **request 21:**  $p = (2, 1)$ . Requested coefficient 1.  $X_1 \in A_{(1,1)}$ ,  $X_2 \in A_{(2,1)}$ ,  $X_3 \in A_{(-3,-2)}$ .

| Line  | Element       | Root       | Applications                            |
|-------|---------------|------------|-----------------------------------------|
| $z_1$ | $X_1$         | $(1, 1)$   | initial                                 |
| $z_2$ | $X_2$         | $(2, 1)$   | initial                                 |
| $z_3$ | $3X_1X_2$     | $(3, 2)$   | $3z_2z_1$                               |
| $z_4$ | $X_3$         | $(-3, -2)$ | initial                                 |
| $z_5$ | $X_2X_3$      | $(-1, -1)$ | $z_4z_2$                                |
| $z_6$ | $X_2^2X_3$    | $(1, 0)$   | $z_4z_2^2$                              |
| $z_7$ | $X_1X_2^2X_3$ | $(2, 1)$   | $z_5z_3$ ; $2z_6z_1$ ; $\gcd(3, 2) = 1$ |

$G_2$ , **request 22:**  $p = (2, 1)$ . Requested coefficient 2.  $X_1 \in A_{(2,1)}$ ,  $X_2 \in A_{(-1,0)}$ ,  $X_3 \in A_{(-1,-1)}$ .

| Line  | Element       | Root       | Applications                            |
|-------|---------------|------------|-----------------------------------------|
| $z_1$ | $X_1$         | $(2, 1)$   | initial                                 |
| $z_2$ | $X_2$         | $(-1, 0)$  | initial                                 |
| $z_3$ | $2X_1X_2$     | $(1, 1)$   | $2z_2z_1$                               |
| $z_4$ | $3X_1^2X_2$   | $(3, 2)$   | $3z_2z_1^2$                             |
| $z_5$ | $X_3$         | $(-1, -1)$ | initial                                 |
| $z_6$ | $2X_1X_3$     | $(1, 0)$   | $2z_5z_1$                               |
| $z_7$ | $X_1^2X_2X_3$ | $(2, 1)$   | $z_5z_4$ ; $2z_6z_3$ ; $\gcd(3, 8) = 1$ |

$G_2$ , **request 23:**  $p = (2, 1)$ . Requested coefficient 1.  $X_1 \in A_{(2,1)}$ ,  $X_2 \in A_{(-1,0)}$ ,  $X_3 \in A_{(0,-1)}$ .

| Line  | Element         | Root      | Applications                            |
|-------|-----------------|-----------|-----------------------------------------|
| $z_1$ | $X_1$           | $(2, 1)$  | initial                                 |
| $z_2$ | $X_2$           | $(-1, 0)$ | initial                                 |
| $z_3$ | $2X_1X_2$       | $(1, 1)$  | $2z_2z_1$                               |
| $z_4$ | $3X_1^2X_2$     | $(3, 2)$  | $3z_2z_1^2$                             |
| $z_5$ | $X_3$           | $(0, -1)$ | initial                                 |
| $z_6$ | $2X_1X_2X_3$    | $(1, 0)$  | $z_5z_3$                                |
| $z_7$ | $3X_1^2X_2X_3$  | $(3, 1)$  | $z_5z_4$                                |
| $z_8$ | $X_1^2X_2^2X_3$ | $(2, 1)$  | $z_2z_7$ ; $2z_3z_6$ ; $\gcd(3, 8) = 1$ |

$G_2$ , **request 32:**  $p = (3, 2)$ . Requested coefficient 3.  $X_1 \in A_{(1,0)}$ ,  $X_2 \in A_{(3,2)}$ ,  $X_3 \in A_{(-2,-1)}$ .

| Line  | Element          | Root       | Applications |
|-------|------------------|------------|--------------|
| $z_1$ | $X_1$            | $(1, 0)$   | initial      |
| $z_2$ | $X_2$            | $(3, 2)$   | initial      |
| $z_3$ | $X_3$            | $(-2, -1)$ | initial      |
| $z_4$ | $X_2X_3$         | $(1, 1)$   | $z_3z_2$     |
| $z_5$ | $3X_1X_2^2X_3^2$ | $(3, 2)$   | $3z_1z_4^2$  |

$G_2$ , **request 33:**  $p = (3, 2)$ . Requested coefficient 1.  $X_1 \in A_{(1,0)}$ ,  $X_2 \in A_{(3,2)}$ ,  $X_3 \in A_{(-3,-1)}$ .

| Line  | Element             | Root       | Applications |
|-------|---------------------|------------|--------------|
| $z_1$ | $X_1$               | $(1, 0)$   | initial      |
| $z_2$ | $X_2$               | $(3, 2)$   | initial      |
| $z_3$ | $X_3$               | $(-3, -1)$ | initial      |
| $z_4$ | $X_1^3 X_3$         | $(0, -1)$  | $z_3 z_1^3$  |
| $z_5$ | $X_2 X_3$           | $(0, 1)$   | $z_3 z_2$    |
| $z_6$ | $X_1^3 X_2 X_3$     | $(3, 1)$   | $z_2 z_4$    |
| $z_7$ | $X_1^3 X_2^2 X_3^2$ | $(3, 2)$   | $z_5 z_6$    |

$G_2$ , **request 34:**  $p = (3, 2)$ . Requested coefficient 1.  $X_1 \in A_{(0,1)}$ ,  $X_2 \in A_{(3,2)}$ ,  $X_3 \in A_{(-1,-1)}$ .

| Line  | Element           | Root       | Applications |
|-------|-------------------|------------|--------------|
| $z_1$ | $X_1$             | $(0, 1)$   | initial      |
| $z_2$ | $X_2$             | $(3, 2)$   | initial      |
| $z_3$ | $X_3$             | $(-1, -1)$ | initial      |
| $z_4$ | $X_2 X_3^3$       | $(0, -1)$  | $z_2 z_3^3$  |
| $z_5$ | $X_2^2 X_3^3$     | $(3, 1)$   | $z_2 z_4$    |
| $z_6$ | $X_1 X_2^2 X_3^3$ | $(3, 2)$   | $z_5 z_1$    |

$G_2$ , **request 35:**  $p = (3, 2)$ . Requested coefficient 1.  $X_1 \in A_{(3,1)}$ ,  $X_2 \in A_{(3,2)}$ ,  $X_3 \in A_{(-2,-1)}$ .

| Line  | Element           | Root       | Applications |
|-------|-------------------|------------|--------------|
| $z_1$ | $X_1$             | $(3, 1)$   | initial      |
| $z_2$ | $X_2$             | $(3, 2)$   | initial      |
| $z_3$ | $X_3$             | $(-2, -1)$ | initial      |
| $z_4$ | $X_2 X_3^3$       | $(-3, -1)$ | $z_2 z_3^3$  |
| $z_5$ | $X_2^2 X_3^3$     | $(0, 1)$   | $z_2 z_4$    |
| $z_6$ | $X_1 X_2^2 X_3^3$ | $(3, 2)$   | $z_5 z_1$    |

$G_2$ , **request 36:**  $p = (3, 2)$ . Requested coefficient 3.  $X_1 \in A_{(3,2)}$ ,  $X_2 \in A_{(-1,0)}$ ,  $X_3 \in A_{(-1,-1)}$ .

| Line  | Element            | Root       | Applications |
|-------|--------------------|------------|--------------|
| $z_1$ | $X_1$              | $(3, 2)$   | initial      |
| $z_2$ | $X_2$              | $(-1, 0)$  | initial      |
| $z_3$ | $X_3$              | $(-1, -1)$ | initial      |
| $z_4$ | $X_1 X_3$          | $(2, 1)$   | $z_3 z_1$    |
| $z_5$ | $3X_1^2 X_2 X_3^2$ | $(3, 2)$   | $3z_2 z_4^2$ |

$G_2$ , **request 37:**  $p = (3, 2)$ . Requested coefficient 1.  $X_1 \in A_{(3,2)}$ ,  $X_2 \in A_{(-1,0)}$ ,  $X_3 \in A_{(0,-1)}$ .

| Line  | Element             | Root       | Applications |
|-------|---------------------|------------|--------------|
| $z_1$ | $X_1$               | $(3, 2)$   | initial      |
| $z_2$ | $X_2$               | $(-1, 0)$  | initial      |
| $z_3$ | $X_3$               | $(0, -1)$  | initial      |
| $z_4$ | $X_1 X_3$           | $(3, 1)$   | $z_3 z_1$    |
| $z_5$ | $X_2^3 X_3$         | $(-3, -1)$ | $z_3 z_2^3$  |
| $z_6$ | $X_1 X_2^3 X_3$     | $(0, 1)$   | $z_5 z_1$    |
| $z_7$ | $X_1^2 X_2^3 X_3^2$ | $(3, 2)$   | $z_4 z_6$    |

$G_2$ , **request 38:**  $p = (3, 2)$ . Requested coefficient 1.  $X_1 \in A_{(3,2)}$ ,  $X_2 \in A_{(0,-1)}$ ,  $X_3 \in A_{(-3,-1)}$ .

| Line  | Element         | Root       | Applications |
|-------|-----------------|------------|--------------|
| $z_1$ | $X_1$           | $(3, 2)$   | initial      |
| $z_2$ | $X_2$           | $(0, -1)$  | initial      |
| $z_3$ | $X_1 X_2$       | $(3, 1)$   | $z_2 z_1$    |
| $z_4$ | $X_3$           | $(-3, -1)$ | initial      |
| $z_5$ | $X_1 X_3$       | $(0, 1)$   | $z_4 z_1$    |
| $z_6$ | $X_1^2 X_2 X_3$ | $(3, 2)$   | $z_5 z_3$    |

$G_2$ , **request 39:**  $p = (3, 2)$ . Requested coefficient 3.  $X_1 \in A_{(3,2)}$ ,  $X_2 \in A_{(-1,-1)}$ ,  $X_3 \in A_{(-2,-1)}$ .

| Line  | Element          | Root       | Applications |
|-------|------------------|------------|--------------|
| $z_1$ | $X_1$            | $(3, 2)$   | initial      |
| $z_2$ | $X_2$            | $(-1, -1)$ | initial      |
| $z_3$ | $X_1 X_2$        | $(2, 1)$   | $z_2 z_1$    |
| $z_4$ | $X_3$            | $(-2, -1)$ | initial      |
| $z_5$ | $X_1 X_3$        | $(1, 1)$   | $z_4 z_1$    |
| $z_6$ | $3X_1^2 X_2 X_3$ | $(3, 2)$   | $3z_5 z_3$   |

## F Artifacts and reproduction

The public repository is <https://github.com/JUrban/kour1>. Its research snapshot is

`bbfac1ac810117da37f01716d4594dc2b2e96980`.

The owner filtered the history with `--strip-blobs-bigger-than 90M`. Its commit identifiers therefore differ from those of the original unfiltered archive, including `cff2c37` in Appendix A. The unfiltered identifiers in the historical timelines are provenance identifiers, not public checkout instructions. On 17 September we compared the two final research trees: all 4,146 retained file blobs have identical Git object IDs; the sole omitted blob at that snapshot is the 1.27 GB compressed Problem 20.100 certificate discussed below. This comparison does not assert equivalence of the full histories. `paper/data/public-artifacts.json` records both snapshots, the omitted file and byte comparisons of selected public artifacts. The public baseline manuscript commit is `315e6c4ddf3b67aca8c97506f078cdc98d5413c1`; its PDF is byte-identical to the reviewed version preserved with this revision.

Recomputing Git measurements by the original scripts requires the unfiltered history, or an explicit mapping to the rewritten commits. The included frozen measurements retain their original identifiers. Typesetting and the portable carpet checks require neither history.

The main entry points in the archive are `reports/FINAL_REPORT.md`, `reports/REVIEW_GUIDE.md`, and `research/complete-candidate-ledger.json`. The last file records the transcribed question, covered parts, proof paths, review paths, and evidence entry points for every candidate. The original PDF remains authoritative if an extraction differs from its typography.

The later `paper/` directory contains manuscript source, generated measurements, and a review ledger. From the repository root, the measurement command is

```
python3 paper/scripts/analyze_experiment.py --check
```

It uses Python’s standard library and Git, reads the supplied HTML and text exports, and recomputes the statistics from the frozen history. It checks agreement with committed outputs rather than modifying them. Without `--check`, it regenerates the five measurement files.

| File in paper/data/                         | Meaning                                                                                        |
|---------------------------------------------|------------------------------------------------------------------------------------------------|
| <code>experiment-metrics.json</code>        | Input hashes, event counts, cumulative counters, Git totals, and measurement limitations       |
| <code>session-tokens.csv</code>             | All retained token observations, including repeated and unitemized records                     |
| <code>session-visible-messages.jsonl</code> | Visible assistant, final, and human messages with event identifiers; excludes directive bodies |
| <code>research-commits.csv</code>           | Commit identifiers, UTC timestamps, subjects, and inclusion in the research interval           |
| <code>proof-artifact-timeline.csv</code>    | First committed principal proof artifact for each problem family on the final candidate list   |

The full JSONL is analyzed separately:

```
python3 paper/scripts/analyze_rollout.py --check
```

This standard-library parser reads the single supplied `session/rollout-*.jsonl`, whose size is 540,278,152 bytes and whose SHA-256 digest is

```
f316fb63957f3ec50895088537786a4946a15601e54da5dac86ec0e4fcdd21d8.
```

It checks unique response and completed-item identifiers, timestamp order, usage identities, the exact compaction reconciliation, every HTML token observation, and agreement with `ccusage-supplied.json`. Its six outputs are `rollout-metrics.json`, `rollout-usage.csv`, `rollout-observations.csv`, `rollout-tasks.csv`, `rollout-hourly.csv`, and `rollout-visible-messages.jsonl`. These retain aggregate metrics, request usage, completed-operation metadata, task events, hourly activity and timestamped visible messages, respectively. Without `--check`, the parser regenerates them.

The phase boundary uses the logged request timestamp. For example, a 90,535-token request is recorded at 12 September 20:56:42.957 UTC, before the deadline, but its cumulative notification appears at 20:56:49.929 UTC after a tool completes. Using only the last pre-deadline notification would omit that request from research usage. The separately identified compaction requests likewise cannot be reconstructed by summing the legacy “Last” display fields.

The raw session inputs remain separate from both source bundles. Their hashes and sizes identify the evidence for an authorized holder; the included projections are not substitutes for the full trace. The additional `extract_retrievals.py` parser exports `retrieval-observations.jsonl`, `retrieval-urls.csv` and `retrieval-metrics.json`. These index 1,324 completed web-tool observations (1,293 during research and 31 during manuscript preparation), preserving timestamps, action types, queries and returned links. They distinguish a requested page action from a URL appearing among search results. Shell downloads are outside this index; neither kind of record establishes that a source was used in an argument. The exact URL of the contemporaneous 21.106 repository has zero literal occurrences in the supplied raw JSONL. That narrow check cannot exclude exposure through other URLs, sources or training data, and the derived index alone cannot independently establish an absence in the raw trace. No internal reasoning, compaction-summary or system/developer instruction bodies are included in the derived data. Mathematical scripts have different purposes: some generate examples, some check certificates, and some summarize existing results. Reproducing a summary is not rerunning a proof check. Each result’s archive entry identifies its commands and the limits of the computation.

The  $n = 7$  certificate for Problem 20.100, `results/20.100-n7-certificate.g.gz`, is not distributed in the filtered public repository or either paper source bundle. Its compressed size is 1,271,256,410 bytes and its SHA-256 digest is

e1b116fa80a4b77a03256d69480b996318522c227ef48e6be7fdf62cbd8bcd8c.

The public archive retains the checker, generator and shard-verifier logs, and final report. They document the historical verification; without the certificate they do not permit a fresh complete replay. The early `20.100-n7-integrity.json` receipt predates completion of verification and must not be read as the final status. No separate large-file deposit or new  $n = 7$  verification is claimed. A later deposit could supply this missing input without rewriting the research record.

The portable source includes the complete integer derivations for Problems 19.61–19.62, their target-coverage checkers, and the integral  $G_2$  matrix-identity check. In the arXiv source archive these are under `anc/`; in the full manuscript project they are under `ancillary/`. Their README gives four standard-Python commands, which require neither GAP nor network access. A manifest binds the 24 original files to their byte-identical research versions. The 19 short representative derivations are also printed above.

The main file is `main.tex`. The source archives include `main.bbl` and the compiled chronology figure, so ordinary XeLaTeX/BibTeX or Tectonic can typeset the paper without regenerating the data or running mathematical computations. The full-source archive additionally contains the aggregate measurements, the frozen candidate ledger, figure source, build scripts, and manuscript review records. Recomputing the original session/Git measurements still requires the separately retained experimental archive and session inputs.

## G Changes following external review

This appendix compares the reviewed 15 September version with the first review revision of 17 September (local commit `f9e1956`). The later reorganization of the paper is recorded at the end of this appendix. Quotations and file locations refer to their respective historical versions, before that reorganization. The first review revision’s PDF is preserved as `paper/versions/kourovka-experiment-2026-09-17-review.pdf`. The 15 September baseline PDF is preserved as `paper/versions/kourovka-experiment-2026-09-15.pdf`; its SHA-256 is

ec9a5482cdfb9ee2b3937fd1980eed6448ab5f954454c9d5287d06aa8a4c3c25.

The local baseline manuscript commit is `8a59409`; its public filtered counterpart is `315e6c4ddf3b67aca8c97506f078cdc98d5413c1`. Before/after quotations below are exact source excerpts, typeset with this paper’s macros. Their file locations are relative to `paper/`; section numbers can change when material moves. The structured change record and source copies of both versions are in `paper/reviews/revision-2026-09-17/`. A validation script checks each quotation against its version. The deadline candidate ledger, research artifacts and usage measurements have not been rewritten.

### G.1 14.72: reduced and doubled divisors

**Review point.** First report, section 6; follow-up, section 2.

**Location.** `appendices/candidates/14-72.tex`.

**Original text.**

It is therefore a smooth effective Cartier divisor, of the codimension one required in the question.

### Revised text.

It is therefore a smooth effective Cartier divisor  $D$ . The equation  $x = 0$  defines  $D$  set-theoretically but defines  $2D$  as a Cartier divisor. The reduced divisor is not globally principal on  $X$ : on the zero section its class is the nonzero order-two class of  $(0, 0) - \infty$  on the elliptic curve. The restriction below supplies a global equation.

**Reason.** Separate the original Cartier assertion from the stronger principal assertion.

**Validation.** Local equation  $x=y^2/(x^2-1)$ ; the zero-section restriction detects the nonzero two-torsion class.

## G.2 14.72: a globally principal fixed divisor

**Review point.** Follow-up, section 2.

**Location.** `appendices/candidates/14-72.tex`.

### Original text.

The full fixed divisor does not control this additional stabilizer.

### Revised text.

Thus  $U$  answers the question negatively even when “regular hypersurface” requires a single global equation for the reduced fixed scheme. This restriction was obtained during revision; it is not part of the deadline construction.

**Reason.** The invariant open removes the obstruction while preserving the singular quotient. This is a post-deadline strengthening.

**Validation.** Exact ideal and invariant-localization argument in the revised proof; symbolic verification recorded in the revision audit.

## G.3 15.92: the prime is selected by the diagram

**Review point.** Follow-up, sections 2 and 3.

**Location.** `appendices/candidates/15-92.tex`.

### Original text.

These cycle and parity facts are read from the construction, not inferred from its headline generation theorem.

### Revised text.

The displayed choice of 11 or 13 is the length of the unique exceptional prime cycle in the respective diagram; it is not a free choice. These cycle and parity facts are read from the construction, not inferred from its headline generation theorem. The finite transcription, diagram reconstruction and 150 cycle controls are indexed in `results/15.92-summary.json` in the public snapshot (Appendix F); the external follow-up reports replaying both controls.

**Reason.** Explain the residue-class split and make the finite input locatable.

**Validation.** Checked the retained source-data paths and the reviewer’s reported 150-row replay; the finite controls do not replace the all-size argument.

## G.4 21.106: contemporaneous work and artifact chronology

**Review point.** First report, section 5; follow-up, sections 1 and 2.

**Location.** `sections/representative.tex`.

### Original text.

The experimental controls checked commutator arithmetic and finite interpretations, while the proof above establishes the infinite statement.

### Revised text.

Contemporaneous work recorded in a public repository credits Lily Zhang with a counterexample and proof and Evan Li with a Heisenberg simplification; it also supplies a Lean formalisation[115]. Our first local proof commit is dated 10 September 2026, 21:05:23 UTC; the commit adding those Lean files is dated 11 September, 10:54:21 UTC. These are artifact timestamps, not discovery or verified upload times, and do not establish priority. We have not built that Lean development.

**Reason.** Credit the parallel construction and formalisation without assigning discovery priority.

**Validation.** Inspected the public repository and commit metadata; no claim of a local Lean build.

## G.5 10.62/4.75: the prescribed relation and initial scale

**Review point.** First author reply, item 6.

**Location.** `appendices/candidates/10-62.tex`.

### Original text.

All choices are made before choosing  $p$ . For a sufficiently large odd prime  $p$ , use the exponent family  $\{p\}$  and include  $w$  in the first maximal admissible family of primitive loxodromic elements. Proposition 6.1 then kills  $w^p$ .

### Revised text.

All choices are made before choosing  $p$ . Explicitly, the source's Remark 6.8 illustrates the extra condition  $2\lambda \leq L_S\delta_1$ ; the word  $abc$  requires  $3\lambda \leq L_S\delta_1$  instead. The permission to decrease  $\lambda$  is the part we use. For a sufficiently large odd prime  $p$ , use the exponent family  $\{p\}$ . The primitive cyclic subgroup  $\langle abc \rangle$  is eligible for the first family; start with it and extend to a maximal admissible selection, choosing one representative for each eligible maximal loxodromic subgroup up to conjugacy. Proposition 6.1 kills  $w^p$ . Neither the extra relation nor the scale  $3\lambda$  is asserted as part of the headline statement of Theorem 6.7.

**Reason.** Identify the precise use of the source construction shared by the two candidates.

**Validation.** Rechecked arXiv:2509.11958v2, Proposition 6.1 and the initialization of Theorem 6.7 with Remark 6.8.

## G.6 10.32: retain the correct lower-bound attribution

**Review point.** First report, section 8.

**Location.** `appendices/candidates/10-32.tex`.

**Original text.**

Rosser–Schoenfeld [93, Theorems 4 and 9] give

**Revised text.**

Rosser–Schoenfeld [93, Theorem 4, (3.14), and Theorem 9] give

**Reason.** The suggested formula (3.16) is a different inequality. Add the correct formula number while retaining Theorem 4.

**Validation.** Visually checked printed page 70 of Rosser–Schoenfeld: (3.14) has denominator  $2 \log x$  and threshold 563; (3.16) has denominator  $\log x$  and threshold 41.

**G.7 13.42: foundational reference and the version used**

**Review point.** First report, section 8.

**Location.** `appendices/candidates/13-42.tex`.

**Original text.**

Here an  $A$ -group satisfies the power, conjugation, and commuting-product axioms in Myasnikov–Remeslennikov [83, Definitions 1–2].

**Revised text.**

The Notebook refers to Myasnikov–Remeslennikov’s foundational paper [82]. Here we use the power, conjugation, commuting-product axioms and tensor-completion universal property as restated in [83, Definitions 1–2].

**Reason.** Add the original reference while preserving the precise definitions actually used.

**Validation.** Checked the bibliographic record of the 1994 paper and the retained definitions in Exponential groups 2.

**G.8 Usage: explain the one-cent difference**

**Review point.** First report, section 8.

**Location.** `sections/methods.tex`.

**Original text.**

Thus the full-session estimate including compactions is USD 969.78, of which USD 910.61 is assigned to the 48-hour research window.

**Revised text.**

Thus the full-session estimate including compactions is USD 969.78, of which USD 910.61 is assigned to the 48-hour research window. Adding the independently rounded phase totals gives USD 969.79; the one-cent difference comes from rounding after aggregation.

**Reason.** Make the effect of independent rounding explicit; no counters or rates change.

**Validation.** The exact usage outputs and their hashes are unchanged.

**G.9 Distinguish deadline review from later external review**

**Review point.** Follow-up, sections 3 and 4.

**Location.** `sections/discussion.tex`.

**Original text.**

Finally, manuscript review is still review by the same agent that produced the research. Repeated rereading and fresh calculations can find errors, but they do not provide authorial independence. The deadline archive contains zero outside reviews. Neither a preserved PASS message nor the absence of a finding in an internal review substitutes for specialist assessment.

**Revised text.**

The initial manuscript review was performed by the same agent that produced the research. The deadline archive contains zero outside reviews. A later external review and follow-up checked arguments and finite inputs, identified scope and packaging issues, and withdrew several initial objections. We preserve the correspondence in Appendices I–L, with a change record in Appendix G. The reviewer’s scripts and logs are separately versioned; their reported outcomes are distinguished from checks rerun during this revision. The revised 14.72 construction makes the fixed divisor globally principal, and the portable bundle now includes the missing small certificate. These are later changes, not deadline achievements. Neither a preserved PASS message nor a reviewer’s acceptance recommendation substitutes for specialist assessment or establishes priority.

**Reason.** Retain the historical zero-review count and add the subsequent review process.

**Validation.** Frozen candidate and usage ledgers remain byte-identical; the new correspondence has separate provenance.

**G.10 4.55: version the character-table indices**

**Review point.** First report, section 8.

**Location.** `appendices/candidates/4-55.tex`.

**Original text.**

All other rows are zero in these columns.

**Revised text.**

These indices use `CharacterTable("3.A7")` and its 5-modular table in GAP 4.16.1 with the character-table library `CTblLib` version 1.3.11. All other rows are zero in these columns.

**Reason.** Tie the row indices to the library used for the manuscript check.

**Validation.** The revised GAP checker prints the GAP and `CTblLib` versions and checks the decomposition rows, Galois orbits and projective dimensions.

**G.11 15.76(b): identify the contrasting subvariety**

**Review point.** First report, sections 4 and 8.

**Location.** `appendices/candidates/15-76.tex`.

**Original text.**

for every proper subvariety of soluble groups: the proper metabelian variety studied by Fernandes–Tsurkov provides a counterexample

**Revised text.**

for every proper subvariety of soluble groups: the proper metabelian subvariety of exponent four studied by Fernandes–Tsurkov provides a counterexample

**Reason.** State the exponent restriction on the cited counterexample.

**Validation.** Checked the retained Fernandes–Tsurkov source; the full soluble varieties remain the scope of the proposition.

## G.12 20.90: reconcile CA and CN terminology

**Review point.** First report, section 8.

**Location.** `appendices/candidates/20-90.tex`.

**Original text.**

It is not necessary that finite quotients be CN-groups.

**Revised text.**

Thus the example is a CA-group (its nonidentity centralizers are abelian), hence a CN-group (those centralizers are pronilpotent). It is not necessary that finite quotients be CN-groups.

**Reason.** Explain why the inventory description and section title both apply.

**Validation.** The proof already establishes abelian centralizers; no additional computation is needed.

## G.13 Public filtered and local unfiltered snapshots

**Review point.** Follow-up, section 2; repository clarification from the organizers.

**Location.** `appendices/reproduction.tex`.

**Original text.**

The configured repository address is <https://github.com/JUrban/kour1>. Anonymous access could not be established during manuscript preparation: unauthenticated API requests returned HTTP 404. A local Git query obtained a remote head different from the frozen experimental revision. These observations do not establish that either revision is anonymously available or that the two histories are equivalent. Readers need the frozen archive identified in Appendix A to reproduce repository-based measurements. The mathematical exposition in the paper does not require executing that archive.

**Revised text.**

The public repository is <https://github.com/JUrban/kour1>. Its research snapshot is

`bbfac1ac810117da37f01716d4594dc2b2e96980`.

The owner filtered the history with `--strip-blobs-bigger-than 90M`. Its commit identifiers therefore differ from those of the original unfiltered archive, including `cff2c37` in Appendix A. The unfiltered identifiers in the historical timelines are provenance identifiers, not public checkout instructions. On 17 September we compared the two final research trees: all 4,146 retained file blobs have identical Git object IDs; the sole omitted blob at that snapshot is the 1.27 GB compressed Problem 20.100 certificate discussed below. This comparison does not assert equivalence of the full histories.

`paper/data/public-artifacts.json` records both snapshots, the omitted file and byte comparisons of selected public artifacts. The public baseline manuscript commit is `315e6c4ddf3b67aca8c97506f078cdc98d5413c1`; its PDF is byte-identical to the reviewed version preserved with this revision.

Recomputing Git measurements by the original scripts requires the unfiltered history, or an explicit mapping to the rewritten commits. The included frozen measurements retain their original identifiers. Typesetting and the portable carpet checks require neither history.

**Reason.** Replace the obsolete access failure and separate public checkout references from historical local hashes.

**Validation.** Compared all 4,146 retained Git blob IDs; byte-checked five selected public files and the baseline PDF.

## G.14 A dated retrieval index

**Review point.** First author reply, item 4; follow-up, section 1.

**Location.** `appendices/reproduction.tex`.

### Original text.

The raw session inputs remain separate from both source bundles. Their hashes and sizes identify the evidence for an authorized holder; the included projections are not substitutes for the full trace.

### Revised text.

The raw session inputs remain separate from both source bundles. Their hashes and sizes identify the evidence for an authorized holder; the included projections are not substitutes for the full trace. The additional `extract_retrievals.py` parser exports `retrieval-observations.jsonl`, `retrieval-urls.csv` and `retrieval-metrics.json`. These index 1,324 completed web-tool observations (1,293 during research and 31 during manuscript preparation), preserving timestamps, action types, queries and returned links. They distinguish a requested page action from a URL appearing among search results. Shell downloads are outside this index; neither kind of record establishes that a source was used in an argument. The exact URL of the contemporaneous 21.106 repository has zero literal occurrences in the supplied raw JSONL. That narrow check cannot exclude exposure through other URLs, sources or training data, and the derived index alone cannot independently establish an absence in the raw trace.

**Reason.** Expose structured source-access observations without treating retrieval as mathematical use.

**Validation.** Reparsed the identified 540,278,152-byte input; all output records come from completed web observations. Raw reasoning and directive bodies are excluded.

## G.15 20.100: name the undistributed input

**Review point.** Follow-up, section 2; repository clarification from the organizers.

**Location.** `appendices/reproduction.tex`.

### Original text.

Large computations need substantial time and storage. In particular, the  $n = 7$  certificate for Problem 20.100 is distinct from smaller diagnostic and intermediate files. The retained report and manifest specify the complete input and all completed checker shards. The manuscript's review does not silently replace that historical verification with a fresh abbreviated run.

**Revised text.**

The  $n = 7$  certificate for Problem 20.100, `results/20.100-n7-certificate.g.gz`, is not distributed in the filtered public repository or either paper source bundle. Its compressed size is 1,271,256,410 bytes and its SHA-256 digest is

`e1b116fa80a4b77a03256d69480b996318522c227ef48e6be7fdf62cbd8bcd8c`.

The public archive retains the checker, generator and shard-verifier logs, and final report. They document the historical verification; without the certificate they do not permit a fresh complete replay. The early `20.100-n7-integrity.json` receipt predates completion of verification and must not be read as the final status. No separate large-file deposit or new  $n = 7$  verification is claimed. A later deposit could supply this missing input without rewriting the research record.

**Reason.** State exactly what the public archive does and does not permit a reader to replay.

**Validation.** Public tree comparison confirms the omitted 1,271,256,410-byte blob; the retained generator and final verification records identify it.

**G.16 Inventory: scope, dependencies and prior-work status**

**Review point.** Follow-up, sections 1 and 5.

**Location.** `appendices/inventory.tex`.

**Original text.**

Several entries share imported ingredients. The mathematical discussions give the relevant scope and dependencies; the frozen ledger supplies the original statement and evidence paths.

**Revised text.**

The additional review column records selected scope qualifications and imported dependencies; an em dash means no additional note here, not that the proof has no imports. Priority remains unestablished for all 46 candidates.

**Reason.** Preserve all historical descriptions and covered parts while adding neutral qualifications. A separate table makes 21.132, 21.42 and 20.33 visible under prior-work status.

**Validation.** The generator checks exactly the same 46 keys; annotations are stored separately. No count is inferred from proof length or presumed proposer intent.

**G.17 19.56 as a fourth main example**

**Review point.** First author reply, item 6.

**Location.** `sections/representative.tex`.

**Original text.**

The following arguments illustrate three different kinds of output.

**Revised text.**

The following arguments illustrate four different kinds of output.

**Reason.** Move the existing 19.56 exposition from the candidate appendix into Section 4. The hyperfocal, minimal-simple and Baer–Suzuki dependencies remain explicit.

**Validation.** The 19.56 source is byte-identical; generated input lists include it once, giving four main examples and 42 further candidate expositions.

## G.18 Portable carpet bundle: supply the fourth checker input

**Review point.** Follow-up, sections 2 and 5.

**Location.** `appendices/reproduction.tex`.

### Original text.

Their README gives three standard-Python commands, which require neither GAP nor network access. A manifest binds the 23 original files to their byte-identical research versions.

### Revised text.

Their README gives four standard-Python commands, which require neither GAP nor network access. A manifest binds the 24 original files to their byte-identical research versions.

**Reason.** Add `results/19.62-g2-monomial-certificates.jsonl` (2,194,486 bytes) to the ancillary directory and both source bundles; retain and document the fourth checker. The first reply overstated the completeness of the bundled data.

**Validation.** Check all four commands after extraction outside the repository, and verify each original file against the ancillary manifest.

## G.19 Withdrawn objections and retained scope

The follow-up withdraws the algebraic-closedness objection to 16.28(a): the original field was already the algebraic closure of  $\mathbb{F}_5(t)$ . That proof is unchanged. We retain characteristic five in part (b), without asserting the identical conjugacy-class argument over every odd characteristic. The suggested unrestricted extension is unnecessary; in characteristic three its trace-one class does not have the same semisimple description.

For 12.40, the follow-up accepts that nonexistence of any finite-valued bound is a negative answer to the printed question. No proof change is needed. The estimated count of “roughly seven” unintended readings, the speculation about proposer intent, and the priority inference from artifact dates are withdrawn in the follow-up and are not adopted here. Proof length and speed of formalisation are not measures of discovery difficulty. The original reports retain those earlier statements solely as part of the correspondence record.

For 15.65, the source-index discrepancy discussed in its proof remains explicit. The revision does not claim a new resolution of that discrepancy. For 21.114, the external follow-up checks the retained order-512 inputs, their central quotients and derived series. The order-8,192 construction in the paper still uses the subsequent central-product argument; the report is not described as a direct enumeration of that final group. The 46 historical candidate classifications and their unresolved priority status remain unchanged.

## G.20 Correspondence, packaging and validation records

The four documents reproduced next were added to the manuscript without rewriting their words. The lightly updated first report is retained as `external-reviews/review-update1.md`, with its exact diff in `external-reviews/review-update1.diff`. The final reply corrects our first reply’s claims about bundle completeness and large-certificate availability. The digest manifest binds the raw documents and their rendered counterparts. Markdown headings and tables have print formatting; this does not change the review text or its historical references.

Both source bundles include the correspondence and the missing small certificate. The full source additionally includes the retrieval index, structured change record, baseline excerpts and PDF, and revision checks. The build receipt, structure audit and portable-validation receipt bind the resulting sources, PDF and ancillary inputs. The latter records all four documented ancillary executions after archive extraction, and a standalone typesetting comparison. The supplemental symbolic check of the 14.72 restriction is an exact algebra check, not a formalisation of the geometric proof. None of these checks is a fresh replay of the undistributed 20.100 certificate or of all the reviewer’s programs.

## G.21 Subsequent editorial reorganization and reviewer credit

After the first review revision, the organizers identified the external reviewer as Anthropic’s Claude Opus[1m], model `claude-opus-5[1m]`, with `xhigh` reasoning through Claude Code. This identification is now credited on the opening page and in Section 3; it is not inserted into the historical reports or replies.

The abstract and introduction now summarize both the mathematical results and the independent review. A new early section describes Claude’s checks, the exchange with Codex, and its confirmations and corrections. The original instruction is summarized in the account of the research process. Detailed configuration, accounting, chronology and study-design material are in Appendices A–D.

The former mathematical appendices now appear in Section 2 and Sections 6–8. The 42 further candidate proofs are grouped by subject, and the opening table points to selected results. Their hypotheses, arguments, imported dependencies and historical classifications are preserved. The contents lists major sections so that technical indexing does not delay the mathematical narrative.

The 18 before/after pairs above remain tied to the actual first review revision, rather than being silently replaced by later wording. Source copies bind both historical sides; the preserved PDF fixes the then-current layout. The subsequent changes to organization and exposition are recorded here and in the Git revision, with a second reading pass and validation notes in `paper/reviews/polish-2026-09-17/`.

In a final editorial pass, the full inventory moved directly after the introduction, before the proof sections, with updated reading routes. A title-page footnote records Michael Kinyon’s research visit to the AI4REASON institute. Urban’s title-page footnote credits ERC NextReason (Grant Agreement No. 101200949) and links to the institute’s sponsors; it replaces the separate acknowledgements paragraph.

The abstract was subsequently shortened by removing a repeated assessment of review evidence and the deadline count. The introduction now states the result categories and the nine partial-result questions explicitly, and places the deadline and requested local resource limits alongside the experiment’s objective. Its motivation relates this evaluation on open problems to the MPTPCchallenge and timed human mathematical contests.

The introduction now also mentions the parallel Fable experiment by the same organizers under the same local conditions. A new Appendix M, after the historical correspondence, provides a preliminary review and comparison, with a pinned source snapshot and three targeted computational checks. It records overlap, prior-work status and complementary results, and sharpens the threshold in Fable’s construction for 2.78 from 86 to 78. This later comparison does not alter either experiment’s historical result count.

## G.22 Version 2: contemporary solutions and attribution

The following changes were made on 19 September after the editor identified further contemporary solutions. They compare the complete pre-v2 paper at local commit `5fcd9df` with

version 2. The preserved PDF is `paper/versions/kourovka-experiment-2026-09-17-v1.pdf`. Source hashes and literal excerpts are bound in `paper/reviews/v2-2026-09-19/`. Each pair identifies the old statement and its replacement or the additional qualification.

**The task input Before.**

48 hours to investigate previously unsolved problems in the Kourovka Notebook

**Version 2.**

48 hours to investigate questions listed as open in the supplied Kourovka Notebook

State what the supplied edition listed; later attribution does not establish that every question was still open at the time of the run.

**The abstract now reports the comparison Before.**

finite certificates, with their hypotheses and imported ingredients.

**Version 2.**

We compare eleven entries with contemporary solutions, identifying shared arguments, different constructions and stronger alternatives.

Add the principal outcome of the second version immediately after the mathematical summary.

**Coverage and novelty Before.**

In particular, we acknowledge contemporaneous work on 21.106, and retain known results and deductions under their prior-work status.

**Version 2.**

We assign no discovery priority from the available chronology, and do not infer that the other 35 entries are new.

Replace the single-overlap discussion with an eleven-entry comparison while retaining 46 as a historical classification.

**16.9: prior qualitative computability Before.**

novelty is a separate question.

**Version 2.**

the qualitative computability assertion in 16.9 is now explicitly identified as a consequence of older work.

The new attribution note credits Dahmani-Guirardel for the decision procedure and Brandenbursky-Gal-Kedra-Marcinkowski for the interval recurrence, following the contemporary paper's source discussion.

**Specialist confirmation has a specific scope Before.**

Its independence concerns the reviewing model and the separately written checks; it is not a proof of discovery independence, human specialist endorsement or formal verification.

**Version 2.**

He reported that group theorists had confirmed alternative solutions for 21.68, 16.28(a) and 15.89; many other drafts remained unverified.

Add the editor's report about particular alternatives without changing the scope of Claude's earlier review.

**Individual attribution and evidence.** Eleven new notes accompany the affected proofs. They credit the papers listed in Appendix N, including both Rizzoli and Jayadevan for 21.68 and the retained Zhang–Li work for 21.106. The inventory identifies every overlap. The new appendix gives the comparison, chronology and checks; the Fable appendix now explicitly retains the historical meaning of our answer count in that comparison. The original proof expositions, deadline ledger and four exchanged review documents remain preserved. The v2 comparison and algebraic identities are later work, separate from the deadline results and Claude's original review. The raw editor email is not reproduced.

## H Review correspondence: provenance

The following four appendices reproduce the supplied documents in order, with Markdown layout converted to print layout and no editorial changes to their wording. Original section, theorem and appendix references point to the 15 September paper, not to this revision. Assertions later withdrawn remain in the original documents. The follow-up supersedes its author’s contradictory earlier assessments; the final reply corrects the first reply’s claims about completeness of the bundle and future certificate availability. These are review records, not additional mathematical premises.

The original first report is undated in the supplied file. Both author replies and the follow-up are dated 17 September 2026. The separately supplied `review-update1.md` is a partial update of the first report; it still contains some statements withdrawn in the follow-up. We retain its raw text alongside an exact unified diff, rather than repeat the near-duplicate report here. The originals, diff, and digest manifest are under `paper/external-reviews/` in the repository and the full source archive.

The reviewer was Anthropic’s Claude Opus[1m], model `claude-opus-5[1m]`, with `xhigh` reasoning through Claude Code. The organizers supplied this identification during the subsequent editorial revision. Claude is credited with the independent mathematical review and its computational and source checks; Codex wrote the author replies. Section 3 describes the process and its mathematical outcomes.

The reviewer artifacts are in a separate repository, <https://github.com/JUrban/kouricl/tree/0d76dc7fc947a89a6f02e37b81da2320450efab6/review-other1>. That pinned tree contains 83 files under `checks/` and five under `checks2/`, together with the working notes. We inspected the artifact index and follow-up logs; we have not rerun all the reviewer’s scripts. The reports distinguish external checking from the deadline record. Claude’s production account describes additional agent instances whose findings it treated as leads and checked itself. This is separate-model review, rather than an assertion of human specialist endorsement.

# I First external report (original version)

## Referee report on *Forty-eight hours with the Kourovka Notebook: an experiment in agentic mathematical research*

Paper: `review-other1/kourovka-experiment.pdf` (Codex gpt-6-astra, M. Kinyon, J. Urban; 15 September 2026; 96 pp.). Scripts and logs written for this report are in `review-other1/checks/`; `review-other1/notes_progress.md` is the working log. No file outside `review-other1/` was created or changed, and no git command was used.

### 1. Summary judgement

The mathematical core of the paper is in much better shape than the genre usually is. I checked all 46 entries of the deadline ledger: 3 in the main text and 43 in Appendix C. Of these I verified a majority in full detail, by reading the proofs line by line and, where a finite object is involved, by recomputing it independently in GAP or Python. I found no false theorem among the claims I was able to settle, and the computations I reran all reproduced the paper's numbers exactly, including the large search-coverage figures quoted in Section 3.

The problems I do see are of five kinds, and all but the third concern the framing rather than the algebra:

1. **Literal versus intended readings.** A handful of entries answer the printed sentence in a way that the proposer almost certainly did not intend, and the paper does not always say so loudly enough (18.92 is the clearest case; 16.28(a), 14.22, 16.20 and 12.40 are milder).
2. **Novelty.** The paper is careful to say that priority is unestablished, but at least one flagship result (Theorem 4.1, Problem 21.106) was duplicated publicly, with a Lean 4 proof, within the run window itself.
3. **One hypothesis-reading problem with substance.** In the surface example for Problem 14.72 the fixed locus is a smooth divisor of codimension one but is not cut out by a single equation, so whether it meets the hypothesis depends on how "regular hypersurface" is read (Section 6).
4. **Under-claiming in the other direction.** Three entries in the appendix of rediscoveries look like answers to their printed questions and are nevertheless excluded from the count. So the number 46 is not a like-for-like count in either direction.
5. **Dependence on unprinted certificates.** Two entries (19.61, 19.62) rest on machine-generated integer certificates that are not in the paper. I re-derived all of them independently, so in this instance the dependence turned out to be harmless, but a reader without that effort cannot check them.

### 2. How this report was produced

I read the whole paper, then for each of the 46 ledger entries I pulled the corresponding statement out of the supplied Notebook file and compared it with what the paper proves. Proofs were read step by step. Wherever a claim involves a finite object, a finite computation, or an explicit numerical constant, I recomputed it independently rather than trusting the paper's own scripts (which are not part of the submission anyway: the archive is not publicly reachable, see Section 7). The scripts I wrote are in `review-other1/checks/` with their logs.

The independent computations were: the order-2592 non-monomial group and its character degrees; the order-605 holomorph example; the p-Jordan estimate on all subgroups of the two smallest groups of the family; the 3.A7 5-modular decomposition data; the order-36864 triple-intersection family; the system-normaliser family for three parameter pairs; the 2-modular data for the dicyclic group of order 12; the  $SL_2$  images over  $F_4[t]/t^k$ ; the numerical constants in the classical-group argument; the 2-adic integer facts for  $PSL_6(5^a)$ ; the orthogonal-basis criterion against brute force; the palindromic-length algorithm against brute force; the common-centraliser construction; the order-120 subgroup over  $F_9$ ; the Chevalley commutator constants for A2, B2 and G2, and with them all 19 printed carpet derivations plus, by an independent saturation search, all 104 requests for inclusion (5) and all 300 certificates behind Lemma C.13. Later I added: the smoothness, invariants and defining relation of the surface for 14.72 by Groebner elimination; the two Baker-Campbell-Hausdorff formulas

for 13.42 in an exact word algebra; the interpolation polynomials for 14.26; the coefficient identity and degree bookkeeping for 20.92(a); the Chebyshev inequalities for 10.32; the Golod-Shafarevich arithmetic for 19.93; the cycle-length arithmetic for 15.92; the quasigroup identity for 9.4 by brute force over all quasigroups of order at most four; the closing counterexample for 16.38; the twisted indicator identity for 16.60 over every automorphism class of seven small groups; the subgroup ranks for 20.8 by Stallings folding; the class-two linearisation for 21.115 over the Heisenberg group mod three; the order-256 subdirect example for 13.19; and the automorphism bound used in 21.121(b) over 344 cases. I also reproduced the paper's search-coverage numbers (91,774 nonabelian groups of order at most 511; 4,722 transitive groups of degrees 2 to 20; 3,933,931,043 binary necklaces of lengths 1 to 36), checked the SHA-256 digest of the Notebook file against the one printed in the paper, and checked the internal arithmetic of the token and cost tables.

Four blocks of appendix sections were additionally read by separate agent instances working from the same text. Those are not independent referees in the sense that matters, and I treated what they returned as leads: every finding of theirs recorded below I checked myself first, in several cases by going back to the cited sources. Three of the four returned; the fourth failed, so I read appendices D and E myself.

### 3. The experimental claims

I could check the parts of Sections 2 and 3 that are checkable from outside the archive, and they hold up.

- The SHA-256 digest printed for the input Notebook matches the file.
- Issue 21 does contain exactly 150 entries; a crude parse of the main body gives 1309 numbered entries against the paper's 1308, which is parsing noise on my side, not a discrepancy.
- Four coverage numbers quoted in Section 3.3 are exactly right: 91,774 nonabelian groups of order at most 511, 4,722 transitive groups of degrees 2 to 20, 3,933,931,043 binary necklaces of lengths 1 to 36, and the 2,750 ordinary character tables behind the 21.113 screen. The control set of 86 transitive groups in degrees 2 to 8 quoted for Problem 21.99 is also exactly right, and so is the finer breakdown of the 21.113 screen: I count 9,850 table/prime cases, 7,573 with the modular table available and 2,277 without, matching the paper to the unit.
- The token and cost tables are internally consistent, and the flat-rate reconstruction reproduces the quoted USD 901.42 to the cent.
- Appendix B has 46 rows with 46 distinct cross-references, covering C.1 to C.43 and 4.1 to 4.3 exactly once, and none of the 46 problems is starred as solved in the supplied Notebook file. Problem 19.63, which the ledger groups with 19.62, is starred there, exactly as the paper says.

The paper's epistemic framing is, in my view, its strongest feature. The repeated insistence that a "complete solution candidate" is an observable outcome and not an accepted theorem, the separation of generation from checking for the large certificate, the explicit statement that a second implementation by the same agent is not independent review, and the admission that the deadline archive contains zero outside reviews are all correct and unusually candid. Section 2.6 and Section 5.2 should be kept as they are.

Two framing points I would still press. First, the abstract's "46 entries labeled complete solution candidates" is qualified properly in the body, but the number will be quoted out of context; it would help to state in the abstract that several of those entries answer the printed sentence under a reading the proposer probably did not intend (see Section 6 below). Second, the claim that the agent "chose its own targets" is used to explain the portfolio, but the selection effect deserves a sharper statement: the entries that survived are, with few exceptions, those where a small explicit object or a short reduction settles the question, which is also why so many of them are checkable in an afternoon.

### 4. Entry-by-entry verdicts

The table records what I was able to establish. "Verified" means I rebuilt the object or the computation independently; "checked" means I read the proof line by line and found it correct; a caveat in the last column is explained in Section 6 or in the notes below the table.

| Problem   | Section | Verdict                                                                                                                                                                                                                                              |
|-----------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 21.106    | 4.1     | Checked. The truth set really is $\{z, z^{-1}\}$ and the group is residually finite. Priority: see Section 5.                                                                                                                                        |
| 21.68     | 4.2     | Verified in GAP: order 2592, not monomial, six non-monomial irreducibles of degree 8, split chain as printed. The auxiliary group of order 96 is monomial, consistent with Kida's own computation for orders at most 240.                            |
| 20.108    | 4.3     | Verified in GAP: centreless,                                                                                                                                                                                                                         |
| 21.121(a) | C.1     | Checked, and the key inequality verified on every subgroup class of the two smallest members of the family.                                                                                                                                          |
| 16.87     | C.2     | Checked. The endomorphism is well defined because $p$ -th powers are central, and it is not surjective.                                                                                                                                              |
| 10.35     | C.3     | Checked, including the lemma on 2-by-2 matrices and the claim that the group does embed in $GL_3(Q)$ .                                                                                                                                               |
| 16.28(a)  | C.4     | Checked; scope caveat (needs an element of infinite multiplicative order).                                                                                                                                                                           |
| 14.22     | C.5     | Checked; scope caveat (coefficient group not finitely generated).                                                                                                                                                                                    |
| 16.20     | C.6     | Checked under the stated convention; the variant convention is handled only by a sketched second example.                                                                                                                                            |
| 17.33     | C.7     | Checked in full: the presentation, torsion-freeness, finiteness of the abelianisation, the embeddings of the proper-subspace preimages, and the variable-count argument.                                                                             |
| 17.101    | C.8     | Checked. The Bass-Serre argument really does give $V$ embedded in $W$ , and the realising automorphism is the expected one; the freeness of $kH$ over $kG$ and the fact that the image of $\delta$ is a submodule are used silently.                 |
| 18.76     | C.9     | Checked, including the four commutation equations and the final cancellation, which needs that the division closure of $kF$ centralises $kL$ .                                                                                                       |
| 18.92     | C.10    | Checked; the literal reading makes the problem trivial (Section 6).                                                                                                                                                                                  |
| 20.90     | C.11    | Checked; elementary (Section 6). Finite quotients verified in GAP.                                                                                                                                                                                   |
| 11.116    | C.12    | Checked; the digit encoding, the fibre description and the "d chains implies dimension at most d" step are all correct.                                                                                                                              |
| 14.72     | C.13    | Checked and recomputed: the surface is smooth (empty singular locus), the invariants satisfy exactly the relation $V^2 = (A-1)D$ , and the quotient is singular at the image of $(1,0,0)$ . One caveat about the word "hypersurface", see Section 6. |
| 21.40     | C.14    | Checked. The key step, that the inverse automorphism turns $\alpha(h) = h^r$ into a tower of $r^k$ -th roots inside the group, is correct, as is the trace-tuple index bound.                                                                        |
| 21.60     | C.15    | Checked; the 2-modular data verified in GAP, and the failure of semiperfectness confirmed independently by a rank argument on idempotents.                                                                                                           |
| 4.55      | C.16    | Checked; the 5-modular decomposition rows, Galois orbits and projective-cover degrees for 3.A7 verified against the character table library.                                                                                                         |
| 21.107    | C.17    | Checked. The subgroup filter, the dense partition and the parity argument against expansive sequences all hold.                                                                                                                                      |
| 15.89     | C.18    | Checked: the operator identities $U^2 = U + 2I$ and $A^{-1} = (U-I)B(U-I)/4$ hold, and the Cayley graph is simple, 4-regular, connected and vertex-transitive.                                                                                       |
| 16.9      | C.19    | Checked, and the dynamic program verified against brute force on all 485 reduced words of length at most 5.                                                                                                                                          |
| 18.18     | C.20    | Checked in full; the common-centraliser construction verified in GAP for three groups, with and without the extra orbit.                                                                                                                             |
| 19.56     | C.21    | Checked in full. This is the strongest entry in the paper: a short, correct solution of a 2018 problem.                                                                                                                                              |

| Problem  | Section | Verdict                                                                                                                                                                                                                                                                                                                                 |
|----------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 20.92(a) | C.22    | Checked. The coefficient extraction needs the hypothesis on the order of 2, and it works; the degree bookkeeping uses $E = k^{(2k-2)}$ . An exhaustive scan of all 1625 three-dimensional braces over $F_5$ , 800 of them with nonlinear $\lambda$ , reproduces the printed operation with no exception.                                |
| 9.47     | C.23    | Checked; contingent on the E-topology being the full Vietoris topology, which the paper states and which the Notebook never pins down.                                                                                                                                                                                                  |
| 9.45     | C.24    | Checked, and the criterion verified against brute-force search over 1264 pairs $(n, m)$ .                                                                                                                                                                                                                                               |
| 10.62    | C.25    | Checked; the elementary deductions are right, but the entry leans on the internals of an unrefered 2025 preprint, and one scale condition is extrapolated. See Section 6.                                                                                                                                                               |
| 10.32    | C.26    | Checked. The covering arithmetic and the prime-counting inequalities hold, verified numerically well past the stated threshold.                                                                                                                                                                                                         |
| 15.65    | C.27    | Analytic argument checked; all constants verified numerically and all inputs matched against the source memoir.                                                                                                                                                                                                                         |
| 12.40    | C.31    | Checked; the 2-adic arithmetic verified.                                                                                                                                                                                                                                                                                                |
| 15.92    | C.28    | Checked; the lemma and the arithmetic hold (720720 is the least common multiple of 1 to 16, the printed exponents are $E/2$ , and $w^2 = xyxy$ ). But the proof depends on cycle-structure data read out of a 1980 doctoral thesis, which I could not verify, and the choice of 11 or 13 by residue class is asserted without a reason. |
| 4.75     | C.29    | Checked; a short corollary of the 10.62 construction, inheriting its dependence on the 2025 preprint.                                                                                                                                                                                                                                   |
| 9.4      | C.30    | Checked, and the quasigroup identity verified by brute force over all quasigroups of order at most four. The cited source is about quasivarieties of automata, and the paper itself raises the resulting convention question.                                                                                                           |
| 13.42    | C.32    | Checked, with both Baker-Campbell-Hausdorff formulas recomputed here in an exact word algebra.                                                                                                                                                                                                                                          |
| 14.26    | C.33    | Checked; the interpolation polynomials have exactly the degree the choice of the parameter $m$ provides, verified in three parameter sets.                                                                                                                                                                                              |
| 15.76(b) | C.34    | Checked; the coefficient identity has exactly the two integer solutions the argument uses. The cited contrasting example is a proper subvariety of the metabelian variety, of exponent four, which the paper could say in half a sentence.                                                                                              |
| 16.38    | C.35    | Checked, including the closing counterexample, which I reproduced.                                                                                                                                                                                                                                                                      |
| 19.93    | C.36    | Checked; the Golod-Shafarevich inequality is used in the form the source states, and the value at two thirds is exactly the printed $-115/729$ .                                                                                                                                                                                        |
| 20.1     | C.37    | Checked; the coprime-order lemma is correct and the order of the Janko group is coprime to 13. The entry rests on the imported theorem that that group is totally 2-closed.                                                                                                                                                             |
| 19.62    | C.38    | Checked, including the direct proof of the square criterion; all 104 requests re-derived independently.                                                                                                                                                                                                                                 |
| 19.61    | C.39    | Checked; all 300 certificates re-derived independently.                                                                                                                                                                                                                                                                                 |
| 21.76    | C.40    | Checked; the rank-one argument is correct and the auxiliary group of order 120 over $F_9$ verified in GAP.                                                                                                                                                                                                                              |
| 11.115   | C.41    | Checked; correct under both readings of the hypothesis.                                                                                                                                                                                                                                                                                 |
| 20.122   | C.42    | Verified in GAP: the full family of 10 distinct intersections, and both invariants equal to $A$ , outside $F(G)$ .                                                                                                                                                                                                                      |
| 17.39    | C.43    | Verified in GAP for three parameter pairs: $p^{d+1}$ system normalisers and minimum $d+1$ with trivial intersection.                                                                                                                                                                                                                    |

**Appendices D and E** These are outside the ledger, and the paper is careful to say so: appendix D collects partial results, appendix E collects rediscoveries and deductions from prior work. I checked the following myself.

- **20.100 (D.1).** The reduction to finite abelian groups and the size bound are stated honestly as a decidability statement for each fixed  $n$ , not as a proof of the conjecture. The hypothesis control is right: over the cyclic group of order five no ordering of the four nonidentity elements makes the prescribed powers distinct, so allowing an element of order five at  $n = 4$  really does break the conclusion.
- **20.89 (D.2).** The remark that bounded finite subgroups are not automatic in positive characteristic is immediate, since a diagonal matrix conjugates one unipotent to another with a scaled parameter, giving elementary abelian subgroups of unbounded rank.
- **21.121(b) (D.3).** Both auxiliary bounds hold. The bound on abelian prime-to- $p$  subgroups of the automorphism group survived 344 test cases, and the Chermak-Delgado step, computed by hand from the measure, gave an abelian normal subgroup of index at most the square of the abelian index in all 132 nonabelian test groups. The exponent for the infinite alternating example lies strictly between two and three, as claimed.
- **20.124 (D.4).** The Rota-Baxter identity follows from the two displayed maps being mutually inverse; I checked the derivation. The paper is explicit that the method needs infinite rank.
- **17.118 (D.5).** With the cited bound on cube roots, four independent hyperplanes already cover more than the permitted proportion, so the rank is at most three and the index bound 27 follows; the case  $p = 2$  gives 4.
- **21.114 and the rest of D.6.** The order-8192 central product rests on finite permutation models kept in the archive, so its numbers cannot be checked from the paper.
- **13.19 (E.1).** The small example is exactly as described: order 256, kernel of order 32, dihedral quotient of order eight, both subgroups subdirect in the fourth direct power, diagonal inside the kernel.
- **17.34 (E.2).** The separation module behaves as claimed in the smallest case: the subalgebra is exactly the kernel of the displayed map.
- **21.115 (E.3).** The class-two linearisation really is an abelian group operation, and subgroups stay closed under it, on the Heisenberg group over the field of three elements.
- **16.60 (E.4).** The twisted indicator identity and the bound by the sum of the degrees hold for every automorphism class of seven small groups.
- **20.8 (E.8).** The two subgroups have the stated ranks three and four by Stallings folding, and the endomorphism that differs from the inclusion does fix the four generators of the smaller one.
- **20.89 and 19.83 (D.2).** The strongest of the partial results, and honestly bounded. The local reduction is correct: each generator is Engel on a finite-index normal subgroup avoiding the nonidentity sink, Gruenberg's theorem then places the commutators in the Hirsch-Plotkin radical, and the rest is bookkeeping. The characteristic-zero closure theorem, the bounded-torsion case, the triangular characterisation, and the case of normal subgroups of unit groups of finite-dimensional algebras all hold. I checked the last one in detail: the units  $1 + ta$  commute with  $a$  and permute the minimal sink, and comparing two parameters forces every sink element to commute with  $a$ , so the sink is trivial. The unresolved positive-characteristic case is stated as such.
- **21.121(b) (D.3).** The soluble theorem and its two auxiliary bounds are correct, and the infinite alternating example really does have exponent exactly the logarithm ratio, strictly between two and three, which is the interesting part: it shows the conjectured bound of three cannot be improved to two in general. The permanence statements are correct and carefully hedged. The universal bound of three is left open, as the paper says.
- **17.118 (D.5).** The class-bounded results and the two optimal small-prime bounds are right, and the lower bound construction of index  $p$  to the  $p$  is correct, including the intrinsic characterisation of its distinguished subgroup. The paper is explicit that the class parameter is not known to depend only on the prime, so the question itself is untouched for primes at least five.
- **D.6 and D.7.** These are correctly labelled. The order-8192 example for 21.114 depends on permutation models in the archive and cannot be checked from the paper. The restricted theorems for 19.108 and 18.120, and the class-two case of 6.47, are stated with their hypotheses in the open. The bounded searches in D.7 carry explicit coverage boundaries, and every number in them that I could recompute was exact.
- **E.5, E.6 and E.7 (21.132, 21.42, 20.33).** These three deserve a remark in the paper's favour. Each appears to answer its printed question: a finitely generated residually finite  $p$ -group

with nontrivial finite centre; a negative answer to whether there are three-generated torsion-free nilpotent groups of class three that fail to be self-similar; and the relativised embedding statement in both directions. All three are nevertheless excluded from the ledger because the ingredients are prior work. That is under-claiming, and it cuts against reading the number 46 as a like-for-like count in either direction.

- **E.9 (19.9, 16.14, 19.48).** Correctly classified as prior answers. The 16.14 derivation really is a one-step specialisation of an earlier centraliser bound, and saying so is right.
- **12.69 (E.10).** The literal counterexample is correct, and the paper is right to record it as a formulation problem rather than a solution.

## 5. Priority

The paper says priority is unestablished and does not claim novelty; that is the right stance, and the following should be added to it.

**Problem 21.106 (Theorem 4.1) was duplicated publicly during the run.** A GitHub repository, `TPColor/kourovka-notebook-problem-21.106`, contains a complete Lean 4 formalisation of a counterexample using the same group (the integral Heisenberg group with the same multiplication) and a formula of the same shape: an existential over a commutator, a centrality predicate, and conditions forcing the two generators to span. The repository was created on 10 September 2026 at 19:32 UTC, about 84 minutes before the experiment's research window opened, but with an empty initial commit; the Lean files carrying the mathematics were pushed on 11 September at 10:54 UTC, roughly 14 hours after the agent had already committed its own proof file for 21.106. The credits name a human author for the counterexample and a second person for the Heisenberg simplification, and mention a planned arXiv preprint, which I could not find. So this is parallel discovery in which the agent was narrowly first, not a case of the agent reproducing something already public. It does show that the problem was, at that moment, within reach of at least two independent efforts, and the paper should say so: a flagship example that a student formalised in Lean within two days of the same week is weak evidence of research-level difficulty.

I checked the two most plausible collision candidates in the recent literature and found no overlap: arXiv:2607.17477 (July 2026) settles Kourovka 3.46, 18.50, 19.25, 20.125, 21.8, 21.24, 21.147, 21.150, and arXiv:2608.29219 (August 2026, itself LLM-assisted) settles 14.85, 19.94, 16.11, 17.47 and 17.32. None of these is among the 46.

I also confirmed the two prior-work statements that matter most for the carpet entries. Nuzhin's 2026 Siberian Mathematical Journal paper does give the complete positive answer to Problem 19.63 (the square criterion), so the ledger's grouping of 19.63 with 19.62 must indeed not be read as a new solution, exactly as the paper states; and Sambale's September 2026 preprint on complements of unions of cosets does cover Problem 21.115 as printed, because a right coset  $Hg$  is the left coset  $g\langle g^{-1} \rangle Hg$ , so the paper's decision to exclude 21.115 from the count is right.

## 6. Where the printed sentence and the intended question part company

This is my main substantive criticism. Several entries are correct answers to the printed words but almost certainly not to the question that was asked. The paper usually flags this in a sentence at the end of the section; the flag is too quiet given that the entries are then counted.

- **18.92 (C.10).** Under the definition printed in the Notebook, a "complete lattice of formations" is any nonempty intersection-closed family with a largest element. The paper answers both parts with the formations defined by sets of primes: an infinite family for (a), and five members forming a pentagon for (b). This is correct and takes a page, but it makes a long-standing question of Skiba's trivial, which is strong evidence that the intended object is the lattice of all formations closed under some operation, not an arbitrary family. I would not count this entry without stating that explicitly.
- **16.28(a) (C.4).** The construction needs an element of infinite multiplicative order in the field, so it lives over  $F_5(t)$  rather than over an algebraically closed field. Under the usual convention that a linear algebraic group is considered over an algebraically closed field, the question is untouched; over the algebraic closure of  $F_p$  every unit is a root of unity and the trick fails. The paper notes the ingredients used but does not say that the natural reading is left open.

- **14.22 (C.5).** The counterexample uses the additive rationals, a coefficient group that is not finitely generated; the cited positive result for free groups is in the finitely generated setting. Again correct as printed, and again a different question from the one the proposers were probably asking.
- **16.20 (C.6).** The construction depends on allowing the ambient group  $A$  to lie outside the quasivariety  $N$  whose dominions are taken. The paper says a second, larger example handles the opposite convention, but that example is only sketched, so the entry is fully verified only under one of the two conventions.
- **14.72 (C.13).** The printed question supposes that the fixed-point variety is "a regular hypersurface of  $X$  (of codimension 1)". The paper's fixed locus is a smooth codimension-one closed subvariety and a Cartier divisor, and the paper says so; but it is not the zero set of a single global function. The surface fibres over the affine elliptic curve  $y^2 = x^3 - x$  minus its point at infinity, as the total space of a line bundle, and the fixed locus is the fibre over the 2-torsion point  $(0,0)$ . Restricting a putative global generator to the zero section would produce a function on that affine curve with divisor exactly one copy of the 2-torsion point, which would force that point to be trivial in the group law. So the hypothesis of 14.72 fails under the reading "hypersurface = one equation" and holds under the reading the parenthetical suggests, namely "closed subvariety of codimension one". The paper should settle which it means; the example looks repairable by deleting a suitable orbit of 4-torsion fibres, which kills the obstruction while keeping both the fixed divisor and the singular point. Separately, the problem allows arbitrary characteristic and the construction is over the complex numbers only, so the wild case is untouched.
- **10.62 (C.25) and 4.75 (C.29).** Both rest on a single geometric existence theorem from a September 2025 preprint, and not only on its statement. I read the source. Its Theorem 6.7 gives what the paper says it gives: elliptic subgroups embed, every finite-order element of the quotient is either killed to exponent twice one of the prescribed odd integers or comes from an elliptic element, and infinitely many elements are new. But two things the paper needs sit outside that statement. The ability to put a chosen primitive loxodromic element into the first family of imposed relations lives in the proof, which builds the quotient by adding relations of that form step by step. And the scale condition is read off a remark that states the principle "a smaller rescaling parameter is available at the cost of a larger critical exponent" but fixes the constant for translation length two, whereas the word  $abc$  used here has translation length three. Both extrapolations look safe, and the paper does say it uses the construction and not only the theorem. One thing that looks like an inconsistency in this section is not one, and I record it so that a later referee does not report it as an error: the vanishing of the overlap invariant coexists with the choice of a smaller scale, because that invariant is computed in the original space and is multiplied by the scale factor under rescaling, while admissibility is measured against a fixed universal constant. The source itself works with hyperbolicity constant zero and vanishing overlap invariant. Still, two ledger entries stand on one unrefereed import, and that belongs in the main text rather than only in the appendix.
- **12.40 (C.31).** The problem asks for the best possible estimate of the  $p$ -part of a Brauer character degree in terms of the  $p$ -part of the group order. The paper shows no finite bound exists. That is a legitimate answer to the printed request, but "find the best possible estimate" suggests the proposer had a restricted class in mind.
- **20.90 (C.11).** The example is correct, but it is elementary: every closed subgroup of  $SL_2$  over a characteristic-two power series ring has abelian centralisers, because centralisers of non-scalar 2-by-2 matrices are commutative and scalars are trivial in characteristic two. The paper says the centraliser argument is classical and claims only the application; that is honest, and the entry should be weighted accordingly.
- **11.115 (C.41).** The counterexample is three lines long. It is correct under both readings of "not maximal" (as a subgroup and as a normal subgroup), and the paper covers the variant where  $T$  is also not maximal. Such a short refutation of a 1990 problem is a priority risk rather than a mathematical one.

None of these is an error. But a reader counting 46 solved problems should be told that roughly a sixth of them are answers to the letter of a question whose spirit is still open.

## 7. Reproducibility and the unprinted certificates

Appendix G is honest that the configured repository address was not anonymously reachable during manuscript preparation, and that a local query found a remote head different from the frozen revision. That means a referee has the paper and nothing else. For most entries this is fine, because the constructions are printed in full and can be rebuilt from the text, which is exactly what I did. Two entries are different in kind.

For 19.62 the paper prints 19 representative derivations in Appendix F and asserts that an ancillary checker verifies all 104 requests; for 19.61 it asserts 300 certificates with 3,191 nodes, none of which is printed. I checked both claims independently. I computed the absolute Chevalley commutator constants for A2, B2 and G2 from a Chevalley basis in the adjoint representation, confirmed that every application in the 19 printed derivations uses a genuine rule with the stated constant and that the greatest-common-divisor steps are right, and confirmed that the displayed destination roots carry exactly the stated numbers of requests. I then wrote my own saturation search over the same proof calculus and re-derived all 104 requests for inclusion (5) and all 300 merged requests behind Lemma C.13, obtaining the paper's counts exactly (6, 20 and 78 requests in types A2, B2, G2; 12, 48 and 240 merged certificates from 600 raw substitution terms). So these two entries are sound. But this took a working day; the right fix is to print the request tables and the certificate grammar, or to ship the ancillary files with the paper as the appendix says they are shipped in the arXiv source.

The same applies to the largest computation in the paper, the  $n = 7$  certificate for Problem 20.100, which is not an entry in the ledger but is the headline of Section 3.3. Nothing about it can be checked from the paper. The paper is careful to say so, and to distinguish generation from verification, which is the right instinct.

## 8. Small things

- Section 2.5: the phase estimates  $846.09 + 64.52$ ,  $0.56$  and  $54.77 + 3.85$  add to USD 969.79, not the 969.78 stated. The paper does say estimates are rounded independently, so this is a rounding artefact, but it reads as an error.
- Appendix C.4 asserts that part (b) of 16.28 also follows from the same class, and then excludes it as prior work. Since the paper does count part (a), it should state that (b) holds for the same  $Y$  over any field, which is what its own sentence implies.
- C.16 states the four projective covers have dimensions 90, 60, 90, 90 and lists GAP ordinary indices 10, 16, 18, 20, 22. Both match the character table library exactly; it would cost one sentence to say which table version was used, since the indices are library-dependent and the paper already warns about this.
- The ledger description for 20.90 in Appendix B says "CA-group" while the section title says "CN-groups". Both are true (the example has abelian centralisers, hence pronilpotent ones), but the inconsistency looks like a slip.
- The two prime-counting inputs in C.26 are correct, but the attribution deserves a look: the upper bound with the constant 1.01624 is indeed Theorem 9 of the cited paper, while the lower bound valid from 563 onwards is usually cited as formula (3.16), not Theorem 4.
- C.28 leans on unpublished cycle-structure data from a 1980 thesis (one  $p$ -cycle in a specific permutation, all cycle lengths dividing 720720), and on an unexplained assignment of the prime 11 or 13 to residue classes of  $r$ . Both should be stated as explicit hypotheses with a precise pointer, since nothing in the paper lets a reader check them.
- C.32 cites "Exponential groups 2" for the definition of an A-group and of tensor completion, whereas the Notebook entry for 13.42 points the reader to the 1994 "Exponential groups 1" paper. Worth aligning.
- C.27 identifies the analytic obstruction at the first zero of  $A(t)$ , namely the golden-ratio conjugate. I verified the inputs against the Fulman-Neumann-Praeger memoir: the paper's A, B, E, D and P are exactly the memoir's quantities, and its  $J_d$  is the memoir's  $N^*(q; 2d)$ . The one item I could not confirm from the memoir text is the exact index in the  $K_d$  exponents, where the paper itself records a discrepancy in the source. The conclusion is therefore correct modulo a transcription the paper has already flagged as delicate.

## 9. Recommendations

On the mathematics:

1. Separate the ledger into columns the reader can see at a glance: answers the printed question under the reading the proposer intended; answers the printed sentence under a reading the proposer probably did not intend; and answers a subpart. On my reading roughly seven of the 46 belong in the middle column. While doing that, move the three appendix entries that do answer their printed question into the same table with their prior-work status recorded, rather than leaving them invisible to anyone counting.
2. Print the request tables and certificate grammar for 19.61 and 19.62, or state plainly that those two entries are unverifiable from the paper alone. The same holds for the  $n = 7$  certificate discussed in Section 3.3.
3. For each entry, name the imported theorems in the entry itself and say what would survive if the import failed. The paper is good at disclosing shared dependencies, for instance that 10.62 and 4.75 rest on one geometric existence theorem, but it never says which parts of an entry would be lost if a cited result were withdrawn or turned out to be weaker than the use made of it.
4. Promote 19.56 in the exposition. It is a complete, short, self-contained solution of a 2018 problem whose proof uses only standard machinery, and it is much stronger evidence for the experiment's thesis than the three examples currently chosen for Section 4.
5. Say explicitly, for each of the elementary entries (20.90, 11.115, 21.76, 18.92), that the mathematics is short. A reader who sees 46 entries will otherwise assume uniform difficulty, and the paper's own honesty elsewhere makes this omission conspicuous.

On the experiment:

6. Record what the agent read. The single most valuable missing datum is the list of pages retrieved during the run. Problem 21.106 was solved publicly and formally, by other people, inside the same 48 hours; without a retrieval log one cannot distinguish independent discovery from a lucky search, even though in this instance the commit times favour the agent.
7. Freeze and publish the archive before the manuscript, not after. Appendix G currently tells the reader that the repository could not be reached anonymously, which undercuts every claim that rests on it.
8. Keep the separation of generation from checking, and extend it: state for each large computation who checked what, with which independent implementation, and what a failure would have looked like.

## 10. Overall

Accept after revision. The mathematics I could settle is sound: no entry I checked is false, the finite objects are what the paper says they are, the large certificate machinery for the two carpet problems survives an independent re-derivation, and the reported search-coverage numbers are exact. The two appendices outside the ledger hold up as well. The partial results are bounded honestly, with the open cases named in each section, and the appendix of rediscoveries is if anything too cautious: three of its entries look like answers to their printed questions and are still excluded from the count. The revisions I would insist on are editorial rather than mathematical: a visible separation of literal from intended readings in the ledger, an explicit statement of which entries are short arguments, publication of the archive and of the two certificate families, and the priority note about Problem 21.106.

## J First author reply

Reply to the referee report on *Forty-eight hours with the Kourovka Notebook*

17 September 2026

Thank you for the detailed report and, especially, for independently rebuilding the finite examples and re-deriving the carpet certificates. The checks and the distinctions between mathematical validity, statement coverage, and priority are useful. We agree that the paper should make its scope and dependencies easier to assess. Below we clarify several points and describe proposed revisions; these revisions have not yet been incorporated into the manuscript.

1. **Repository and certificates.** Josef Urban will publish the full repository and send you the link. The frozen experimental revision is `cff2c37b9bf6b737e8ad5f7ead12291a551b5201`; subsequent manuscript work is separate from that snapshot.

The complete certificates for 19.61 and 19.62, their inputs, and their checkers already accompany the manuscript source. They are under `paper/ancillary/` in the repository and under `anc/` in `paper/dist/kourovka-experiment-arxiv-source.tar.gz`. The README gives the checking commands. We inspected the archive and confirmed that these files are included. Your report indicates that the source bundle was not supplied with the PDF; we agree that the review materials should include it explicitly. The larger 20.100 certificate belongs to the research archive and requires separate handling. Public access will make it available for examination, but will not itself constitute a fresh verification.

2. **Problem 16.28(a).** The objection concerning algebraic closedness appears to overlook the bar in the first line of C.4. The field used there is  $K = \text{algebraic closure of } F_5(t)$ , not  $F_5(t)$  or the algebraic closure of  $F_5$ . It is algebraically closed, and its transcendental element  $t$  has infinite multiplicative order. The construction therefore does meet the usual algebraically closed-field convention. It uses a reducible closed set and a field not algebraic over a finite field; the printed question excludes neither. We would consequently not classify this entry as failing the algebraically closed-field reading.

We will also check the field restrictions when clarifying part (b). The suggestion that the identical class works over any field needs qualification: in characteristic two, for example, the argument that its square omits  $-I$  no longer applies.

3. **Problem 14.72.** We agree that the distinction between a smooth effective Cartier divisor and a globally principal divisor should be made explicit. There is, however, a further distinction between a set-theoretic equation and a scheme-theoretic equation. In the displayed surface,  $x = 0$  forces  $y = u = 0$ , so it already cuts out the fixed locus set-theoretically. On the chart containing that locus,  $x = y^2/(x^2 - 1)$ : this equation defines twice the fixed divisor, rather than its reduced scheme.

Thus the issue is whether “regular hypersurface” requires the reduced fixed divisor itself to have a single global defining equation. The current proof establishes the smooth Cartier-divisor reading. We propose clarifying this and investigating the suggested strengthening to the globally principal reading. A construction over the complex numbers is sufficient to disprove the unrestricted assertion over fields of arbitrary characteristic; treating the wild case would be a further result.

4. **Problem 21.106 and priority.** Thank you for identifying the related Heisenberg construction and Lean formalisation. We checked the [repository](#), which credits Lily Zhang with the original counterexample and proof and Evan Li with the Heisenberg simplification. This work should be acknowledged in the discussion of Theorem 4.1.

Our local proof file was first committed on 10 September at 21:05:23 UTC. The other repository's initial README contained only its title; the [commit adding the Lean files](#) is dated 11 September at 10:54:21 UTC. These are facts about the recorded artifacts, not proof of discovery priority. We therefore would not adopt the report's stronger conclusion that the agent was first. We propose reporting the chronology and acknowledging the contemporaneous work without assigning priority.

The exact repository URL does not occur in the supplied session JSONL. This limited check does not establish the absence of exposure through another source. The JSONL retains web

queries, actions, and results, from which a dated source-access index can be prepared. Such an index should distinguish recorded retrieval from evidence that a source was actually used in an argument.

5. **The ledger and intended readings.** We agree that scope and conventions should be visible in the inventory. We suggest separate fields for covered parts, convention-sensitive hypotheses, prior-work status, and imported dependencies. The 46-entry deadline ledger is a historical record; a subsequent assessment can supplement it without altering what was classified at the deadline. Likewise, the complete answers and deductions in Appendix E can be made more visible while retaining their prior-work classification.

We would avoid labeling approximately seven entries as answers the proposers probably did not intend without firmer evidence. The objection to 16.28 above does not apply. For 12.40, showing that no finite-valued bound of the requested form exists is a direct negative answer to the printed request. For 18.92 and 14.22, the stated definitions and hypotheses permit the constructions, although identifying any intended additional restrictions would be valuable. We agree that 16.20's convention and the scope of its alternative construction should be especially clear.

Proof length, discovery difficulty, and novelty should also remain separate. A short proof or a rapidly completed formalisation does not by itself establish that the question was easy or was misread.

6. **Dependencies and choice of examples.** We agree that the finite input used for 15.92 should be easier to inspect. C.28 already states the required cycle and parity properties and points to Chapter 5, pages 117–120 of Conder's thesis; exposing the relevant finite construction data would strengthen that presentation. The scale and prescribed-relation steps used in 10.62/4.75 also merit a more explicit parameter justification tied to the exact source version.

Promoting 19.56 is a useful suggestion. Its adaptation to coprime prime-power inputs gives a substantial example of the work performed. Its exposition should retain the imported hyperfocal theorem, minimal-simple structure, and Baer–Suzuki theorem rather than describe the proof as self-contained without qualification.

7. **Smaller corrections and review records.** We will check the cited equation/theorem numbers, the exponential-group reference, the characteristic assumptions in C.4, and the character-table version; clarify the CA/CN terminology; and make the one-cent rounding difference explicit. These are useful presentation and attribution checks rather than grounds for changing the exact usage totals.

Please also send `review-other1/checks/` and `notes_progress.md`, which were not included with the copy of the report we received. They will let us preserve the external checks with their exact inputs, implementations, and outcomes. We will distinguish these later checks from the historical statement that there were no outside reviews at the research deadline.

## K External follow-up report

### Response to the authors' reply

17 September 2026. This responds to `review1-reply/reply.md`, records corrections to my first report, and reports a second round of checking made possible by the material the authors supplied and by the now-public repository `github.com/JUrban/kour1`. Scripts and logs for this round are in `review-other1/checks2/`; the first round is in `review-other1/checks/` and `review-other1/revi`  
`ew.md`.

#### 1. Corrections to my report

**Problem 16.28(a): my objection was wrong, and I withdraw it.** The authors are right. Section C.4 sets  $K$  equal to the algebraic closure of the rational function field in one variable over the field of five elements, and I rendered a page image to confirm the bar is there. My text extraction dropped it, and I built a criticism on the result without checking the typeset page. That field is algebraically closed, so the example meets the usual convention for a connected reductive algebraic group, and its transcendental element still has infinite multiplicative order, which is all the argument needs. Nothing in the entry depends on a non-closed field. Please delete that bullet from Section 6 of my report; I have marked it retracted there.

A second error of mine follows from the same place. In Section 8 I suggested the paper should say that part (b) holds for the same conjugacy class over any field. The authors' qualification is correct: in characteristic two the step that the square omits minus the identity collapses, since minus the identity is the identity and the trace condition no longer separates. The suggestion should read "over any field of odd characteristic", if it is made at all.

**The count of convention-sensitive entries.** I withdraw the figure of roughly seven entries, and with it the phrase about readings the proposer probably did not intend. Two of my examples do not survive: 16.28 for the reason above, and 12.40, where showing that no finite-valued bound of the requested shape exists is a direct negative answer to the printed request, as the authors say. What I should have written is narrower and I still hold it. Three entries turn on a stated convention that a reader must notice to judge the result: 18.92 on the printed definition of a complete lattice of formations, 14.22 on the coefficient group not being finitely generated, and 16.20 on whether the ambient group must lie in the quasivariety whose dominions are taken. Naming the convention in the entry is enough; speculating about intent was not my business.

**Proof length.** I accept the point. A short proof, or a formalisation completed quickly by others, says nothing about how hard the question was or whether it was misread. My recommendation should be read only as asking that the reader be told which entries are short, because a bare count of 46 invites the opposite assumption.

**Problem 21.106.** I adopt the authors' formulation. The artifact record shows their proof file committed on 10 September at 21:05:23 UTC and the Lean files pushed on 11 September at 10:54:21 UTC, and artifact times are not discovery times. My report should not have said the agent was first, only that this is the recorded order. I also cannot check their statement that the repository URL is absent from the session record: no session log is public. The repository's `state/session.json` is 155 bytes and no rollout file appears anywhere in the tree.

#### 2. Where I hold the position, with one refinement

**Problem 14.72.** We agree on the mathematics and the authors have sharpened it. Their observation is the right one: the function  $x$  vanishes on the fixed locus to order two, so it cuts out twice the fixed divisor, not its reduced scheme. In divisor-class terms the class of the reduced fixed divisor is a nonzero two-torsion class, which is exactly why twice it is principal while it is not. So the entry establishes the smooth Cartier-divisor reading and leaves the globally principal reading open, and the distinction between a set-theoretic equation, a scheme-theoretic equation for twice the divisor, and a global equation for the reduced divisor is worth one sentence in the paper. The repair I suggested still looks available: deleting a suitable orbit of four-torsion fibres kills the class while keeping both the fixed divisor and the singular quotient point.

**The remaining archive gap is small and concrete.** Three of the four ancillary scripts run and pass here, and those three are the ones the README documents. The fourth, `verify_19_62_m`

onomial\_certificates.py, aborts immediately because results/19.62-g2-monomial-certificates.jsonl is not in the bundle. It is absent from the supplied directory, from the thirty entries under anc/ in the arXiv tarball, and from the twenty-eight entries under ancillary/ in the full-source zip; neither archive contains any file whose name matches "monomial-certificates". Either ship the data or drop the script.

**Problem 20.100.** Unchanged, and I read the authors as agreeing. The public repository holds 88 files for this problem totalling 7.5 megabytes. The  $n = 7$  certificate is not among them; the largest file is the  $n = 6$  certificate at 6.4 megabytes. What is public is the generator log, which records the certificate's digest, and the shard verifier logs, which record their PASS lines. The claim is therefore documented but not independently checkable, which is what the paper already says.

**Problem 15.92.** I downgrade my flag rather than withdraw it. The data is now inspectable, and I checked it. See the next section: the choice of prime is forced by the transcribed cycle structures, not arbitrary. What the paper should add is one clause saying so, because as printed the assignment of 11 or 13 to residue classes looks like an unexplained stipulation.

### 3. Second round of verification

Everything below is new since my first report.

- **The carpet certificates now check out from the supplied bundle as well as from my own re-derivation.** The three documented commands pass: rank-two certificates for the three rank-two types with 6, 20 and 78 targets; square-completion certificates with 12, 48 and 240 targets and 72, 380 and 2,739 nodes; and the integer-constant check reporting 156 rule entries for the exceptional rank-two type. Every one of those counts agrees with what I computed independently in the first round, including the 156 rule entries, which I had obtained from a Chevalley basis without seeing their table.
- **Problem 21.114 is now verified.** Working from the archive's own permutation models, GAP confirms both retained models have order 512 with derived series orders 512, 32, 2, 1, centre cyclic of order four with a unique cyclic central subgroup of that order, quotient of order 128 whose abelianisation has order 16, that quotient weakly ab-maximal, and derived length three. These were the numbers I could not check from the paper.
- **Problem 15.92 is largely resolved.** The archive's two replay scripts run here: the cycle controls finish with their sentinel over 150 rows, and the diagram reconstruction completes under GAP with exit code zero. I then checked their receipt myself: in all 150 rows the chosen prime occurs exactly once as a cycle length, no other cycle length is divisible by it, every length divides 720720, and dividing that number by the prime kills every other cycle. The prime is 11 exactly for the residue classes 7, 8, 11 and 13 exactly for 9, 10, 12. The thesis itself is in the archive with its page images, and its digest matches the receipt.
- **The ledger matches the paper.** The frozen ledger records 46 candidates, zero outside reviews, priority unestablished on every entry, and the same problems in the same order as Appendix B.
- **The handoff is internally consistent.** The two archives and the PDF match the supplied manifest and validation digests, and the PDF I reviewed is the one they validated.

### 4. On the authors' requests

The external checks are already committed. They are on branch review/kourovka-other1, commit 2b5bfe4, in review-other1/checks/ with review-other1/notes\_progress.md; this round adds review-other1/checks2/. Each log names the entry it belongs to, and the index at the end of the working notes maps logs to entries. I agree these are later external checks and not part of the deadline record, and that they do not change the historical statement that there were no outside reviews at the deadline.

### 5. Revised judgement

Unchanged in direction and slightly stronger. Accept after revision. The mathematical picture is better than at my first pass: three items I had recorded as unverifiable are now verified, one

criticism of mine was wrong and is withdrawn, and the remaining defects are editorial. The revisions I would still ask for are the convention fields in the ledger, the one missing certificate file, the clause explaining the choice of prime in 15.92, the reduced versus non-reduced distinction in 14.72, and the acknowledgement of the contemporaneous work on 21.106.

## L Reply accompanying the revision

### Reply to the follow-up report

17 September 2026

Thank you for the second round of checks and the corrections to the first report. We have revised the paper as follows.

- 1. 14.72: the suggested repair works.** Let  $X$  be the original surface and restrict to the invariant affine open  $U$  defined by  $x^2 + 1 \neq 0$ . Put  $f = y - (1+i)x$ . In its coordinate ring,  $f(y + (1+i)x) = x(x-i)^2$ .  
Since  $x-i$  is a unit,  $x$  belongs to  $(f)$ , hence so does  $y$ . The surface relation  $u = x^2 u - yv$  then gives  $u$  in  $(f)$ . Thus  $(f) = (x, y, u)$ , and the fixed scheme is the smooth reduced principal divisor with coordinate ring  $C[v]$ . The quotient is the original nodal surface localized at  $a+1$ , where  $a=x^2$ . Its singular point  $(a, v, d)=(1, 0, 0)$  survives. This gives the counterexample under the globally principal reading too. We credit your suggestion and identify this restriction as a post-review strengthening. The explanation of the original surface now distinguishes the reduced Cartier divisor  $D$  from the principal double divisor  $2D$ .
- 2. The missing ancillary input is included.** Both source bundles now contain `results/19.62-g2-monomial-certificates.jsonl` (2,194,486 bytes). Its SHA-256 is 29550181fd180ecf906395614398c8500f85d5fed950fa72efc8e0e424abbf68. The README documents all four checks, including the 2,712-request monomial check. Our first reply overstated the completeness of the bundle: the three documented checks had their inputs, but this fourth script did not. The omission is corrected.
- 3. The public repository is filtered.** The paper now gives the accessible research commit `bbfac1ac810117da37f01716d4594dc2b2e96980` separately from the original local commit `cff2c37b9bf6b737e8ad5f7ead12291a551b5201`. The owner used `--strip-blobs-bigger-than 90M`. All 4,146 retained blobs at the final research snapshot have the same object IDs; the omitted blob is the 1,271,256,410-byte 20.100 certificate. It is not supplied with this revision. The available logs document historical verification but do not enable a complete new replay. Our first reply's suggestion that publishing the repository would make this certificate available was therefore incorrect. No separate deposit is claimed.
- 4. Scope and dependencies are visible in the inventory.** We retained the 46 deadline entries and their covered parts, adding a separate scope/dependency column. The annotations identify the conventions in 18.92, 14.22 and 16.20 without attributing intentions to their proposers. They also distinguish shared imported constructions and the published 19.63 criterion. A companion table highlights 21.132, 21.42 and 20.33 under their prior-work classification. The 19.56 argument is now the fourth main example, with its imported theorems still credited.
- 5. 15.92 and 10.62/4.75.** The text now explains that the diagram's unique exceptional prime cycle determines the choice of 11 or 13, and points to the retained finite transcription and replay controls. We have made the use of Amelio's smaller initial scale and prescribed odd relation more explicit. The geometric quotient construction remains an imported dependency shared by 10.62 and 4.75.
- 6. 21.106.** We acknowledge the contemporaneous work credited to Lily Zhang and Evan Li, including the Lean formalisation. The dates quoted are commit timestamps, not verified upload times or discovery times; we assign no priority. We have not built that Lean development. The source bundle now includes a dated index of the structured web-tool observations in the supplied JSONL. It distinguishes requested page actions from returned search links and makes no inference of mathematical use. The exact repository URL has no literal occurrence in the raw input; this remains a narrow observation, not evidence excluding other exposure. The raw session is still separately retained.
- 7. Smaller points and retained conclusions.** We identify GAP 4.16.1 and CTblLib 1.3.11 for the character-table indices; explain  $CA$  implies  $CN$ ; specify the exponent-four restriction in the contrasting 15.76 example; add the foundational 1994 exponential-groups reference; and explain the one-cent rounding difference. The Rosser–Schoenfeld suggestion needs a correction:

the lower bound used is Theorem 4, formula (3.14), on printed page 70. Formula (3.16) has a different denominator and threshold. We retained Theorem 4 and added (3.14). We retained characteristic five in 16.28, without an unnecessary extension to all odd characteristics. The acknowledged source-index discrepancy in 15.65 remains explicit.

The revised paper contains an issue-by-issue change appendix followed by the original report, our first reply, the follow-up and this reply. Their wording is preserved, including superseded assertions. The lightly updated first report is retained separately with an exact diff. References within the historical correspondence still refer to the old paper's numbering.

We have pinned the reviewer artifacts in `JUrban/kour1c1` at commit `0d76dc7fc947a89a6f02e37b81da2320450efab6`, distinguishing your reported checks from executions performed during this revision. Neither the review nor these changes alter the historical statement that there were no outside reviews at the research deadline. The revision audit records the completed build, algebra and portable-bundle checks separately.

## M The parallel Fable experiment: preliminary review and comparison

### M.1 Scope and common setting

Kinyon and Urban also ran Anthropic’s Claude Fable 5.1 through Claude Code on 13–15 September 2026 [26]. The organizers confirm that this used the same machine and experimental conditions: the Kourovka Notebook, a 48-hour task, access to GAP and further software, and requested local limits of generally 20 CPU cores and 100 GB RAM. This was a separate research run, later than the Codex run of 10–12 September. Fable is distinct from the Claude Opus[1m] agent that reviewed this paper.

The comparison below was prepared by Codex on 17 September from the 29-page Fable paper and selected research reports and programs at [repository commit 290fc829d2be](#). It combines a preliminary reading of the arguments with three targeted computational checks. It is smaller in scope than the external review reported in Section 3; it does not establish priority or certify the whole Fable portfolio. Configuration and large search totals below are taken from that paper, rather than from a new audit of its complete session.

|                         | Codex experiment                                                | Fable experiment                                                                |
|-------------------------|-----------------------------------------------------------------|---------------------------------------------------------------------------------|
| Research window (UTC)   | 10 September 20:56 to 12 September 20:56                        | 13 September 08:03 to 15 September 08:03                                        |
| Agent and harness       | GPT-6 Astra; Codex; <b>xhigh</b>                                | Claude Fable 5.1; Claude Code 2.1.270; <b>xhigh</b> initially, then <b>high</b> |
| LM delegation           | One LM research agent                                           | Six sub-agent calls for Notebook triage                                         |
| Reported answer entries | 46 deadline candidates; prior-work deductions listed separately | Five answer entries, including the main question of 2.78                        |
| Further outcomes        | Nine selected partial-result questions and bounded searches     | Seven partial-result topics and a table of about 45 bounded verifications       |

The bounded-verification table overlaps the Fable partial-result list, so those counts should not be added. Nor are the two answer totals normalized scores: 13.19 and 16.60 appear among Fable’s five but outside our 46, under prior-work status. The common local budget also does not equate model inference expenditure, reasoning settings or delegation. Fable records final consolidation at 09:44 on 15 September, about 101 minutes after its deadline, and reports that it added no new mathematics then; that closeout included further verification and table generation. A strict deadline comparison would need to separate those activities in both archives.

### M.2 Reading the five answer entries

**21.32: finite derived-subgroup witnesses.** Fable gives an explicit bound for a finite witness to  $G \cong H'$ :

$$|H| \leq |\operatorname{Aut} G| |Z(G)|^2 (2 \exp Z(G))^{2 \lceil \log_2 |G| \rceil}.$$

The argument bounds the index of a central subgroup and then replaces that subgroup by its intersection with the derived subgroup. The important step is the latter replacement: the universal coefficient sequence and a Baer-sum argument preserve the isomorphism type of the derived subgroup, not merely its order. The supplied proof addresses this point explicitly and appears sound on this reading. With a multiplication-table input, the bound yields

decidability and a nondeterministic quasi-polynomial upper bound. We have not undertaken a literature search sufficient to determine priority for this result.

**2.78: finiteness of two sets of group-theoretic numbers.** Let  $f(G)$  count the subgroup orders that support a nonnormal subgroup. Fable’s construction gives

$$f(A_5 \times M_{p^n} \times C_{p^r}) = 7n + 9r + 9, \quad p \geq 7, \quad n \geq 3, \quad r \geq 0,$$

where  $M_{p^n}$  is the modular  $p$ -group in its presentation. The coprime-product formula and the subgroup argument for this family give a convincing proof of the claimed finiteness. Our GAP spot checks confirmed the family invariant in seven small parameter choices and the subgroup-order count for  $A_5$ .

The stated threshold  $k \geq 86$  can be sharpened to  $k \geq 78$  without changing the construction. Taking  $n = 3, \dots, 11$ , the base values  $7n + 9$  are

$$30, 37, 44, 51, 58, 65, 72, 79, 86.$$

They represent every residue class modulo nine. In each class the largest integer below its base value is at most  $86 - 9 = 77$ , and adding  $9r$  gives every subsequent value. Thus every  $k \geq 78$  occurs for a nonsoluble, nonsimple group. This small strengthening was noticed during the present comparison. The further question in 2.78 about composite nonsoluble numbers, and the distribution question 3.57, remain open in the Fable report. Its additional classification of the values  $0, \dots, 8$  uses deeper group theory and was not independently verified in this review.

One statement in that classification needs the hypothesis that  $G$  is nonsoluble: the assertion  $f(G) = 7$  if and only if  $G \cong A_5$  is false without it. The soluble dihedral group of order 512 has nonnormal subgroups of exactly the seven orders 2, 4, 8, 16, 32, 64, 128, as also confirmed by our check. The Fable proof already assumes nonsolubility at this step; adding it to the statement preserves the intended conclusion about absolutely simple numbers.

**18.46: a sharp square-root overgroup bound.** The dihedral group  $D_8$  of order eight needs an overgroup of order at least  $128 = 2|D_8|^2$  in which all its elements are squares. We independently programmed a search for generators  $r, s$  with  $|r| = 4$ ,  $|s| = 2$ ,  $r^s = r^{-1}$  and all eight elements of  $\langle r, s \rangle$  in the square set. Unlike Fable’s subgroup enumeration, this uses element conjugacy classes and multiplication. It excludes all 880 catalogue groups whose orders are multiples of eight below 128, reproduces exactly the 14 successful groups of order 128, and checks the witness  $D_8 \wr C_2$ . This confirms the finite calculation, conditional on GAP and the SmallGroups catalogue; the two implementations share that dependency. The diagonal embedding in the wreath product also proves the upper bound directly, since  $((g, 1)\sigma)^2 = (g, g)$ , where  $\sigma$  interchanges the two factors.

**13.19: subdirect quotients.** Fable constructs a group  $Q$  of order 512, with a normal subgroup  $H$  of order 64 and three normal subgroups  $N_i$  of order 16, such that  $\bigcap_i N_i = 1$ ,  $HN_i = Q$  and  $Q/H \cong D_8$ . We rebuilt the multiplication from its bilinear formula over  $\mathbb{F}_2$  in a separate Python program, checked the cocycle identity on all  $64^3$  triples, and verified these subgroup and quotient properties. Our example in Section 8.1 has smaller  $Q$ , of order 256, but uses four factors; Fable’s uses three. These are different size tradeoffs.

Both papers also give universal constructions for finite  $p$ -group quotients. Fable’s tagged-generator argument uses  $c + 1$  factors for a quotient of nilpotency class  $c$ , whereas ours uses  $|P|$  factors for a quotient  $P$ . That factor bound is a useful additional feature. The negative answer to the original regularity question already follows from Kearnes–Mayr–Ruškuc [64],

as explained in Section 8.1; this prior implication does not settle the novelty of Fable’s particular construction or factor bound.

**16.60: the character-degree bound.** The reduction to  $C_G(\alpha^2)$ , followed by the twisted Frobenius–Schur formula and monotonicity of the sum of character degrees on subgroups, is a short, sound argument. An affirmative answer was already stated in May 2026 by Yerrapati–Dixit–Shukla [113, Theorem 3.1]. This is a rediscovery at the level of the problem’s answer, with a proof worth retaining; see also Section 8.4.

### M.3 Overlap, differences and useful corrections

The overlapping questions show progress in both directions. For 19.56, Fable excludes minimal simple groups and leaves a Frattini lifting step; our proof in Section 5.4 handles the general finite-group question. For 20.21, Fable reports the stronger lower bound  $|G| \geq 6144$ , compared with our deadline bound  $|G| \geq 3072$ . We read its structural reductions but did not rerun the large catalogue exclusions behind the stronger bound. For 20.100, its finite search covers  $n = 4$  in 85 groups of order at most 300 coprime to 30; Section 7.1 gives the torsion-free argument, the finite abelian reduction and certificates for  $n = 4, 5, 6, 7$ , with the previously stated public-artifact limitation for  $n = 7$ . Fable also develops substantial partial work on 16.14 and 20.37 that is not part of our selected mathematical portfolio.

Two small corrections would improve the Fable exposition. Its Appendix B definition of  $L(G)$  for 20.37 should include  $|A||B| = |G|$ ; without that condition, taking  $B = G$  makes the definition trivial. In the proof for 21.111, the normalizer of a cyclic subgroup generated by a unipotent element is *contained in* the indicated Borel subgroup, generally not equal to it. The containment suffices for the subsequent involution argument. These are local corrections, rather than objections to the stated results.

The two archives illustrate complementary research choices under the same local constraints: Fable emphasizes finite computations and constructions, while Codex’s larger candidate list includes more structural arguments and reuse of literature. The Fable results on 21.32, 2.78 and 18.46 deserve further review on their own merits. A useful comparison should match exact problem parts, prior status and verification scope, rather than interpret the headline totals as a model ranking. Neither the shared organizers nor this later cross-reading establishes a blind experimental comparison.

**Version 2 attribution update.** The eleven overlaps with further contemporary papers examined in Appendix N concern our deadline portfolio. They are additional to the Fable comparison above. The figure 46 in this appendix retains its historical meaning; it is not a count of new solutions to compare with Fable’s five answer entries. The checks and assessments above remain those of the 17 September comparison.

The pinned source manifest, new check programs and their outputs accompany this paper in

`paper/reviews/parallel-fable-2026-09-17/`.

The large Fable searches are available through its referenced repository.

## N Contemporary solutions: comparison and attribution

### N.1 Scope and sources

This comparison was prepared by Codex on 19 September 2026 following Evgeny Khukhro’s communication of 18 September. Nine overlaps were named in the message and two more, 16.9 and 17.101, appeared in its appended index. Its 17.34 entry is already treated under prior-work status in Section 8.2 and is outside the 46 candidates.

We read the eleven papers at the supplied Zenodo DOIs, all attributed there to Achyuth Jayadevan, and compared them with the arguments in this paper. For 21.68 we also read Aluna Rizzoli’s separate note in the Notebook repository. The two papers must not be conflated. We retain the earlier acknowledgment of Zhang and Li for 21.106. Source URLs, PDF and source-archive hashes, repository metadata, individual review notes and new checking programs are recorded in [paper/reviews/v2-2026-09-19/](#).

The assessment below concerns mathematical statements and arguments. No error was found in the alternatives on this reading, with the imported theorems and character data retaining their stated roles. Our new executions cover the specific algebraic identities and finite character inputs described below. The papers report Lean developments; we downloaded the eleven accompanying source archives but did not build them. This comparison is therefore not a kernel verification or an audit of those projects’ full theorem statements and dependencies. Khukhro reports specialist confirmation of alternative solutions for 21.68, 16.28(a) and 15.89; that report is distinct from our reading.

### N.2 The eleven entries

| Problem  | Relationship to the argument in this paper                                                                                                                                                                                                                                     |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4.55     | Jayadevan’s different example uses $\mathbb{Z}_{(11)}[\mathrm{PSL}_2(\mathbb{F}_{11})]$ , with total rank 88 and indecomposable rank-44 summands. Our rank is 55,440 over $\mathbb{Z}_{(5)}[3.A_7]$ . Both use completion and rational-character obstructions to descent [59]. |
| 14.22    | A different Laurent-polynomial construction gives the radical obstruction over a two-generated torsion-free polycyclic subgroup of $\mathrm{SL}_3(\mathbb{Z})$ . Our coefficient groups are infinitely generated; the alternative removes that feature [55].                   |
| 15.89    | The same Cayley graph and inverse formula occur, with the stronger point-spectrum assertion $\mathbb{C} \setminus \{0\}$ on all functions. Two-dimensional representations supply the additional eigenfunctions [61].                                                          |
| 16.9     | The same palindrome-to-reflection reduction, matching recurrence and cubic operation bound occur. Reconstruction is presented differently. Qualitative computability and the recurrence also have older sources, discussed below [60].                                         |
| 16.28(a) | The alternative uses a triangular $\mathrm{GL}_3$ construction containing the identity. Our trace-one $\mathrm{SL}_2$ class and central marker give a different closed set. Both use an algebraically closed field of characteristic five with a transcendental element [58].  |
| 17.101   | The HNN-induced-module construction, forest injectivity argument and cardinal bound agree. Adding regular modules during the iteration also gives a faithful extension over a nonzero coefficient ring [56].                                                                   |
| 18.76    | The free-factor derivations and commuting-lift obstruction agree. The characteristic-zero constructions are isomorphic by the explicit change below, which inverts the kernel. Our argument also permits positive characteristic [52].                                         |
| 21.40    | The bounded-root argument, trace radical and index bound $(2n+1)^d$ , $d = \dim_{\mathbb{Q}} \mathrm{span}_{\mathbb{Q}} G$ , agree. The alternative separately states the finite-trace-set theorem and a number-field corollary [54].                                          |
| 21.68    | Rizzoli and Jayadevan give the same order-2,592 extension of Kida’s configuration, the same degree-eight representation, and the same index-two obstruction, in separate accounts [92, 53].                                                                                    |

| Problem   | Relationship to the argument in this paper                                                                                                                                                                                                                                           |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 21.106    | Jayadevan gives a different positive formula defining the same two central generators in the integral Heisenberg group; over a commutative ring it defines central units. Zhang–Li attribution remains separate [51, 115].                                                           |
| 21.121(a) | A different cyclotomic construction gives every prime $p$ , a locally finite metabelian subgroup of $\mathrm{GL}_2(\overline{\mathbb{F}_p})$ , and exponent set $(1, \infty)$ . Our quaternion/code construction uses a free product, $p = 2$ , and infimum $\log 24 / \log 8$ [57]. |

### N.3 Mathematical checks and prior ingredients

**The smaller projective example, 4.55.** We reconstructed  $\mathrm{PSL}_2(\mathbb{F}_{11})$  and the displayed subgroups of orders 12 and 60 from matrices. Induction and removal of the Steinberg constituent give the three projective characters in the alternative paper. The fourth follows by subtracting the chosen completed summand. The ranks are 44, and the involution traces 0,  $-4$ ,  $4$  distinguish the required modules. GAP’s character-table library agrees with the displayed decomposition data and the irrational order-five traces of the individual completed projectives. The descent and indecomposability steps were checked by reading their proofs; the computation does not replace them.

**An explicit comparison of extensions, 18.76.** Identify the two division closures and write  $g = (x, y)$ ,  $h = (x', y')$ . Let  $\alpha, \beta$  be the derivations on the first and second free factors. The cocycles in our coordinates and the alternative coordinates are respectively

$$C_o(g, h) = x\beta(y)\alpha(x'), \quad C_a(g, h) = \alpha(x)y\beta(y').$$

For  $f(g) = \alpha(x)\beta(y)$ , expansion of the derivation laws gives

$$(\delta f)(g, h) = gf(h) - f(gh) + f(g) = -C_o(g, h) - C_a(g, h).$$

Consequently  $(r, g) \mapsto (-r + f(g), g)$  is an isomorphism from the first extension to the second, inducing the identity on the quotient and negation on the additive kernel. Our exact polynomial check allows commutation between the two factor algebras only. Thus the difference between the displayed factor sets does not give a different abstract group in characteristic zero.

**Additional formulas and older algorithms.** For 15.89 we checked the characteristic polynomial  $\lambda^2 - 3s\lambda - 4$  and the eigenvector used for every  $\lambda \neq 0$ . For 21.106 we checked the determinant identity that forces the central coordinate to be a unit; the explicit converse witnesses respect the formula’s quantifier order.

For 16.9, Jayadevan’s source discussion led us to check Dahmani–Guirardel[30, Theorem 4] directly. Their twisted-equation decision procedure gives qualitative computability of palindromic length, by testing products of at most  $|w|$  palindromes. The interval recurrence specializes the cancellation-norm algorithm of Brandenbursky–Gal–Kędra–Marcinkowski[15, Section 2.I]; their 2023 erratum establishes word independence in the Coxeter case. The explicit reflection-length reduction and its application to optimal palindrome factors are shared with the contemporary paper. Neither qualitative computability nor the recurrence should be counted as a new discovery of this experiment.

Table 7: Artifact chronology in September 2026 (UTC). The first column of dates is the first local commit containing the proof file, not a verified discovery or upload time. DOI registration is taken from DataCite.

| Problem   | Local proof-file commit | Zenodo date | DOI registered |
|-----------|-------------------------|-------------|----------------|
| 4.55      | 11 Sep 08:37            | 14 Sep      | 14 Sep 14:08   |
| 14.22     | 11 Sep 00:48            | 12 Sep      | 12 Sep 10:15   |
| 15.89     | 11 Sep 11:52            | 12 Sep      | 12 Sep 10:18   |
| 16.9      | 11 Sep 12:13            | 12 Sep      | 12 Sep 21:08   |
| 16.28(a)  | 11 Sep 00:04            | 13 Sep      | 12 Sep 19:39   |
| 17.101    | 11 Sep 02:22            | 12 Sep      | 14 Sep 16:06   |
| 18.76     | 11 Sep 02:59            | 14 Sep      | 14 Sep 14:15   |
| 21.40     | 11 Sep 07:53            | 14 Sep      | 14 Sep 14:43   |
| 21.68     | 10 Sep 23:00            | 14 Sep      | 14 Sep 14:17   |
| 21.106    | 10 Sep 21:05            | 12 Sep      | 12 Sep 21:09   |
| 21.121(a) | 10 Sep 22:03            | 14 Sep      | 14 Sep 14:17   |

#### N.4 Chronology and what it establishes

The local proof-file commits precede the corresponding DOI registrations. This comparison concerns those artifacts only. In particular, the Zenodo publication-date field can differ from record creation and DOI registration: the 17.101 record lists 12 September, while both technical timestamps are on 14 September. The 16.28(a) record lists 13 September but was created and registered on 12 September. Rizzoli’s note is dated 11 September; we have not established its first public upload time. The earlier Zhang–Li artifact dates are discussed in Section 5.1.

These dates do not exclude an earlier draft or discovery, nor do they establish which sources another AI run could access. The eleven DOI records expose first-version metadata at retrieval, but we cannot infer the entire history of the underlying work. The research-repository URL listed in those records returned HTTP 404 during this comparison, so we could not inspect its development history.

Our retained structured retrieval index contains no matching URLs for the eleven supplied DOIs or the compared paper filenames. The exact patterns and input hash accompany the review. That index omits some forms of access, including shell downloads, and says nothing conclusive about training exposure or another run’s sources. It is a limited record, not proof of discovery independence. We accordingly assign no priority direction and make no copying inference from the similarities.

The historical total remains 46 deadline candidates. The eleven overlaps are not a uniform category of prior solutions, and the other 35 entries are not thereby established as new. The original mathematical expositions remain available in version 2, with the individual attribution notes above; the pre-v2 PDF and source baseline are preserved separately.